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Yoshino

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(54) **OPTICAL COMMUNICATION APPARATUS,
OPTICAL COMMUNICATION SYSTEM AND
OPTICAL COMMUNICATION METHOD**

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patent is extended or adjusted under 35
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H04B 10/032 (2013.01)

(52) **U.S. Cl.**
CPC **H04Q 11/0067** (2013.01); **H04B 10/032**
(2013.01)

(58) **Field of Classification Search**
CPC H04B 10/032; H04Q 11/0067

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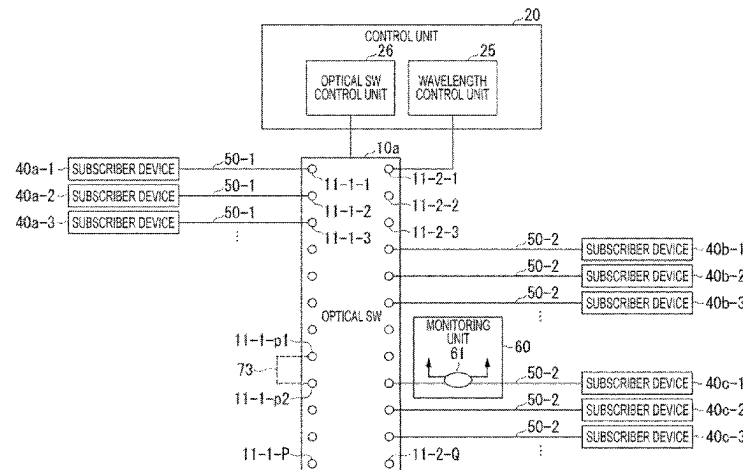
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Pierce, P.L.C.

(57) **ABSTRACT**

An optical communication device include: an optical switch
configured to output an optical signal input from one trans-
mission line to another transmission line among the plurality
of transmission lines; a wavelength management control unit
configured to allocate a wavelength, a time, a polarized
wave, a mode, a code, a frequency, a core, a core wire, a
wavelength that is a combination thereof, or the like to a
subscriber device by using a control signal in accordance
with the subscriber device and a transfer destination on a
path to a communication destination, and transmits the
control signal to the subscriber device by using a carrier
different from a carrier to which the wavelength or the like
is allocated; and an optical switch control unit configured to
control the optical switch such that the input optical signal
is output to the another transmission line corresponding to
the communication destination.

15 Claims, 85 Drawing Sheets



(58) **Field of Classification Search**

USPC 398/66
See application file for complete search history.

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Fig. 1

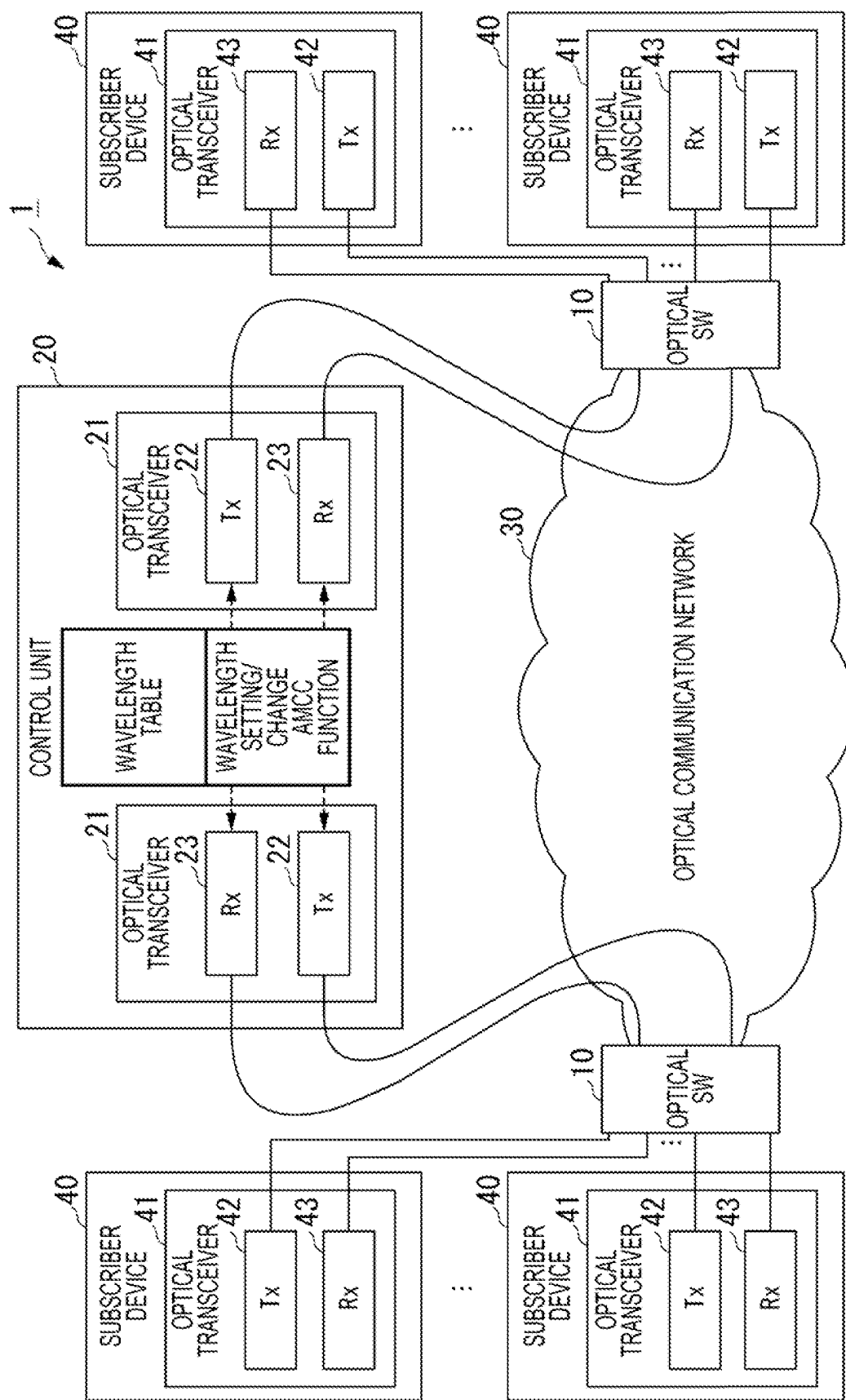


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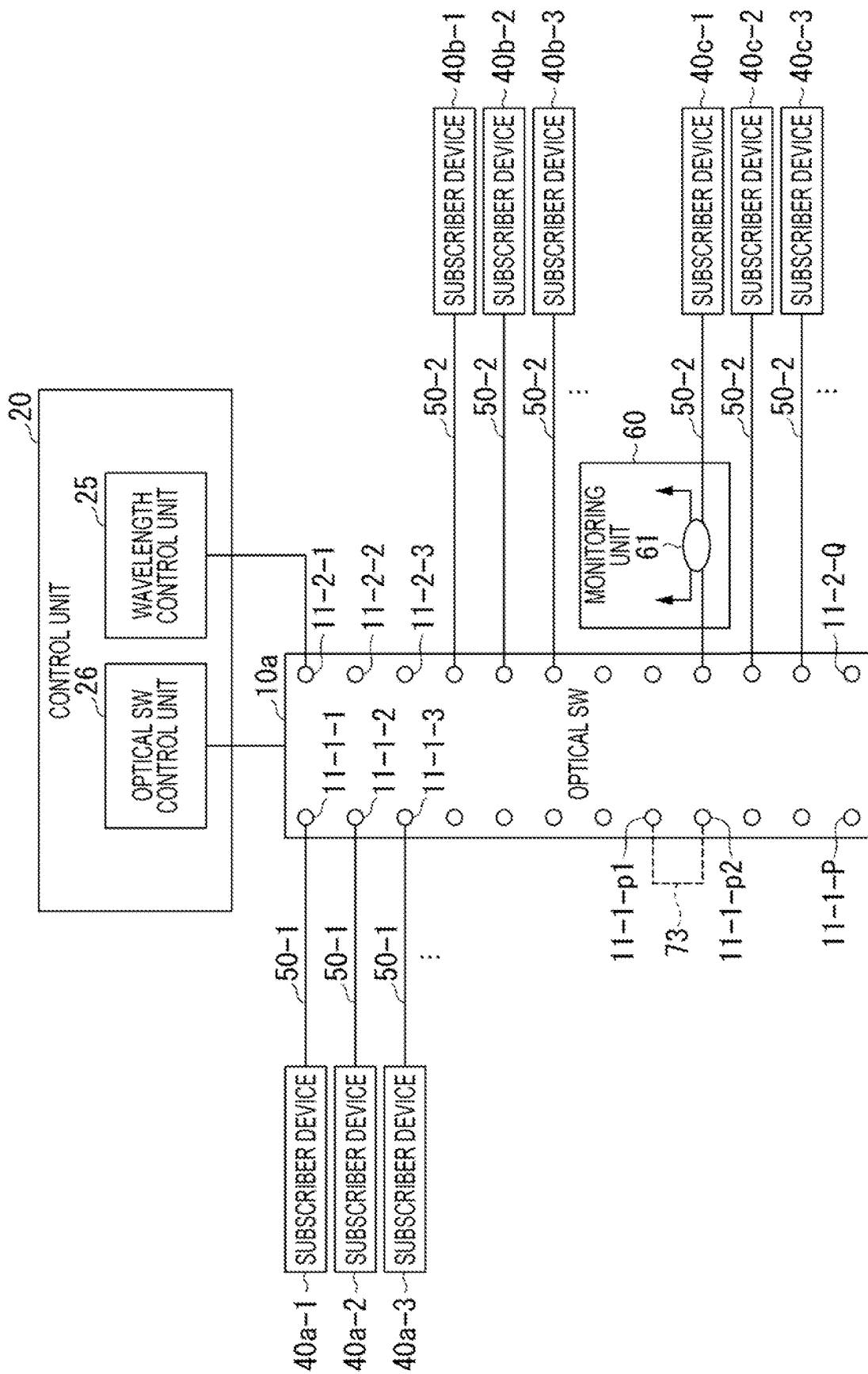


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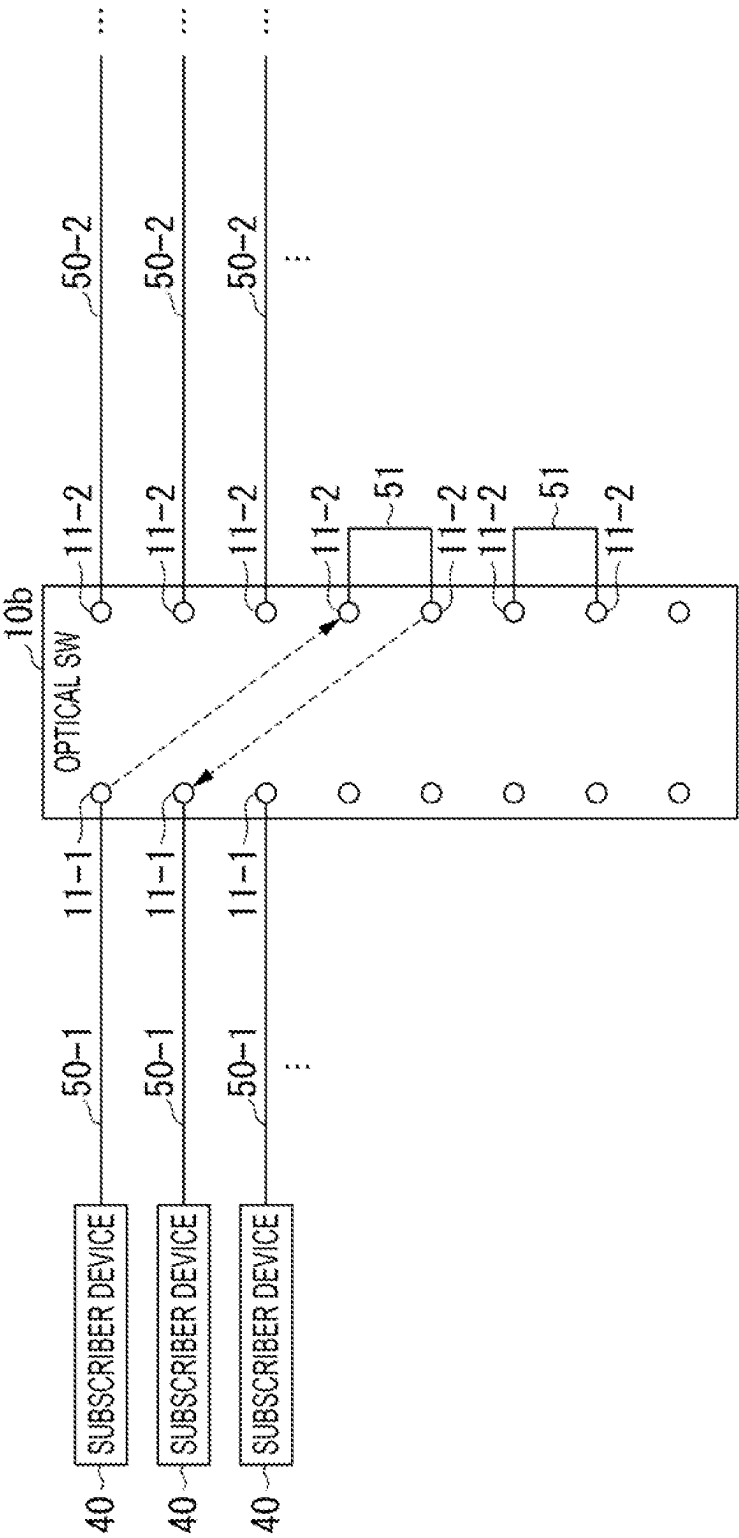


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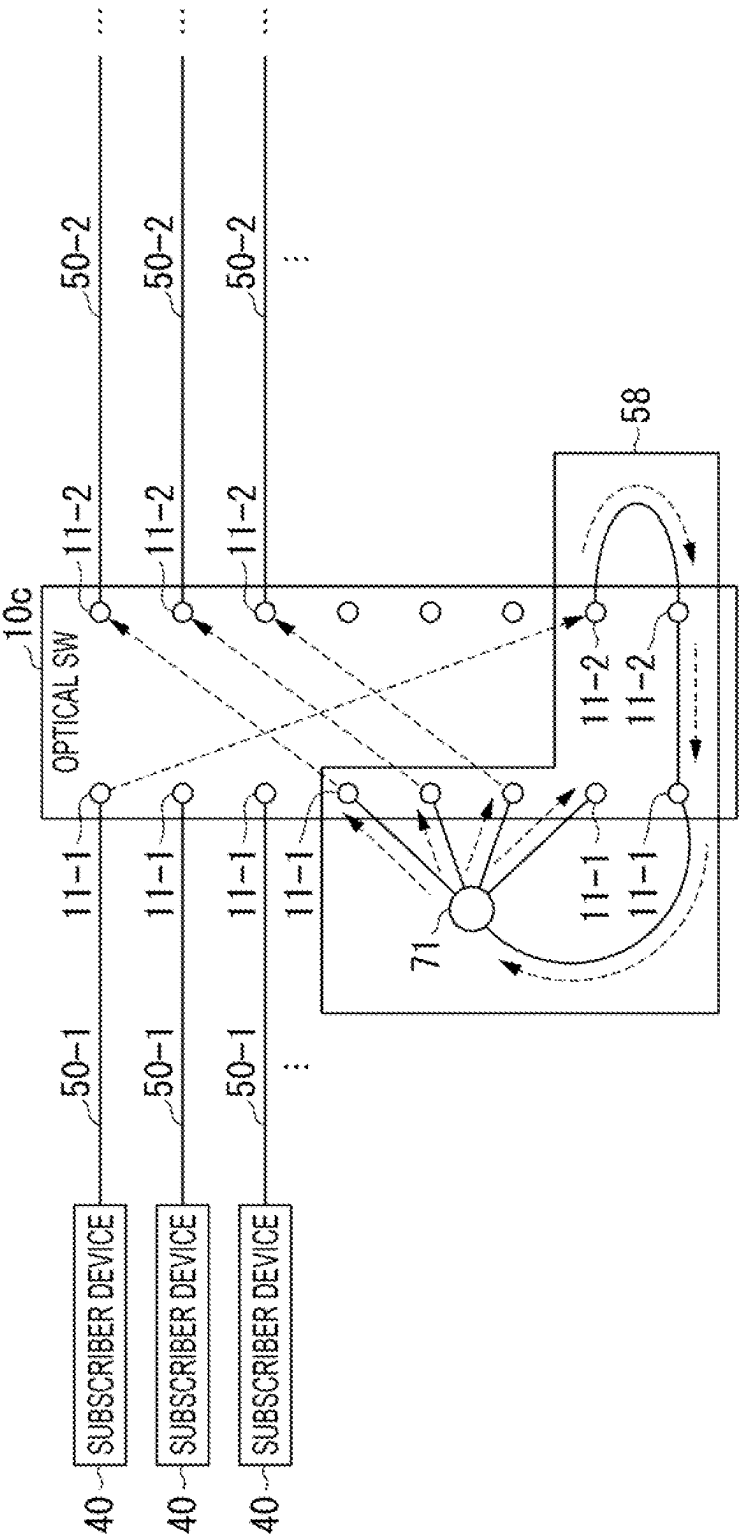


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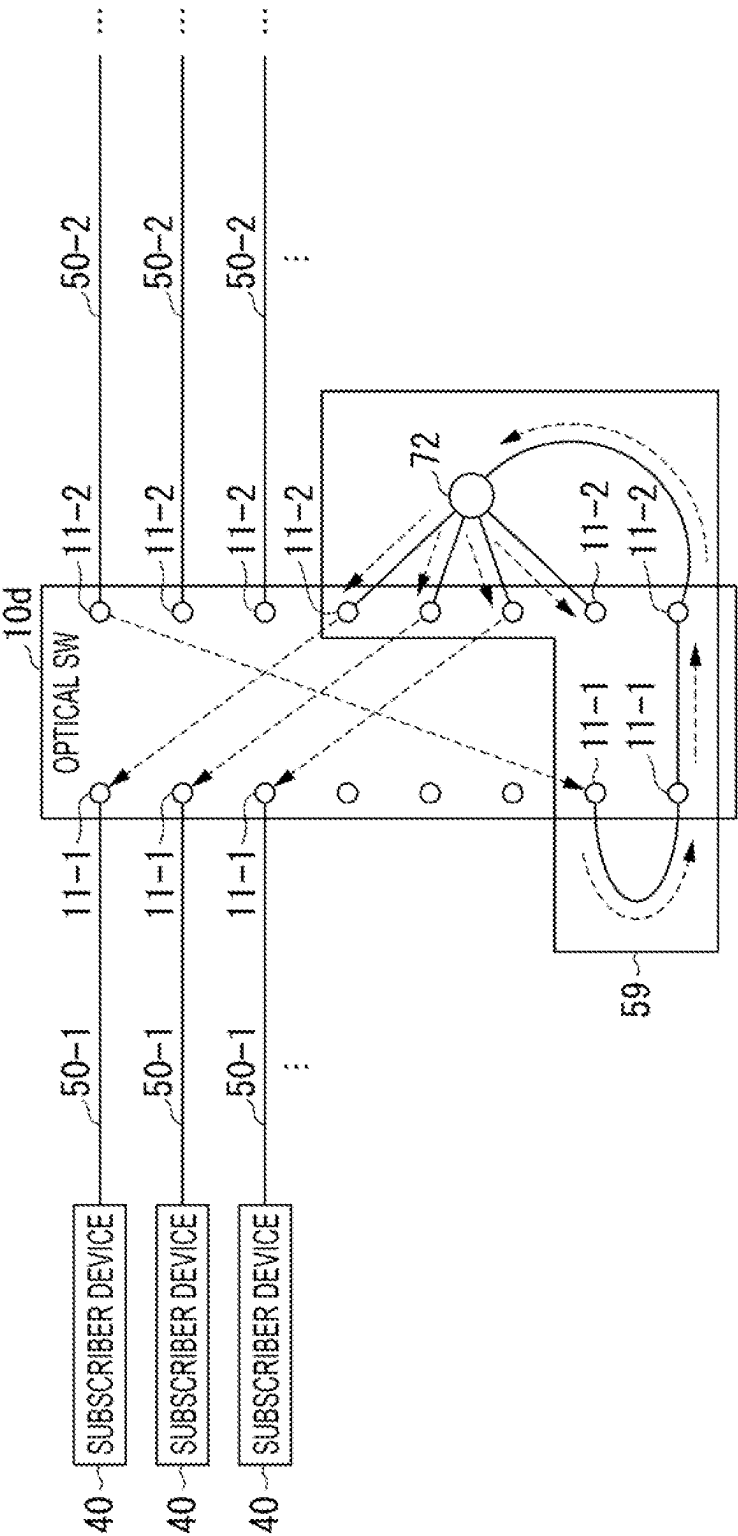


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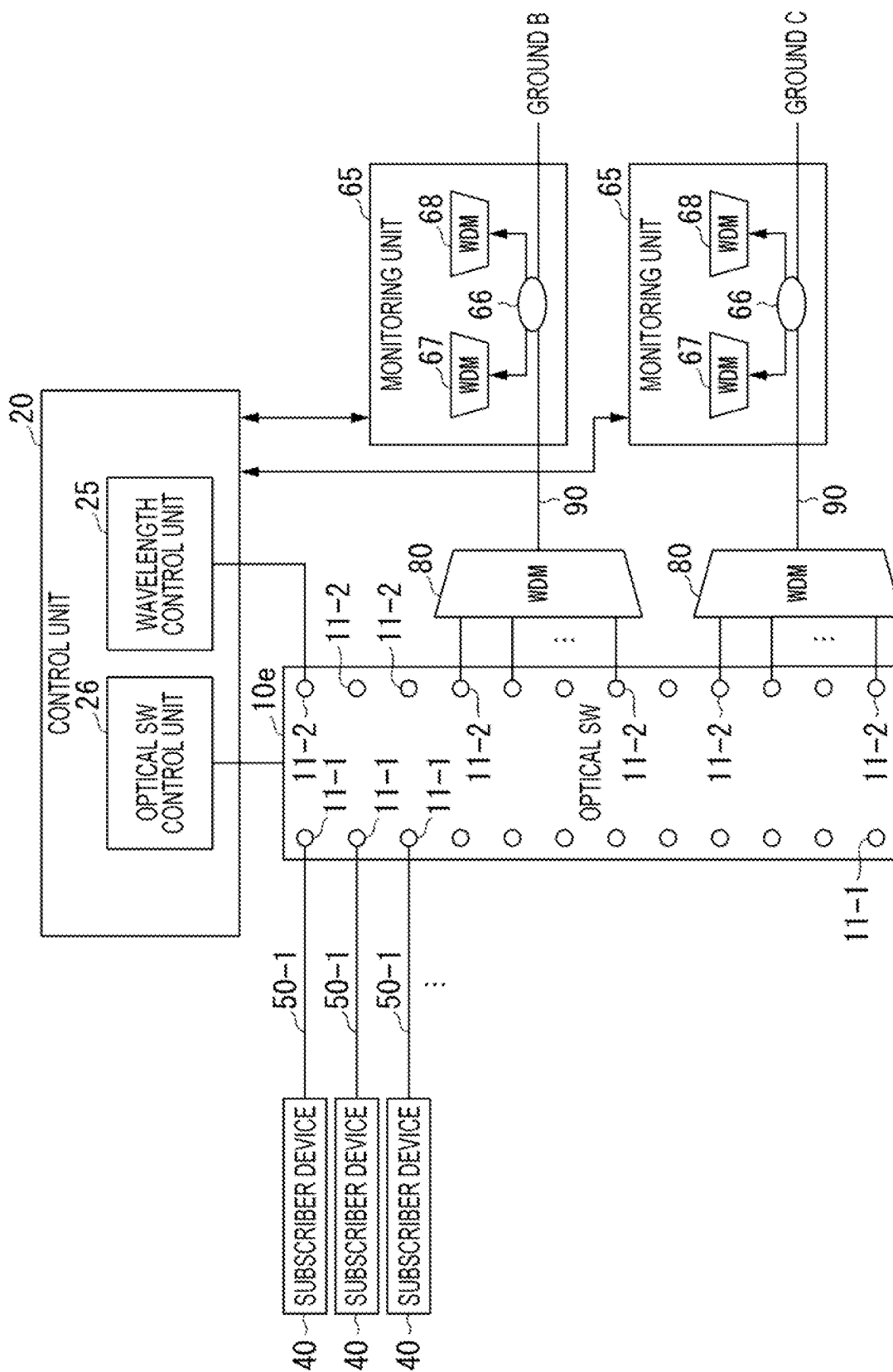


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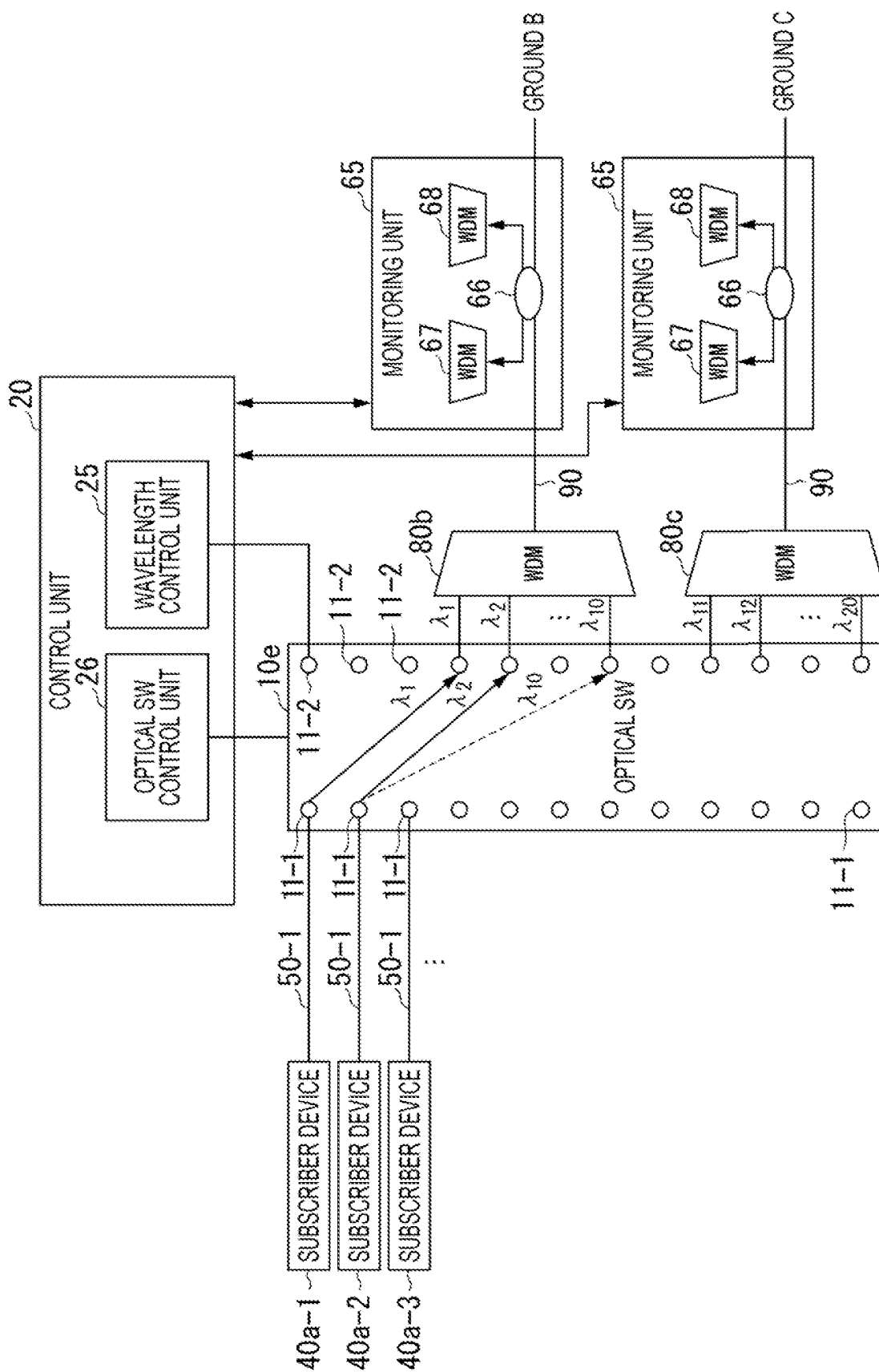


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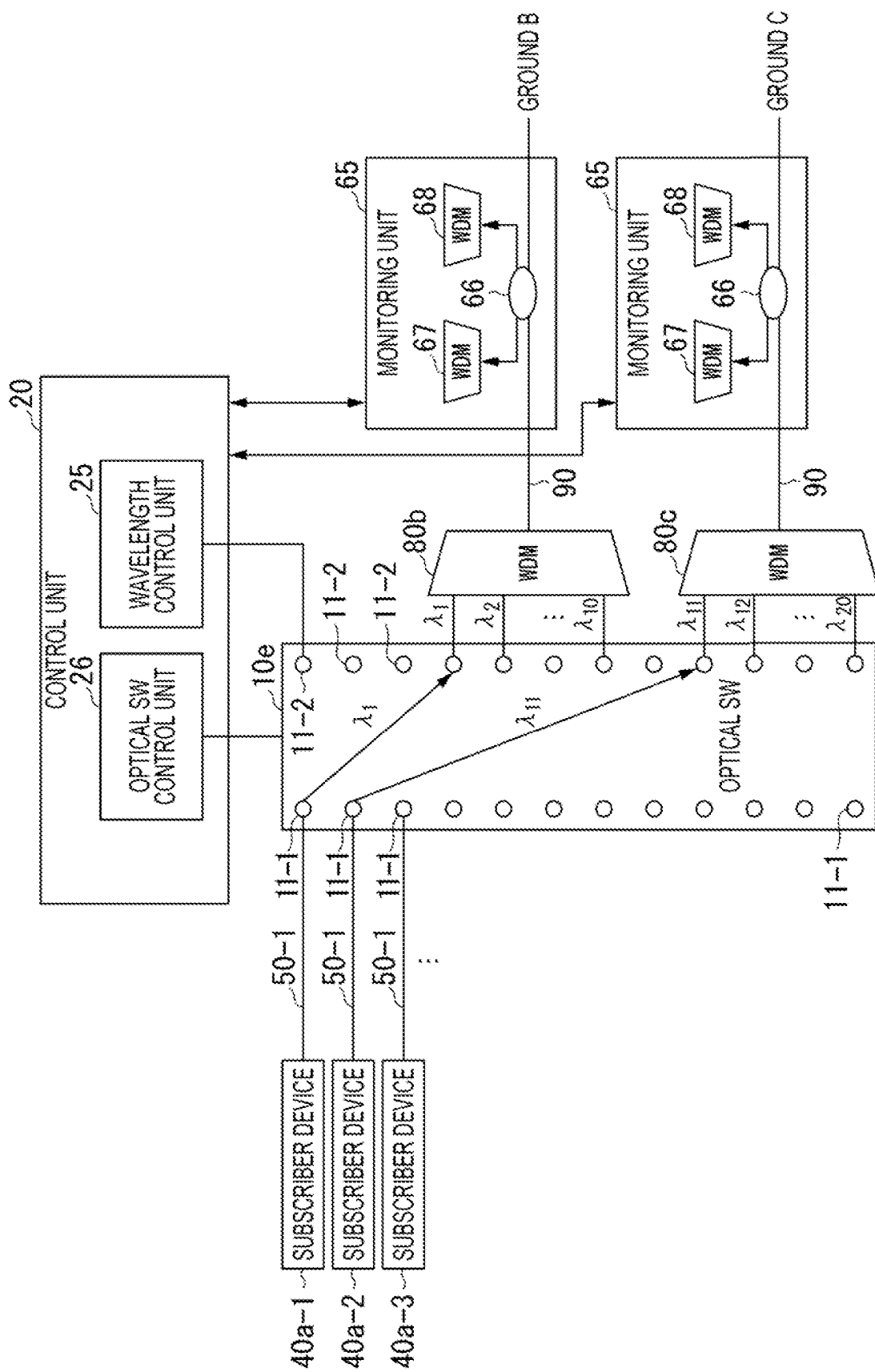


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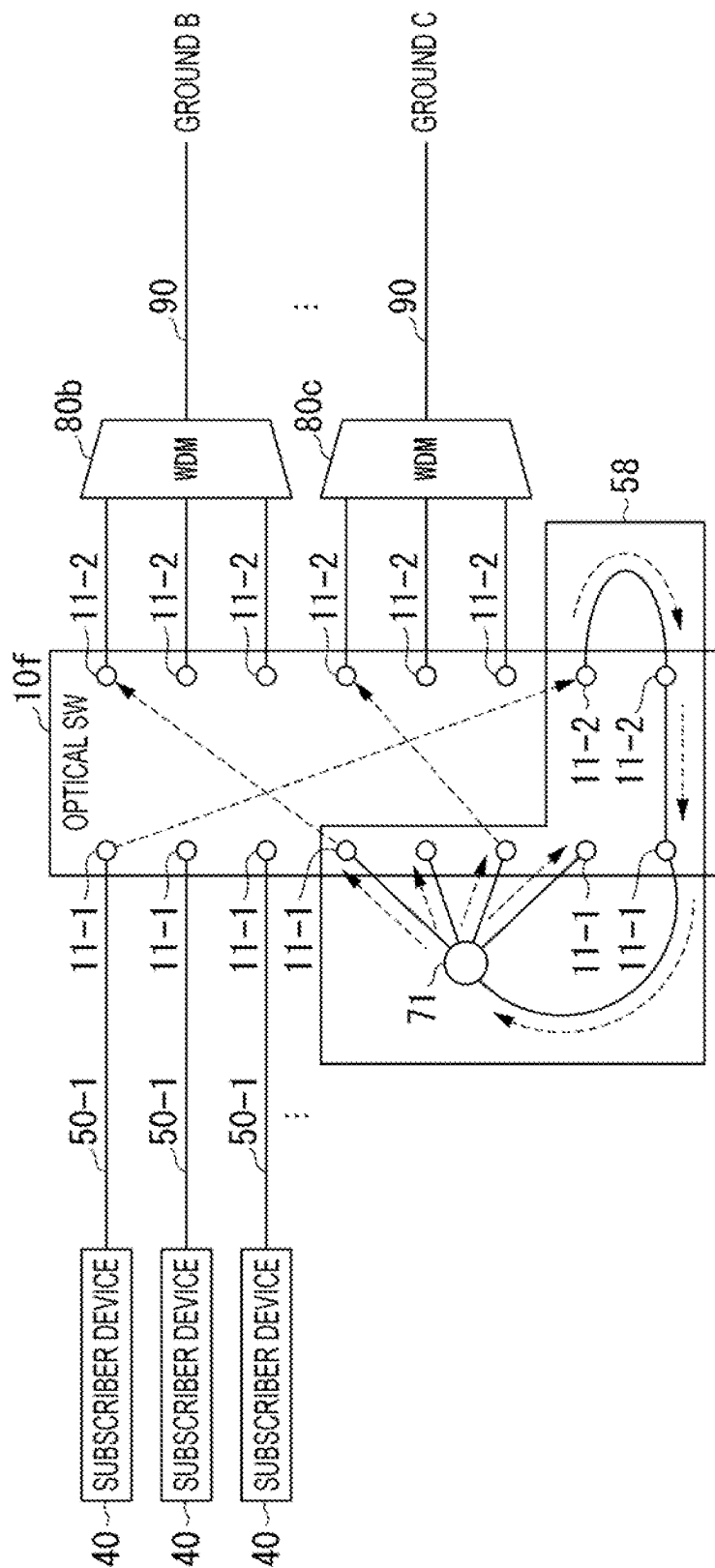


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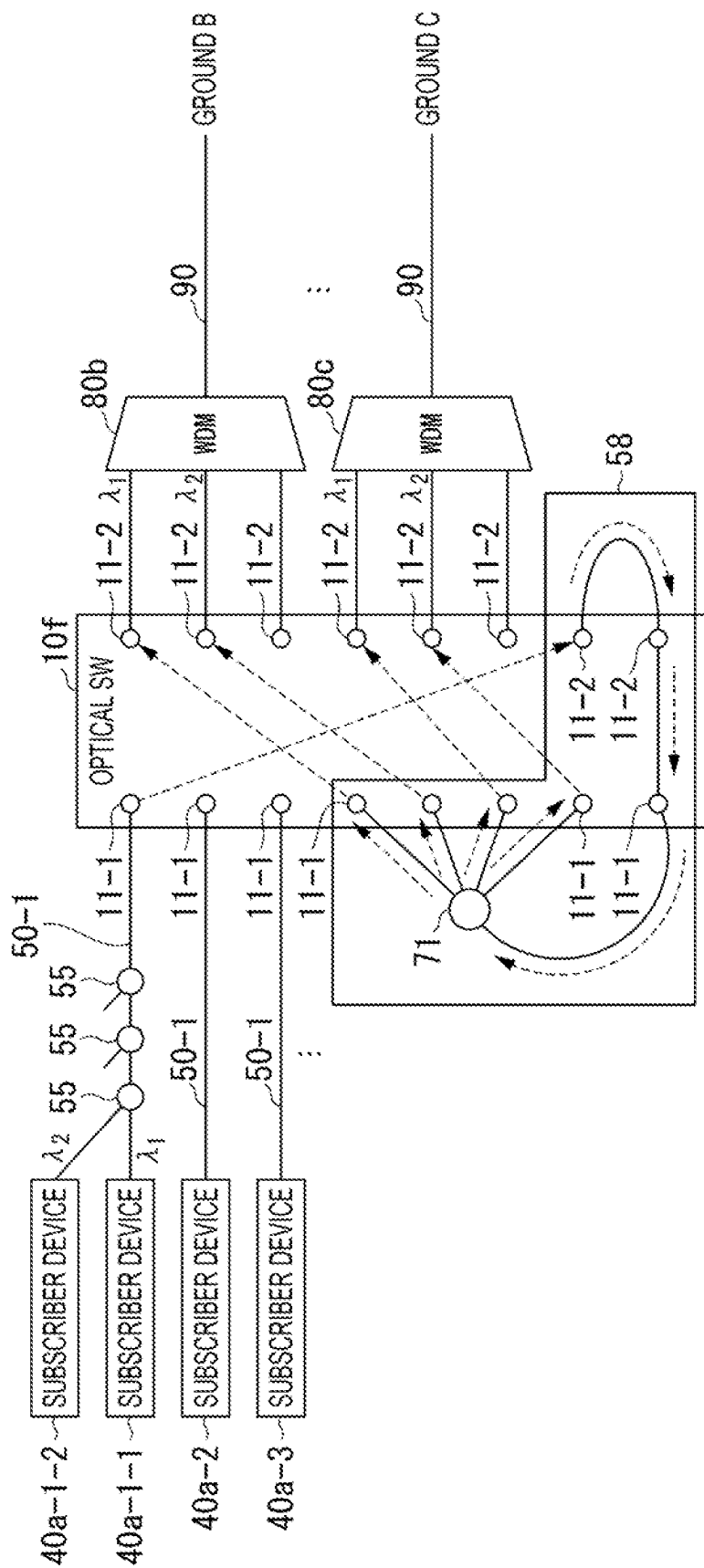


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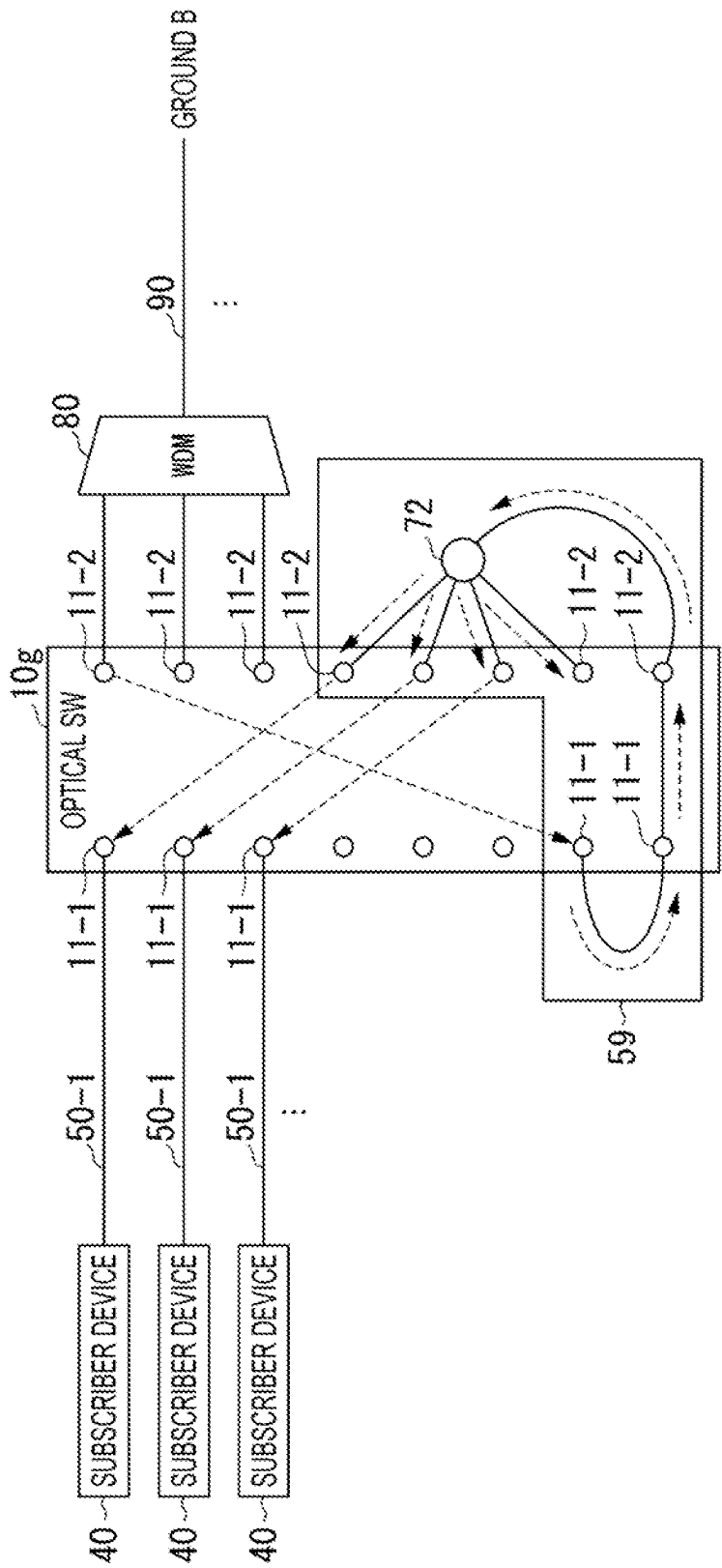


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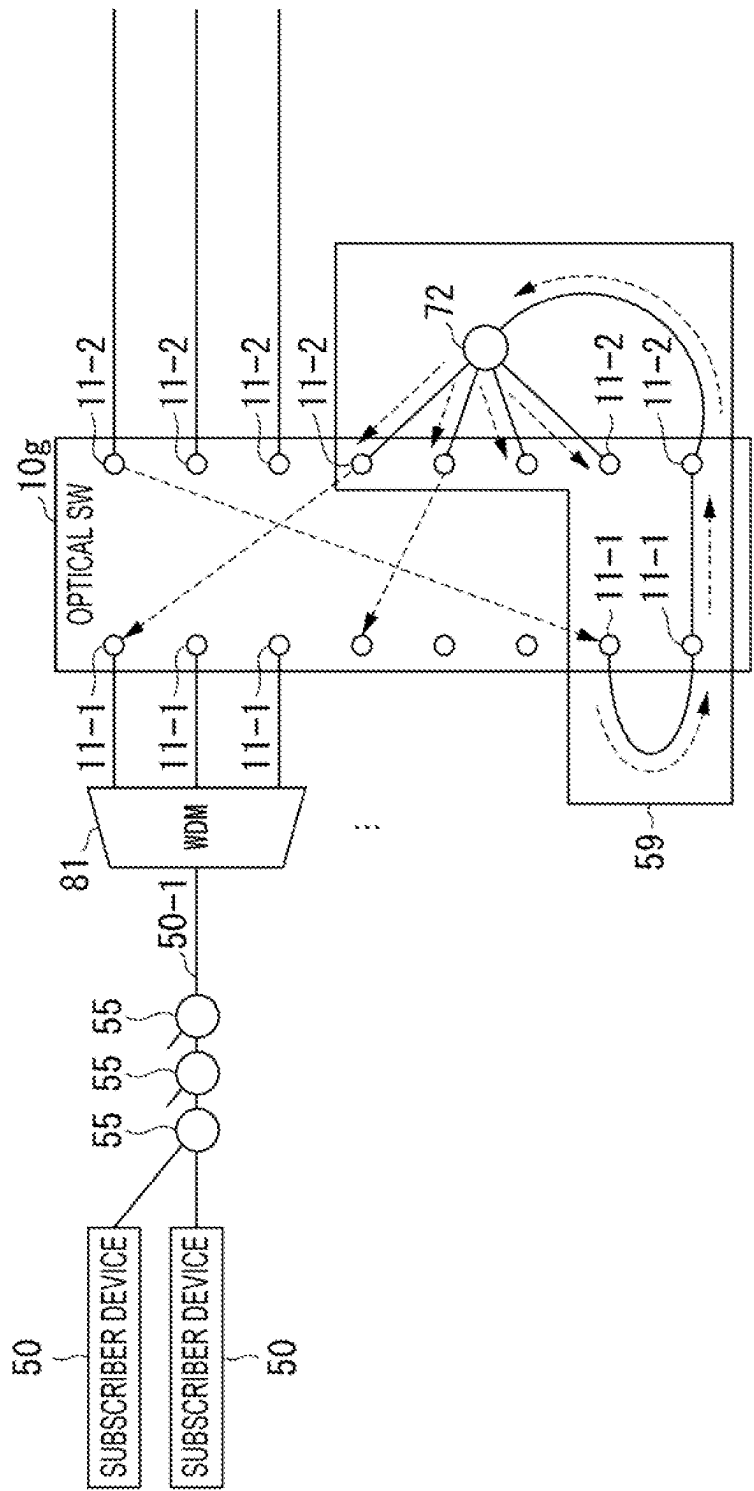


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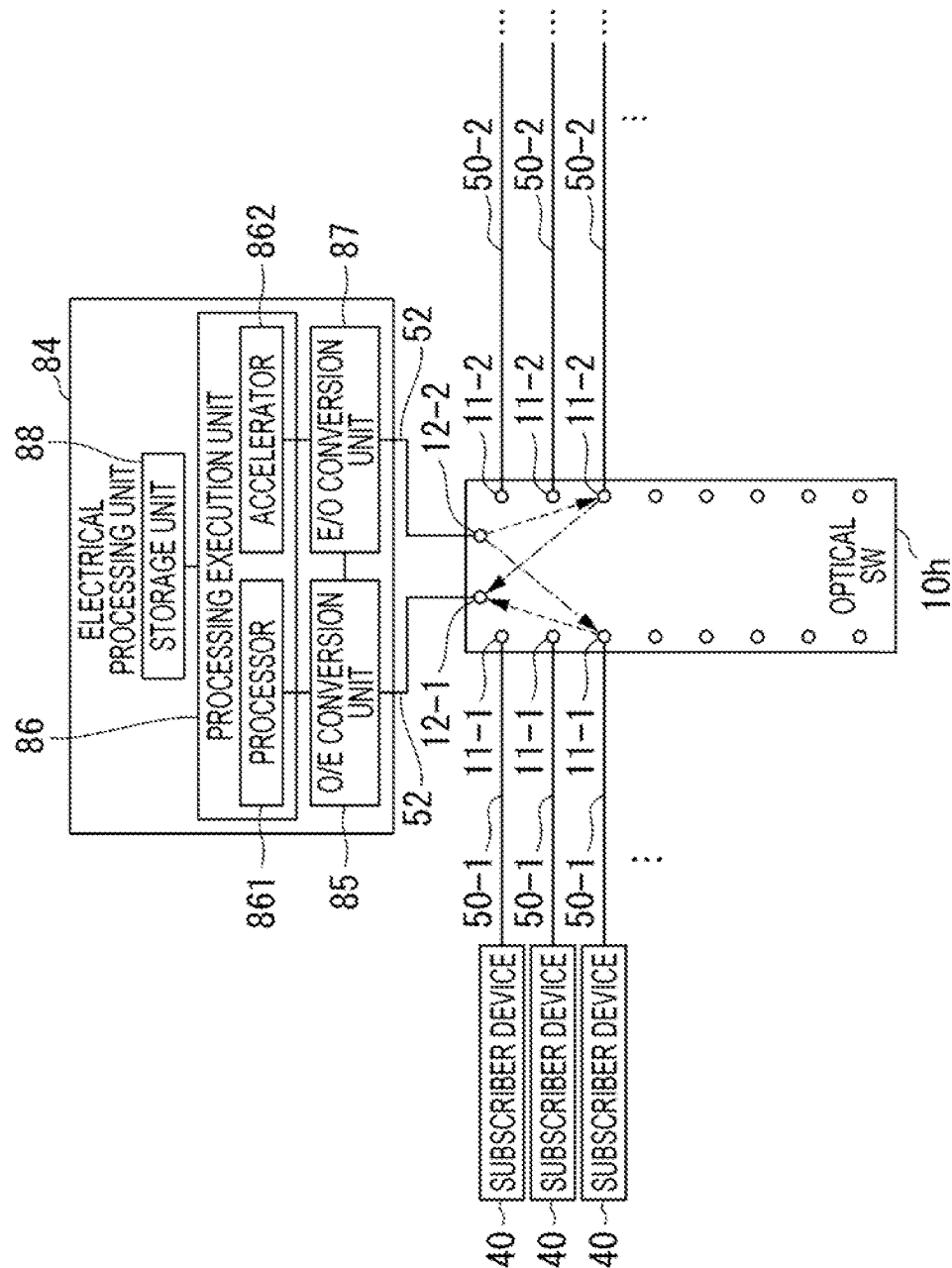


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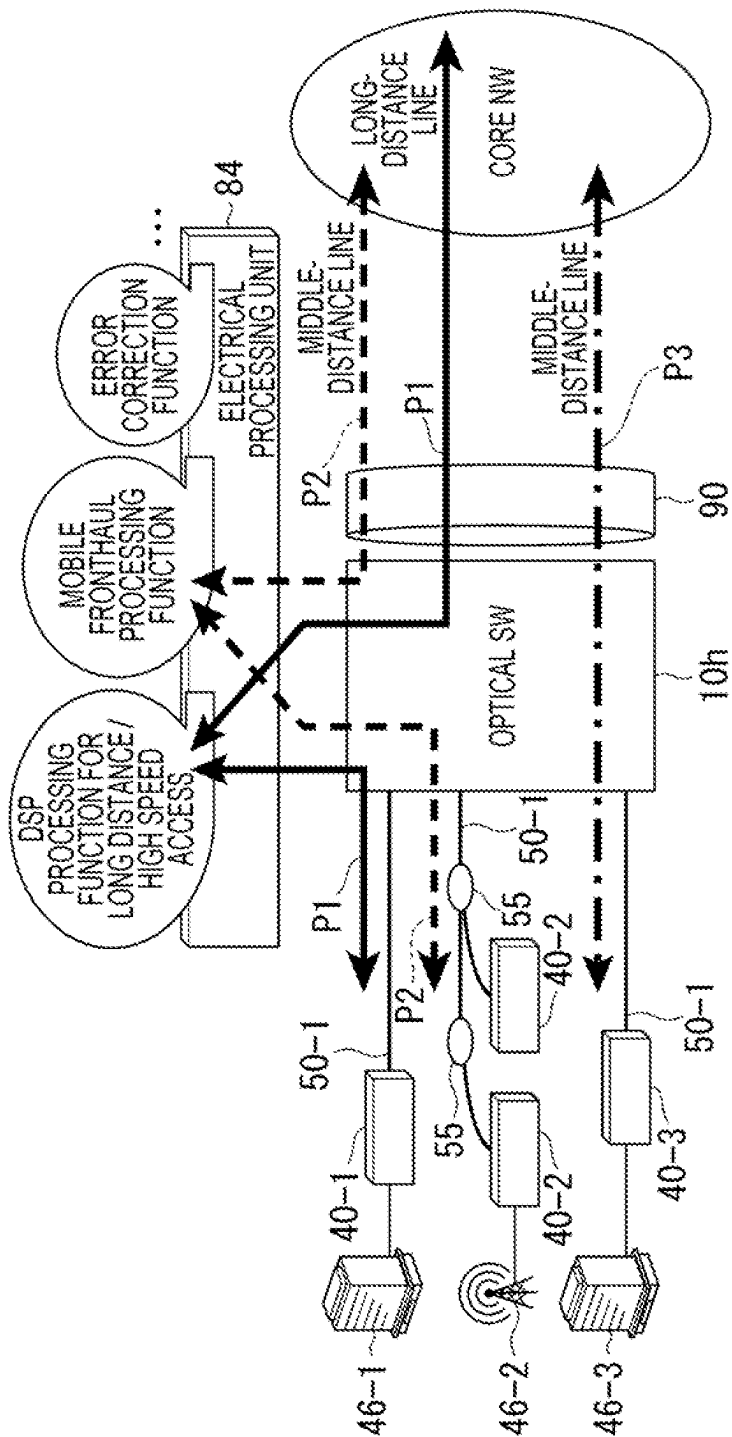


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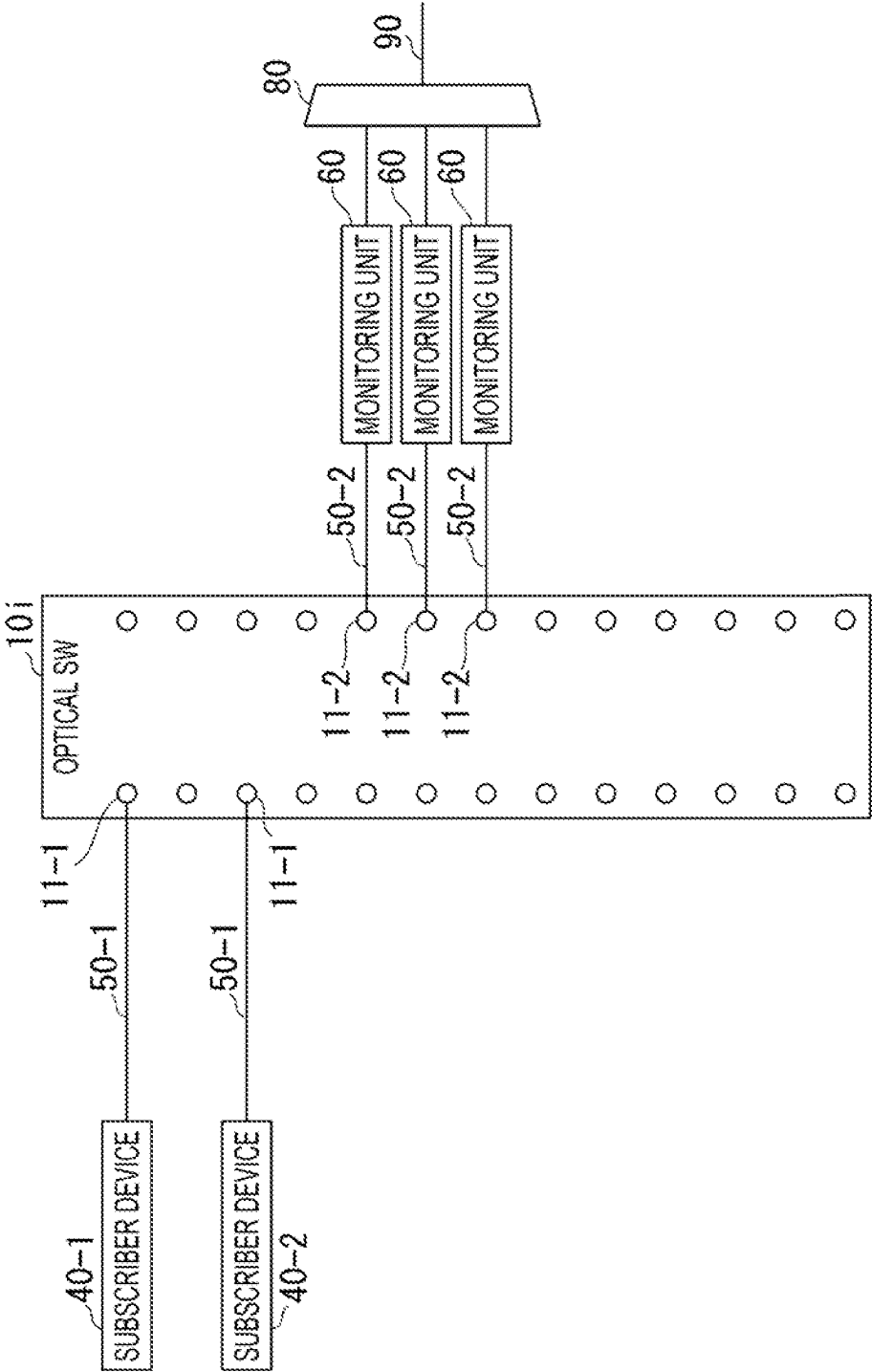


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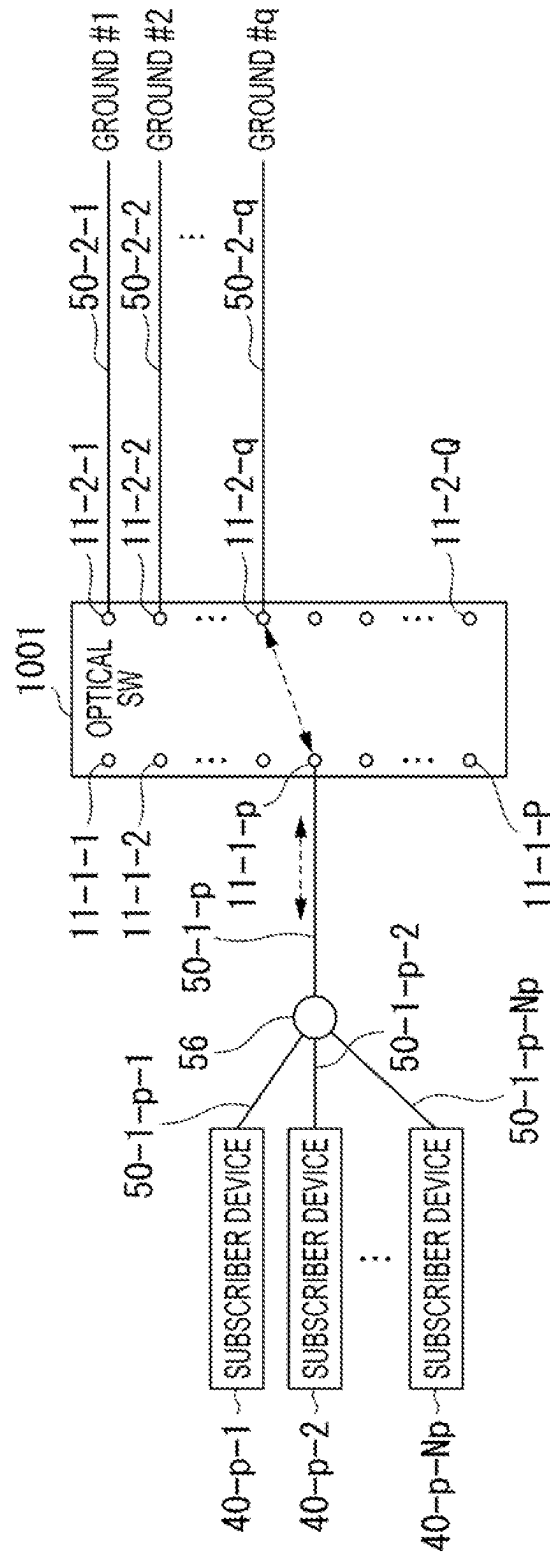


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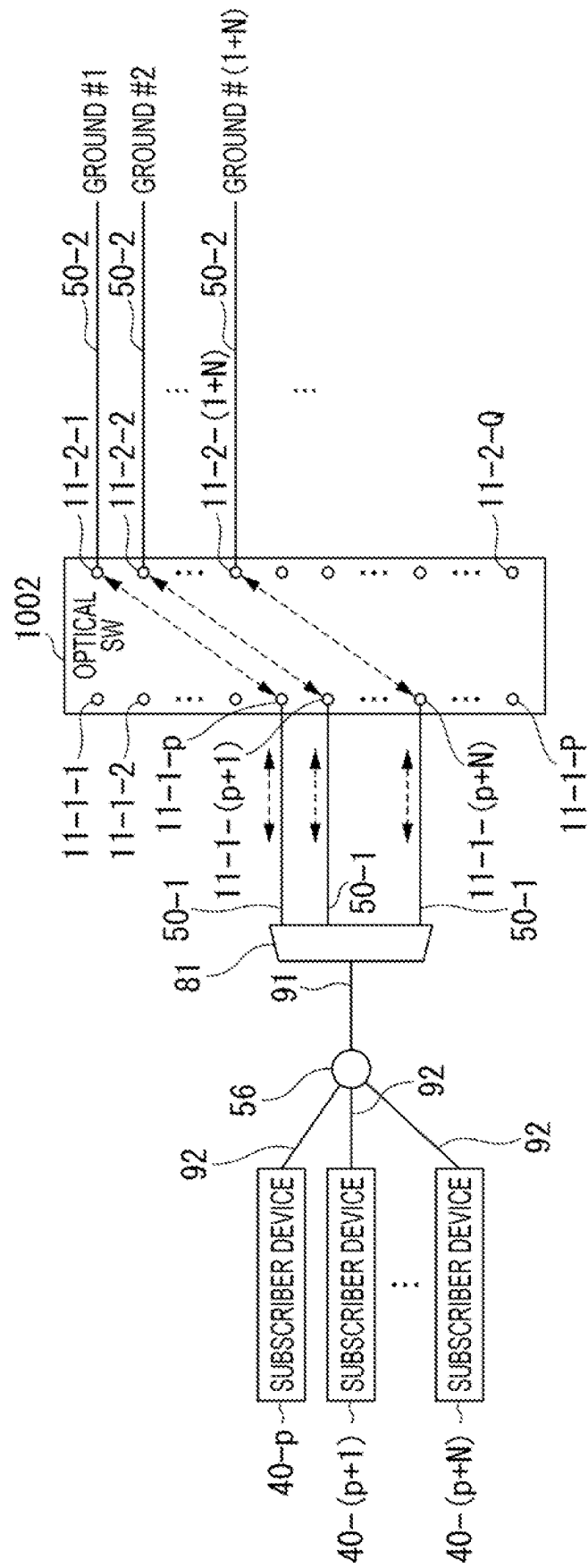


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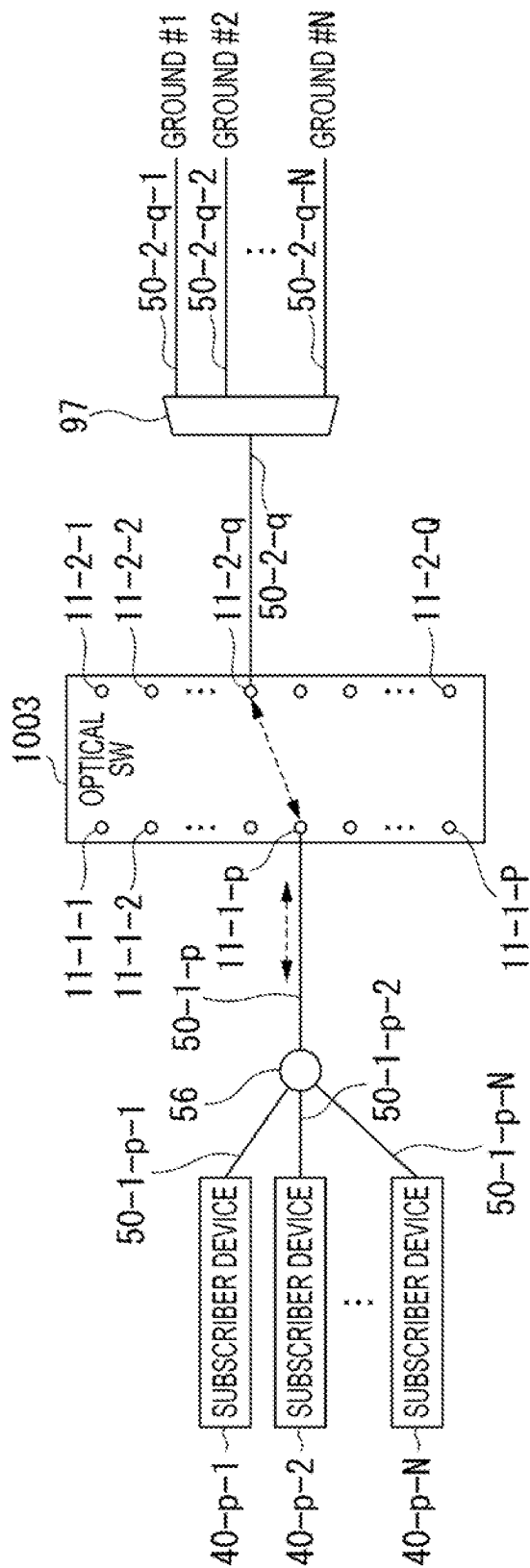


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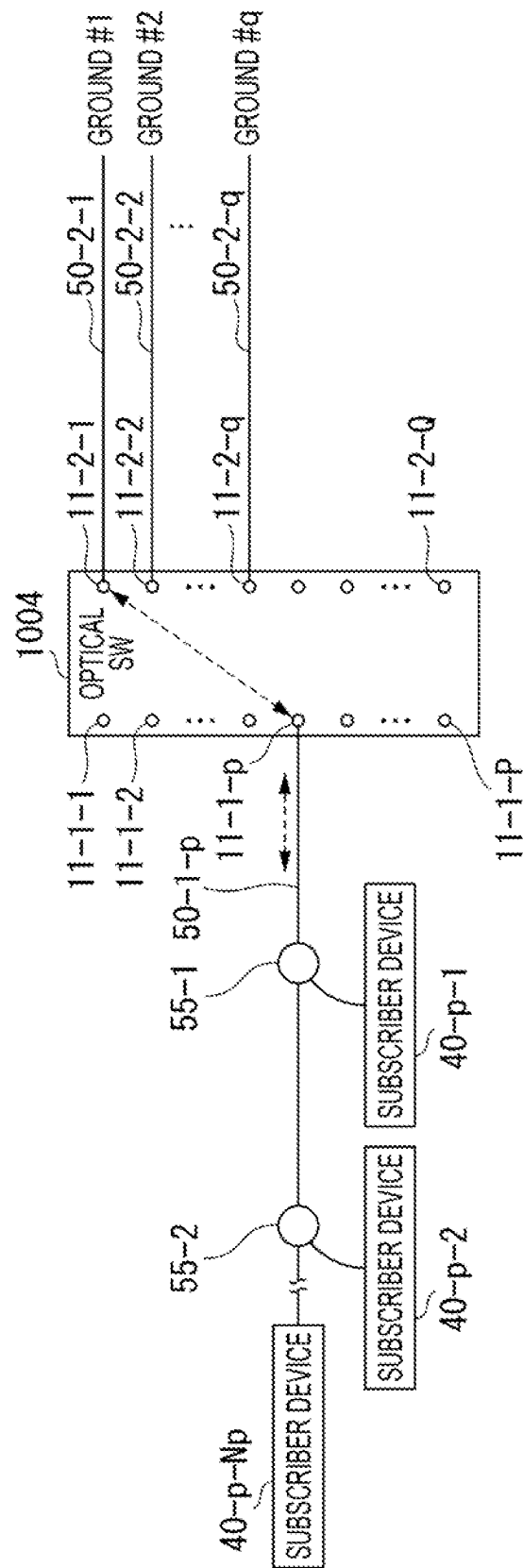


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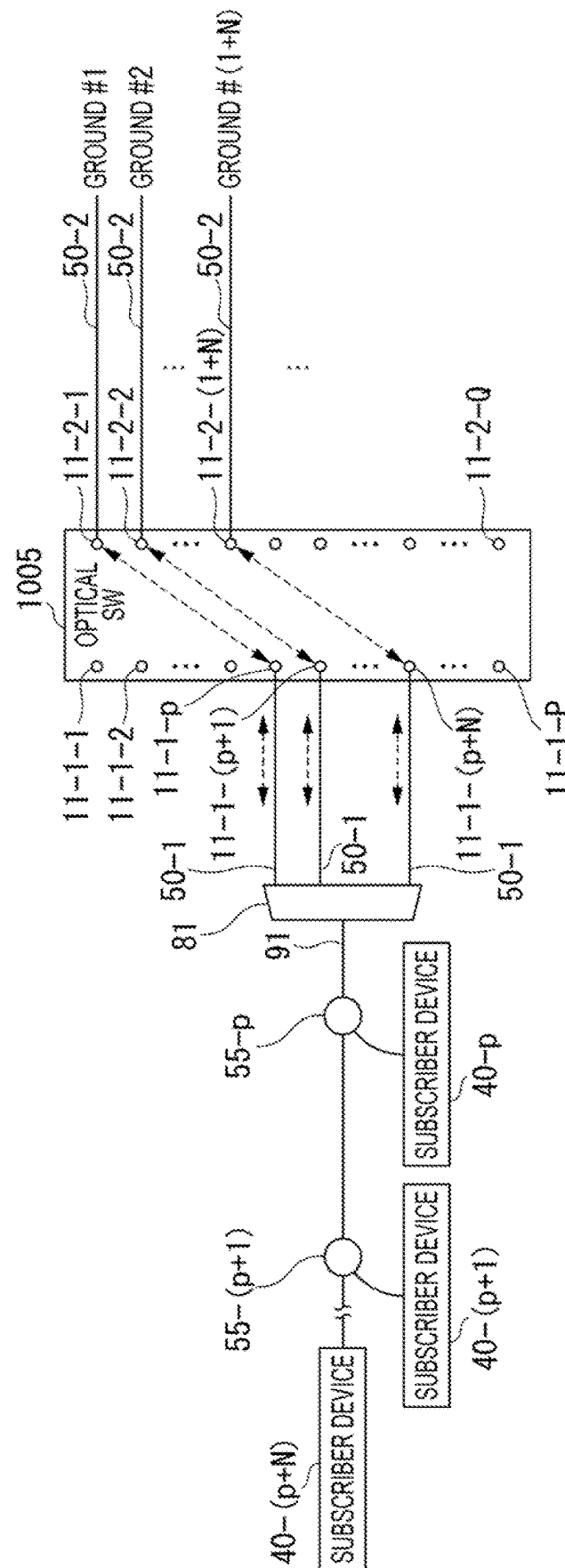


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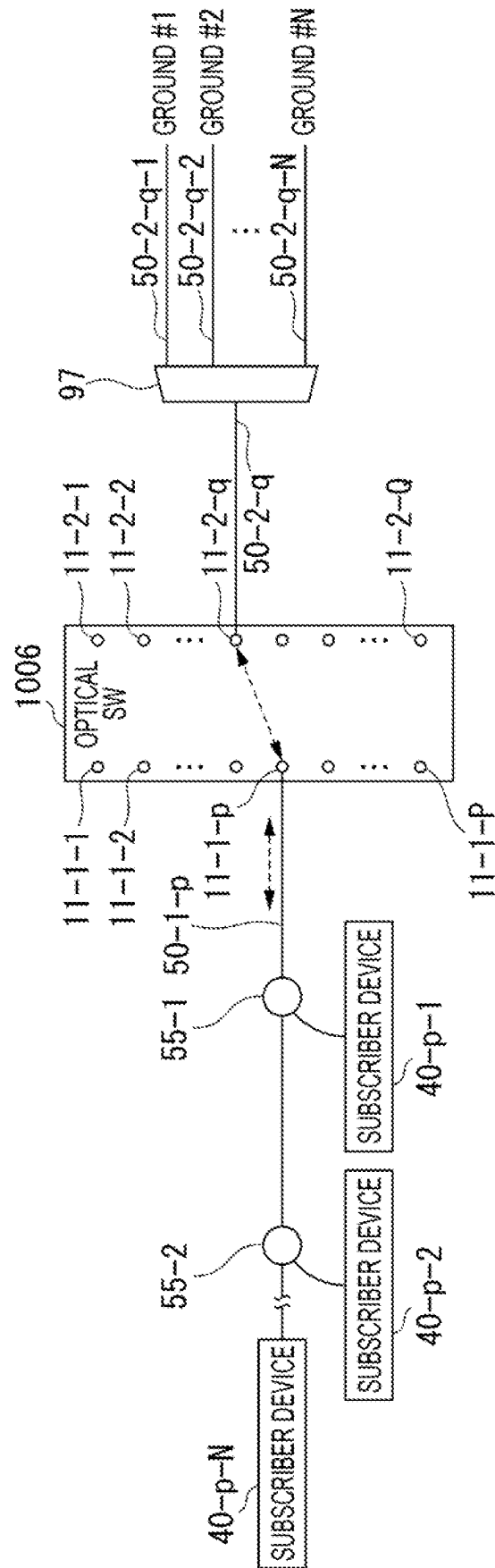


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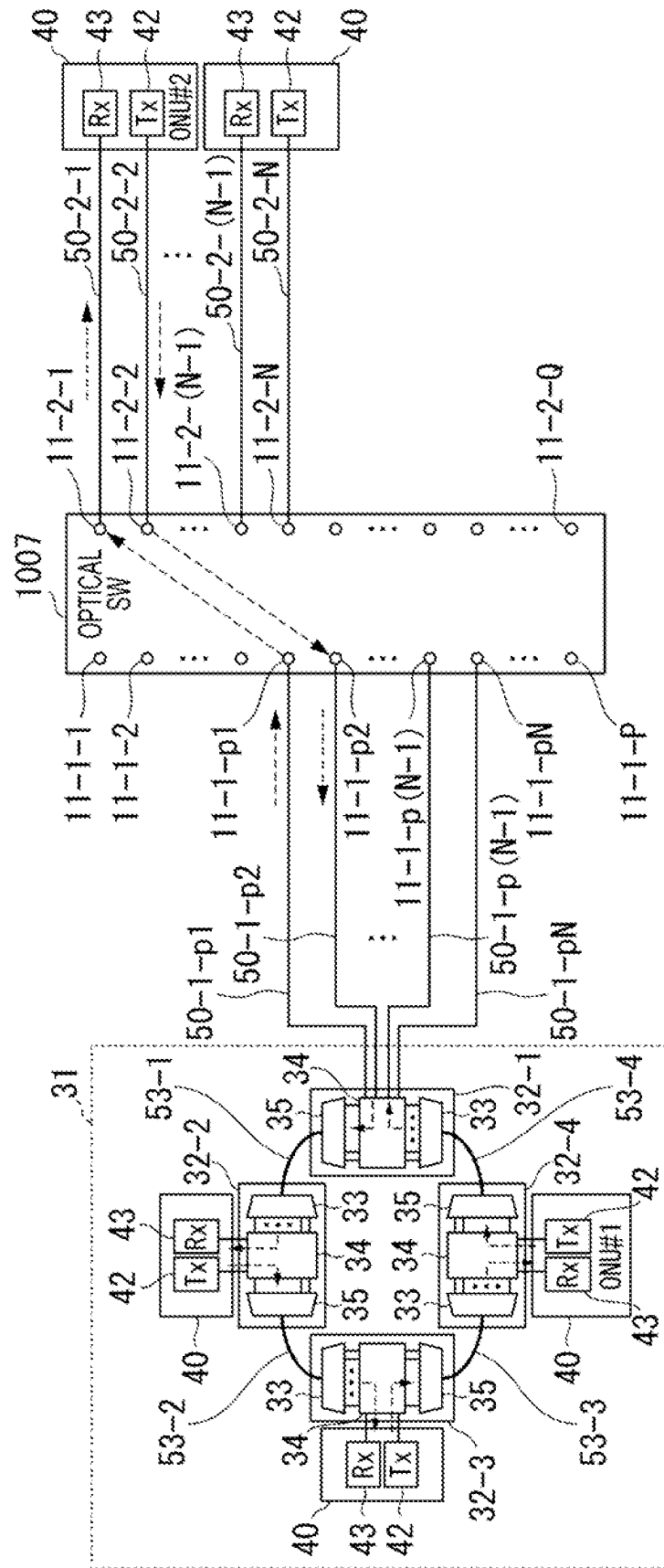


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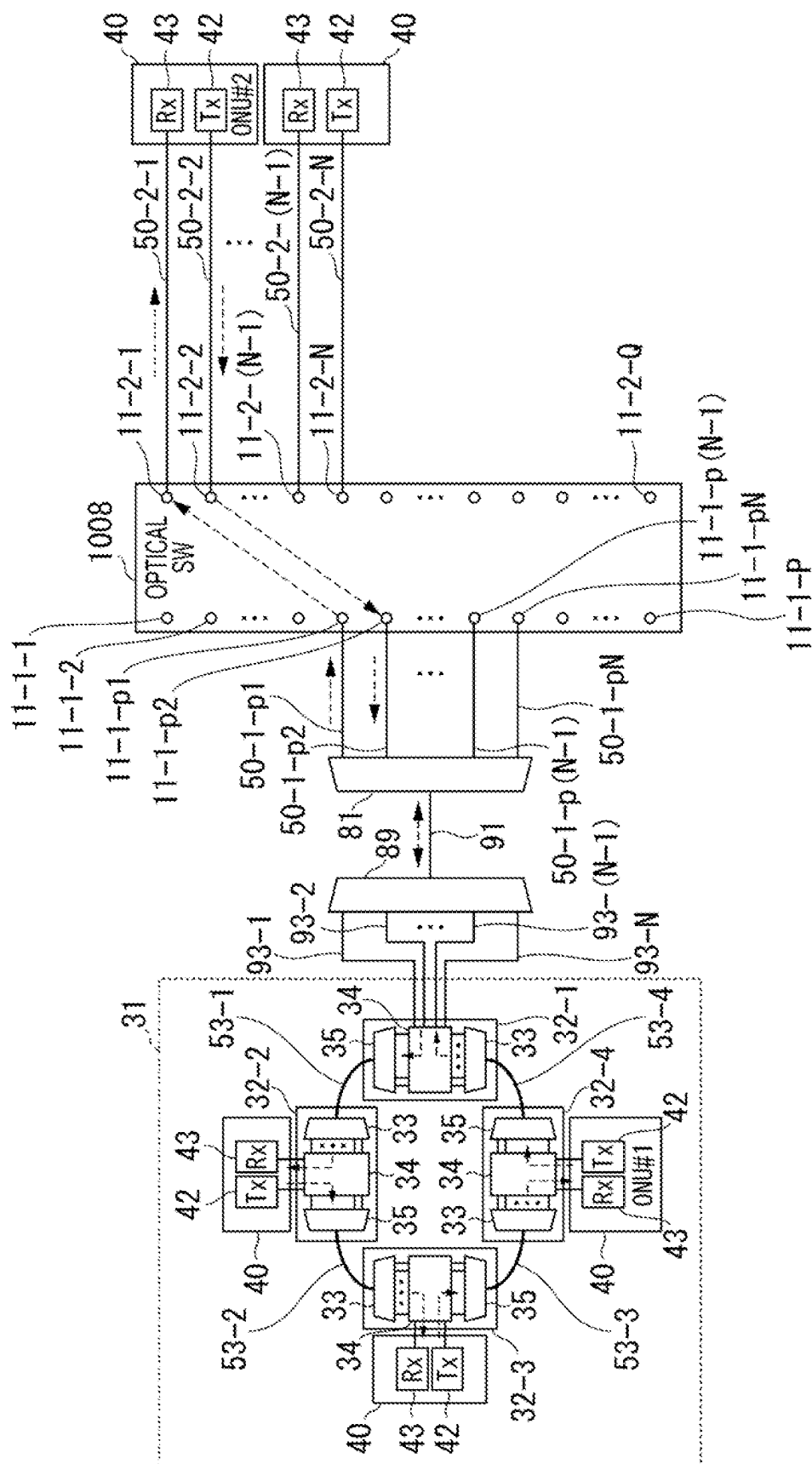


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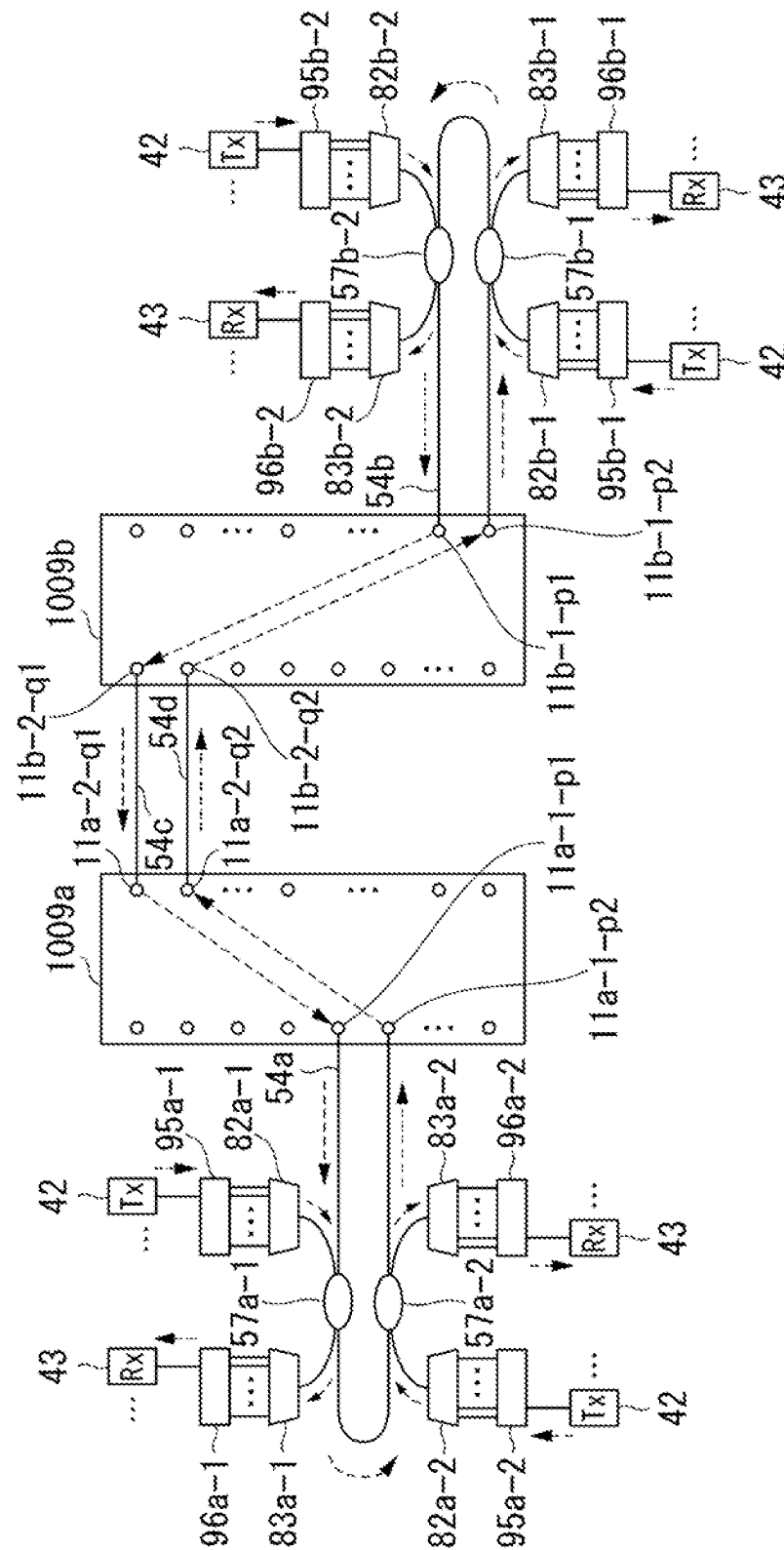


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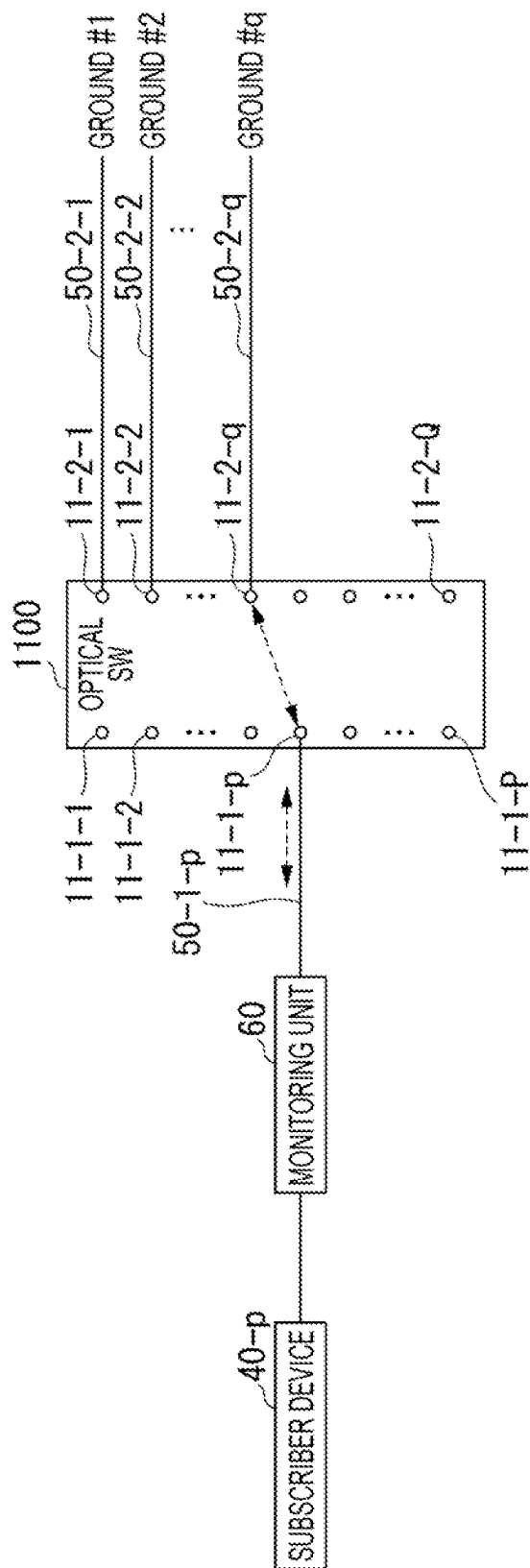


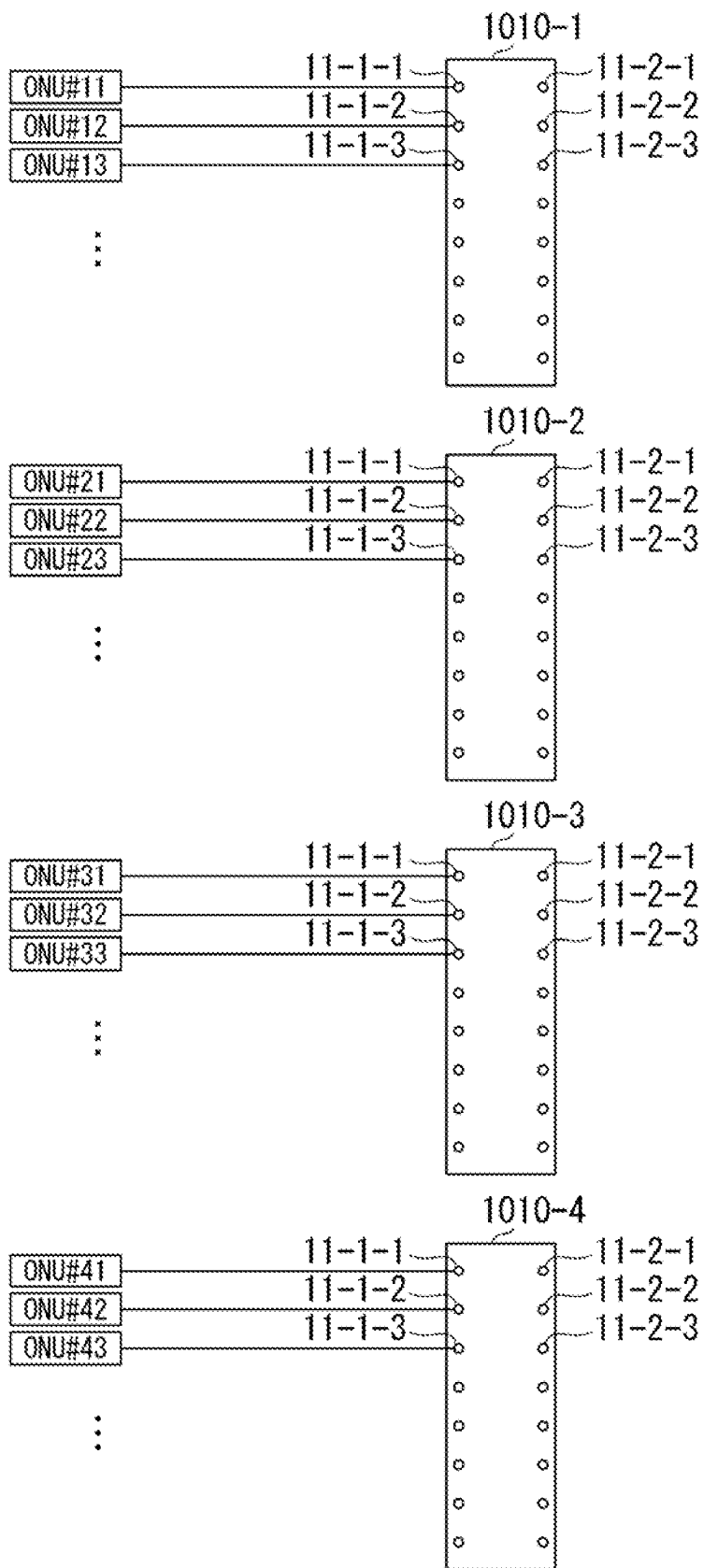
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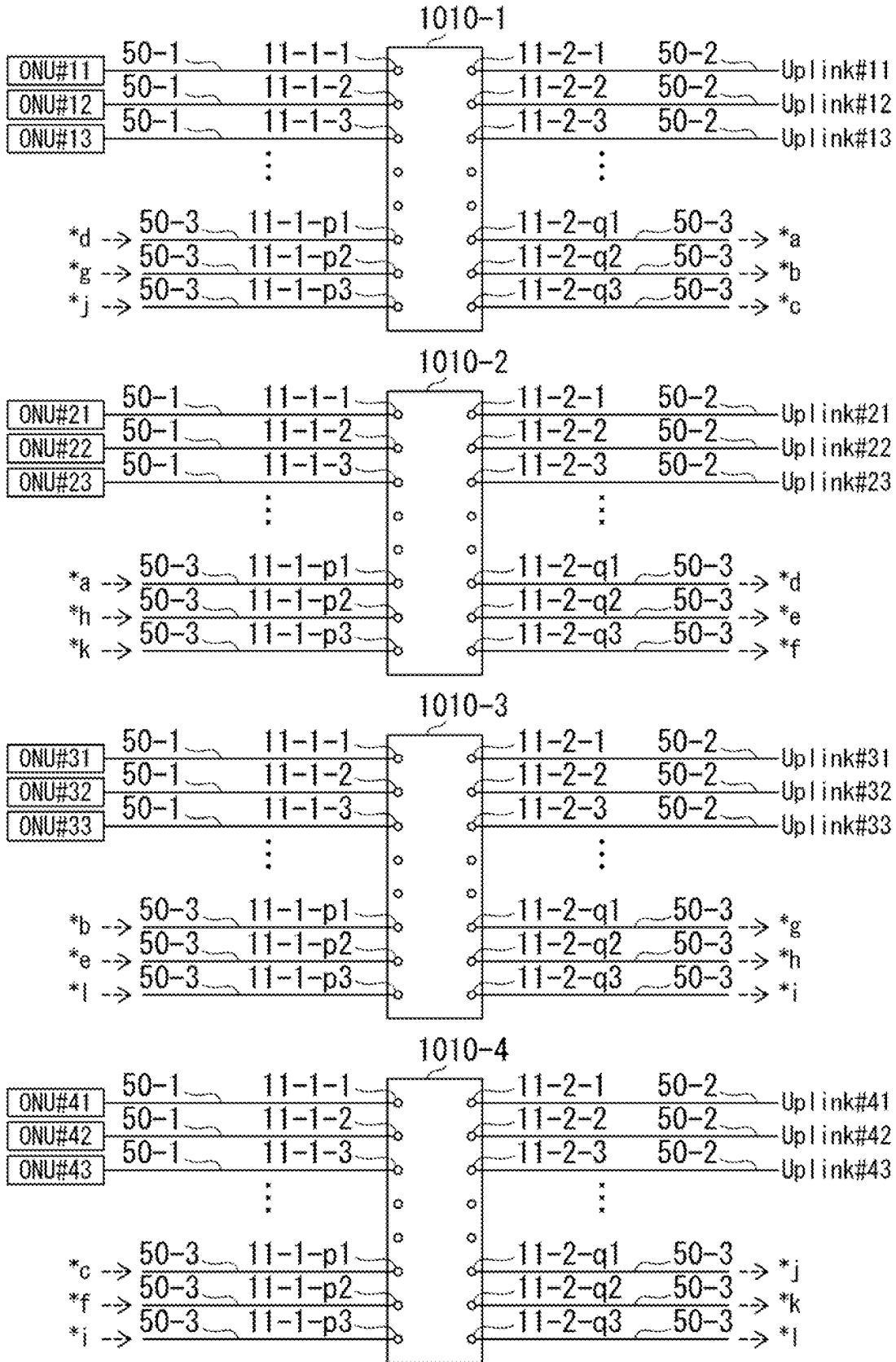
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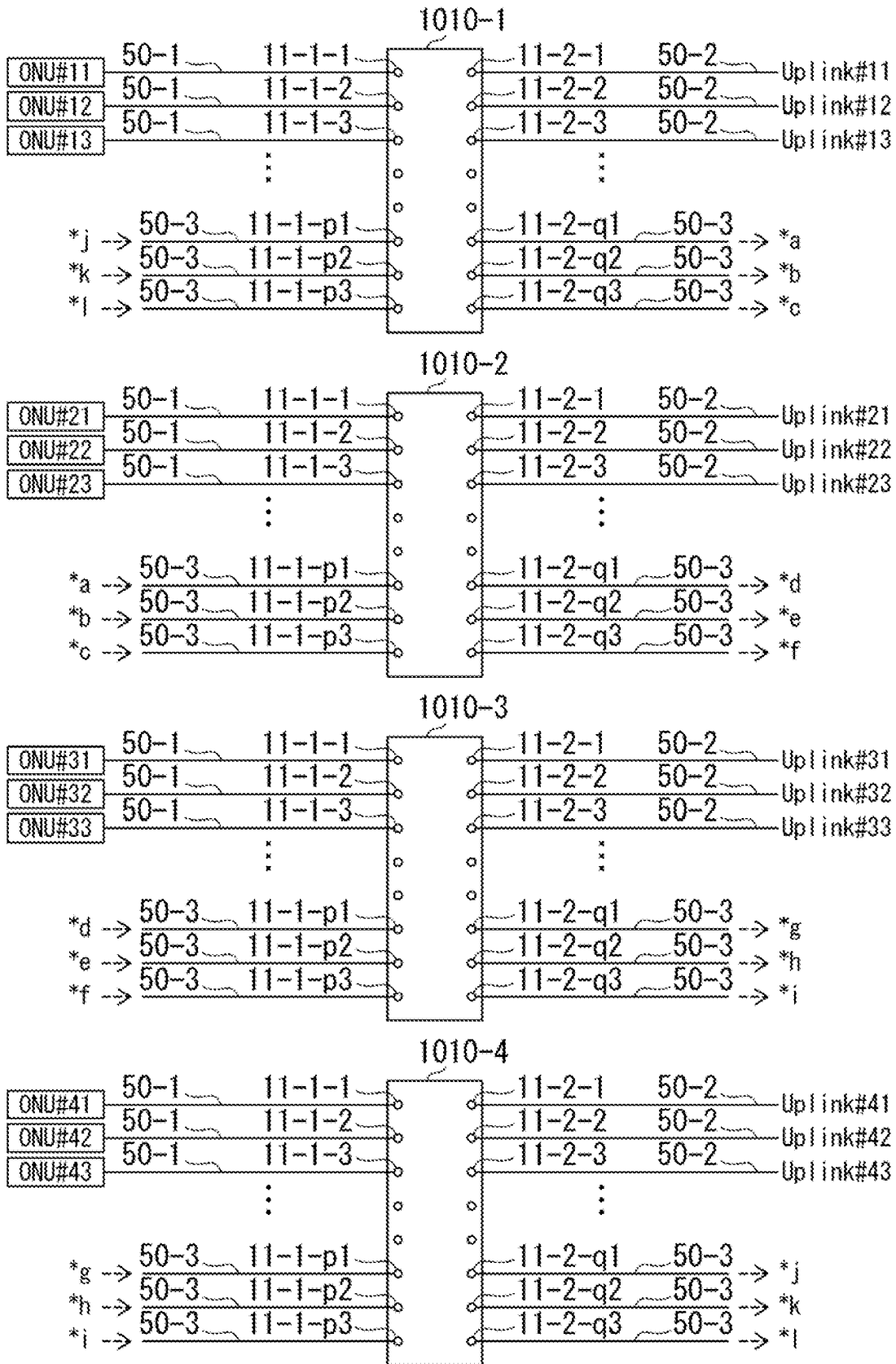
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Fig. 29

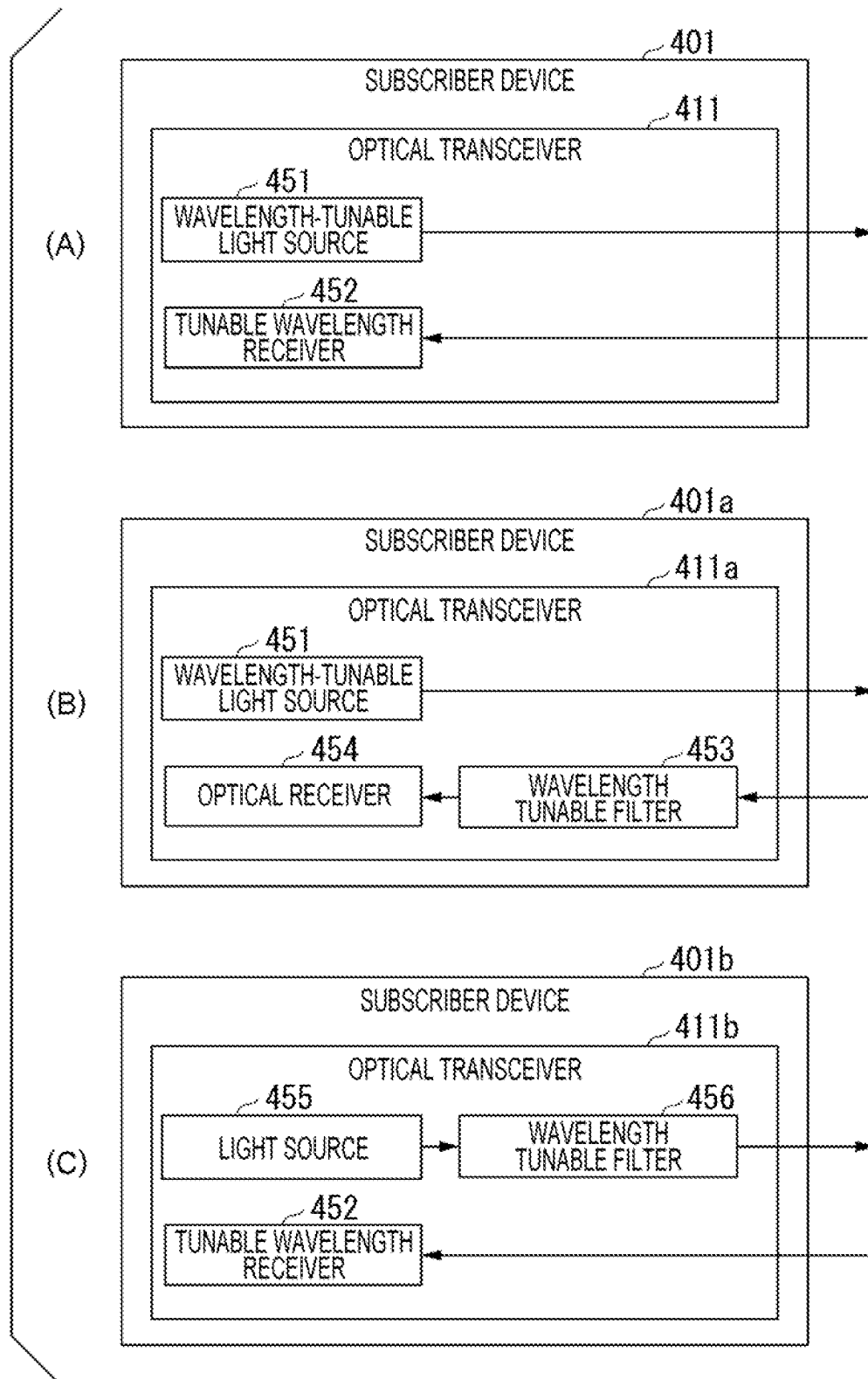


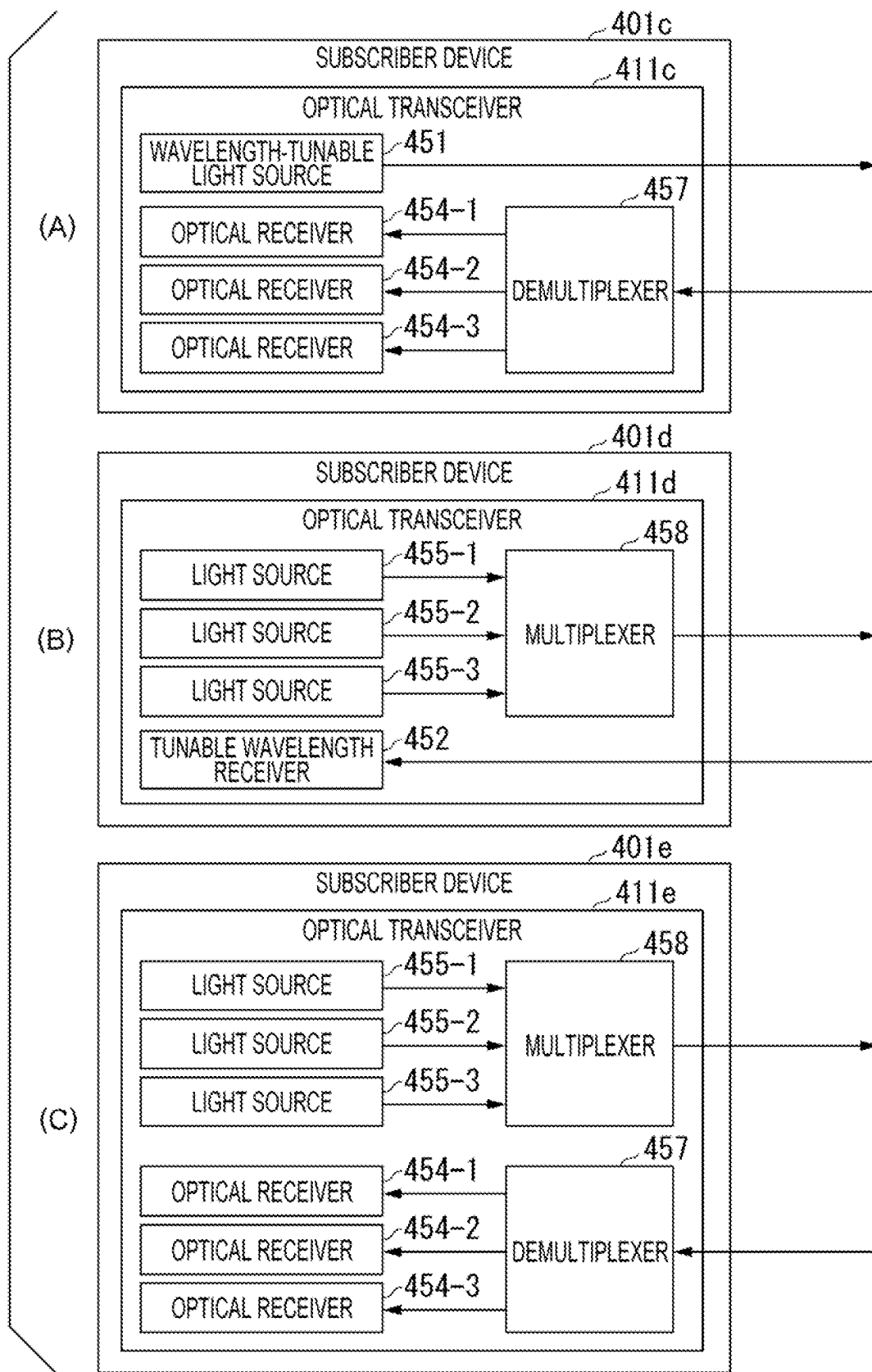
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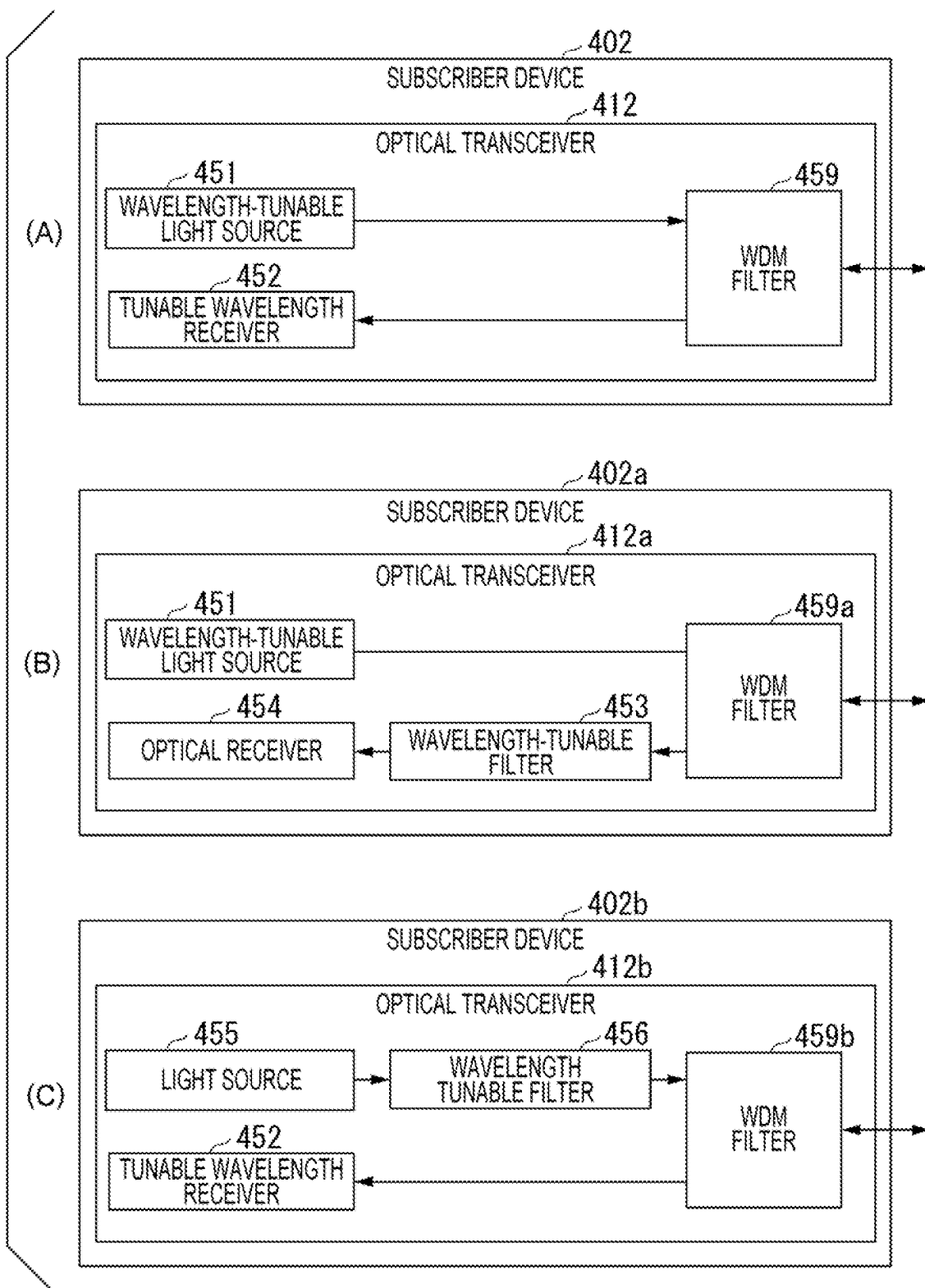
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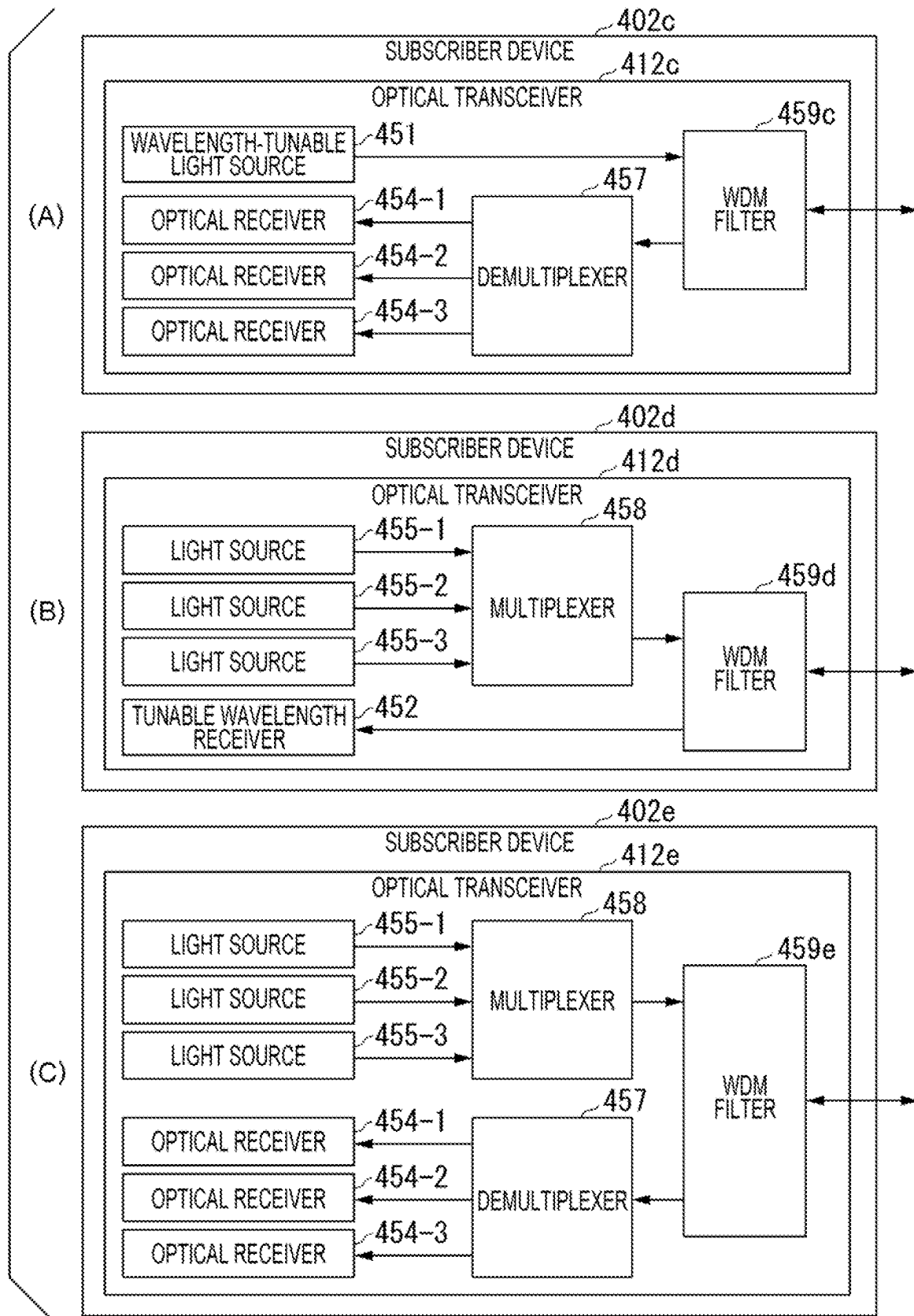
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Fig. 33

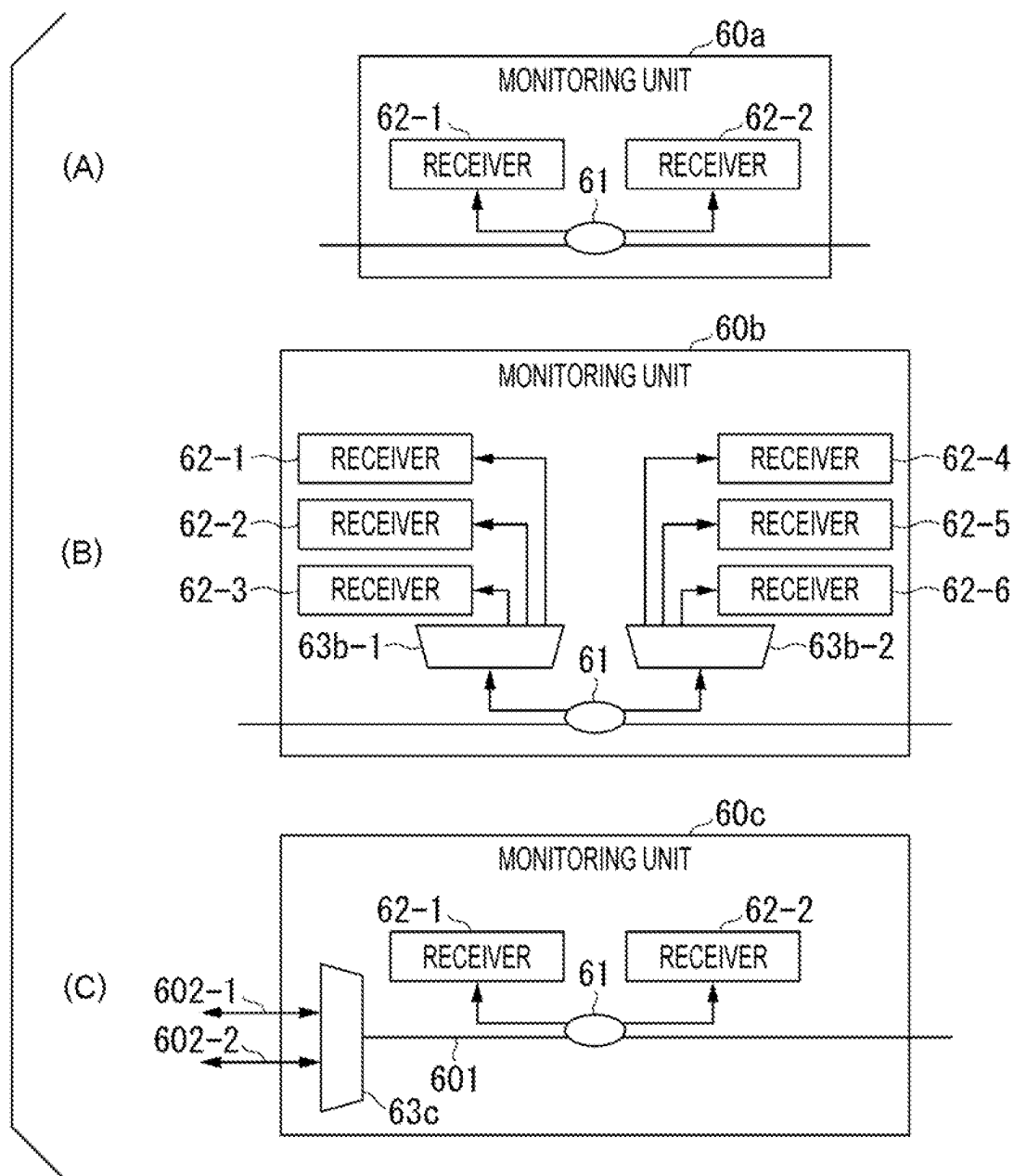


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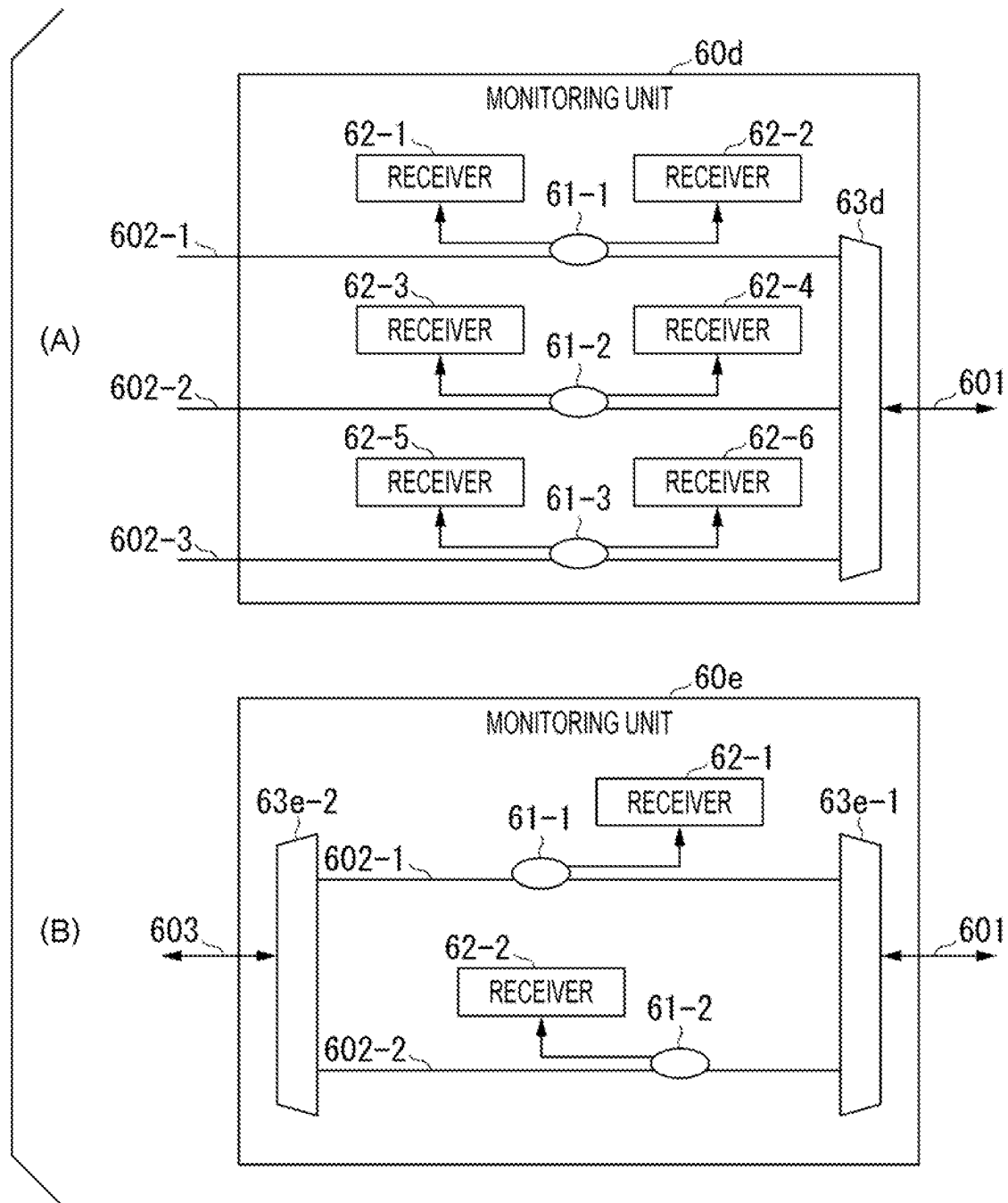


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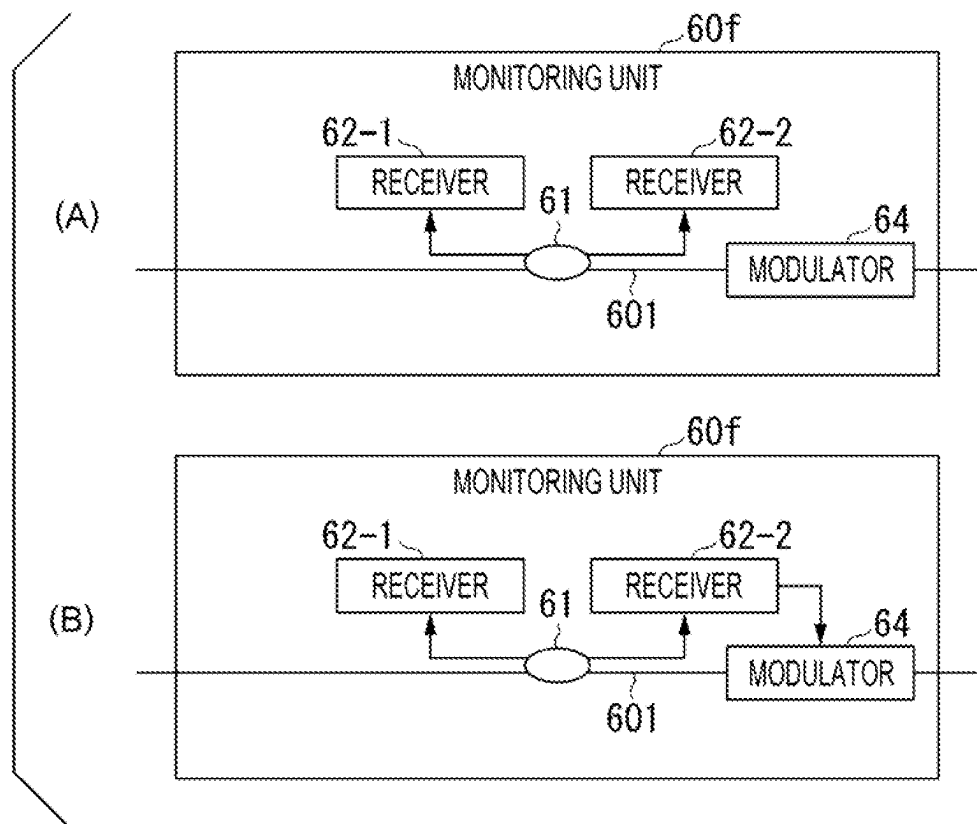


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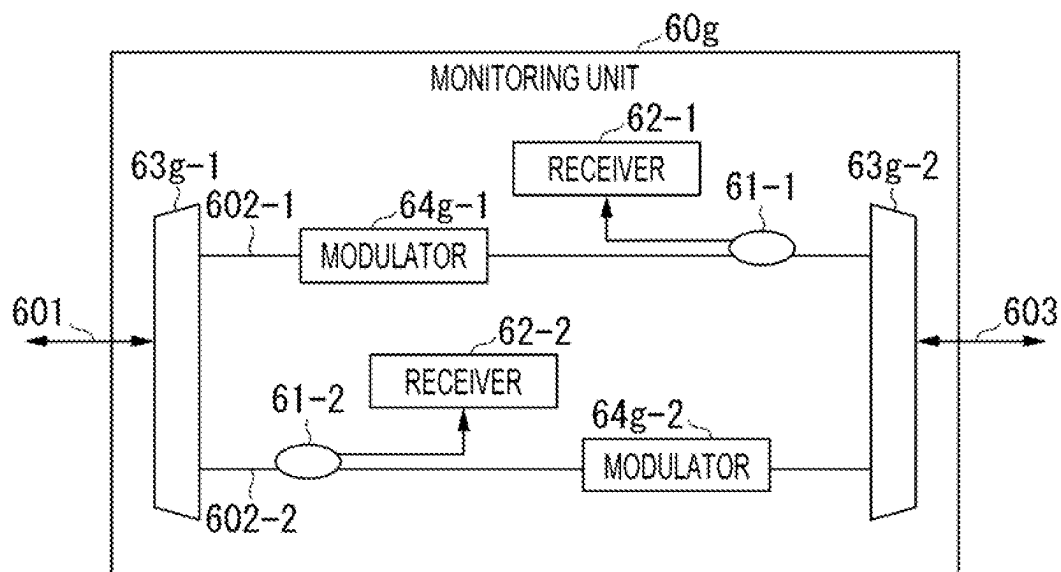


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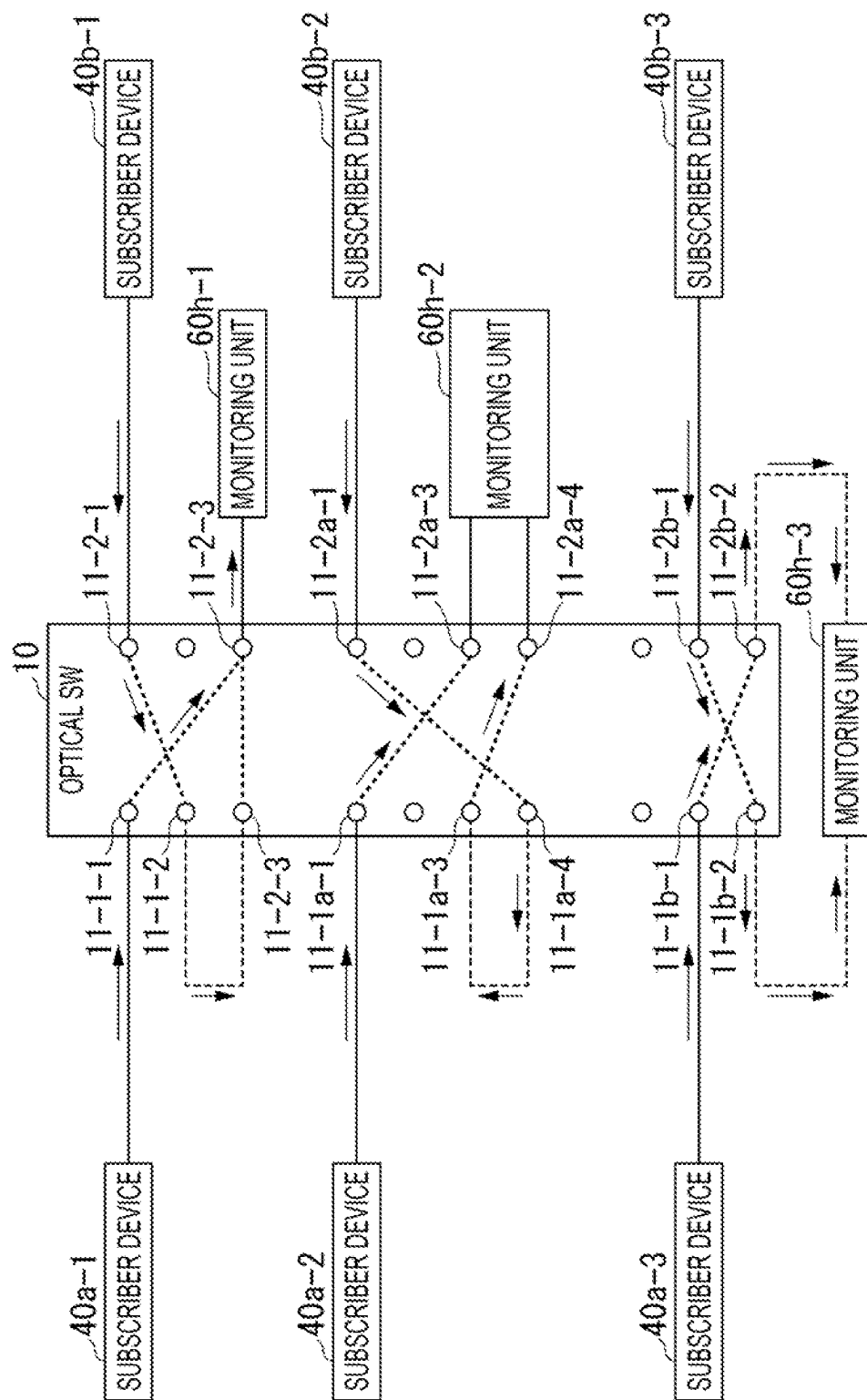


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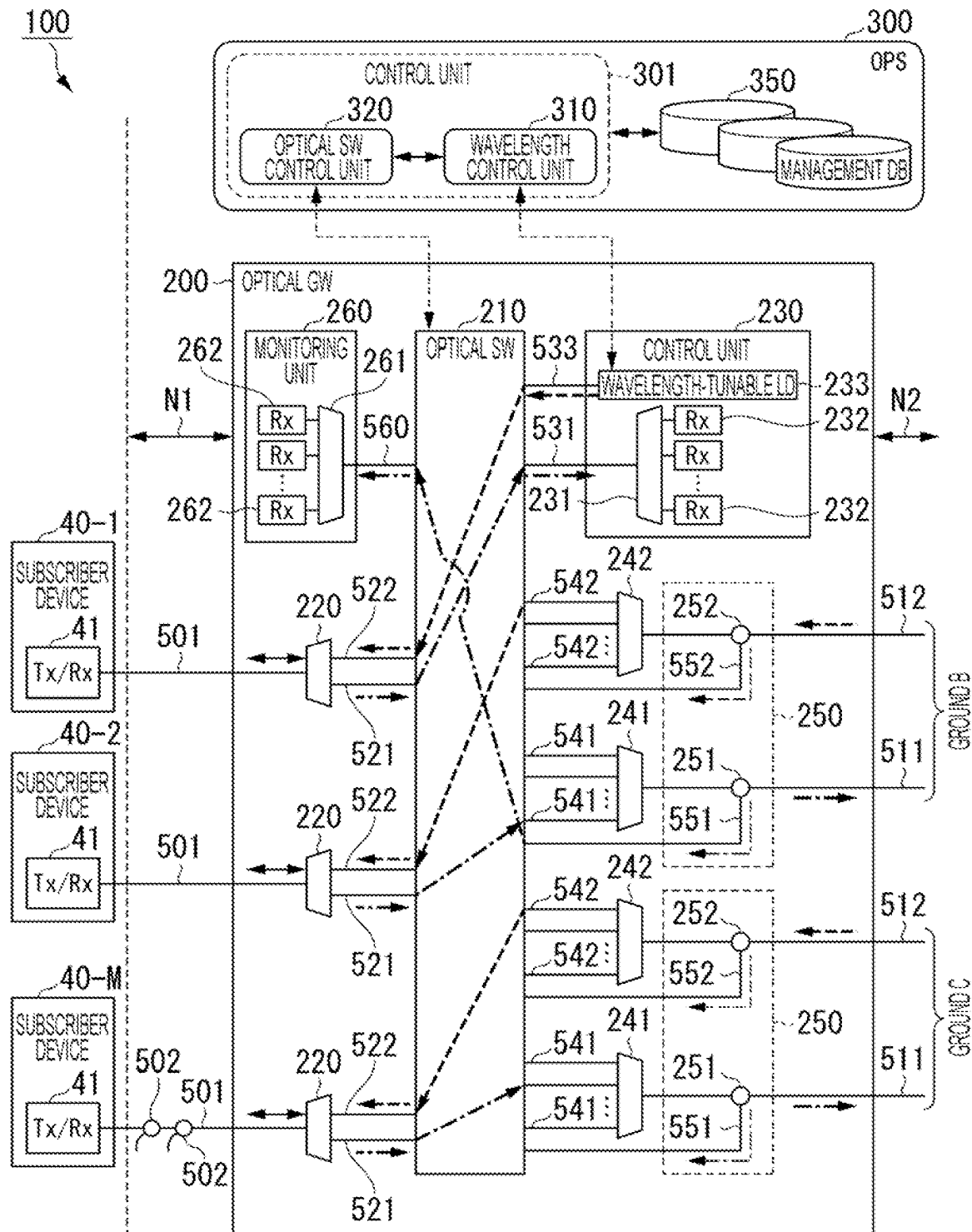


Fig. 39

PORT NUMBER	CONNECTED DEVICE
1	TRANSCEIVER 1x
2	AWG#1 port #3
3	EMPTY
:	:
XXX	YYY

Fig. 40

	$\lambda 1$	$\lambda 2$	$\lambda 3$
USER 1	TRANSMISSION	RECEPTION	FAILURE
USER 2	EMPTY	EMPTY	RECEPTION
USER 3	EMPTY	TRANSMISSION	EMPTY

Fig. 41

	$\lambda 1$	$\lambda 2$	$\lambda 3$
GROUND A	USED	USED	FAILURE
GROUND B	EMPTY	EMPTY	USED
GROUND C	EMPTY	USED	EMPTY

Fig. 42

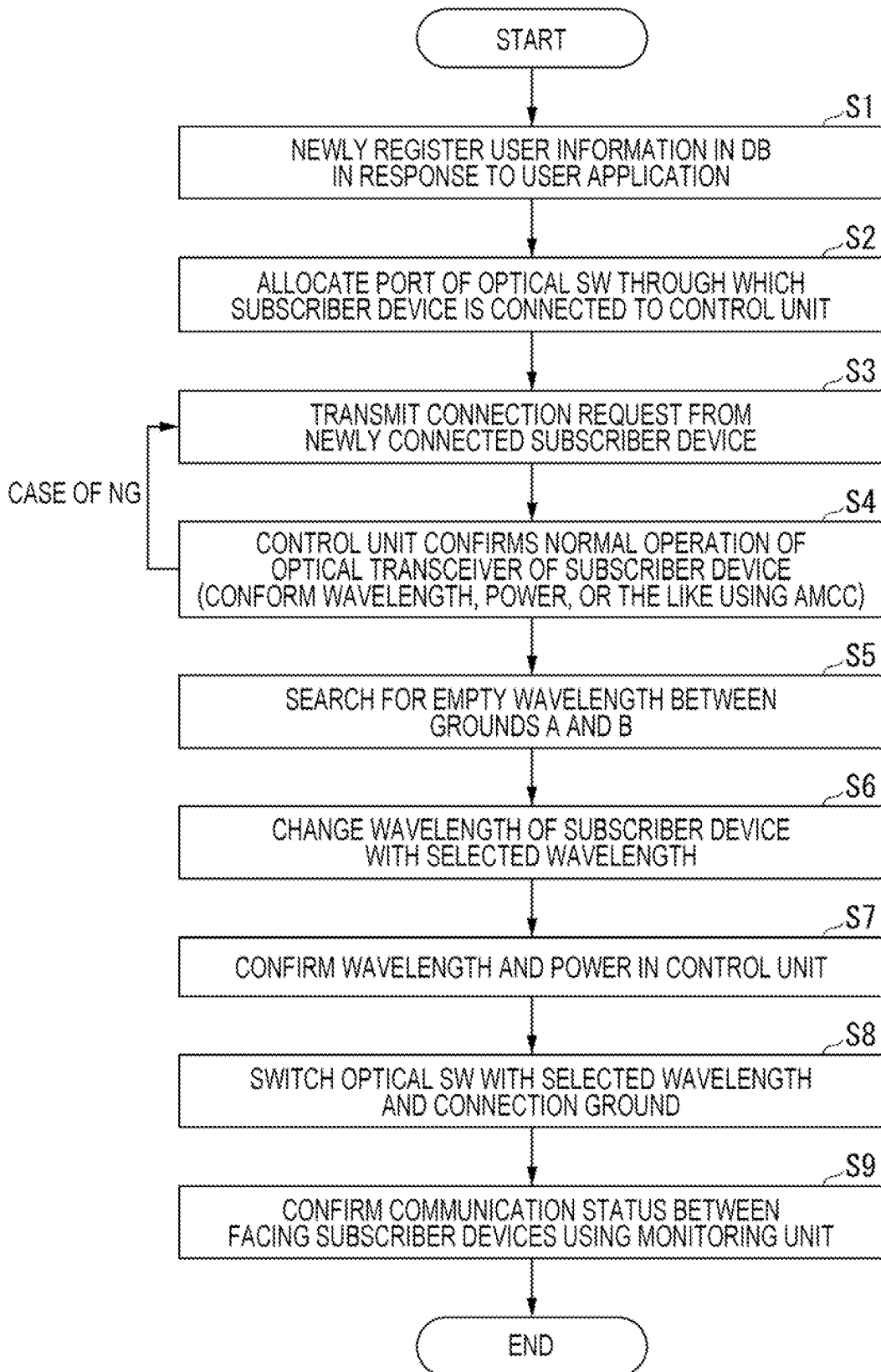


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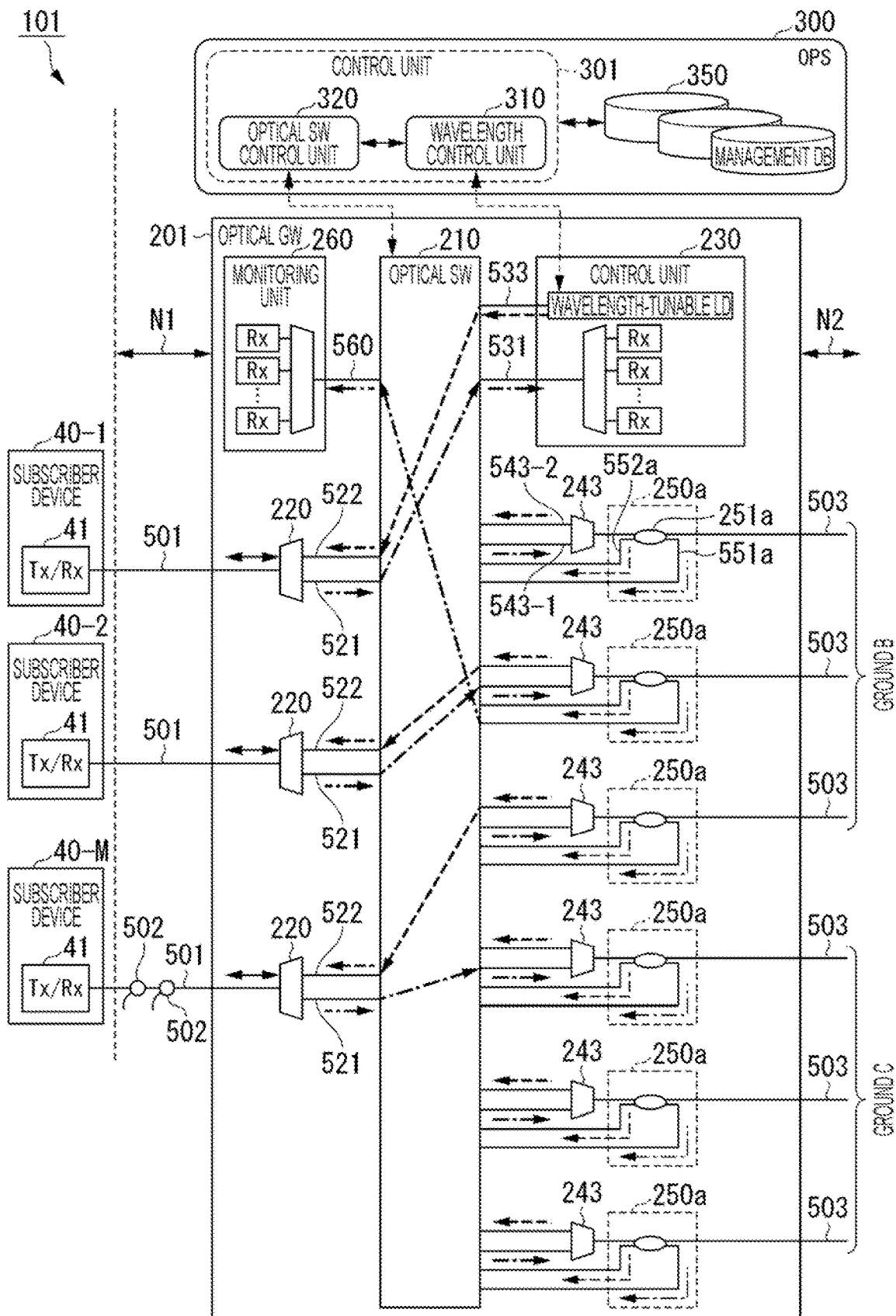


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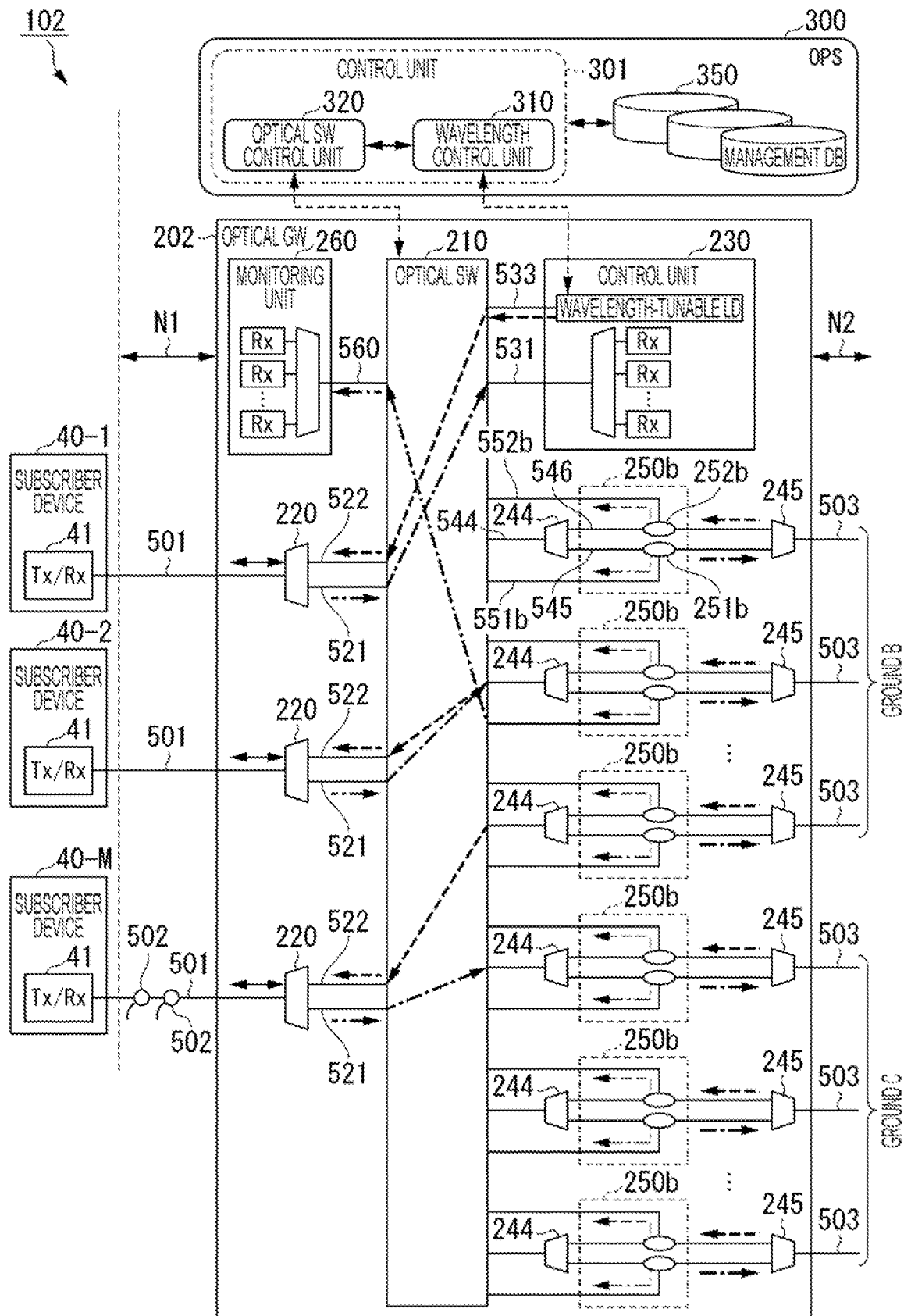


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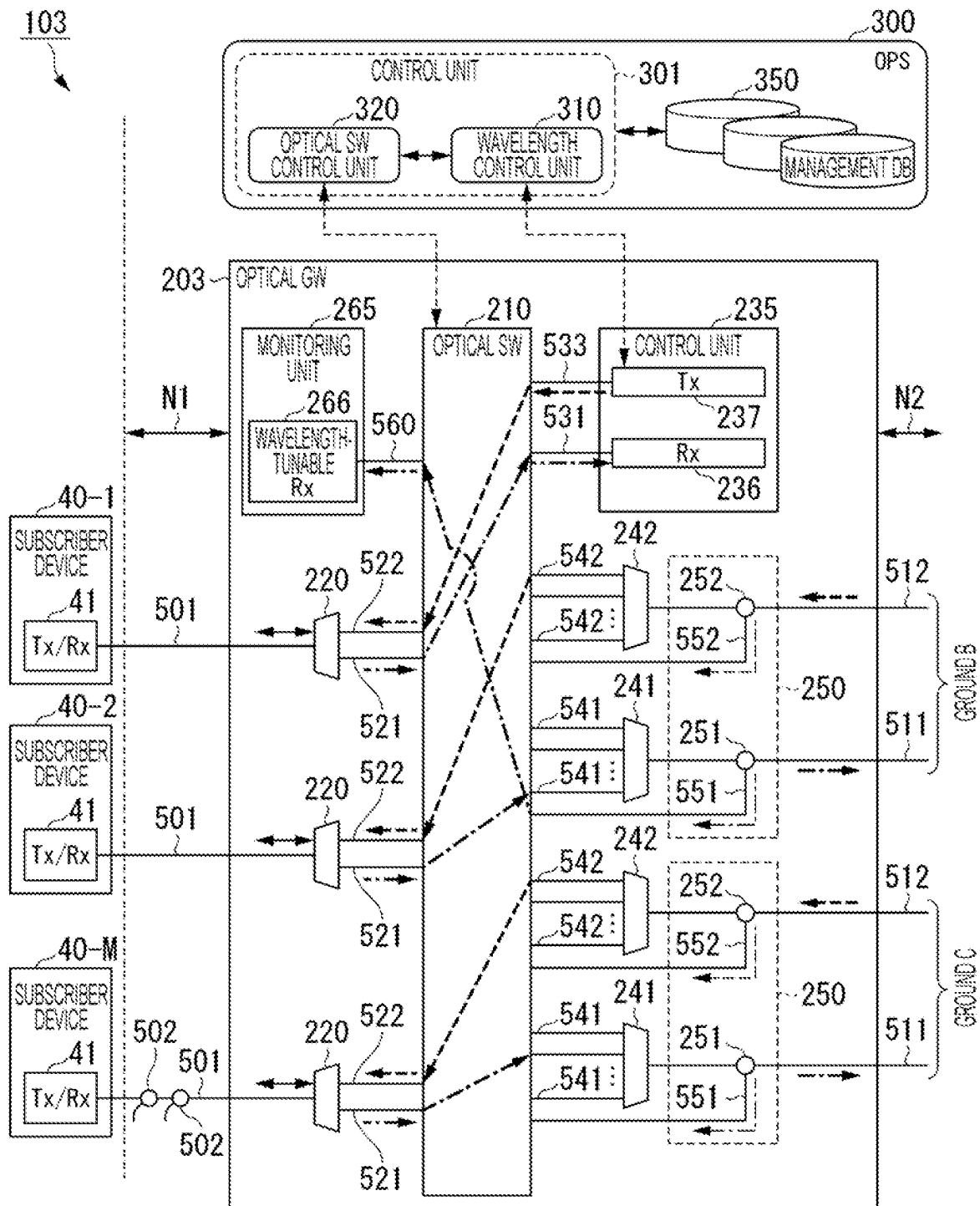


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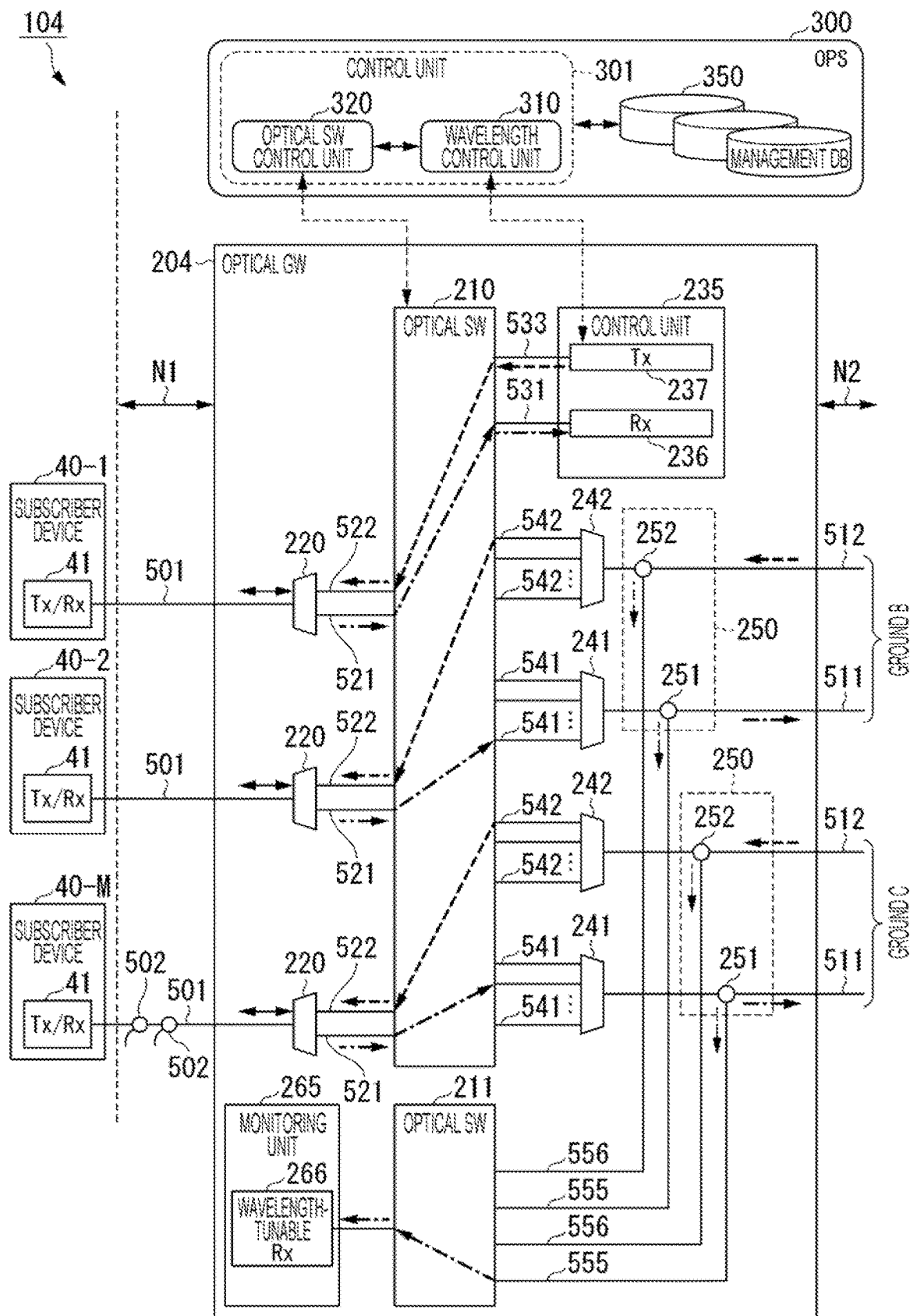


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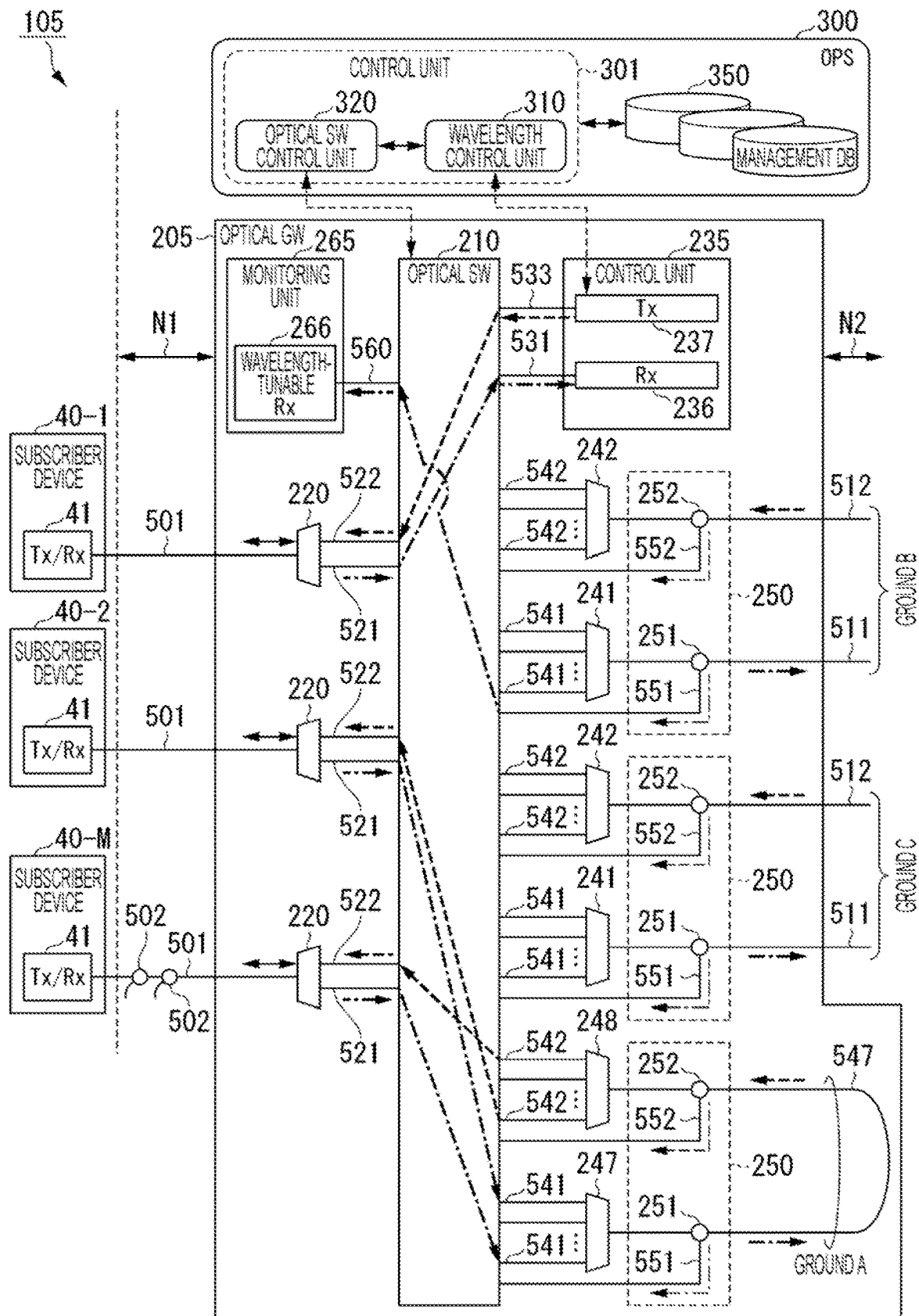


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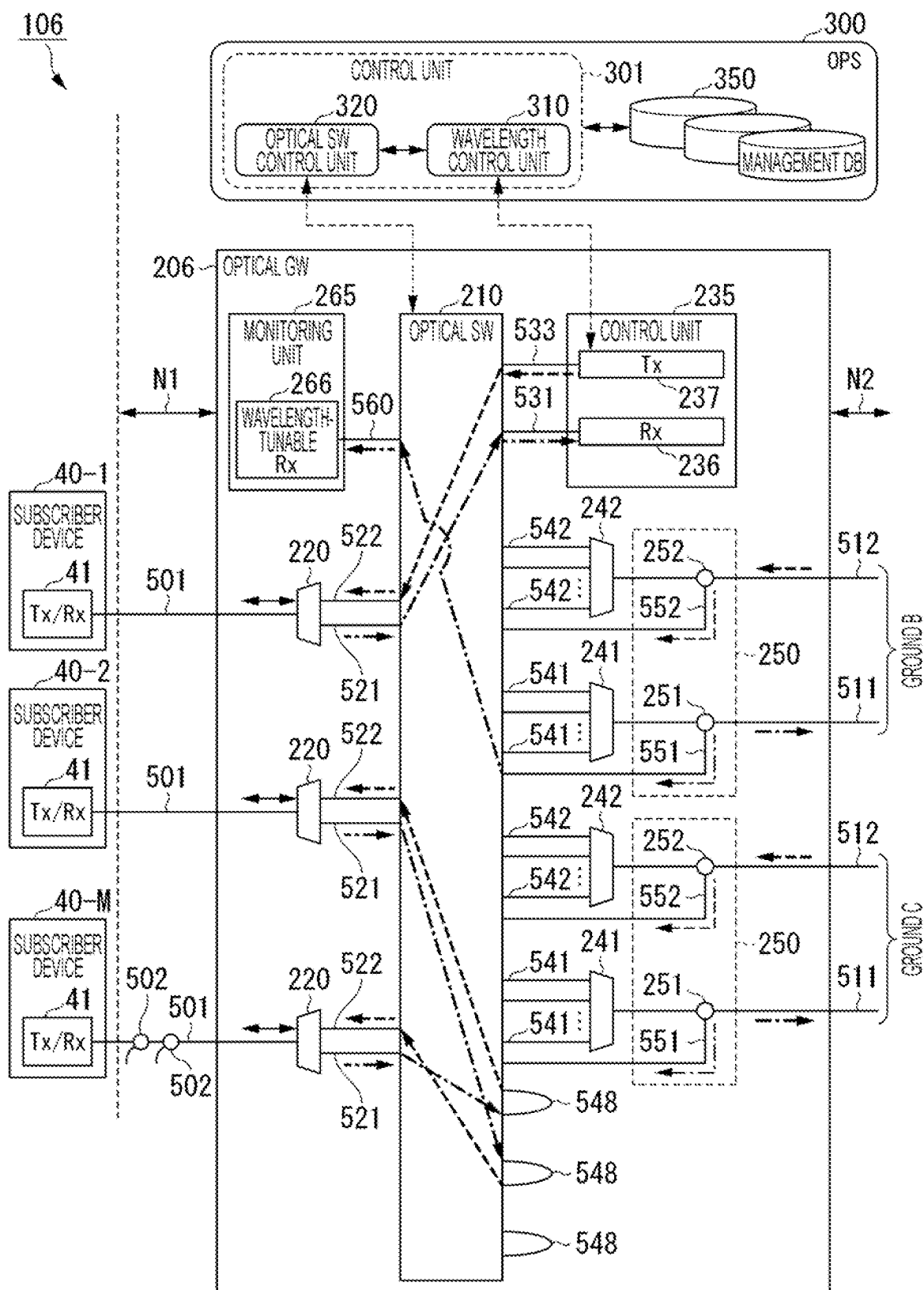


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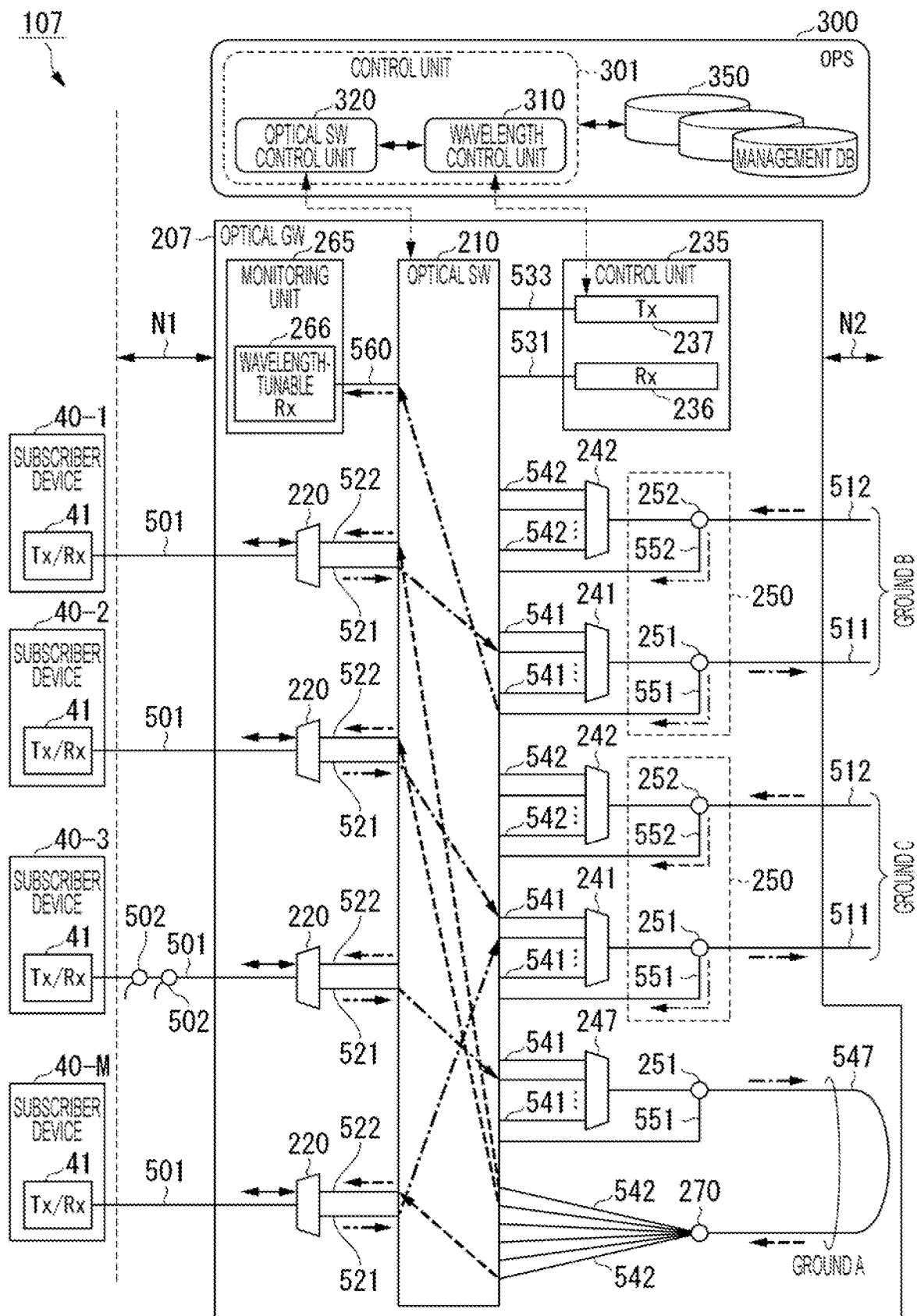


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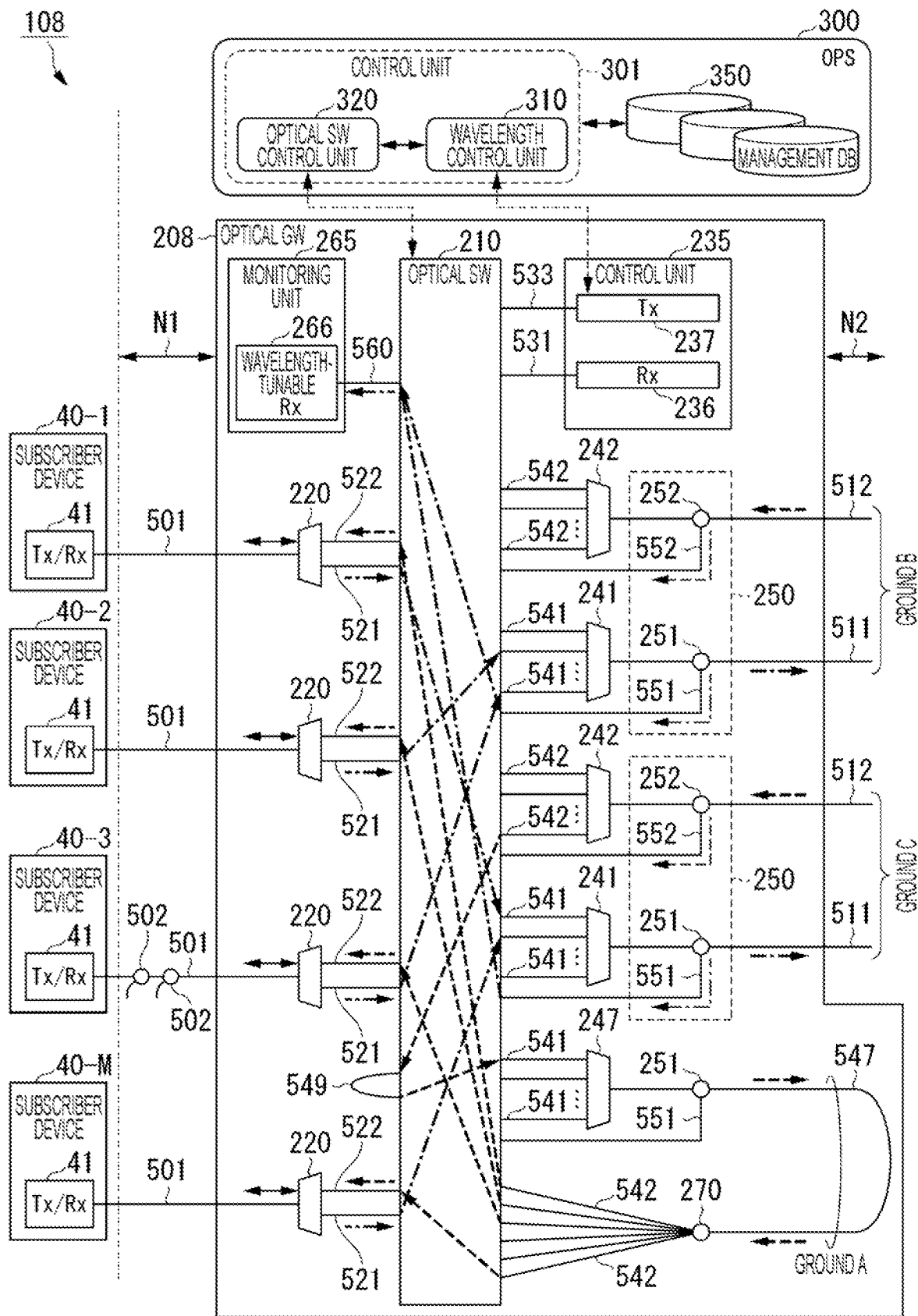


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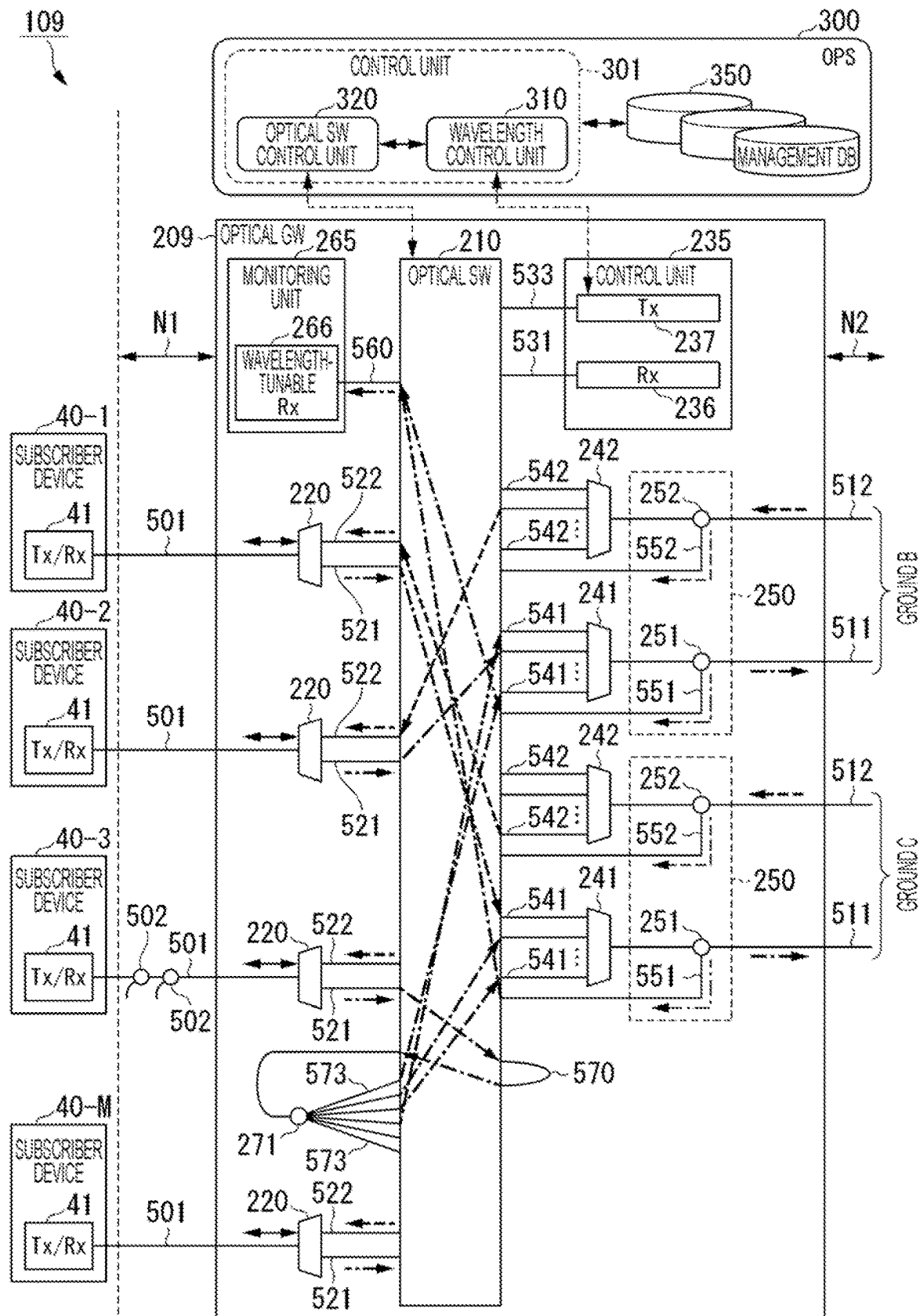


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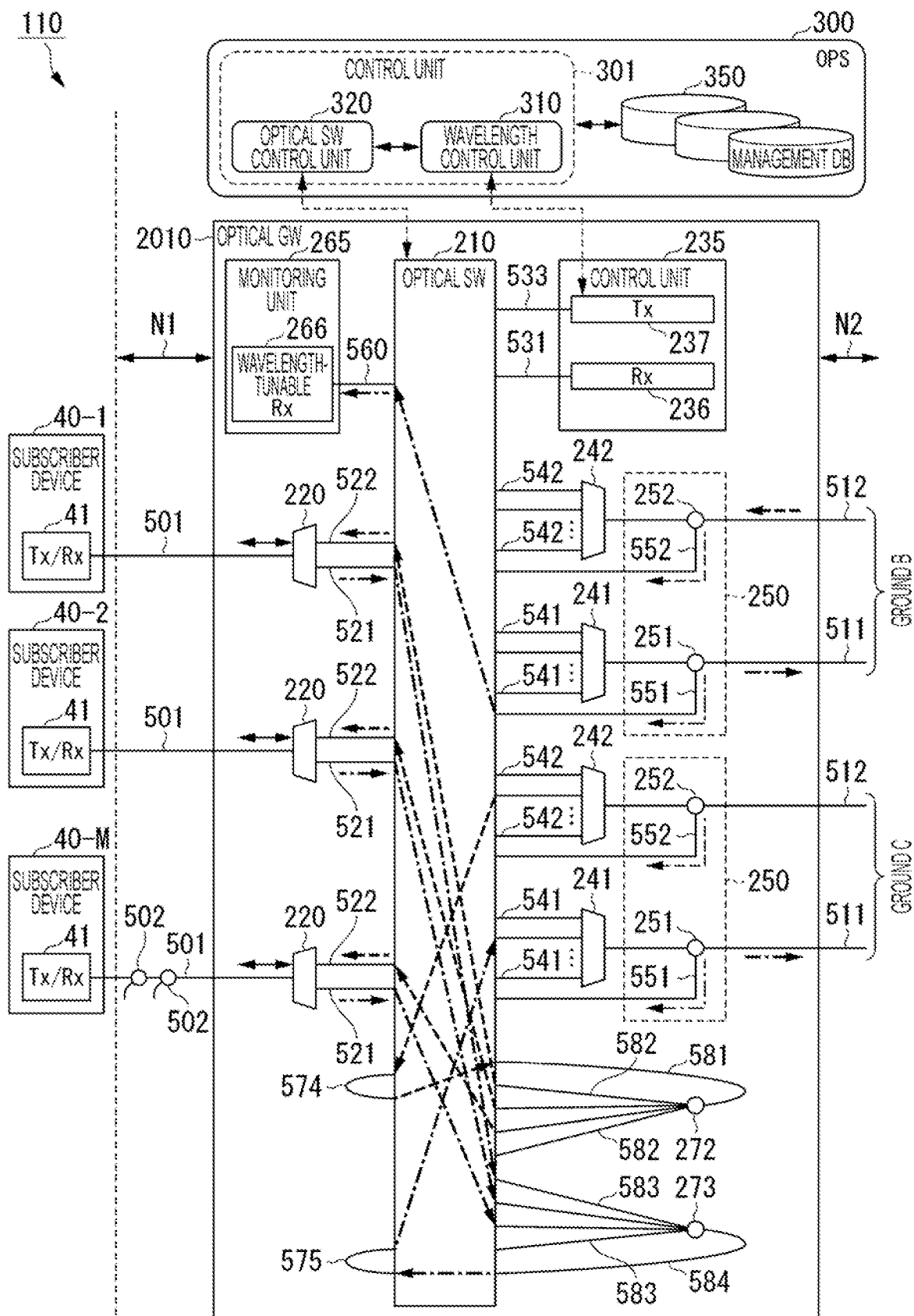


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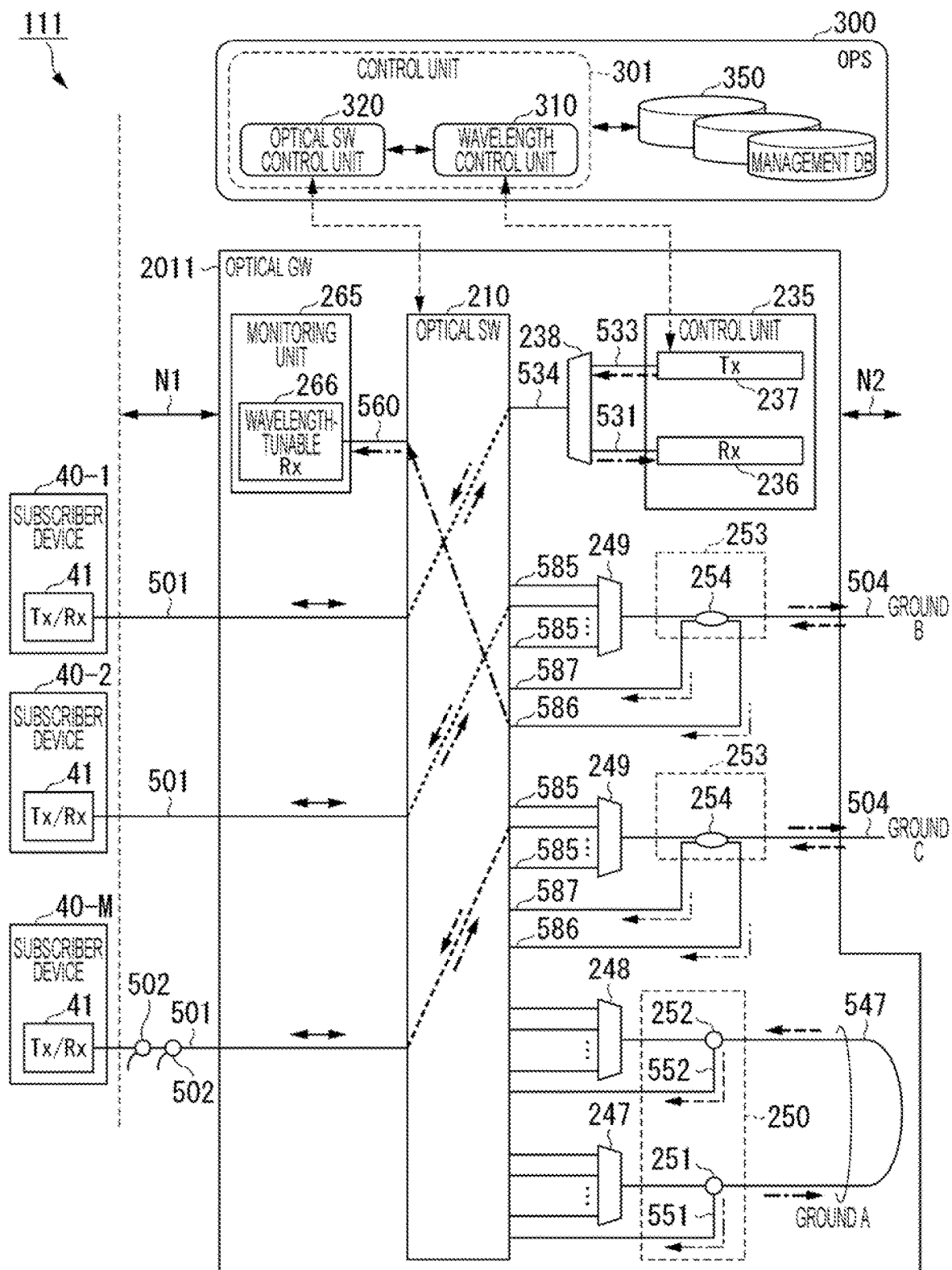


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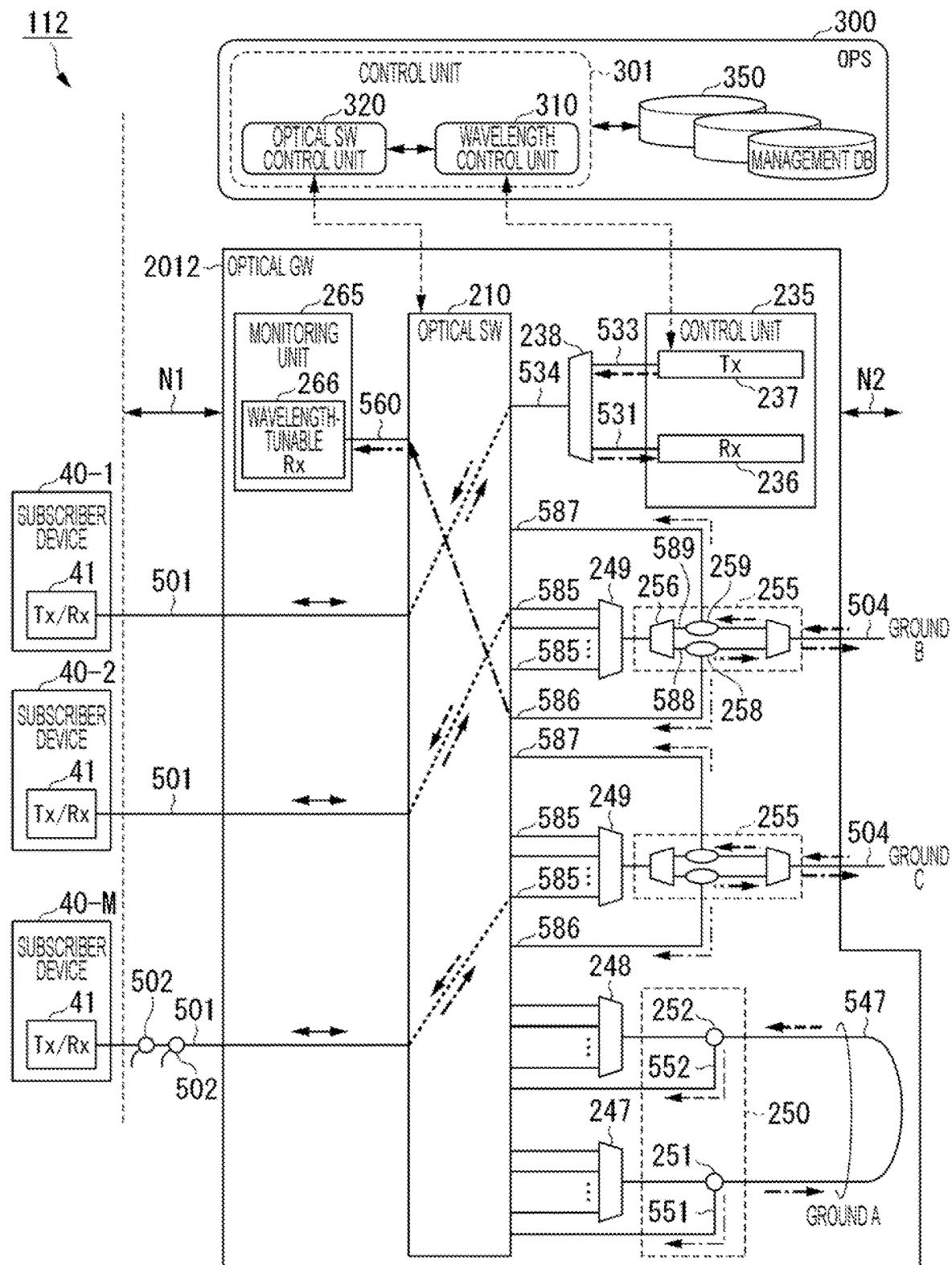


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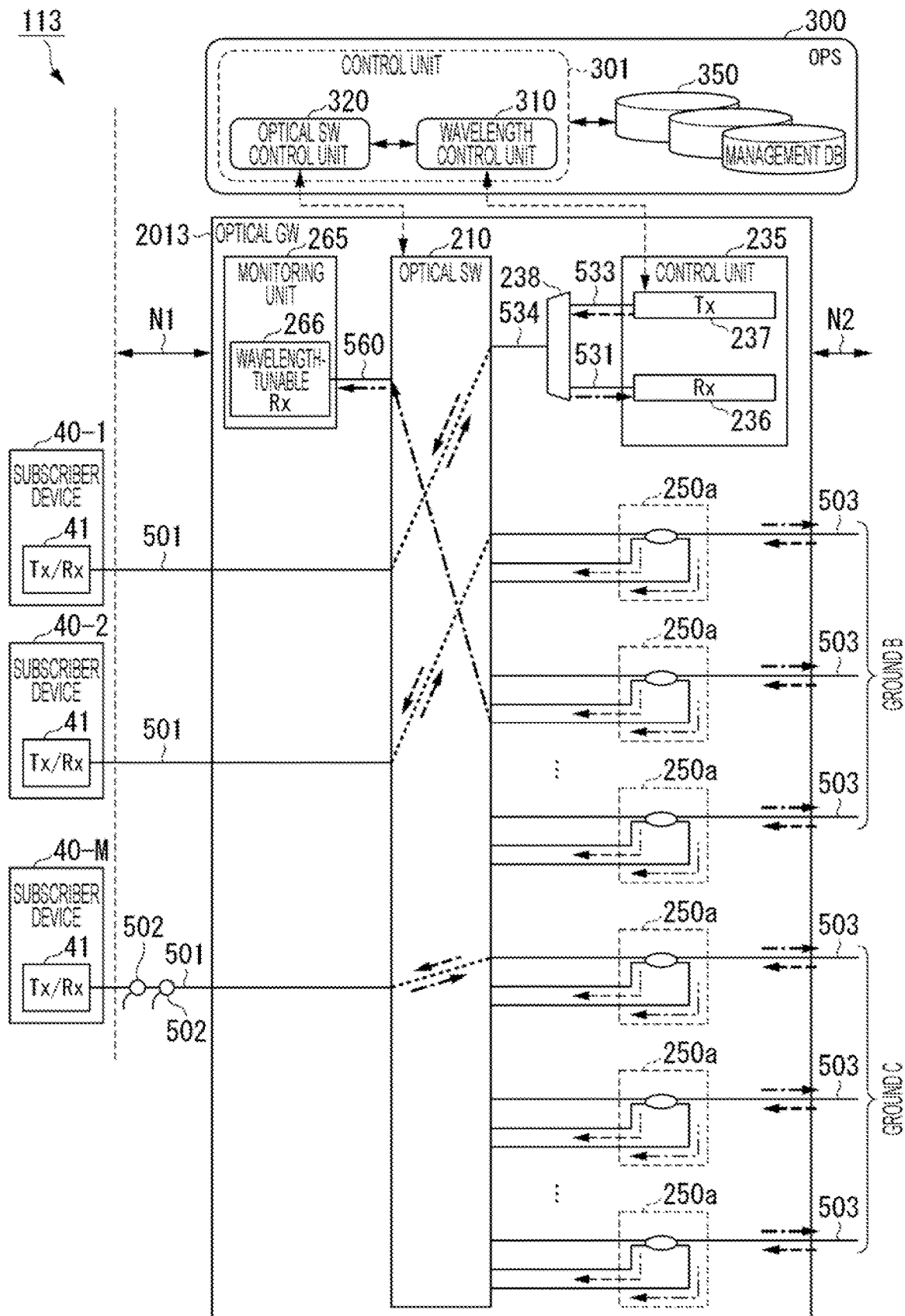


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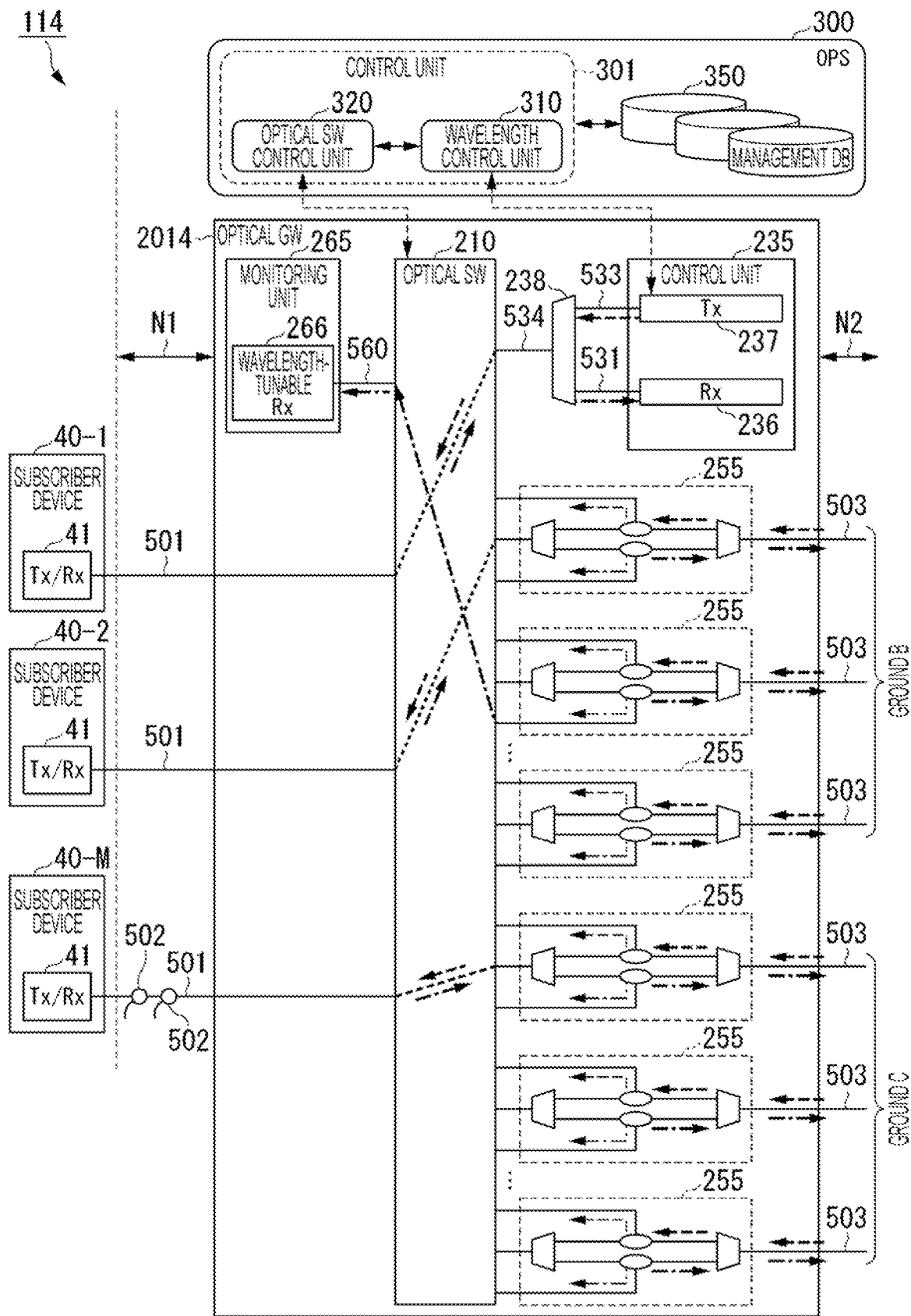


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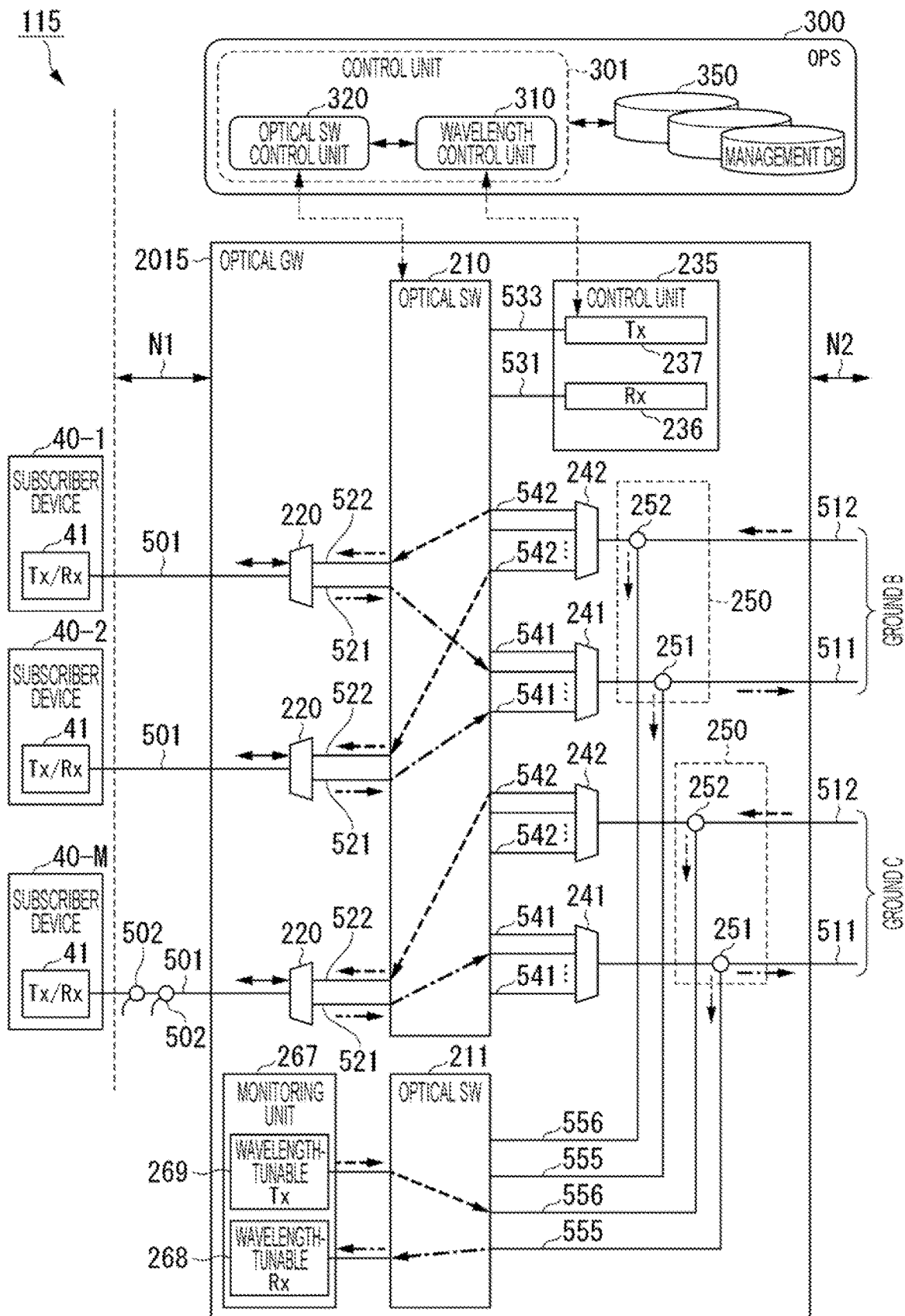


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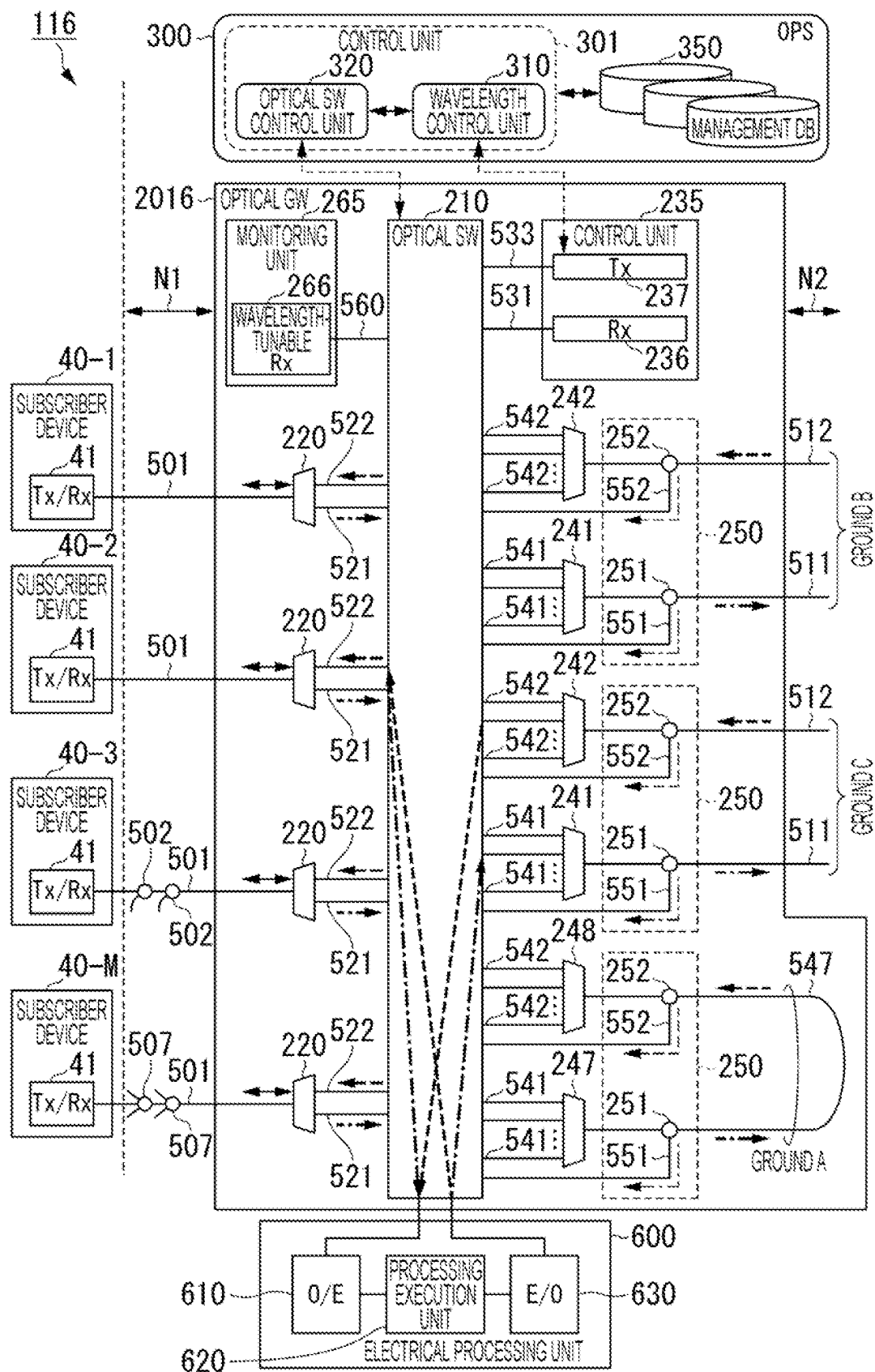


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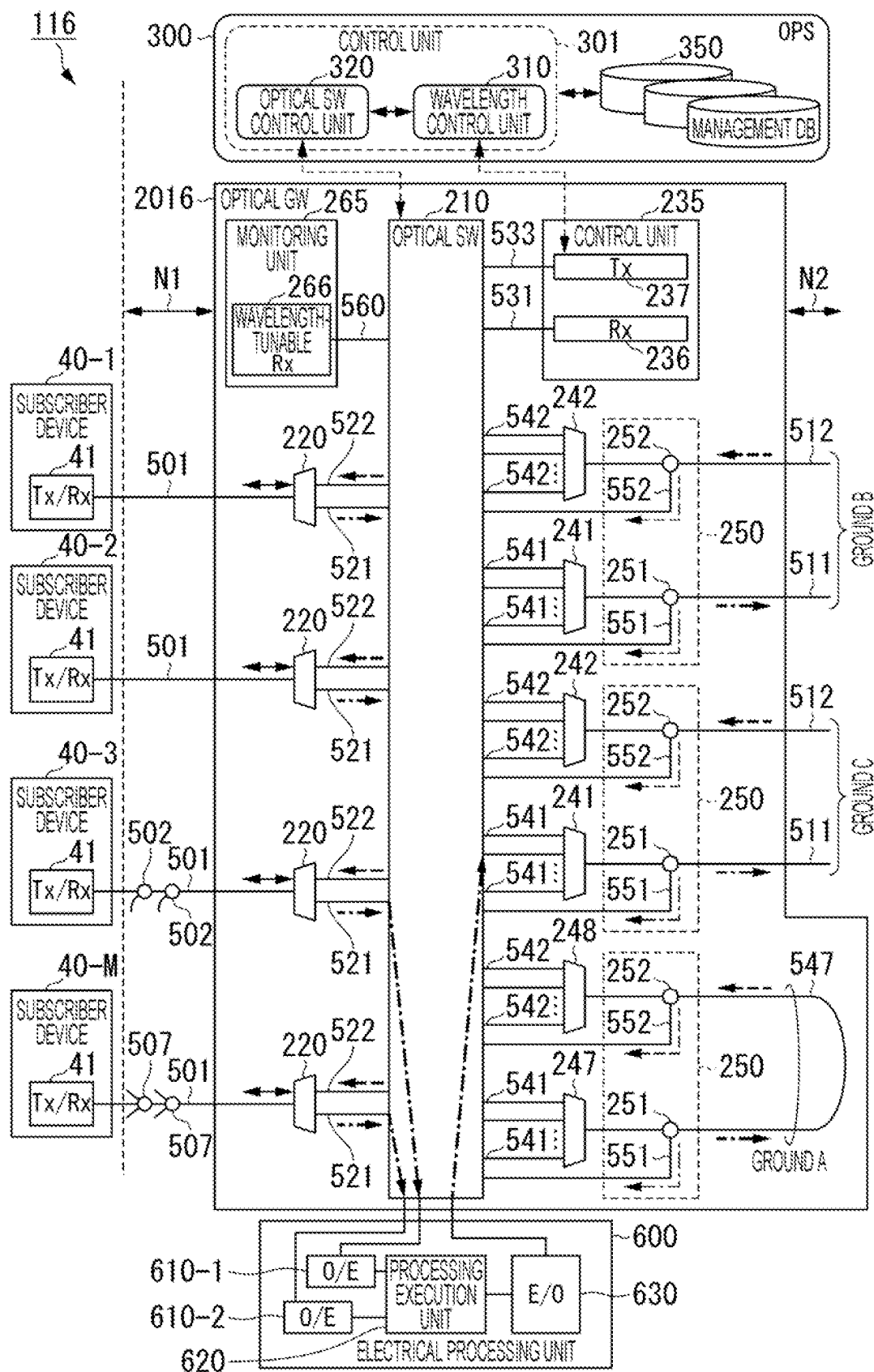


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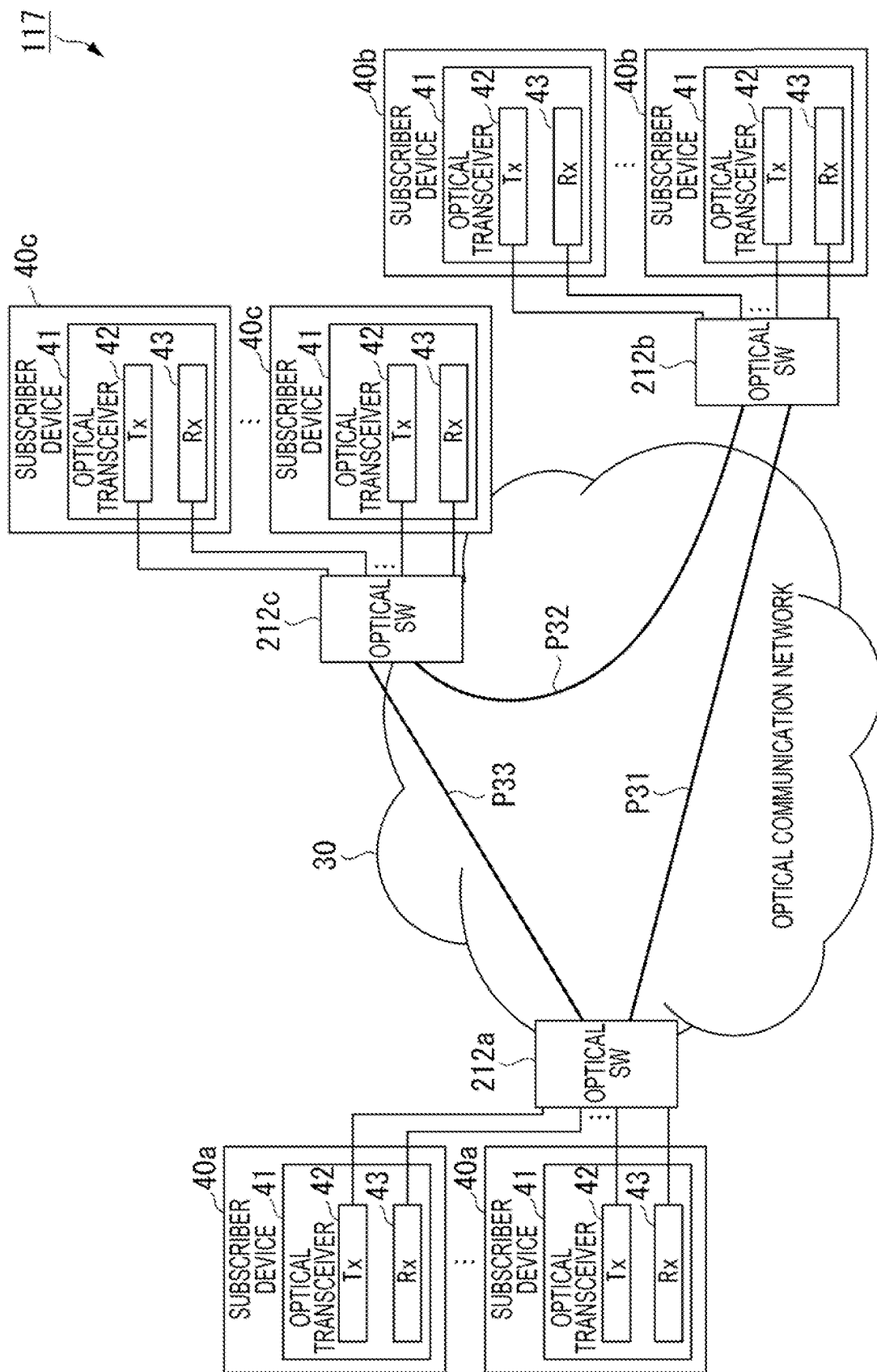


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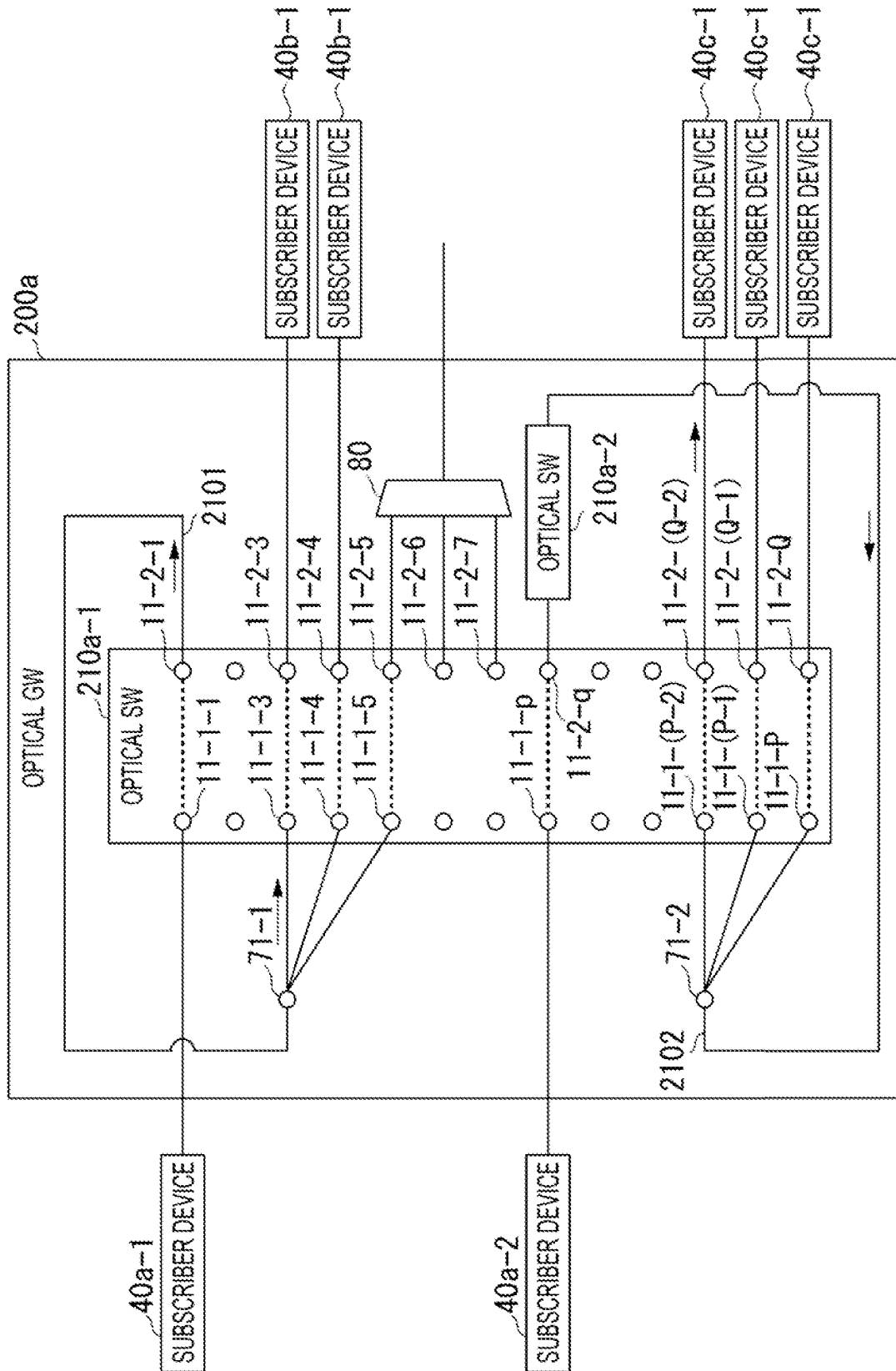


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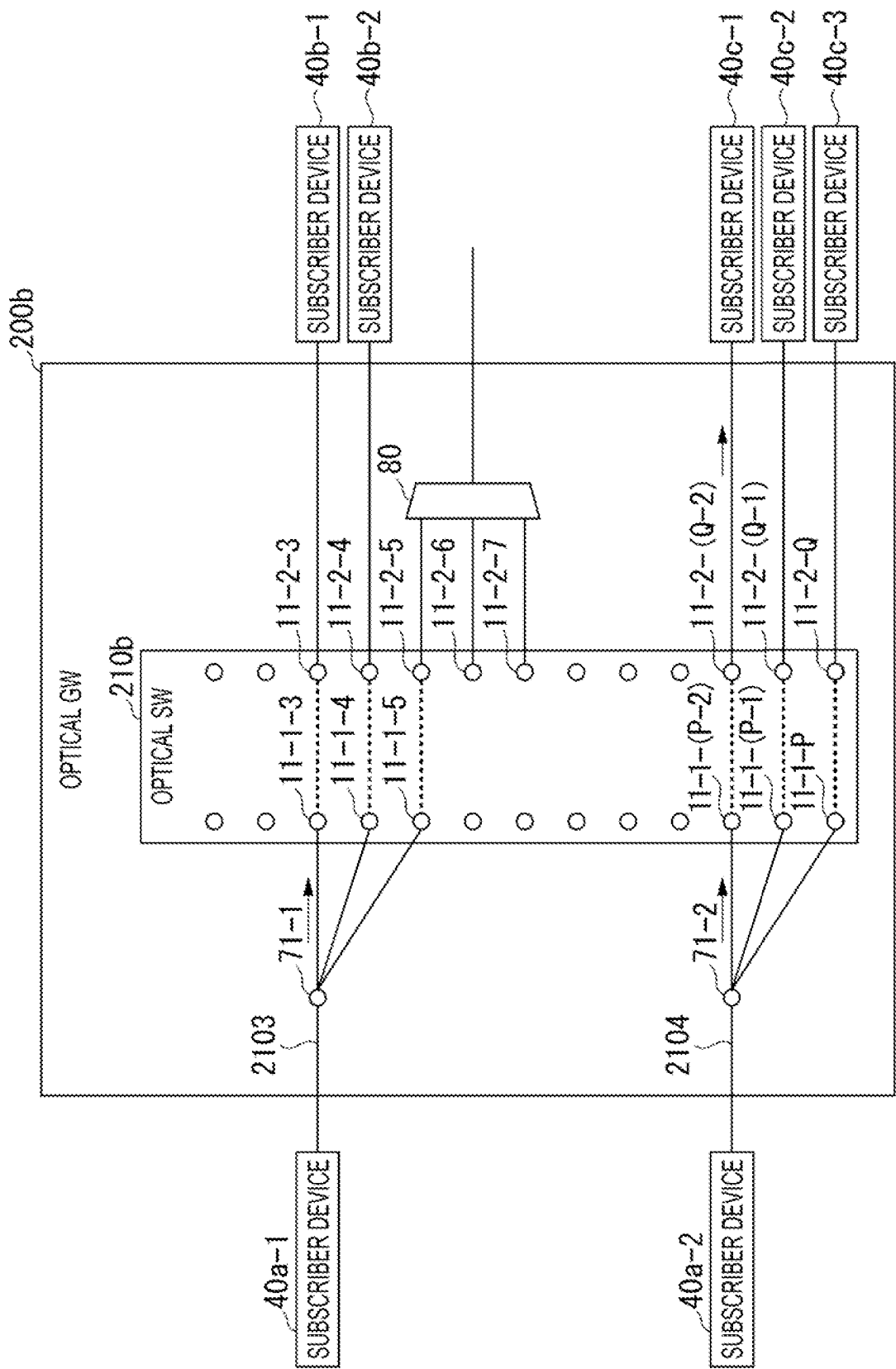


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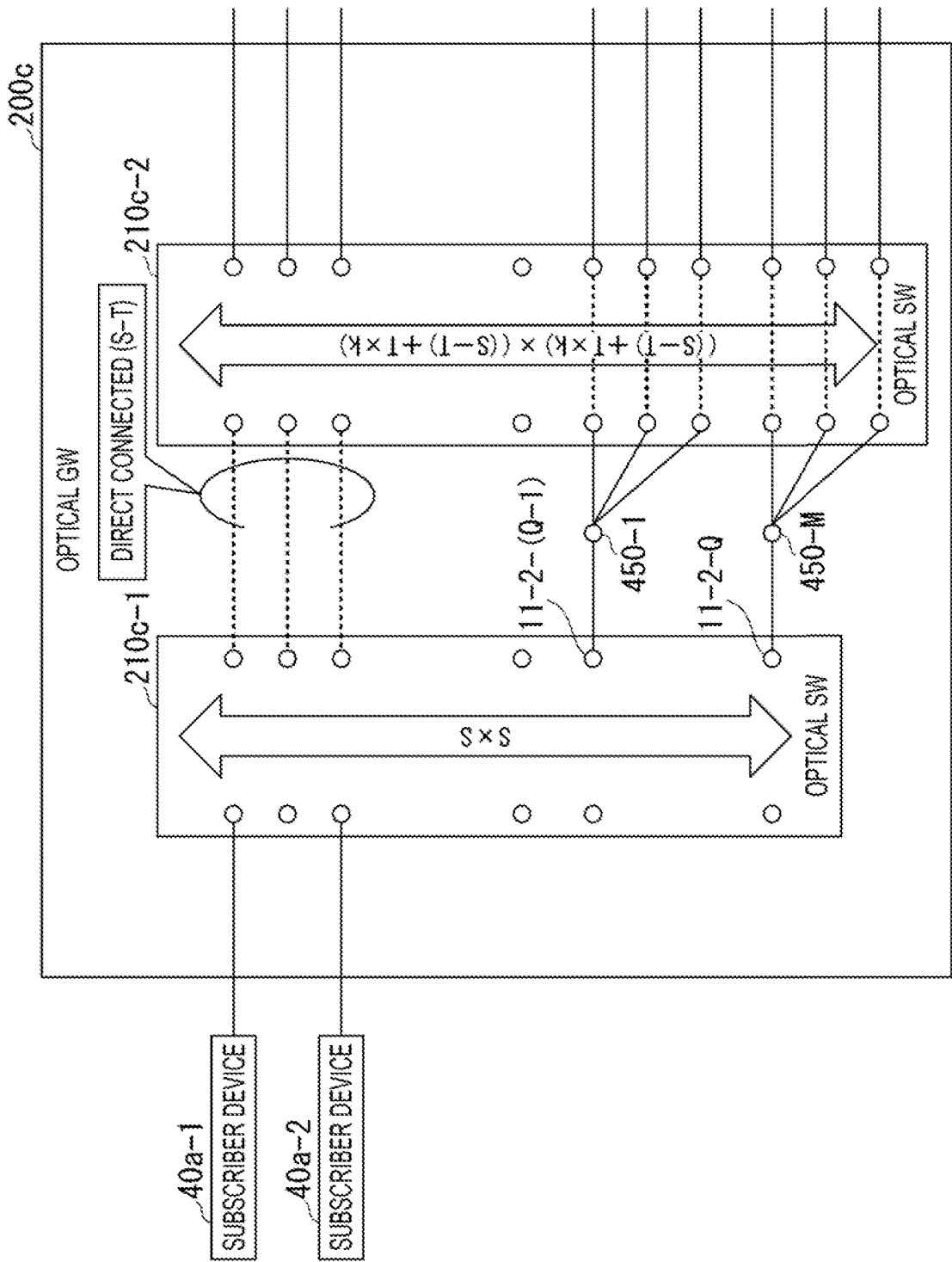


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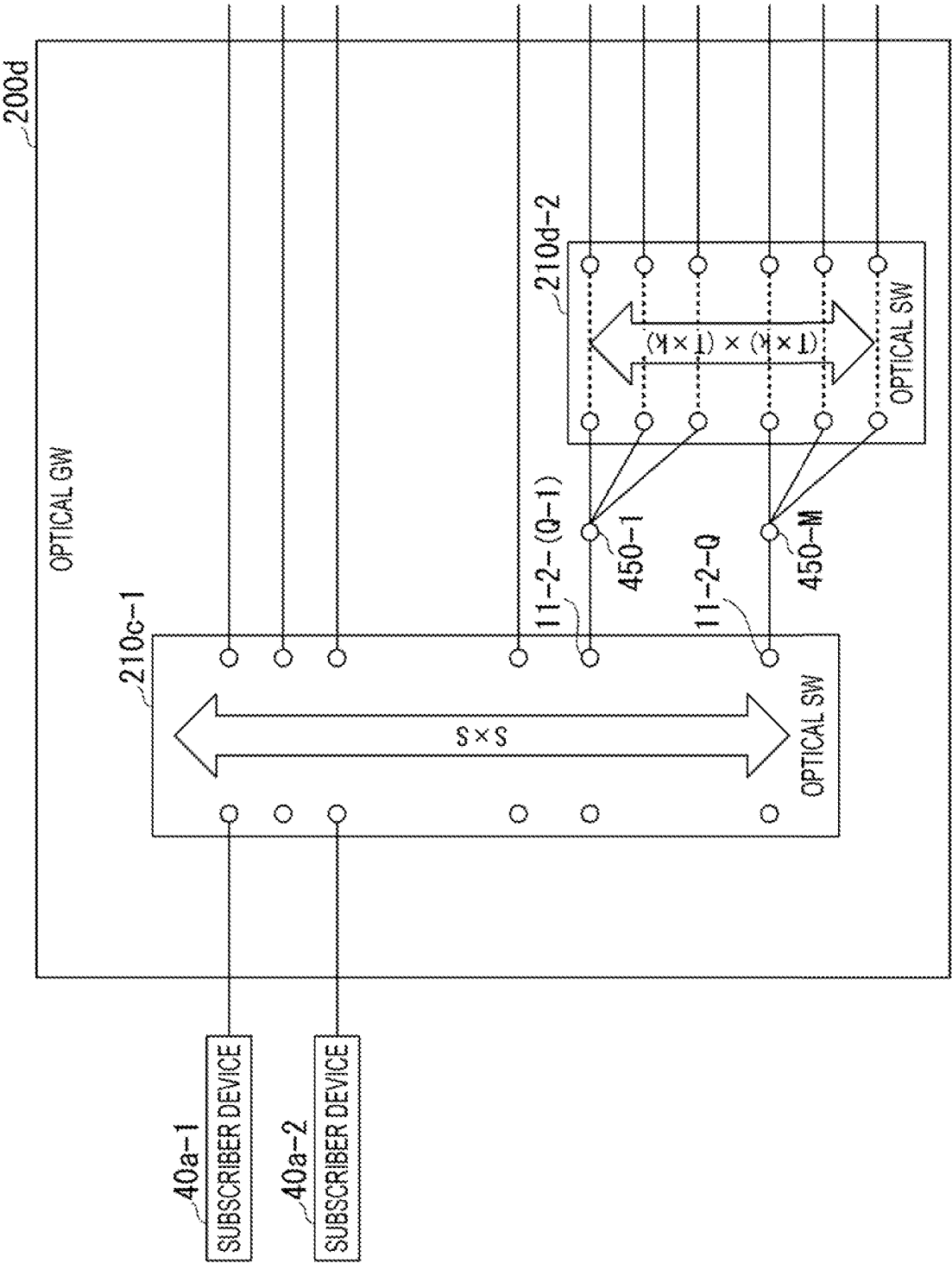


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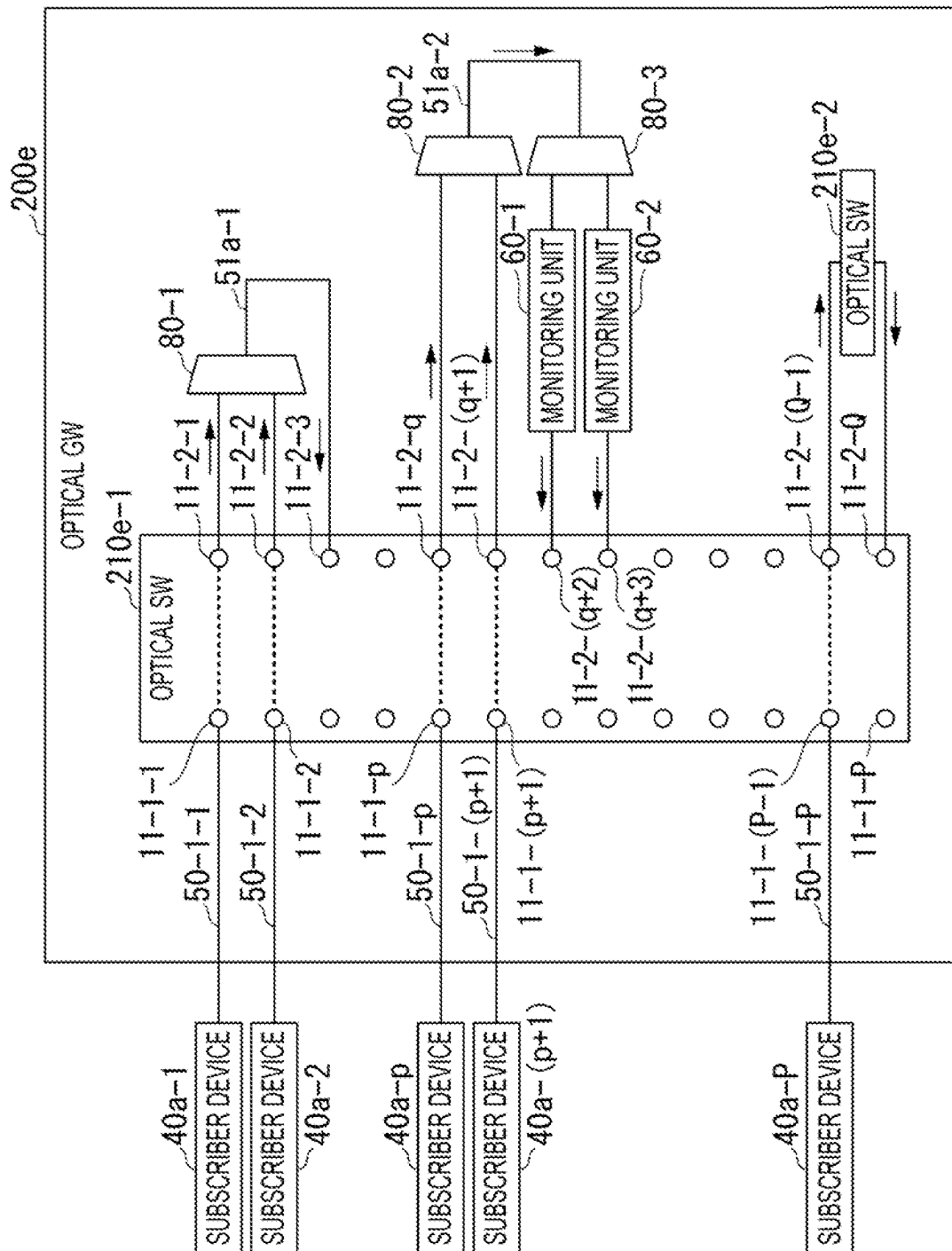


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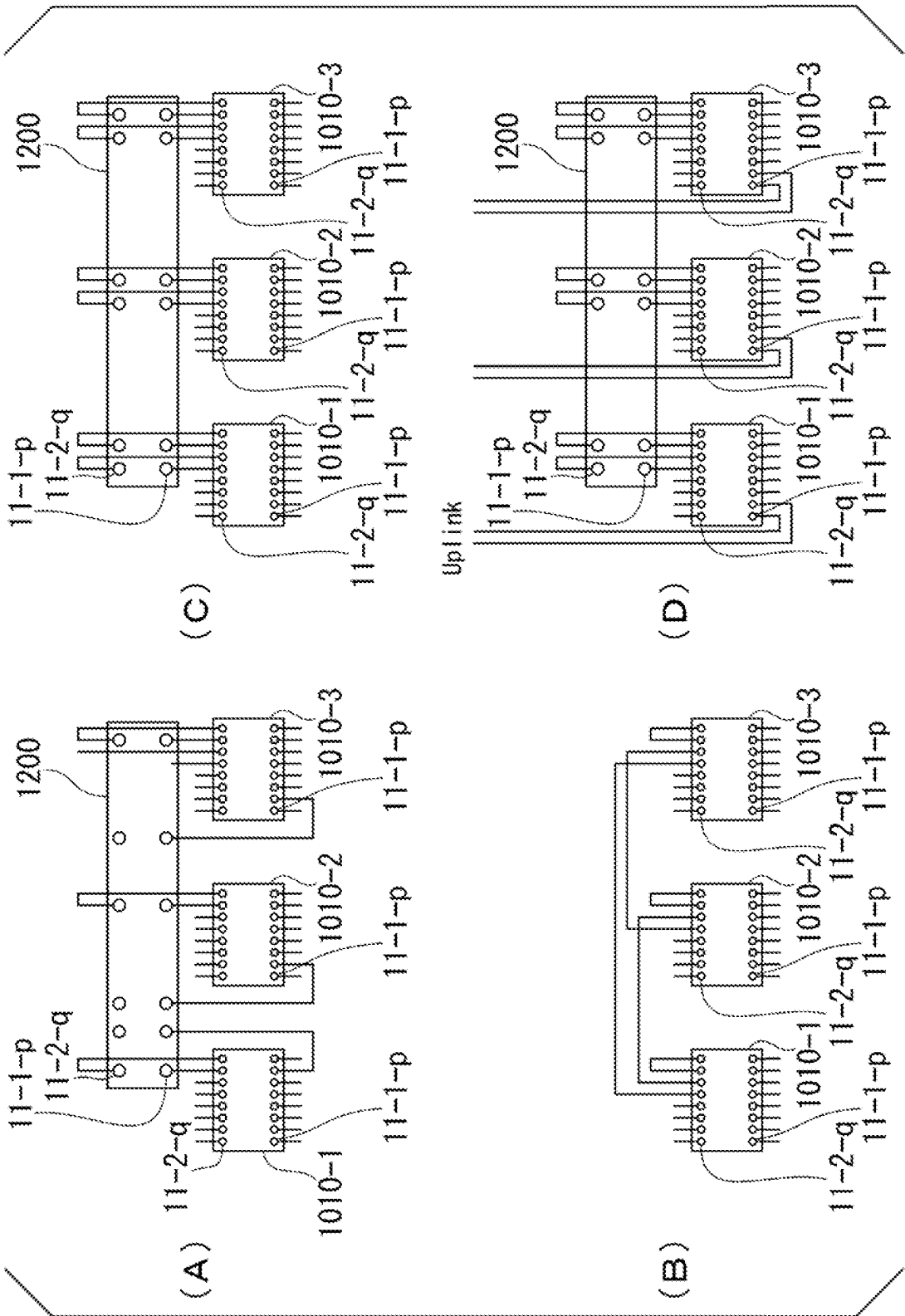


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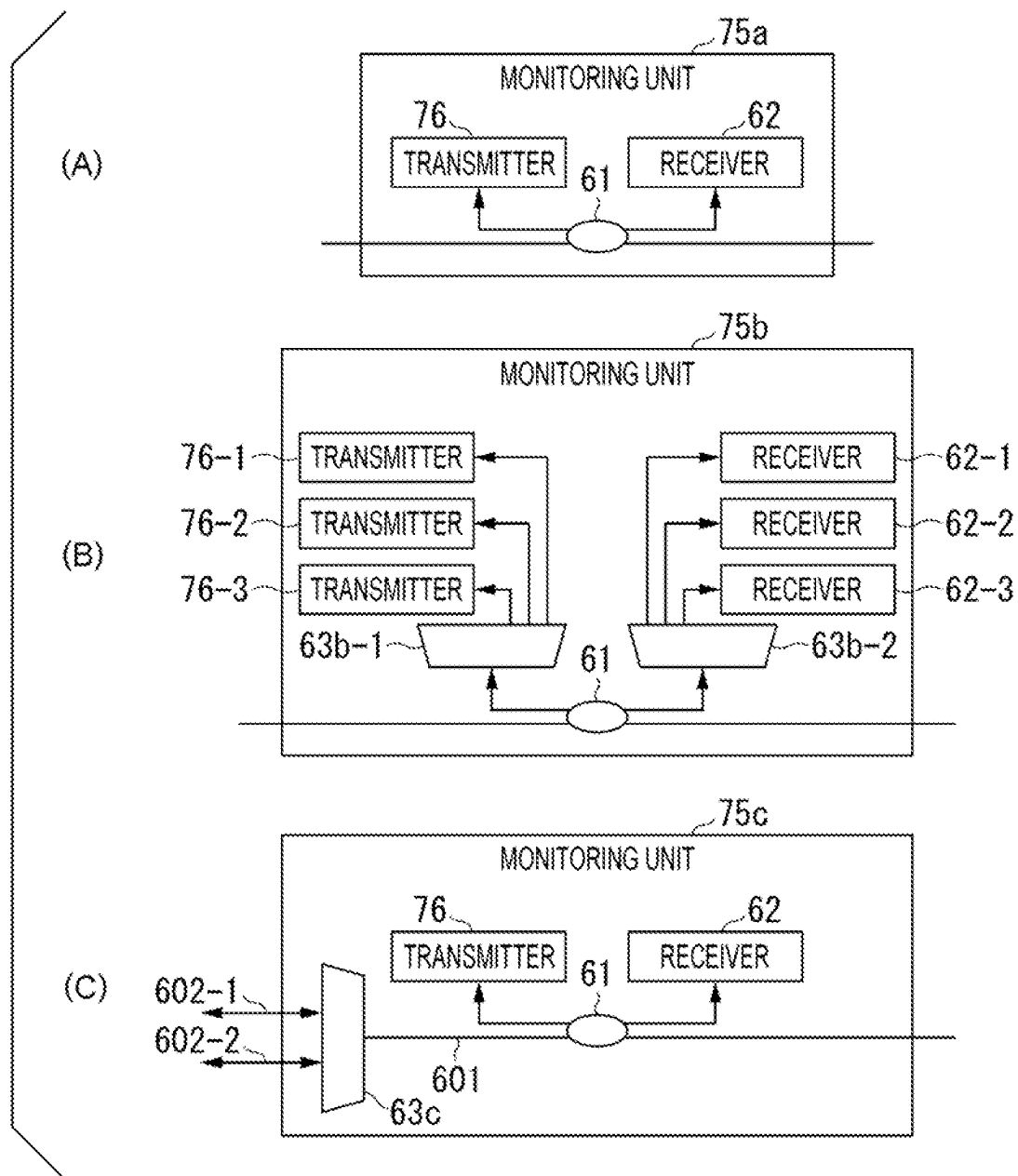


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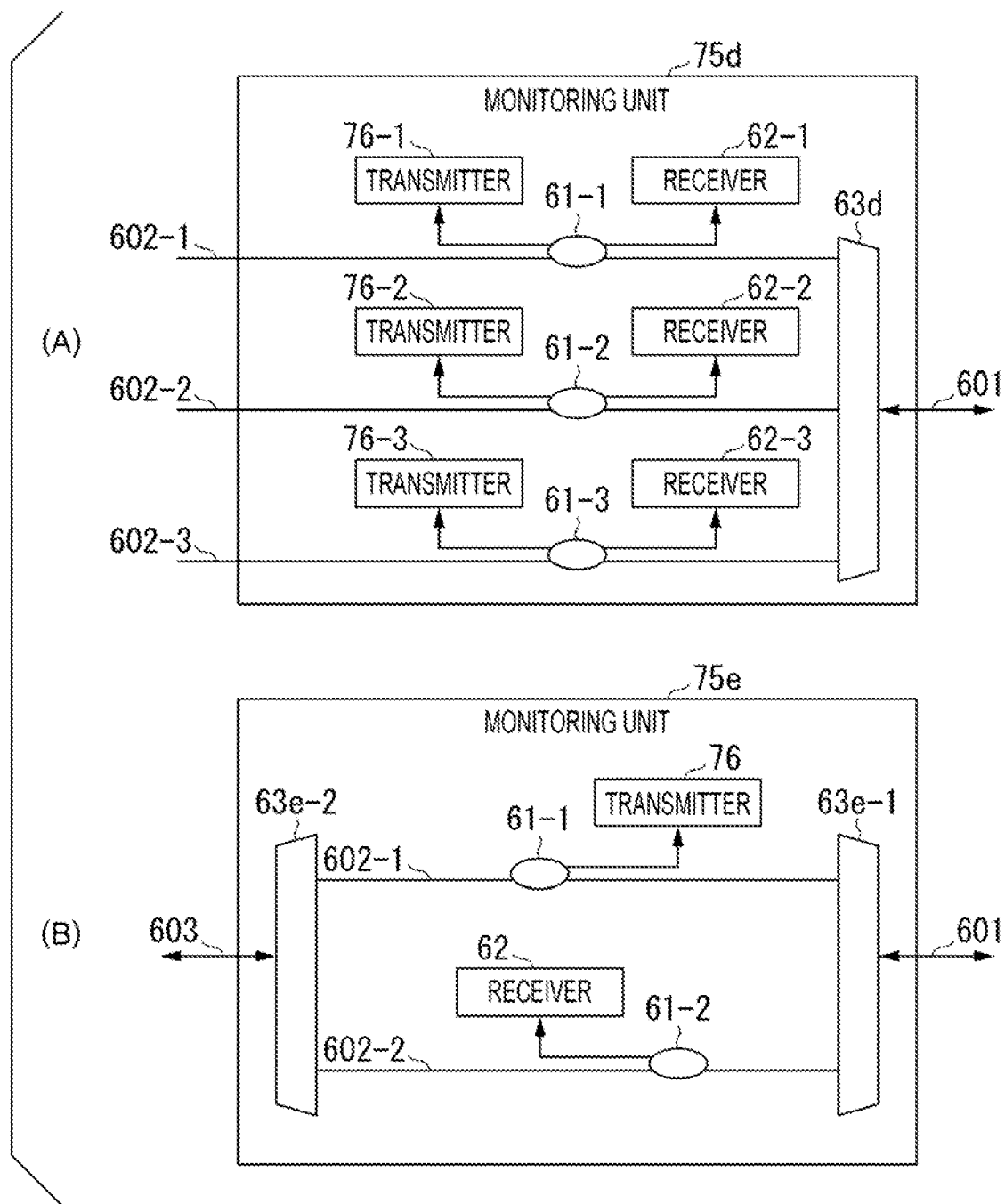


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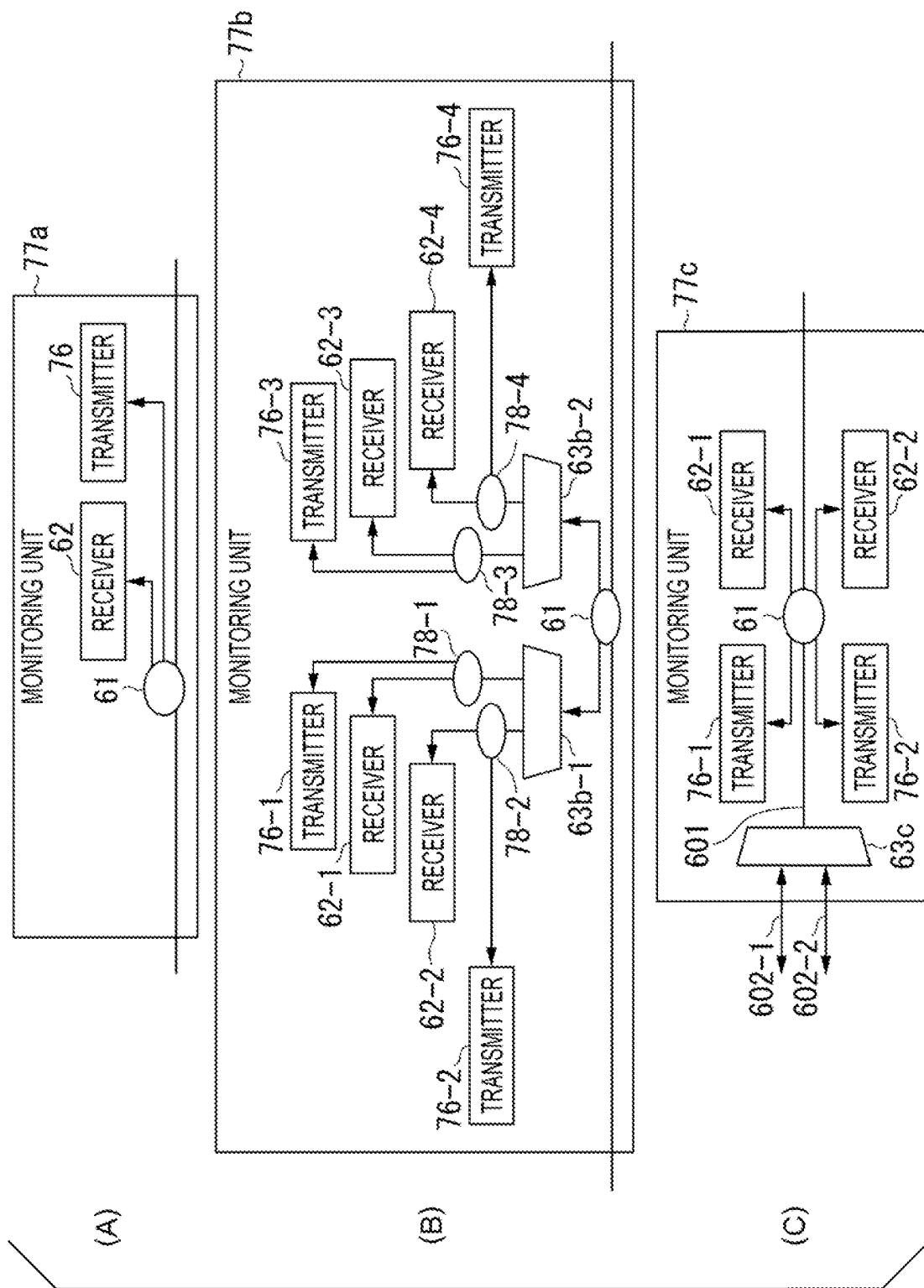


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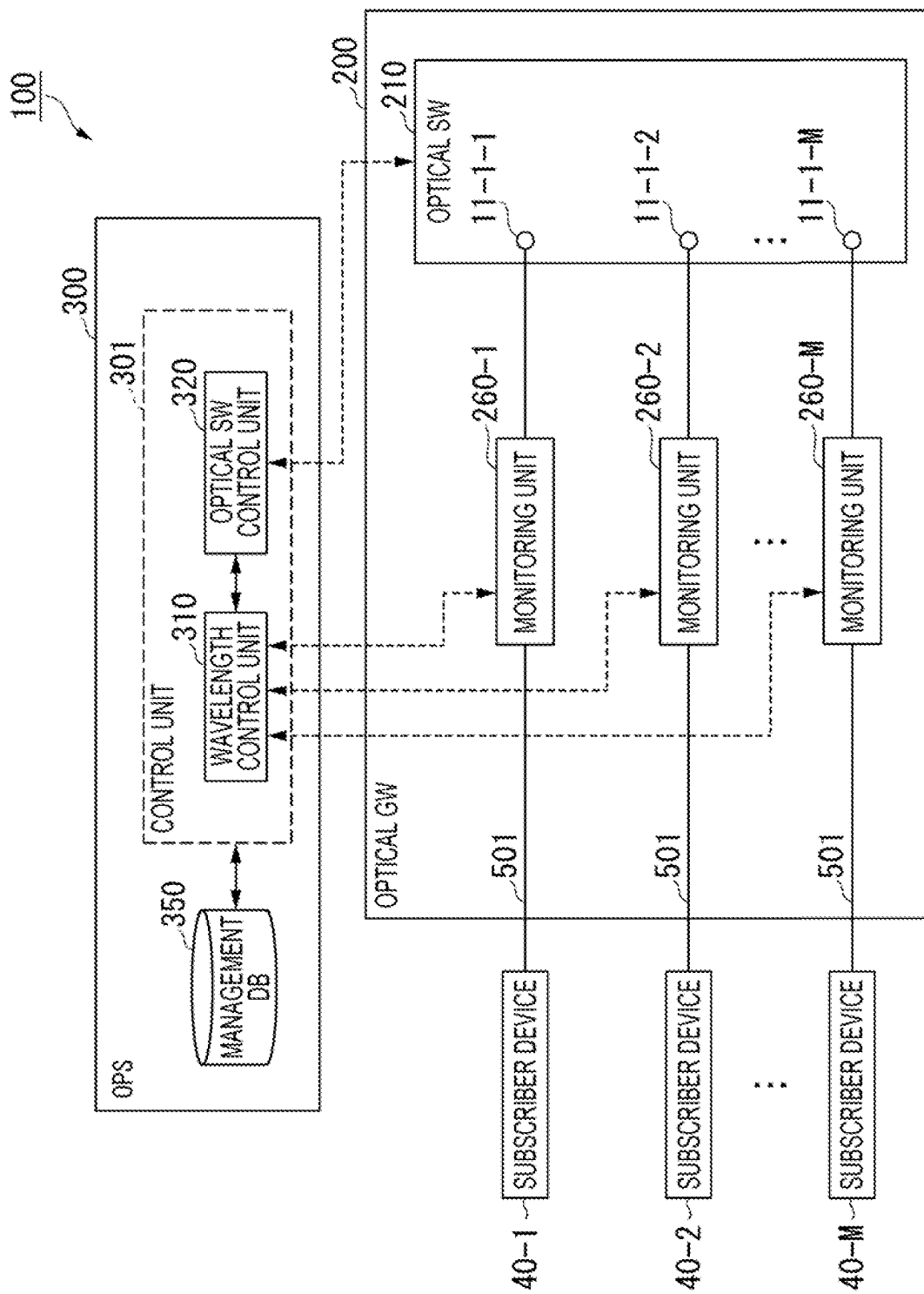


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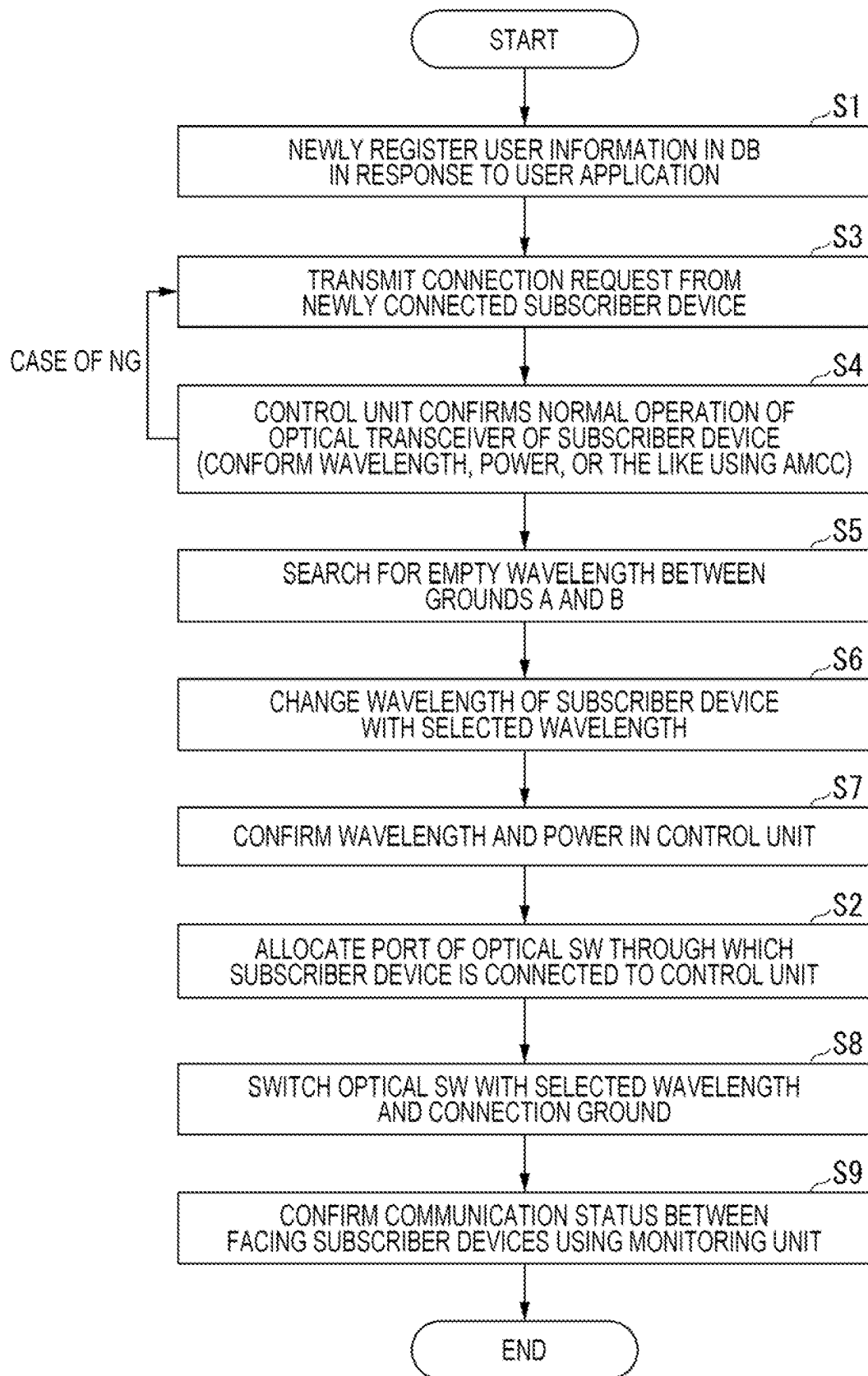


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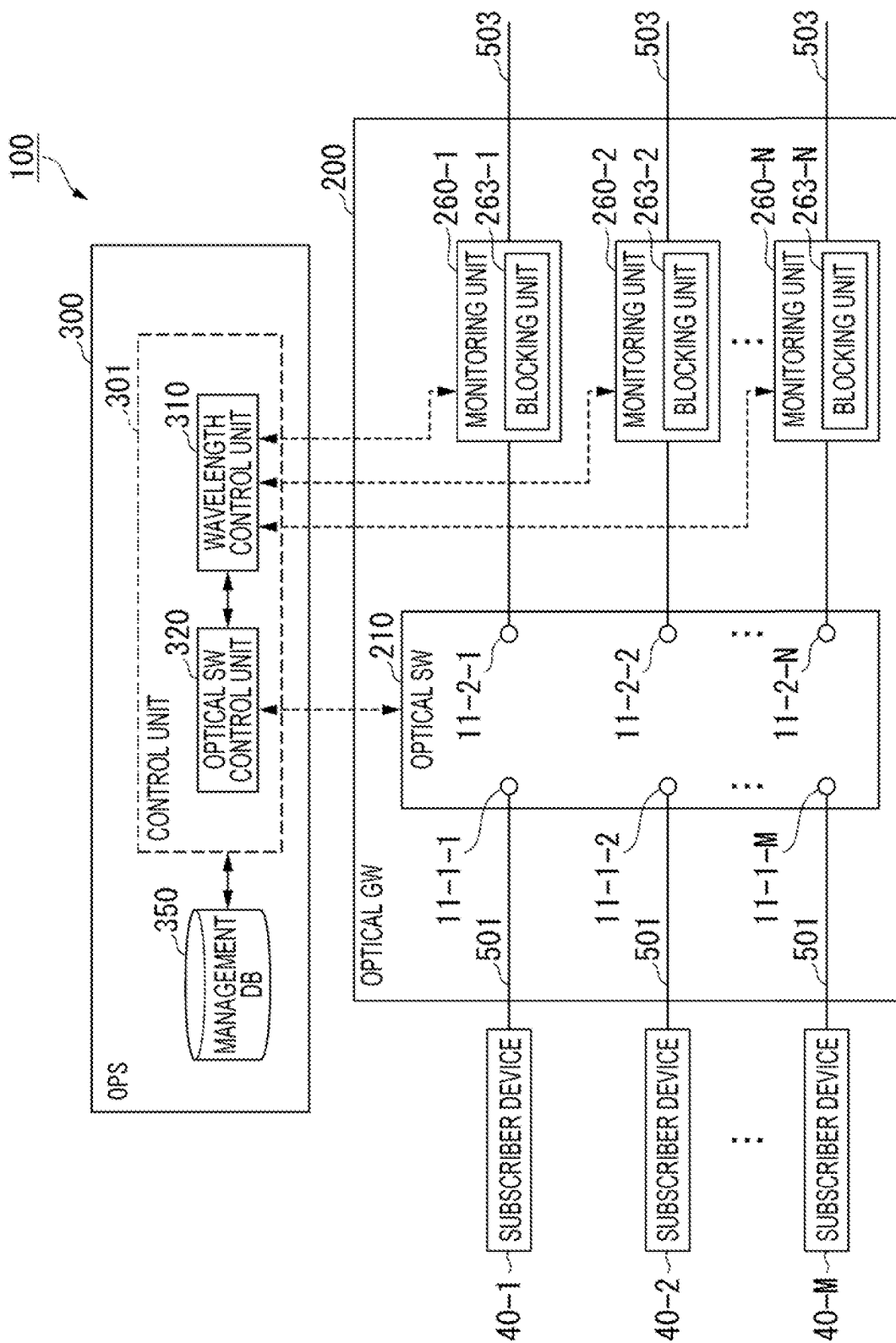


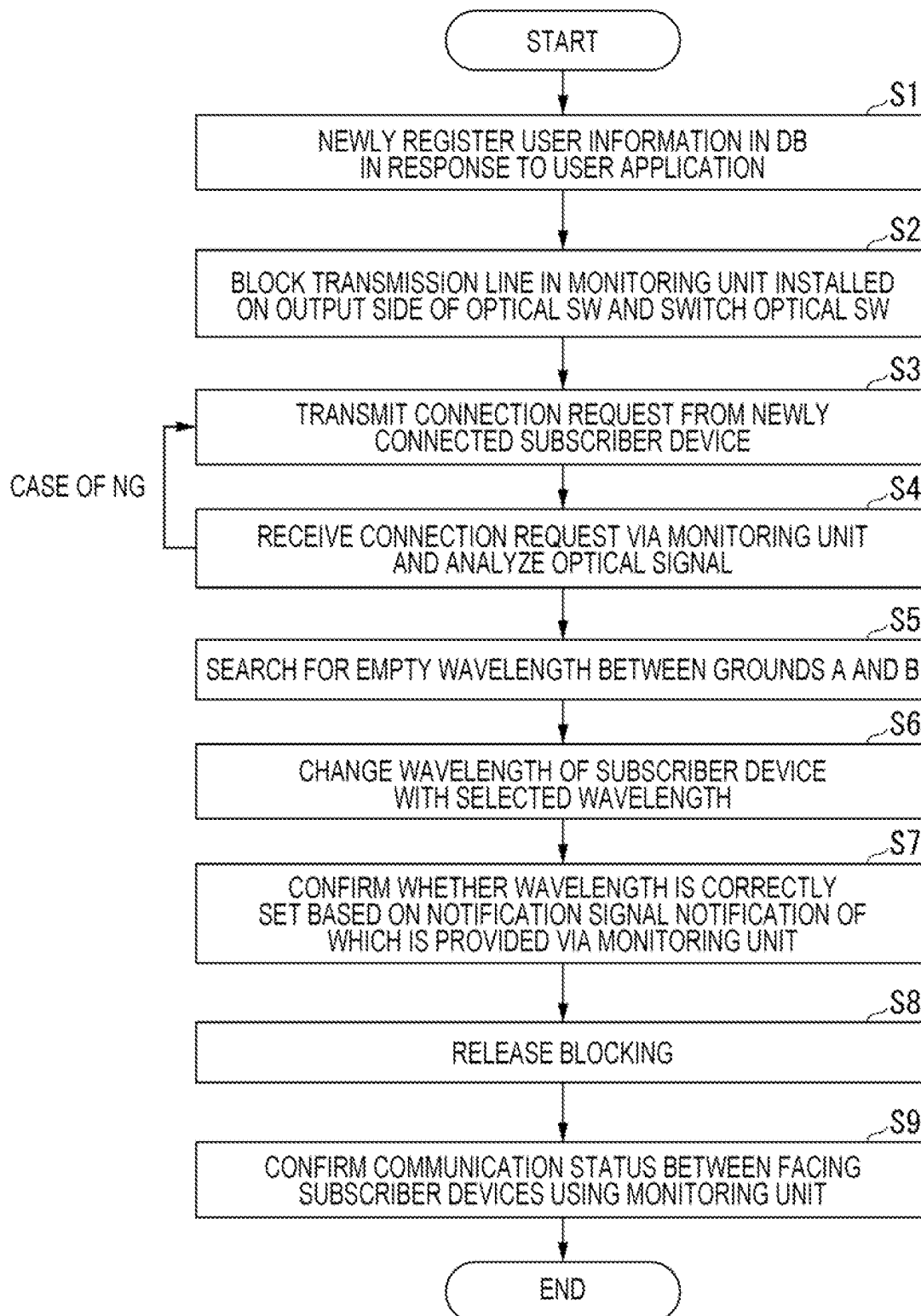
Fig. 73

Fig. 74

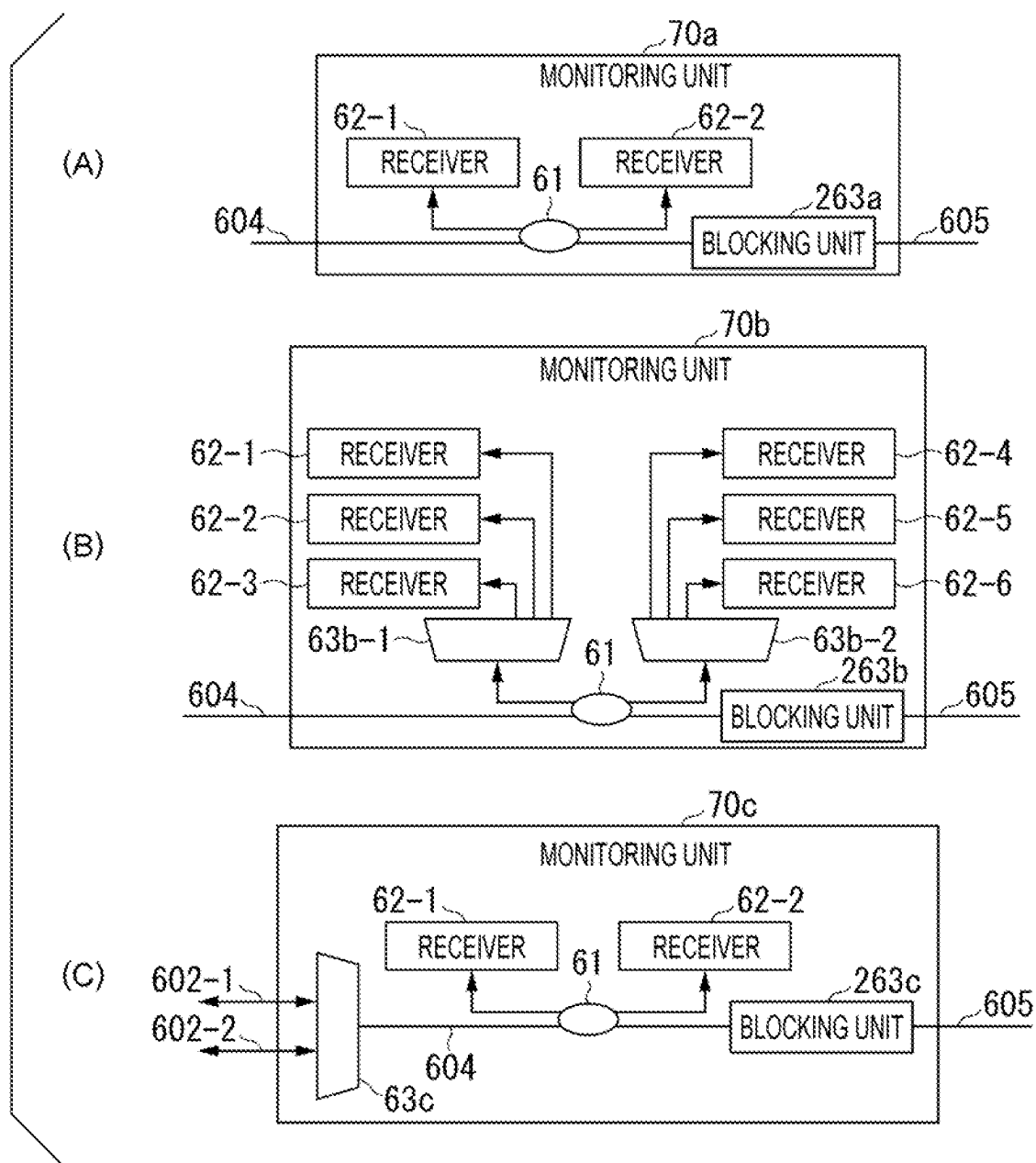


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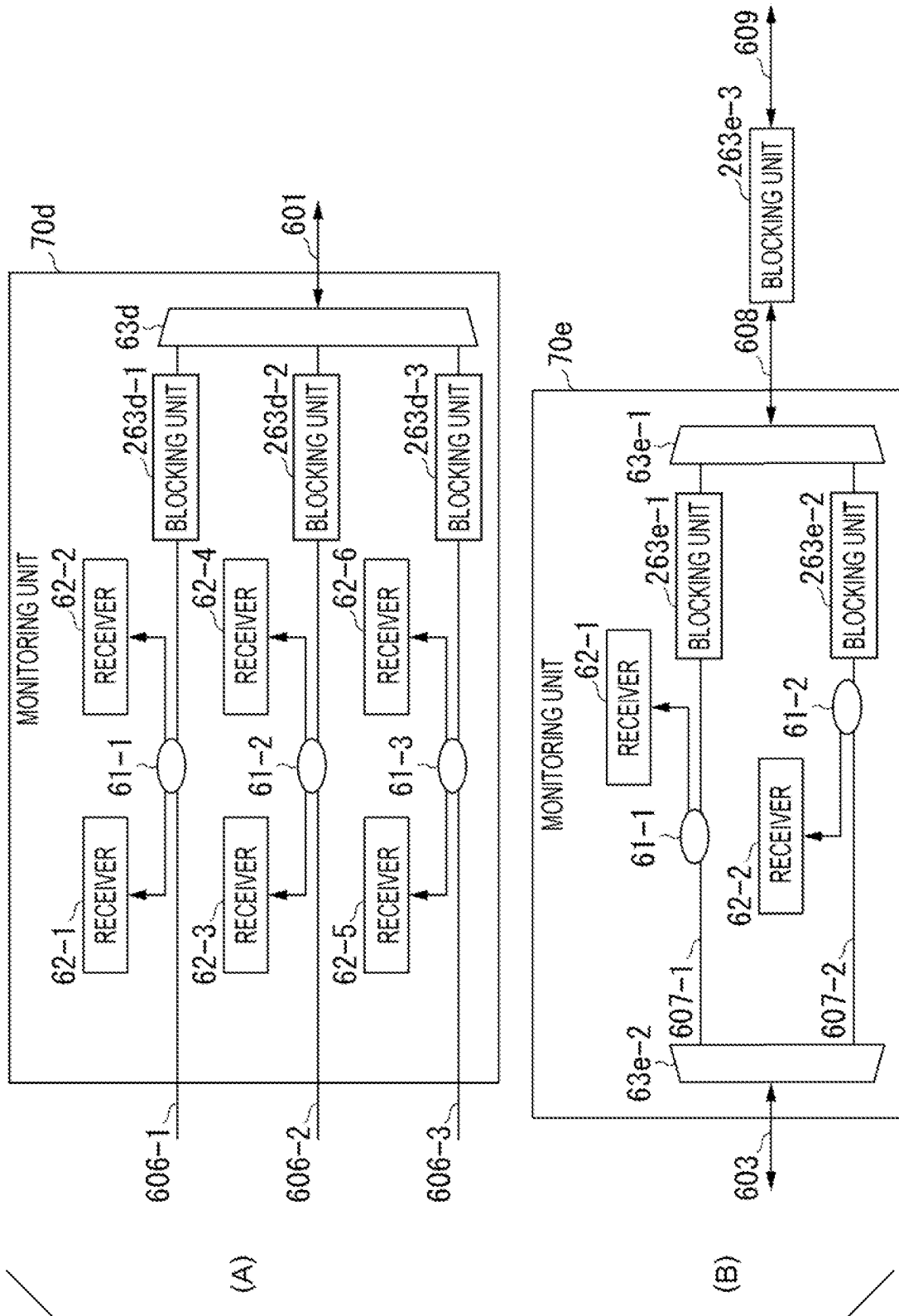


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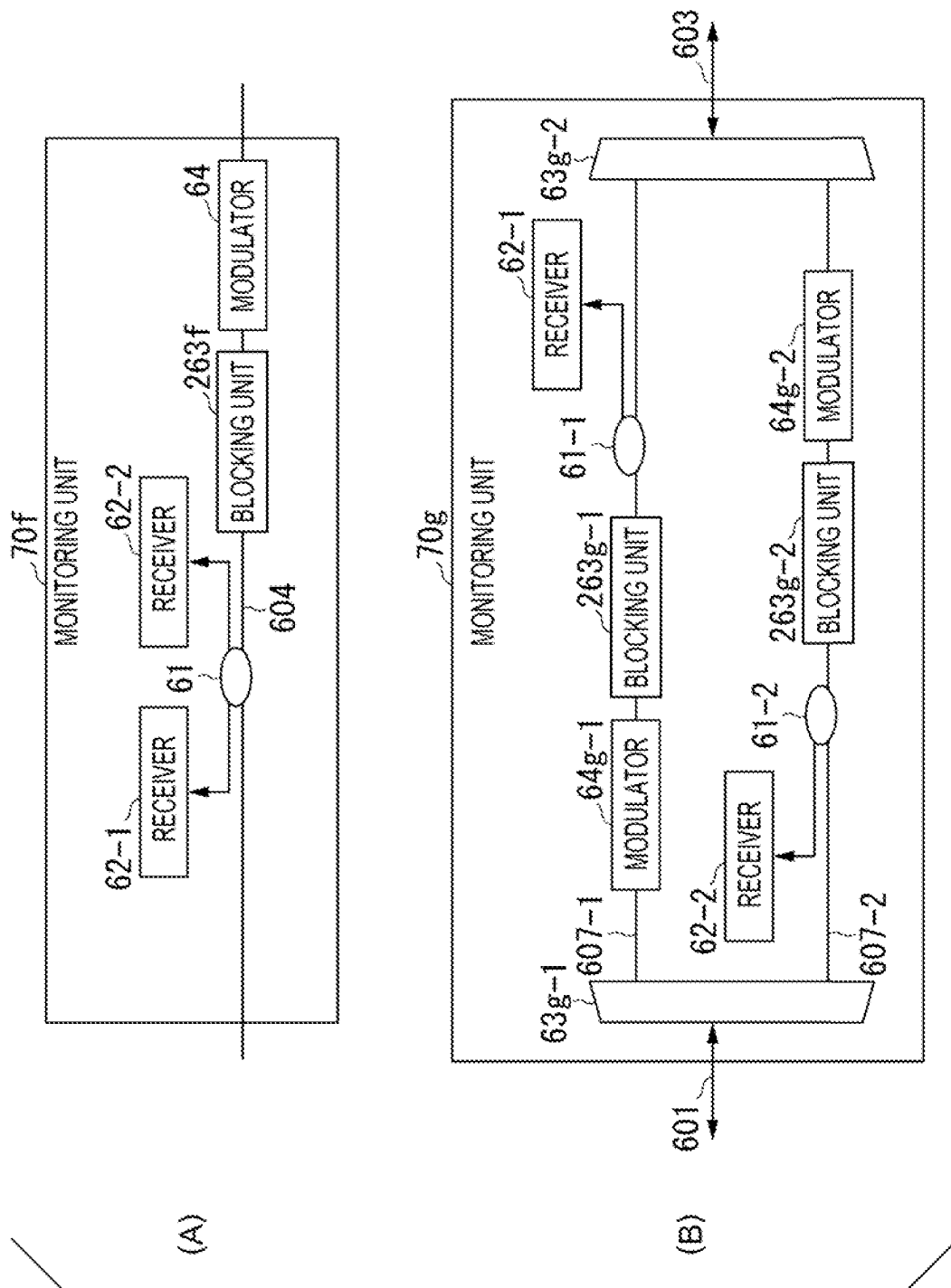


Fig. 77

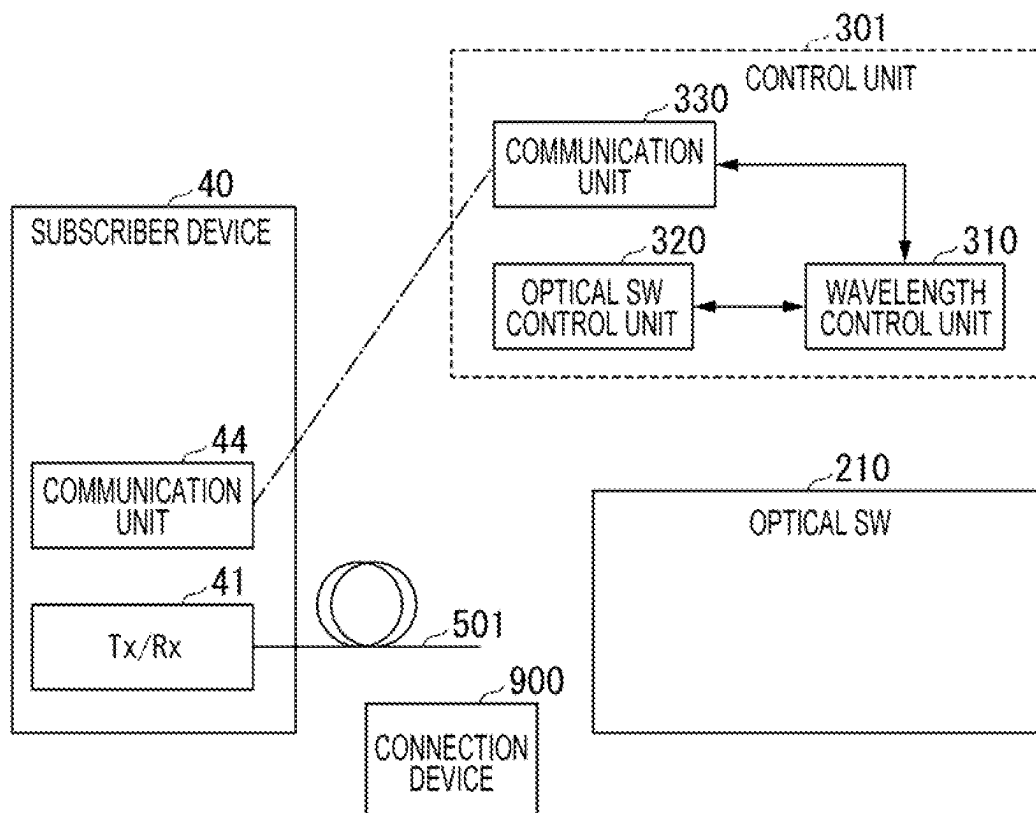


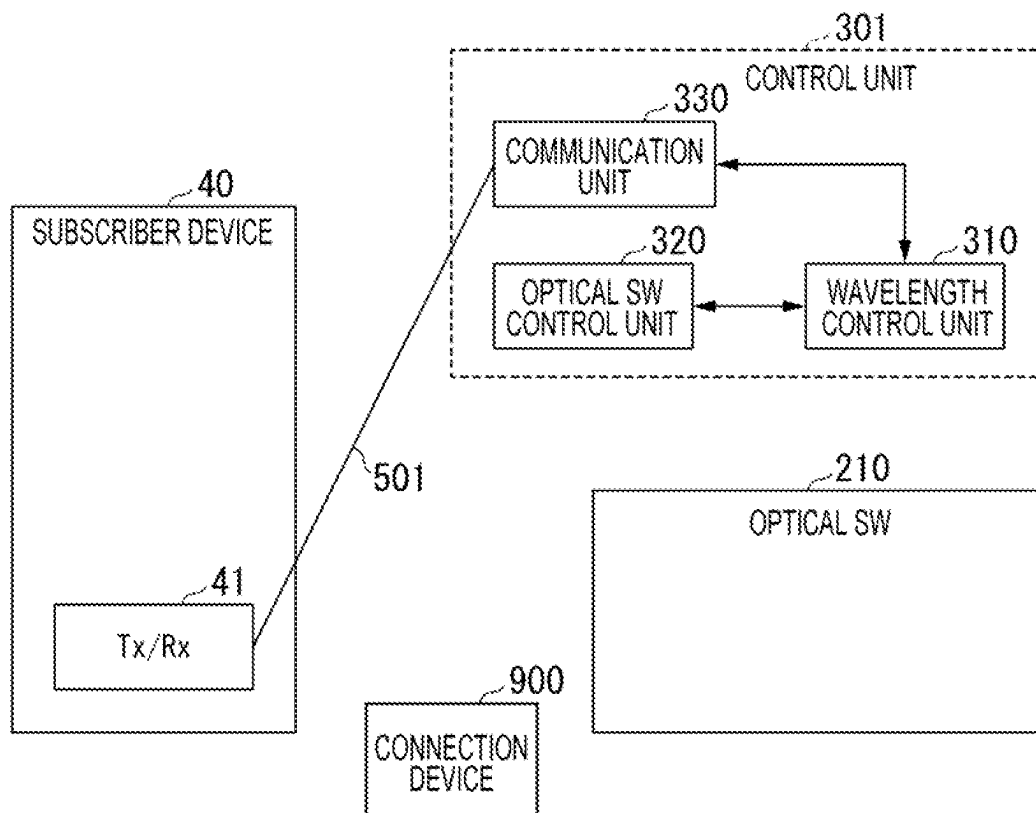
Fig. 78

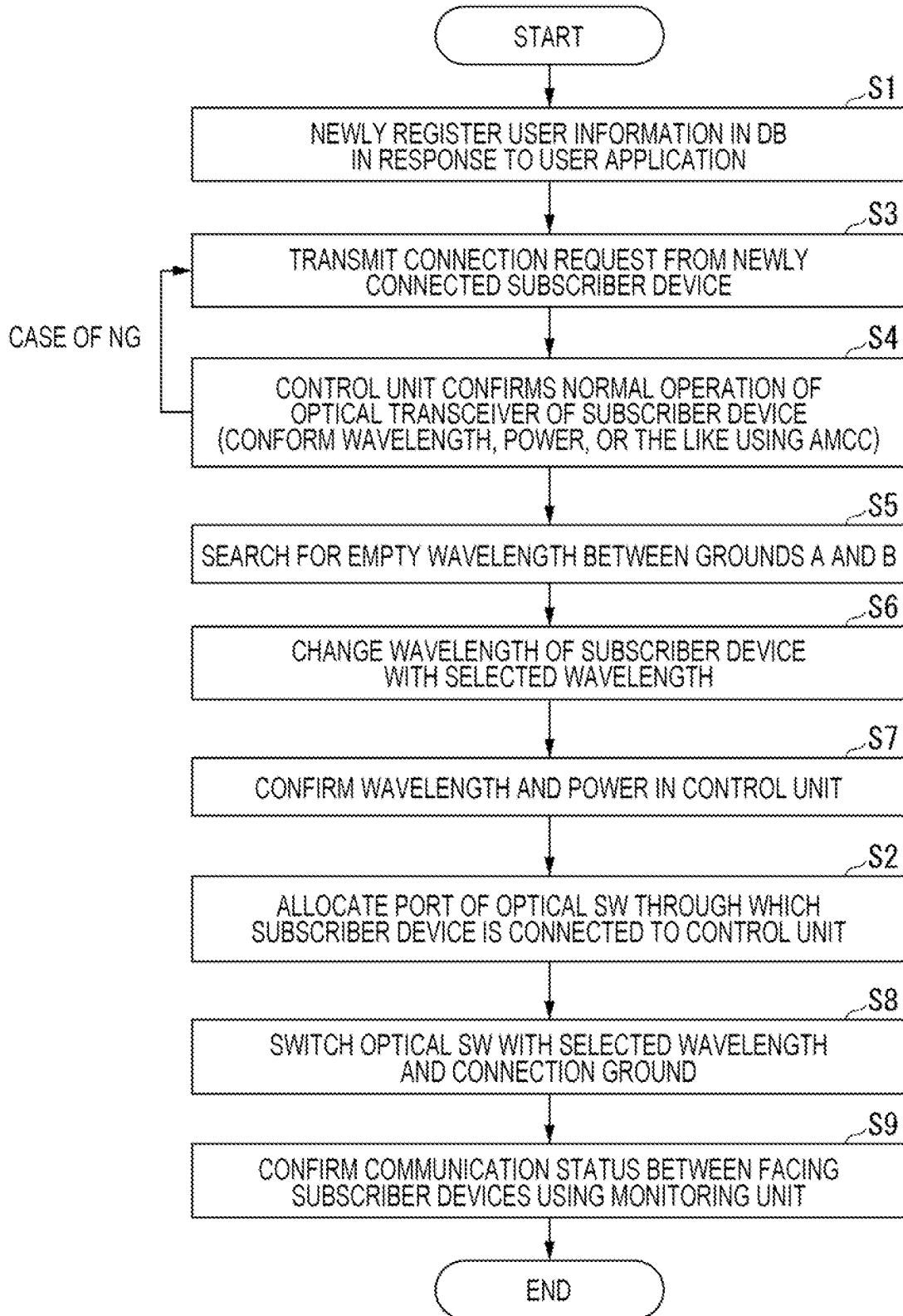
Fig. 79

Fig. 80

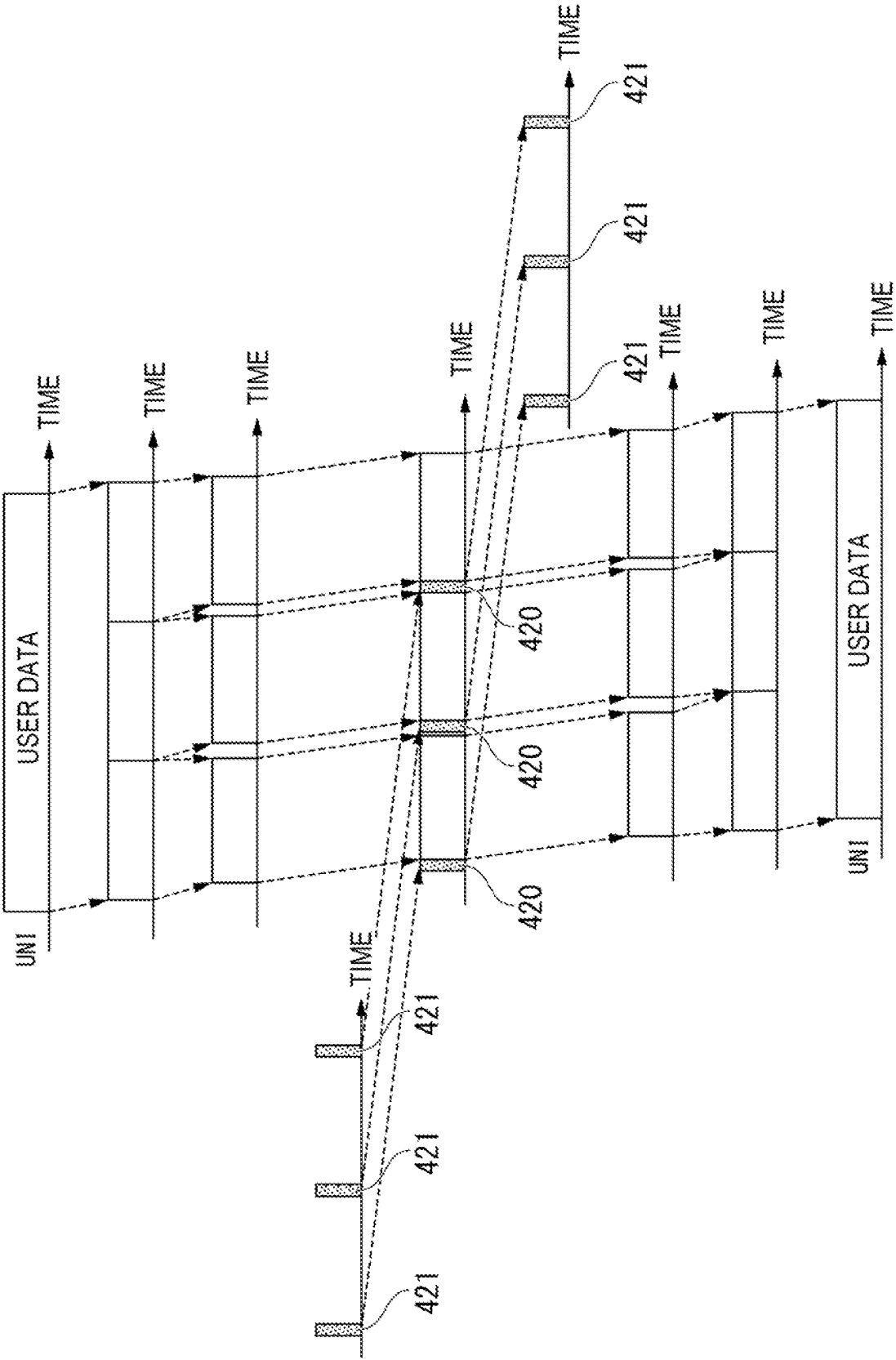


Fig. 81

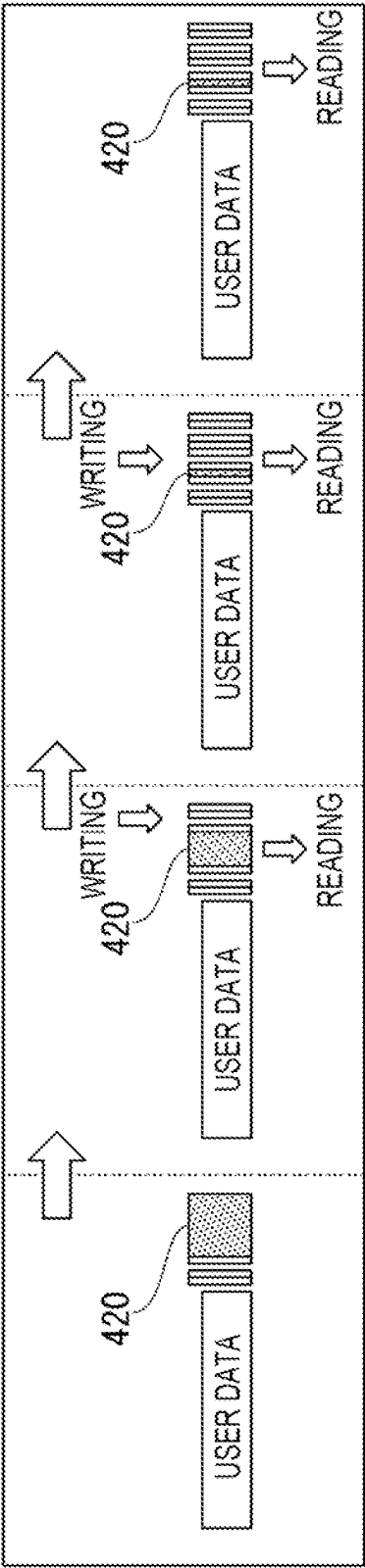


Fig. 82

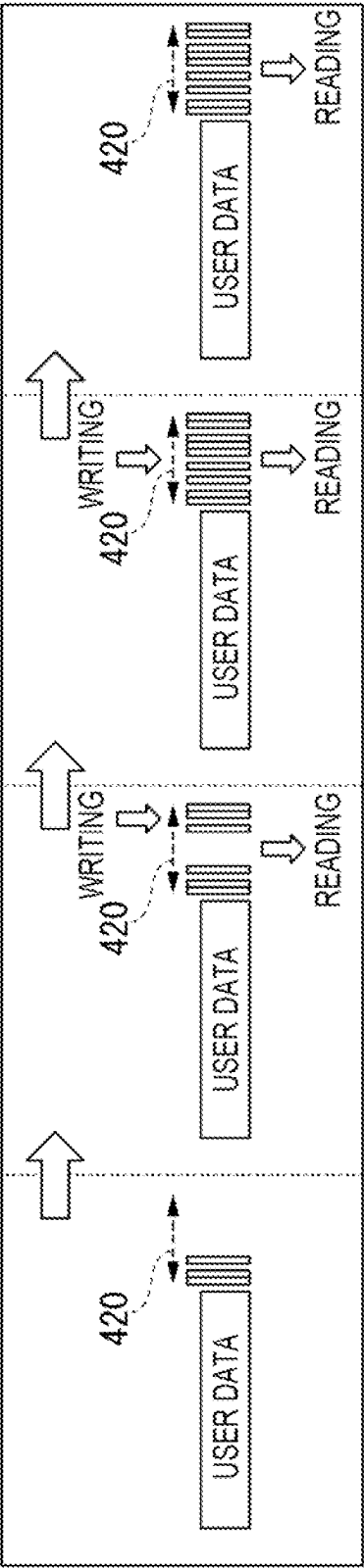


Fig. 83

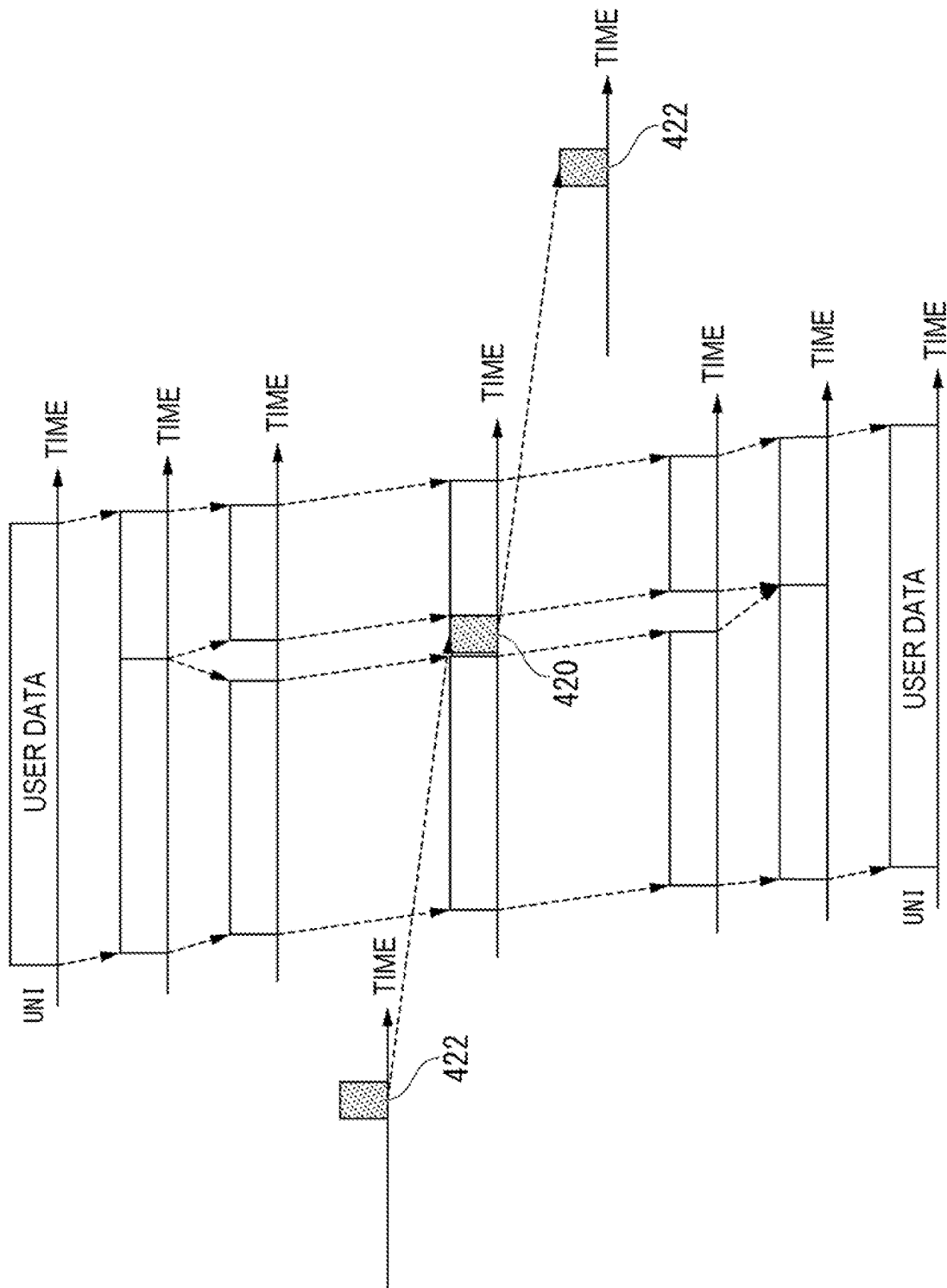


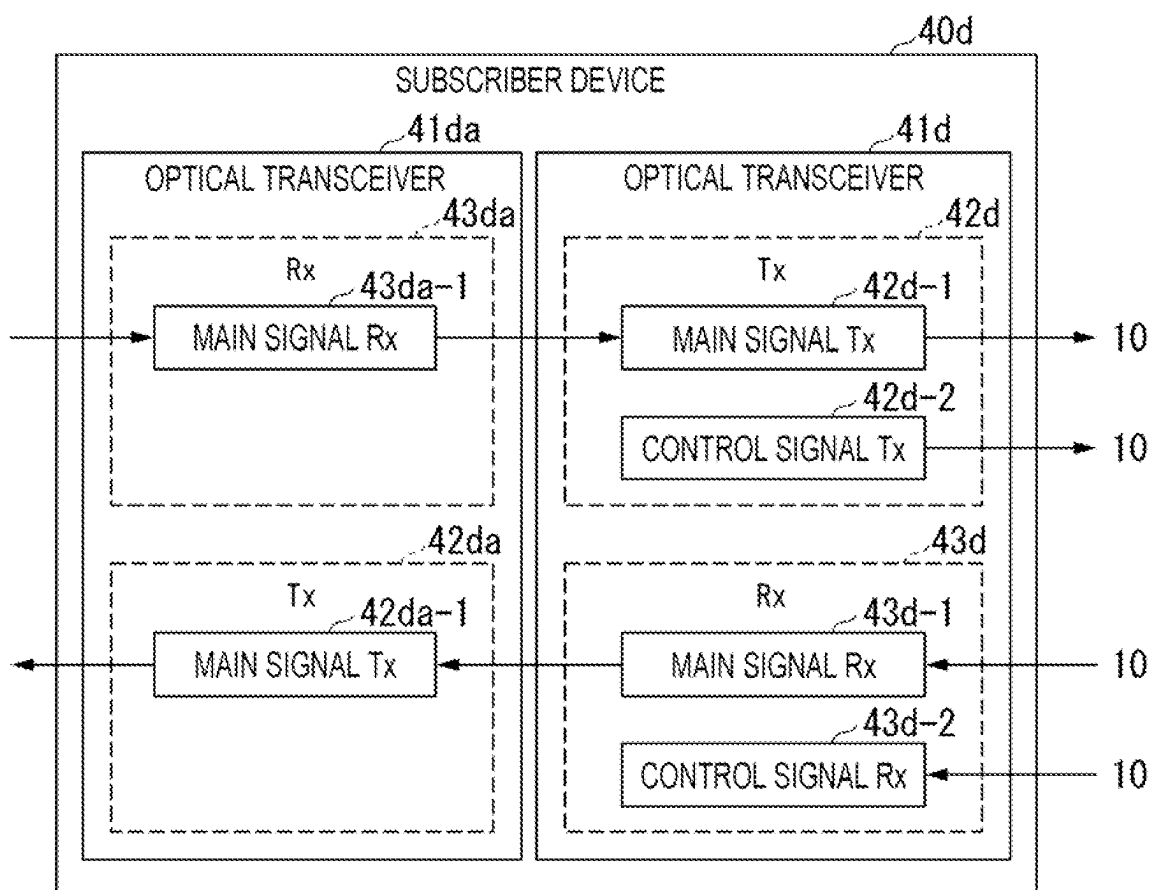
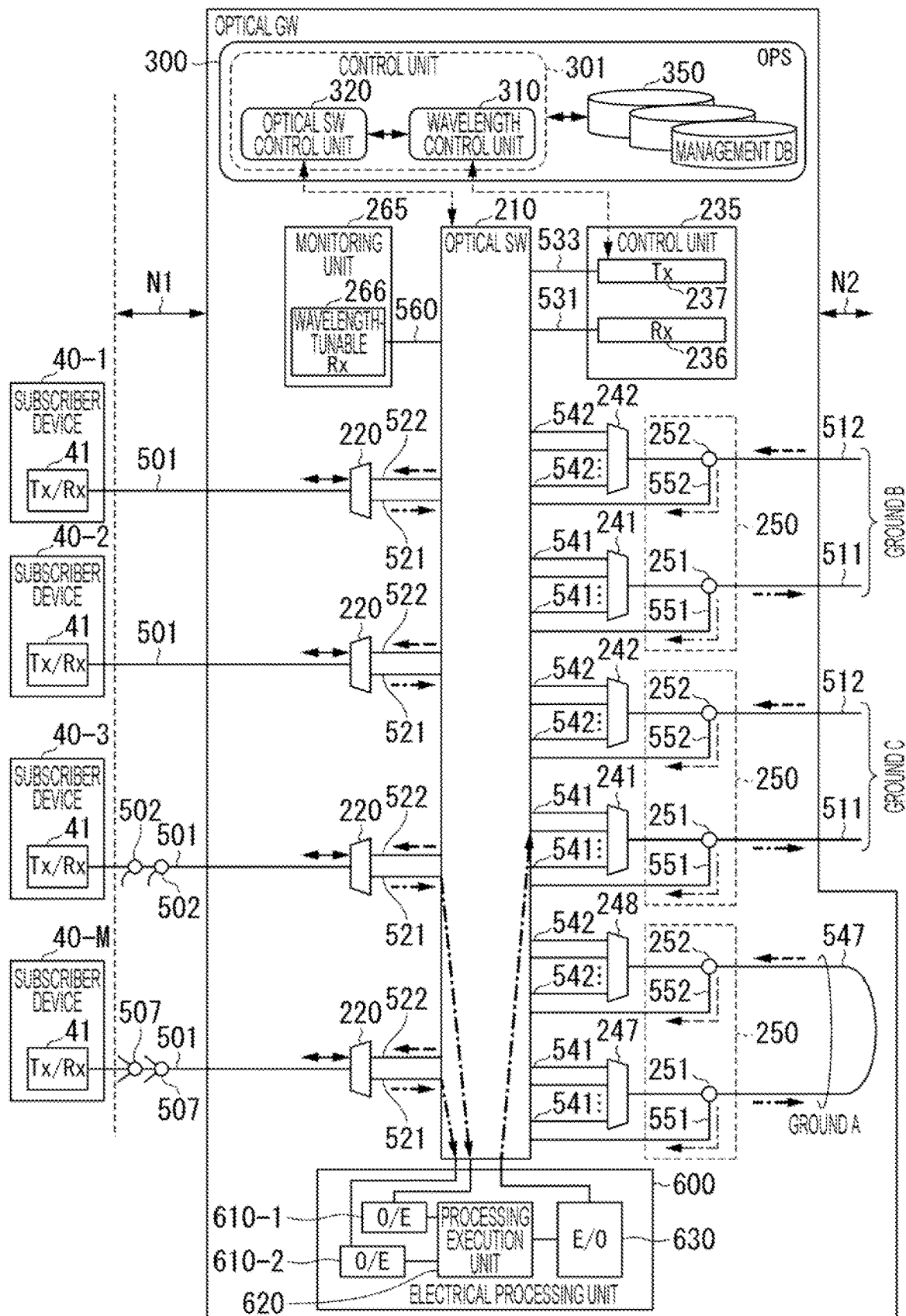
Fig. 84

Fig. 85



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OPTICAL COMMUNICATION APPARATUS, OPTICAL COMMUNICATION SYSTEM AND OPTICAL COMMUNICATION METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a 371 U.S. National Phase of International Application No. PCT/JP2020/040993, filed on Oct. 30, 2020. The entire disclosure of the above application is incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to an optical communication device, an optical communication system, and an optical communication method.

BACKGROUND ART

A network that provides a high-speed Internet by fiber to the home (FTTH) or a mobile service is inefficient in terms of administration because the network is independently constructed for each service. Accordingly, an optical access system that accommodates a plurality of services in one device has been proposed (see, for example, Non-Patent Literature 1). Further, in order to realize an optical access system capable of accommodating multiple services, passive optical network (PON) and wavelength division multiplexing (WDM)-PON using a plurality of wavelengths have been standardized by International Telecommunication Union-Telecommunication Standardization Sector (ITU-T) (see, for example, Non-Patent Literature 2).

On the other hand, an existing optical access system includes a subscriber-side device and a station-side terminal device. Communication between these devices is connected to a host network such as a core network via the terminal device. The subscriber-side device is, for example, an optical network unit (ONU). Connection to the core network is made via a terminal device in the station-side device. The terminal device is, for example, an optical line terminal (OLT). In optical access connected to a core network by packet switching, a process of adding or deleting user information and destination information to or from a signal, a routing process, and the like are performed. When user information and destination information are added or deleted, an optical signal may be converted into an electrical signal temporarily in some cases. In this case, certain delay occurs in communication. Further, when the amount of data increases, signals are stored in a buffer, and priority control or the like may be performed in some cases. Accordingly, the delay further increases. When the delay increases, quality of an optical service greatly deteriorates. Thus, it is important to reduce the delay as much as possible.

To improve quality of an optical service and provide various services in an optical access network, delays that occur need to be reduced. By using an optical switch or the like capable of performing a process such as routing without converting an optical signal into an electrical signal, it is possible to considerably reduce delay.

CITATION LIST

Non Patent Literature

Non Patent Literature 1: "ITU-T G.983.3", International Telecommunication Union, 2001

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Non Patent Literature 2: "ITU-T G.989.1", International Telecommunication Union, 2013

SUMMARY OF INVENTION

Technical Problem

When routing in which an optical switch is used is applied to optical access, it is preferable to reduce the number of ports used for the routing in order to connect many devices. It is preferable to relay an optical signal to a destination using a control (wavelength setting or the like) signal that does not depend on a protocol of a main signal.

In view of the foregoing circumstances, an objective of the present invention is to provide a technology capable of relaying an optical signal to a destination by using a control signal that does not depend on a protocol of a main signal.

Solution to Problem

An aspect of the present invention is an optical communication device including: an optical switch configured to output an optical signal input from one transmission line to another transmission line among the plurality of transmission lines; a wavelength management control unit configured to allocate a wavelength, a time, a polarized wave, a mode, a code, a frequency, a core, a core wire, a wavelength that is a combination thereof, or the like to a subscriber device by using a control signal in accordance with the subscriber device and a transfer destination on a path to a communication destination, and transmits the control signal to the subscriber device by using a carrier different from a carrier to which the wavelength or the like is allocated; and an optical switch control unit configured to control the optical switch such that the input optical signal is output to the another transmission line corresponding to the communication destination.

An aspect of the present invention is an optical communication system including a plurality of subscriber devices and the foregoing optical communication device.

An aspect of the present invention is an optical communication method performed by an optical communication device. The method includes: an optical switching step of outputting an optical signal input from one transmission line to another transmission line among the plurality of transmission lines; a wavelength management control step of allocating a wavelength, a time, a polarized wave, a mode, a code, a frequency, a core, a core wire, a wavelength which is a combination thereof, or the like to a subscriber device by using a control signal in accordance with the subscriber device and a transfer destination on a path to a communication destination, and transmitting the control signal to the subscriber device by using a carrier different from a carrier to which the wavelength or the like is allocated; and an optical switch control step of controlling the optical switch such that the input optical signal is output to the another transmission line corresponding to the communication destination.

Advantageous Effects of Invention

According to the present invention, it is possible to relay an optical signal to a destination by using a control signal that does not depend on a protocol of a main signal.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram illustrating a basic configuration example of an optical communication system according to an embodiment.

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FIG. 2 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 3 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 4 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 5 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 6 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 7 is a diagram illustrating an example of routing before a change in a wavelength in the optical SW in the basic configuration.

FIG. 8 is a diagram illustrating an example of routing after the change in the wavelength in the optical SW in the basic configuration.

FIG. 9 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 10 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 11 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 12 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 13 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 14 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 15 is a diagram illustrating an example of an optical SW in a basic configuration.

FIG. 16 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 17 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 18 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 19 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 20 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 21 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 22 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 23 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 24 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 25 is a diagram illustrating an example of an access topology in a basic configuration.

FIG. 26 is a diagram illustrating an example in which scalability of an optical SW in a basic configuration is required.

FIG. 27 is a diagram illustrating an example in which scalability of an optical SW in a basic configuration is required.

FIG. 28 is a diagram illustrating an example in which scalability of an optical SW in a basic configuration is required.

FIG. 29 is a diagram illustrating a configuration example of a dual-core subscriber device in a basic configuration.

FIG. 30 is a diagram illustrating a configuration example of a dual-core subscriber device in a basic configuration.

FIG. 31 is a diagram illustrating a configuration example of a single-core subscriber device in a basic configuration.

FIG. 32 is a diagram illustrating a configuration example of a single-core subscriber device in a basic configuration.

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FIG. 33 is a diagram illustrating a configuration example of a monitoring unit in a basic configuration.

FIG. 34 is a diagram illustrating a configuration example of a monitoring unit in a basic configuration.

FIG. 35 is a diagram illustrating a configuration example of a monitoring unit in a basic configuration.

FIG. 36 is a diagram illustrating a configuration example of a monitoring unit in a basic configuration.

FIG. 37 is a diagram illustrating a configuration example of a monitoring unit in a basic configuration.

FIG. 38 is a diagram illustrating a configuration example of an optical access system according to a configuration example.

FIG. 39 is a diagram illustrating an example of an SW connection table according to the same configuration example.

FIG. 40 is a diagram illustrating an example of a user wavelength table according to the same configuration example.

FIG. 41 is a diagram illustrating an example of an inter-station wavelength table according to the same configuration example.

FIG. 42 is a flowchart illustrating an initial setting process of the optical access system according to the configuration example.

FIG. 43 is a diagram illustrating a configuration example of an optical access system according to a first configuration example.

FIG. 44 is a diagram illustrating a configuration example of an optical access system according to a second configuration example.

FIG. 45 is a diagram illustrating a configuration example of an optical access system according to a third configuration example.

FIG. 46 is a diagram illustrating a configuration example of an optical access system according to a fourth configuration example.

FIG. 47 is a diagram illustrating a configuration example of an optical access system according to a fifth configuration example.

FIG. 48 is a diagram illustrating a configuration example of an optical access system according to a sixth configuration example.

FIG. 49 is a diagram illustrating a configuration example of an optical access system according to a seventh configuration example.

FIG. 50 is a diagram illustrating a configuration example of an optical access system according to an eighth configuration example.

FIG. 51 is a diagram illustrating a configuration example of an optical access system according to a ninth configuration example.

FIG. 52 is a diagram illustrating a configuration example of an optical access system according to a tenth configuration example.

FIG. 53 is a diagram illustrating a configuration example of an optical access system according to an eleventh configuration example.

FIG. 54 is a diagram illustrating a configuration example of an optical access system according to a twelfth configuration example.

FIG. 55 is a diagram illustrating a configuration example of an optical access system according to a thirteenth configuration example.

FIG. 56 is a diagram illustrating a configuration example of an optical access system according to a fourteenth configuration example.

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FIG. 57 is a diagram illustrating a configuration example of an optical access system according to a fifteenth configuration example.

FIG. 58 is a diagram illustrating a configuration example of an optical access system according to a sixteenth configuration example.

FIG. 59 is a diagram illustrating a configuration example of an optical access system according to the sixteenth configuration example when an electrical processing unit multiplexes signals.

FIG. 60 is a diagram illustrating a configuration example of an optical access system according to a seventeenth configuration example.

FIG. 61 is a diagram illustrating a configuration example of an optical GW in the optical access system according to a first embodiment.

FIG. 62 is a diagram illustrating a configuration example of an optical GW in an optical access system according to a second embodiment.

FIG. 63 is a diagram illustrating a configuration example of an optical GW in an optical access system according to a third embodiment.

FIG. 64 is a diagram illustrating a configuration example of an optical GW in an optical access system according to a fourth embodiment.

FIG. 65 is a diagram illustrating a configuration example of an optical GW in an optical access system according to a fifth embodiment.

FIG. 66 is a diagram illustrating another aspect of scalability of the optical SW.

FIG. 67 is a diagram illustrating another aspect of the monitoring unit.

FIG. 68 is a diagram illustrating another aspect of the monitoring unit.

FIG. 69 is a diagram illustrating another aspect of the monitoring unit.

FIG. 70 is a diagram illustrating a configuration example in a first modification of the subscriber device and a control unit.

FIG. 71 is a flowchart illustrating an initial setting process of the optical access system according to the first modification of the subscriber device and the control unit.

FIG. 72 is a diagram illustrating a configuration example in a second modification of the subscriber device and the control unit.

FIG. 73 is a flowchart illustrating an initial setting process of the optical access system according to the configuration example.

FIG. 74 is a diagram illustrating another aspect of the monitoring unit.

FIG. 75 is a diagram illustrating another aspect of the monitoring unit.

FIG. 76 is a diagram illustrating another aspect of the monitoring unit.

FIG. 77 is a diagram illustrating a configuration example in a third modification of the subscriber device and the control unit.

FIG. 78 is a diagram illustrating a configuration example in a fourth modification of the subscriber device and the control unit.

FIG. 79 is a diagram illustrating a modification of the flowchart.

FIG. 80 is a diagram illustrating an example of writing and reading of a control signal in which an overhead of a periodic frame is used.

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FIG. 81 is a diagram illustrating a first example of details of the writing and reading of the control signal in which an overhead of the periodic frame is used.

FIG. 82 is a diagram illustrating a second example of details of the writing and reading of the control signal in which an overhead of the periodic frame is used.

FIG. 83 is a diagram illustrating an example of writing and reading of a control signal in which an OAM frame is used.

FIG. 84 is a diagram illustrating a configuration example of a subscriber device that acquires a control signal with a carrier different from a main signal.

FIG. 85 is a diagram illustrating a modification of the optical access system.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of the present invention will be described in detail with reference to the drawings. In the drawings, the same units are denoted by the same reference numerals, and the description thereof will be omitted. In the following description, a basic configuration in the present invention will first be described, and then specific embodiments will be described.

(Basic Configuration)

First, a basic configuration of the present invention will be described with reference to FIGS. 1 to 41.

FIG. 1 is a diagram illustrating a basic configuration example of an optical communication system 1 according to the present embodiment. The optical communication system 1 includes one or a plurality of optical SWs (switches) 10. Although two optical SWs 10 are illustrated in the drawing, the number of optical SWs 10 is any number. The optical SW 10 is connected to the control unit 20.

When the optical SW 10 is controlled with an optical signal, the optical SW 10 includes an optical receiver that receives at least an optical signal, but description thereof is omitted in the following description.

The optical SW 10 communicates with another optical SW 10 via the optical communication network 30. For example, in the optical communication network 30, a WDM network including various topologies can be used. One or more subscriber devices 40 are connected to the optical SW 10. The subscriber device 40 is connected to the optical SW 10 by, for example, an optical access network such as PON. The subscriber device 40 includes an optical transceiver 41. The optical transceiver 41 is an example of an optical transmission unit and an optical reception unit in the configuration of the subscriber device. The optical transceiver 41 includes an optical transmitter (Tx) 42 and an optical receiver (Rx) 43. The optical transceiver 41 is a wavelength-tunable optical transceiver. As the optical transceiver 41, for example, a conventional optical transceiver of the related art that has an auxiliary management and control channel (AMCC) function can be used.

The control unit 20 may include an operation system (OpS) and the OpS may include the control unit 20. Hereinafter, the control unit 20 may be similarly connected to any port of the optical SW 10, for example, a port 2. The control unit 20 may be installed in a building different from the optical SW 10 and may be connected to the optical SW 10 or an optical SW control unit (not illustrated in FIG. 1) via a network. The control unit 20 may be connected to a port that is not connected to the subscriber device 40, another optical SW 10, a higher network, a transmission line to another ground, or the like among the ports of the optical

SW 10. The control unit 20 may be installed for each of the optical SW 10 or may be installed for each plurality of optical SWs 10.

When the control unit 20 controls the optical SW 10 by an optical signal, the control unit 20 includes one or a plurality of optical transceivers 21. The optical transceiver 21 is an example of a configuration of an optical transmission unit and an optical reception unit in the control unit 20. The optical transceiver 21 includes an optical transmitter (Tx) 22 and an optical receiver (Rx) 23. The optical transceiver 21 is a tunable wavelength optical transceiver. The case where the plurality of optical transceivers 21 is provided is appropriate for a case where the connectable port is different for each port of the optical SW 10, a case where the port controlling the optical SW 10 is different from that of the subscriber device 40, a case where a control target is at least one of the plurality of optical SWs 10 and the subscriber device connected to the plurality of optical SWs 10 and the connectable ports are different, and the like. In the case where the number of optical SWs 10 is plural, the optical SWs may be connected by a transmission line. In the case where the ports are different, the optical SWs may be connected by a folded connection using a folded transmission line or the like to be described below. When the control unit includes a transmitter and the subscriber device includes a receiver, the transmission side of the control unit 20 is connected to the reception side of the subscriber device 40. When the control unit 20 includes a receiver and the subscriber device 40 includes a transmitter, the reception side of the control unit 20 is connected to the transmission side of the subscriber device 40.

When only the optical transceiver or a modulation unit of the monitoring unit to be described below is used, the optical transceiver may not be provided. When the control unit 20 controls the subscriber device 40 with an optical signal, when the control unit 20 performs a wavelength allocation process or the like on the subscriber device 40 with an optical signal via the optical SW 10, the optical SW 10 is controlled to transmit and receive an optical signal between the subscriber device 40 and the control unit 20. After the wavelength allocation process, the optical SW 10 is controlled to output the optical signal input from the transmission line to a port corresponding to a transmission destination specified by a subscriber, the subscriber device 40, the port of the optical SW 10, the wavelength of the optical signal, a combination of the wavelength of the subscriber device 40 and the wavelength of the optical signal, a combination of the port and the wavelength of the optical signal, or the like.

When the wavelength allocation process or the like is performed via the monitoring unit in the preceding stage of the optical SW 10, the optical SW 10 is controlled to transmit and receive an optical signal between the monitoring unit and the subscriber device 40 and output the optical signal to a port corresponding to a specified transmission destination after the allocation process.

When the wavelength allocation process or the like is performed via the monitoring unit at the subsequent stage of the optical SW 10, the optical SW 10 is controlled to output an optical signal to a port corresponding to a specified transmission destination, the optical signal is transmitted and received between the monitoring unit and the subscriber device 40 in a state where the output from the port is blocked by a blocking unit. After the allocation process, the blocking is released.

When the wavelength allocation process or the like is performed before connection to the optical SW 10, after the

wavelength allocation process, the optical SW 10 is controlled to output an optical signal to a port corresponding to the specified transmission destination.

The control unit 20 has or is connected to a management database (DB) such as an SW connection table and a wavelength table.

When the control unit 20 is connected to the management DB, information regarding a user and a wavelength to be used is exchanged with the management DB. The management DB stores a use wavelength, destination information, and transmission destination information of each user. The destination and the transmission destination are represented by, for example the ground A and the ground B, in the case of the ground A or the like. In the case of the subscriber device 40 or the like, the destination and the transmission destination are represented by an identifier of the subscriber device 40, or an identifier of a transmission line or a port to which a device is connected, an identifier of a device, a component, or a function passed through on the way, or an identifier of a port to which the device or the component or the function is connected. The management DB manages information regarding a user connected to the optical access system.

The SW connection table indicates a connection destination of each port of the optical SW 10. That is, a port to which an optical signal is input and output can be used as information for identifying a subscriber device, a control unit, a monitoring unit, a ground, or the like of a transmission source or a transmission destination of the optical signal.

The wavelength table is data indicating a wavelength allocated to each subscriber device 40 or the like. The wavelength table may be divided into a user wavelength table and an inter-station wavelength table.

The user wavelength table indicates, for example, a wavelength which is used for user transmission, a wavelength which is used for reception, an available wavelength which is not being used for transmission and reception, and a wavelength which cannot be used due to a failure.

The inter-station wavelength table indicates, for example, a wavelength which a certain ground uses for communication with each of the other grounds, an available wavelength which is not used for communication with each of the other grounds, and a wavelength which cannot be used due to a failure in communication with each of the other grounds.

In order to allocate a wavelength corresponding to a transmission destination on a path to a communication destination to the subscriber device 40, an optical transceiver of the subscriber device 40 or the like and an optical transceiver of the control unit 20 first communicate with each other. With reference to the wavelength table, the control unit 20 selects a wavelength to be allocated to the subscriber device 40 or the like according to the transmission destination on the path to the communication destination.

For example, a wavelength control unit of the control unit 20 allocates the wavelength used by the user with reference to information indicating the wavelength used by the user or a service. The wavelength control unit manages and controls information on which user is connected to which port of the optical SW 10 and which wavelength is used by which user by sharing each of pieces of connection information.

When a wavelength is selected, in a case where the wavelength is multiplexed in a section included in the middle of a path and is identified by the wavelength, a free wavelength in the section included in the path is an available wavelength and is selected from available wavelengths.

On the other hand, it is not necessary to consider a free wavelength in a section in which wavelength is not multiplexed in a section included in the middle of a path. In a case where identification and branching are not performed only with the wavelength (for example, time division multiplexing, code division multiplexing, mode division multiplexing, core division multiplexing, core wire multiplexing, space division multiplexing, frequency division multiplexing, polarization division multiplexing, or a combination thereof, or a combination thereof and wavelength division multiplexing), even a wavelength that is not a free wavelength in the section may be selected as an available wavelength when identification and branching are possible in the section. When wavelength conversion is performed in the middle of a path, a free wavelength in a section included in a path until wavelength conversion or a wavelength is not multiplexed in a section included in the middle of the path until wavelength conversion, or identification is performed only with the wavelength. When the wavelength is not branched, a free wavelength may not be selected.

In this way, since the wavelength to be selected does not depend only on a use status of the wavelength in the section included in the middle of the path, it is preferable that the wavelength table indicate available wavelengths in consideration of not only a free wavelength but also a combination with another multiplexing. Since the available wavelength depends on the use status and multiplexing status of the sections included in the middle of the path, the control unit can use an AND (OR of wavelengths that cannot be identified or separated by factors other than the wavelength) as the available wavelength of the section included in the path.

When a destination port of the optical SW 10 is set with the wavelength and the subscriber device 40 for transmission and the ports included in the optical SW 10 are divided into two groups, for example, ports 1 and 2, are connected between the groups and are not connected within the group, available wavelengths may be separate in a direction from the port 1 which is one group to the port 2 which is the other group and a direction from the port 2 which is the other group to the port 1 which is one group.

When the same control unit controls or manages all the sections included in the path, the wavelength allocation process is closed meanwhile. However, when a plurality of control units or an external device controls or manages all the sections, the wavelength allocation process may be controlled by operating in cooperation or may be controlled by receiving a use authority of the wavelength table itself which shows the available wavelengths or values thereof.

The control unit 20 sets the selected wavelength in the subscriber device 40 by the control signal. Thereafter, the control unit 20 sets the optical SW 10 to output the optical signal transmitted from the subscriber device 40 according to a destination or a transmission destination of the optical signal.

When the subscriber devices 40 face each other via the optical SW 10, the facing subscriber devices 40 are accordingly connected. When the connection is made via a device, a component, or a functional unit in the middle of the path connected between the subscriber devices 40, the connection is made after the device, the component, or the functional unit is routed. The optical SW 10 may connect the subscriber device 40, the device, component, or functional unit to be routed, or the facing subscriber device 40 with the same wavelength of light or after performing photoelectric conversion or other processing, or may connect them with different wavelengths in at least a part of the path as light or after performing photoelectric conversion or other processing.

For example, when a device, a component, or a functional unit to be routed in the middle of the path and both subscriber devices are connected to the same optical SW, routing is formed from the subscriber device to the next device, component, or function to be routed, from the device, component, or function to be routed to the next device, component, or function to be routed, or from the last device, component, or function to the facing subscriber device. The routing from the subscriber device which is a communication source to the facing subscriber device which is a communication partner is a setting in a case where the device, component, or function to be routed is not routed. In the routing, identification of multiplexing processes of wavelengths or the like may be added to parameters, in addition to a transmission source, a destination including a transit, or a transmission destination. This is appropriate when a plurality of optical signals with different wavelengths and optical frequencies are aggregated and transmitted to the same stations or the optical SWs or the like on other grounds which are routing destinations or when a plurality of optical signals with different wavelengths and optical frequencies are demultiplexed and distributed to different devices, components, functions, or subscriber devices.

The control unit 20 allocates a wavelength used for communication by the subscriber device 40 using, for example, an AMCC function of using a control signal that is slower than a main signal which is an optical signal between the subscriber devices 40 and can be superimposed on the main signal. Hereinafter, communication between the subscriber device 40 and the control unit 20 will be exemplified by the AMCC function, but the present invention is not limited thereto. In particular, when an initial setting or a setting change of the wavelength or the like is performed in a state where the main signal is not conducted to the facing device, the control signal may not use another AMCC or the like different from the main signal, and may be exchanged as the main signal. The main signal may be transmitted and received in accordance with any modulation scheme as long as the scheme can be realized with functions of the subscriber device 40 and the control unit 20.

For example, in order to allocate a wavelength corresponding to a transmission destination on a path to the communication destination to the subscriber device 40, the optical transceiver 41 of the subscriber device 40 and the optical transceiver 21 of the control unit 20 first perform communication using the AMCC. With reference to the wavelength table, the control unit 20 selects a wavelength to be allocated to the subscriber device 40 according to the transmission destination on the path to the communication destination. As an example, the control unit 20 selects a wavelength from among free wavelengths that are not used for other paths in a link for wavelength multiplexing on the path. The control unit 20 may allocate individual wavelengths to each subscriber device 40. The control unit 20 sets the selected wavelength in the subscriber device 40 by the control signal in which the AMCC is used. Thereafter, the control unit 20 switches the optical SW 10 to output the optical signal transmitted from the subscriber device 40 to the transmission line corresponding to the transmission destination on the path to the communication destination.

The control unit 20 may control the optical switch to perform routing in accordance with the destination information. As the destination information, a subscriber device, a wavelength, an input port, an output port, or a set such as a set of a subscriber device, an input port, or an output port and a wavelength may be used.

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In the following embodiment, a case where a set of a subscriber device and a wavelength is used as destination information will be mainly described.

Accordingly, the facing subscriber devices 40 are connected.

The control unit 20 may perform a wavelength changing process of instructing the subscriber device 40 to allocate the wavelength to which the wavelength has been allocated. For example, a target subscriber device 40 is specified based on monitoring information output from a monitoring unit to be described below, and the wavelength changing process is performed on the specified subscriber device 40.

An instruction from the control unit 20 to the subscriber device 40 may be given in a path at the time of initial setting or may be performed from a monitoring unit or the like.

The control unit 20 controls the optical SW 10 or the blocking unit, if any, such that the optical signal of the target subscriber device 40 is not transmitted during the wavelength change. When the control unit 20 directly controls the subscriber device 40 by an optical signal, the optical SW 10 is controlled such that the optical signal is transmitted and received between the subscriber device 40 and the wavelength control unit.

For example, when there is no other influence on the output side before the change in the wavelength, the output port is switched after the change in the wavelength. When there is no other influence on the output side after the change in the wavelength, the output port is switched before the change in the wavelength. The transmission may be stopped by a device other than the optical SW 10.

For example, the transmission line from the optical SW 10 to the subscriber device 40 or the subscriber device 40 is removed, and the connection is reestablished after the setting. Alternatively, the transmission line, the optical SW 10, a branch combiner/splitter, or a multiplexer/demultiplexer and a blocking unit at a connection point thereof are provided. The blocking unit performs blocking before the change in the wavelength and performs setting before or after switching an output port as necessary to release the blocking.

For example, after the wavelength changing process, the optical SW control unit controls the optical SW 10 such that an optical signal with the changed wavelength from the subscriber device 40 is output from the port 2 in accordance with the communication destination or the transmission destination.

At the time of wavelength switching, for example, when the wavelength switching has an influence on another subscriber device 40, the optical output in the path is blocked. Specifically, the transmission of the optical signal itself is stopped, the connection to the optical SW 10 is released, the connection from the input side to the output side of the optical SW 10 is released, and the optical signal is blocked by, for example, the blocking unit included in the monitoring unit or the like such as a shutter.

In a wavelength allocation process, the optical SW control unit controls the path in the optical SW 10 such that the optical SW 10 does not transmit the optical signal of the subscriber device 40 which is a wavelength allocation target. That is, the optical SW control unit controls the path in the optical SW 10 such that the optical signal transmitted from the subscriber device 40 is not output to a port (another port) other than the port connected to the monitoring unit of the subscriber device 40 which is a wavelength allocation target.

When there is no particular influence despite switching of the output destination of the optical SW 10 before setting of

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the wavelength (before allocation of the wavelength), the optical SW control unit may switch the output destination of the optical SW 10 before the change in the wavelength. In the wavelength allocation process, a functional unit other than the optical SW 10 may stop the transmission of the optical signal transmitted from the subscriber device 40.

When the wavelength is switched in a state of the connection to the path before or after the switching in association with the switching of the wavelength and the switching of the optical SW 10, there is no influence.

The wavelength allocation process (wavelength setting) and a path setting process are performed in this order. The optical SW control unit switches the output destination (path) of the optical SW 10 after the change in the wavelength so that there is no particular influence despite of the switching of the output destination of the optical SW 10.

For example, cases where the first port and a first wavelength "λ1" are used before the switching and the second port and a second wavelength "λ2" are used after the switching may be the following cases.

(1) A case where there is no subscriber device 40 using the first port and the second wavelength "λ2."

There is no influence despite switching from the first wavelength "λ1" to the second wavelength "λ2" in the first port. In this case, it is possible to perform wavelength switching before port switching.

(2) A case where there is the subscriber device 40 using the first port and the second wavelength "λ2."

There is an influence when the wavelength is switched from the first wavelength "λ1" to the second wavelength "λ2" in the first port. In this case, it is not possible to perform wavelength switching before port switching.

(3) A case where there is no subscriber device 40 using the second port and the first wavelength "λ2."

There is no influence despite switching from the first wavelength "λ2" to the second wavelength "λ2" in the second port. In this case, it is possible to perform wavelength switching before port switching.

(4) A case where there is the subscriber device 40 using the second port and the first wavelength "λ1":

There is an influence when the wavelength is switched from the first wavelength "λ1" to the second wavelength "λ2" in the second port. In this case, it is not possible to perform wavelength switching before port switching.

In the foregoing "(1)" and "(3) or (4)," the wavelength can be switched before the port switching.

In the above "(1) or (2)" and "(3)," the wavelength can be switched before the port switching.

In the above "(2)" and "(4)," it is not possible to switch the wavelength before port switching, and it is necessary to stop or block the optical signal.

In the wavelength allocation process, the wavelength control unit performs the wavelength allocation process on the subscriber device 40 which is a wavelength allocation target via the monitoring unit. For example, the wavelength control unit performs the allocation process of a new wavelength that is the same as or different from the wavelength used in detection of the abnormality on the subscriber device 40 which is a wavelength allocation target.

In the path setting process, an optical switch control unit controls the path in the optical SW 10 such that the subscriber device 40 which is a wavelength allocation target is connected to a port (another port) other than the port connected to the monitoring unit of the subscriber device 40 which is the wavelength allocation target. For example, the optical switch control unit controls the optical SW 10 such that the optical signal input from the transmission line of the

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subscriber device **40** which is a wavelength allocation target is output to the port (another port) specified in accordance with the transmission destination. Here, the transmission destination is specified in accordance with the subscriber, the subscriber device **40**, the port of the optical SW **10**, the wavelength of the optical signal, a combination of the subscriber device **40** and the wavelength of the optical signal, a combination of the optical signal of the port **1** and the wavelength of the optical signal, or the like.

The optical SW **10** is included in, for example, an optical gateway (GW). An example of the optical SW **10** included in the optical GW will be described with reference to FIGS. **2** to **15**.

FIG. **2** is a diagram illustrating a configuration example of the optical SW **10a**. The optical SW **10a** is connected to the plurality of transmission lines **50**, and outputs an optical signal input from any of the transmission lines **50** to the other transmission line **50**. The transmission line **50** is, for example, an optical fiber. The optical SW **10a** includes ports **11-1-1** to **11-1-P** (where **P** is an integer equal to or greater than 1) and ports **11-2-1** to **11-2-Q** (where **Q** is an integer equal to or greater than 1 and at least one of **P** and **Q** is equal to or greater than 2). When any of the ports **11-1-1** to **11-1-P** is not specified, the ports **11-1-1** to **11-1-P** are collectively referred to as the port **11-1**. When any of the ports **11-2-1** to **11-2-Q** is not specified, the ports **11-2-1** to **11-2-Q** are collectively referred to as the port **11-2**. The transmission line **50** connected to the port **11-1** is referred to as a transmission line **50-1**. The transmission line **50** connected to the port **11-2** is referred to as a transmission line **50-2**.

Each port **11-1** is connected to the subscriber device **40** via the transmission line **50-1**. Each port **11-2** is connected to the subscriber device **40** via the transmission line **50-2**. The subscriber device **40** is, for example, an ONU. The transmission line **50-2** may be connected to optical communication network **30** that is an upper network. In this case, a direction of the subscriber device **40** connected via the transmission line **50-1** is the downlink direction, and a direction of the upper network connected via the transmission line **50-2** is the uplink direction. The transmission line **50-2** may include an optical communication device such as another optical SW **10**.

The ports **11-1-1**, **11-1-2**, **11-1-3**, . . . are connected to the subscriber devices **40a-1**, **40a-2**, **40a-3**, . . . which are the subscriber devices **40** on the ground A, respectively, via transmission lines **50-1**. Any one of the ports **11-2** (in the drawing, a port **11-2-1**) is connected to a wavelength control unit **25** to be described below. Some of the ports **11-2-i**, **11-2-(i+1)**, **11-2-(i+2)**, . . . are connected to **40b-1**, **40b-2**, **40b-3**, . . . that are the subscriber devices **40** on the ground B, respectively, via a transmission line **50-2** (where **i** is an integer equal to or greater than 2). Some of the ports **11-2-j**, **11-2-(j+1)**, **11-2-(j+2)**, . . . different from the port **11-2** connected to the subscriber device **40** on the ground B are connected to the subscriber devices **40c-1**, **40c-2**, **40c-3**, . . . , which are the subscriber devices **40** on the ground C, respectively, via a transmission line **50-2** (where **j** is an integer equal to or greater than 2). The optical SW **10a** outputs an optical signal input from the port **11-1** to the port **11-2** and outputs an optical signal input from the port **11-2** to the port **11-1**. Here, an optical communication device such as another optical SW or the optical communication network **30** may be interposed between the subscriber device **40** on the ground A, the subscriber device **40** on the ground B, and the subscriber device **40** on the ground C.

The optical SW **10a** is connected to the control unit **20**. The control unit **20** includes a wavelength control unit **25**

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and an optical SW control unit **26**. The wavelength control unit **25** performs a wavelength allocation process of receiving a wavelength allocation request from the subscriber device **40** by an optical signal, allocating a wavelength corresponding to a transmission destination on a path to a communication destination to the subscriber device **40** that has transmitted the request, and notifying the subscriber device **40** of the allocated wavelength by an optical signal. For example, the wavelength control unit **25** may dynamically allocate a wavelength corresponding to a transmission destination on a path to a communication destination to the subscriber device **40** that has transmitted the request. As the optical signal transmitted and received between the wavelength control unit **25** and the subscriber device **40**, it is preferable to use a control signal superposition scheme that does not depend on a communication protocol of the optical signal (a main signal) between the subscriber devices **40**. As an optical signal transmitted and received between the wavelength control unit **25** and the subscriber device **40**, for example, protocol-free AMCC is used. The main signal may be transmitted and received and may be set without using the control signal superimposed by the AMCC or the like.

While the wavelength allocation process is being executed, the optical SW control unit **26** controls the optical SW **10a** such that optical signals are transmitted and received between the subscriber device **40** and the wavelength control unit **25**. After the wavelength allocation process, the optical SW control unit **26** controls the optical SW **10a** such that the optical signal input from the transmission line **50** is output to the transmission line **50-2** corresponding to the transmission destination on the path to the communication destination specified by a combination of the subscriber device **40** that has transmitted the input optical signal and the wavelength of the input optical signal.

Each transmission line **50-2** includes a monitoring unit **60**. In the drawing, only one monitoring unit **60** is illustrated. The monitoring unit **60** includes a power splitter **61**. The power splitter **61** branches an optical signal transmitted through the transmission line **50-2**. The monitoring unit **60** monitors the optical signal branched by the power splitter **61**. The monitoring unit **60** generates monitoring information based on a monitoring result and outputs the generated monitoring information. The monitoring information is information indicating the monitoring result or information obtained from the monitoring result. For example, when an abnormality or the like of the communication status between the subscriber devices **40** is detected by a change request from the subscriber device **40** using a control signal or by monitoring the optical signal, the monitoring information in which the fact that the abnormality of the communication status has occurred and information for specifying the subscriber device **40** in which the abnormality of the communication status has occurred are set is output. Examples of the abnormality in the communication status include a wavelength shift, an increase or decrease in an output, and a communication abnormality (error). An example of an output destination of the monitoring information is the control unit **20**. When the abnormality is detected, the control unit **20** controls the optical SW **10**, the monitoring unit, or the blocking unit such that transmit the optical signal of the target subscriber device **40** is not output, for example. Further, when the control unit **20** directly controls the subscriber device **40** by an optical signal, the optical SW **10** is controlled such that the subscriber device **40** is connected to the control unit **20** again. Then, the control unit **20** performs an allocation process of a new wavelength that is the same as or different from the wavelength at the time of

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detection of the abnormality, similarly to a time at which the subscriber device **40** is newly connected. Accordingly, the optical SW **10** connects the optical signal with the wavelength before or after the change from the subscriber device **40** to the port before the change by the subscriber device **40**. From a case of non-recovery from the abnormal state despite a predetermined process such as reactivation or reallocation performed a predetermined number of times due to a failure, a Rogue ONU, or the like, or an identification number or a behavior of a subscriber device registered in a blacklist or performing a similar operation in the past, the connection in the optical SW may be released, the connection may be blocked by a blocking unit, or setting, connection, or transmission may be stopped. During communication with another subscriber device **40**, the power splitter **61** can branch a control signal transmitted by the subscriber device **40** or superimpose and output the control signal to the subscriber device **40** or the like.

When the subscriber device **40** is connected to the transmission line **50-2**, the control unit **20** may be connected to the port **11-1**. Alternatively, when the subscriber device **40** is connected to the transmission line **50-2**, the subscriber device **40** connected to the transmission line **50-2** may be connected to the control unit **20** via a folded transmission line **73**. The folded transmission line **73** is an optical fiber, an optical switch, an optical branch combiner/splitter, or an optical multiplexer/demultiplexer that inputs an optical signal output from the port **11-1-p1** to another port **11-1-p2** (where **p1** and **p2** are integers equal to or greater than 1 and equal to or less than **P**). In this case, the optical signal transmitted from the subscriber device **40b** or **40c** is input to the optical SW **10a** via the transmission line **50-2**. The optical SW **10a** outputs the optical signal input from the transmission line **50-2** to the port **11-1-p1**, and inputs the optical signal transmitted through the folded transmission line **73** from the port **11-1-p2**. The optical SW **10a** outputs the optical signal input from the port **11-1-p2** from the port **11-2-1** to the control unit **20**. Accordingly, the subscriber device **40b** or **40c** and the control unit **20** are connected.

The wavelength control unit **25** may perform a wavelength changing process of instructing the subscriber device **40** that has performed the wavelength allocation process to change the wavelength or the like. Here, the wavelength changing process is illustrated as an example in which the subscriber device **40** communicates with the control unit using the monitoring information as a trigger. For example, the wavelength control unit **25** specifies the subscriber device **40** which is a wavelength change target based on the monitoring information output from the monitoring unit **60** and performs the wavelength changing process on the specified subscriber device **40**. The optical SW control unit **26** controls the optical SW **10a** such that an optical signal is transmitted and received between the subscriber device **40** and the wavelength control unit **25** during the wavelength changing process. After the wavelength changing process, the optical SW control unit **26** controls the optical SW **10a** such that the optical signal transmitted from the subscriber device **40** at the changed wavelength is output to the transmission line **50-2** corresponding to the transmission destination on the path to the communication destination. For example, after the wavelength changing process, the optical SW control unit **26** controls the optical SW **10a** such that the optical signal with the changed wavelength from the subscriber device **40** is output to the transmission line **50-2** corresponding to the transmission destination on the path to the communication destination used in a combination of the subscriber device **40** which is a transmission source and the

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wavelength before the change. Alternatively, the optical SW **10a** may be controlled to output the optical signal having the changed wavelength to the transmission line **50-2** different from that before the wavelength change. In this case, before and after the wavelength changing process, the subscriber device **40** of the transmission destination on the path to the communication destination is different. The wavelength control unit **25** may receive a wavelength changing request from the subscriber device **40** during communication or after the end of communication and perform the wavelength changing process on the subscriber device **40** which is a request source. Both the wavelength used for transmission and the wavelength used for reception by the subscriber device **40** may be changed through the wavelength changing process, or one thereof may be changed.

The monitoring information used for the wavelength changing process may be, for example, a change request from the subscriber device **40** using a control signal, an abnormality of a communication status such as a wavelength shift, an increase or decrease in an output, or a communication abnormality (error), a deviation from specification, setting, or an allowable range of a wavelength, a modulation scheme, a protocol, or the like to be used, use of an unallocated wavelength, or abnormality detection such as signal interruption. The monitoring information may be a trigger other than the monitoring signal, for example, a change request for a transmission destination or the like via a main signal or via a signal not routed through an optical SW, or may be a request for abnormality detection, stop, or change from a device inside or outside of a network of a transmission line, a management system, or the like. When abnormality detection, stop, or change is requested and the wavelength or the like is not changed to a designated, set, or allowable range, a shutter or the like may interrupt an instruction or a signal for restart or the like in a state the control unit is connected to the subscriber device **40** or a signal may be blocked in a state in which the signal is not output to the transmission line. Exchange between the control unit **20** and the subscriber device **40** may be performed by AMCC or may be performed by a main signal. When the subscriber device **40** is not compatible with AMCC, the exchange may be appropriately performed with the main signal. When the wavelength or the like before the switching does not have an adverse influence on the facing device before the switching, the path with the facing subscriber device **40** may be set by the monitoring unit or the like while being connected without switching to the control unit **20**. When the wavelength or the like after the switching does not have an adverse influence on the facing device after the switching, the path with the facing subscriber device **40** may be set by the monitoring unit or the like after the switching without the connection switching to the control unit **20**.

FIG. 3 is a diagram illustrating a configuration example of an optical SW **10b** having a folding circuit for folding communication. In the drawing, the same portions as those of the optical SW **10a** illustrated in FIG. 2 are denoted by the same reference numerals, and the description thereof will be omitted. In FIG. 3, description of the control unit **20** is omitted. In the following description, description of the control unit **20** is omitted in the drawings when the description is not particularly necessary. The optical SW **10b** is connected to a folded transmission line **51**. The folded transmission line **51** is an optical fiber, an optical switch, an optical branch combiner/splitter, or an optical multiplexer/demultiplexer that inputs an optical signal output from the

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port 11-2 to the other port 11-2. Accordingly, the optical SW 10b enables folded communication.

When a port which is an output destination of the optical signal is set in a combination of the transmission source subscriber device 40 which is a transmission source and the wavelength, the destination may be different between a direction from the port 11-1 to the port 11-2 to which the folded transmission line 51 is connected and a direction from the port 11-2 to which the folded transmission line 51 is connected to the port 11-1.

FIG. 4 is a diagram illustrating a configuration example of an optical SW 10c that performs uplink multicasting using a one-to-one folded transmission line and one-to-other folded transmission lines facing each other. In the drawing, the same portions as those of the optical SW 10a illustrated in FIG. 2 are denoted by the same reference numerals, and the description thereof will be omitted. The optical SW 10c includes a distribution unit 58 that distributes an optical signal output from the port 11-2 into a plurality of signals and inputs the plurality of distributed optical signals to different ports 11-1. In FIG. 4, the optical SW 10c inputs the optical signal output from the port 11-2 to the other port 11-2 via the folded transmission line. The optical SW 10c outputs the input optical signal to the port 11-1 to which the 1×N power splitter 71 is connected. The optical signal output from the port 11-1 is distributed by the power splitter 71 to be input to the plurality of other ports 11-1. The optical SW 10c outputs the optical signals input from the plurality of ports 11-1 to different ports 11-2. Bidirectional communication is also possible. The optical signal in the downlink direction is routed in a direction reverse to the uplink direction.

The optical SW 10c may receive optical signals with a plurality of wavelengths from the port 11-1. In this case, the optical SW 10c distributes the optical signals with the plurality of wavelengths input from the port 11-1 by the distribution unit 58 and outputs the distributed optical signals to each subscriber device 40 connected to the port 11-2 or a transmission line connected to another ground. The subscriber device 40 connected to the port 11-2 selects and receives an optical signal with a predetermined wavelength among the optical signals with the plurality of wavelengths. The transmission line connected to the other ground may transmit the optical signals with the plurality of wavelengths as it is or may transmit the optical signals with a wavelength selected by a WDM device illustrated in FIG. 6 to be described below.

FIG. 5 is a diagram illustrating a configuration example of an optical SW 10d that performs downlink multicasting. In the drawing, the same portions as those of the optical SW 10a illustrated in FIG. 2 are denoted by the same reference numerals, and the description thereof will be omitted. The optical SW 10d includes a distribution unit 59 that distributes an optical signal output from the port 11-1 into a plurality of signals and inputs the plurality of distributed optical signals to different ports 11-2. In FIG. 5, the optical SW 10d inputs the optical signal output from the port 11-1 to the other port 11-1 via the folded transmission line. The optical SW 10d outputs the input optical signal to the port 11-2 to which the 1×N power splitter 72 is connected. The optical signal output from the port 11-2 is distributed by the power splitter 72 to be input to the plurality of other ports 11-2. The optical SW 10d outputs the optical signals input from the plurality of ports 11-2 to different ports 11-1.

The optical signals with the plurality of wavelengths may be input to the optical SW 10d from the port 11-2. In this case, in the optical SW 10d, the optical signals with the

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plurality of wavelengths input from the port 11-2 are distributed by the distribution unit 59, and the distributed optical signals are output to the subscriber devices 40 connected to the port 11-1. Each of the subscriber devices 40 connected to the port 11-1 selects and receives an optical signal with a predetermined wavelength from among the received optical signals with the plurality of wavelengths.

FIG. 6 is a diagram illustrating a configuration example of an optical SW 10e that performs WDM transmission. In the drawing, the same portions as those of the optical SW 10a illustrated in FIG. 2 are denoted by the same reference numerals, and the description thereof will be omitted. The optical SW 10e is connected to one or more WDM devices 80. The WDM device 80 is an example of a multiplexing/demultiplexing device. The WDM device 80 multiplexes optical signals with different wavelengths output from the plurality of ports 11-2 and outputs the multiplexed optical signals to a multiplexing communication transmission line 90. The WDM device 80 demultiplexes the optical signal received via the multiplexing communication transmission line 90 by the wavelength and inputs the demultiplexed optical signals to the plurality of ports 11-2. As described above, the WDM device 80 has functions of a multiplexing device and a demultiplexing device. As a function of the multiplexing device, optical signals with different wavelengths output from the plurality of ports 11-2 of the optical SW 10e are multiplexed and output to the multiplexing communication transmission line 90. As a function of the demultiplexing device, an optical signal received via the multiplexing communication transmission line 90 is demultiplexed by the wavelength, and the demultiplexed optical signals is input to the plurality of different ports 11-2 of the optical SW 10e. The optical SW that performs WDM transmission may connect the folded transmission line 51 illustrated in FIG. 3 to the port 11-2 that is not connected to the WDM device 80.

The multiplexing communication transmission line 90 includes a monitoring unit 65. The monitoring unit 65 includes a power splitter 66 and WDM devices 67 and 68. The power splitter 66 branches an optical signal transmitted through the multiplexing communication transmission line 90. The WDM device 67 demultiplexes the uplink optical signal branched by the power splitter 66. The WDM device 68 demultiplexes the downlink optical signal branched by the power splitter 66. The monitoring unit 65 monitors the optical signals demultiplexed by the WDM devices 67 and 68. The monitoring unit 65 generates monitoring information based on a monitoring result and outputs the generated monitoring information.

The monitoring unit 65 may include a power splitter 69 in each of the transmission lines between the port 11-2 and the WDM device 80. The power splitter 69 branches an optical signal transmitted through a transmission line between the port 11-2 and the WDM device 80 and outputs the branched optical signal to the control unit 20.

The wavelength control unit 25 may perform the wavelength changing process of instructing the subscriber device 40 that has performed the wavelength allocation process to change the wavelength. For example, the wavelength control unit 25 specifies the subscriber device 40 which is wavelength change target based on the monitoring information output from the monitoring unit 65 and performs the wavelength changing process on the specified subscriber device 40. The optical SW control unit 26 controls, for example, reconnects the optical SW 10e via the monitoring unit such that the optical signal is transmitted and received between the subscriber device 40 and the wavelength control unit 25

during the wavelength changing process. When the optical signal with the changed wavelength is input from the subscriber device 40 after the wavelength changing process, the optical SW control unit 26 controls the optical SW 10e such that the input optical signal from the port 11-2 corresponding to the transmission destination on the path is output to the communication destination. The wavelength control unit 25 may receive a wavelength changing request from the subscriber device 40 during communication or after the end of communication and perform the wavelength changing process on the subscriber device 40 which is a request source.

An example of a change in a wavelength in the optical SW 10e will be described with reference to FIGS. 7 and 8. FIG. 7 is a diagram illustrating an example of routing before the change in the wavelength in the optical SW 10e. The optical SW 10e is connected to the subscriber devices 40a-1, 40a-2, 40a-3, . . . which are the subscriber device 40 on the ground A. The WDM device 80 connected to the ground B is referred to as a WDM device 80b, and the WDM device 80 connected to the ground C is referred to as a WDM device 80c. The WDM device 80b transmits and receives optical signals with wavelengths λ_1 to λ_{10} to and from the optical SW 10e, and the WDM device 80c transmits and receives optical signals with wavelengths λ_{11} to λ_{20} to and from the optical SW 10e.

In FIG. 7, before the change in the wavelength, the optical SW 10e outputs an optical signal with the wavelength λ_1 input from the subscriber device 40a-1 and an optical signal with the wavelength λ_2 input from the subscriber device 40a-2 from different ports 11-2 to the WDM device 80b. The subscriber device 40a-2 transmits a wavelength changing request to the wavelength control unit by a control signal during communication or after end of the communication. When the wavelength changing request is received from the subscriber device 40a-2, the wavelength control unit 25 performs the wavelength changing process of instructing the subscriber device 40a-2 to change the wavelength to the wavelength λ_{10} . The optical SW control unit 26, the monitoring unit, or the blocking unit does not output an optical signal with the wavelength λ_{10} received from the subscriber device 40a-2 until the change in the wavelength is completed as necessary. The optical SW control unit 26 controls the optical SW 10e such that the optical signal from the port 11-2 corresponding to the wavelength λ_{10} is output to the WDM device 80b after the switching is completed. The wavelength control unit 25 may further change the wavelength used for reception by the subscriber device 40a-2.

After the wavelength changing process, the optical SW control unit 26 may control the optical SW 10e such that the optical signal transmitted from the subscriber device which is a transmission source using the changed wavelength is output to the WDM device 80 different from that before the change in the wavelength. FIG. 8 is a diagram illustrating an example of routing after the change in the wavelength in the optical SW 10e when the WDM device which is an output destination is changed. Before the change in the wavelength, as illustrated in FIG. 7, the subscriber device 40a-1 communicates using the wavelength λ_1 , and the subscriber device 40a-2 communicates using the wavelength λ_2 or λ_{10} . The subscriber device 40a-2 transmits a wavelength changing request to the wavelength control unit 25 by a control signal during communication or after end of the communication. When the wavelength changing request is received from the subscriber device 40a-2, the wavelength control unit 25 performs the wavelength changing process of instructing the subscriber device 40a-2 to change the wave-

length to the wavelength λ_{11} in order to communicate with the subscriber device 40 on the ground C. The optical SW control unit 26, the monitoring unit, or the blocking unit does not output the optical signal with the wavelength λ_{10} input from the subscriber device 40a-2 until the change in the wavelength is completed as necessary. After the switching is completed, the optical SW control unit 26 controls the optical SW 10e such that the optical signal with the wavelength λ_{11} received from the subscriber device 40a-2 is output from the port 11-2 corresponding to the wavelength λ_{11} to the WDM device 80c. The wavelength control unit 25 may further change the wavelength used for reception by the subscriber device 40a-2.

When the subscriber device 40a-2 does not change the wavelength used for reception, the wavelength control unit may operate as follows. When the wavelength is not used as at least a part of the destination information, the following may not be applied.

(1) The wavelength control unit 25 releases a transmission wavelength used by the subscriber device 40 on the ground B which is a communication destination before the wavelength switching. By releasing the transmission wavelength, the path from the subscriber device 40a-2 to the subscriber device 40 on the ground B is reset. Thereafter, the wavelength control unit 25 reallocates a wavelength that has become a free wavelength by the releasing for reception of a signal addressed to the subscriber device 40a-2 from the subscriber device 40 on the ground C that is a new communication destination. This is performed when a wavelength used for each subscriber device 40 is unique, and a wavelength other than the free wavelength is not allocated.

(2) When the subscriber device 40 connected via the different multiplexing communication transmission line 90 before and after the change in the wavelength of the subscriber device 40a-2 is a communication destination, the wavelength used before the change in the wavelength can be reused as it is. However, although the wavelength is used as the destination information, for example, when different transmission lines are routed or an input port or an output port of the optical switch is different, the same wavelength is treated as a different path. In order to enable reuse as described above, in this case, for example, an "input transmission line," an "output transmission line," or a "combination of all transmission lines included in a path" is added to an argument serving as a condition for determining an output destination of an optical signal. For example, the output destination is determined in a combination of a transmission line or a port that has received the optical signal and a wavelength of the optical signal, or a combination of a transmission line or a port that has received the optical signal, the subscriber device 40 that has transmitted the optical signal, and a wavelength of the optical signal.

Although the wavelength changing process performed by the subscriber device 40 requesting the change in the wavelength has been described above, the same applies to the wavelength changing process performed based on the monitoring information.

An optical SW that performs WDM transmission and multicasting will be described with reference to FIGS. 9 to 11. FIG. 9 is a diagram illustrating a configuration example of an optical SW 10f that performs WDM transmission and multicasting in an uplink direction. In FIG. 9, the optical SW 10f performs multicasting in an uplink direction with a single wavelength. As illustrated in FIG. 9, the optical SW 10f includes a distribution unit 58 similar to that in FIG. 4. In FIG. 9, multicasting is performed on the grounds B and C. The optical SW 10f outputs the optical signal input from

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the port 11-1 connected to the subscriber device 40 from the port 11-2 connected to the folded transmission line and inputs the optical signal transmitted through the folded transmission line from the other port 11-2. The optical SW 10f outputs the input optical signal from the port 11-1 to which the 1xN power splitter 71 is connected. The optical SW 10f inputs the optical signal distributed by the 1xN power splitter 71 from the plurality of ports 11-1, outputs one of the input optical signals to the port 11-2 connected to the ground B, and outputs the other optical signal to the port 11-2 connected to the ground C.

The subscriber device 40 may output the WDM signal. For example, the subscriber device 40 outputs a WDM signal in which the optical signal with the wavelength and the optical signal with the wavelength 2'2 are multiplexed. A plurality of transmission lines between the WDM device 80b and the optical SW 10f transmit and receive optical signals with the wavelengths λ_1 , λ_2 , . . . in order from the upper side. Similarly, in a plurality of transmission lines between the WDM device 80c and the optical SW 10f, optical signals with the wavelengths λ_1 , λ_2 , . . . are transmitted and received in order from the upper side.

In the optical SW 10f, the WDM signals with the wavelengths λ_1 and λ_2 input from the port 11-1 connected to the subscriber device 40 are distributed by the distribution unit 58. The optical SW 10f outputs the distributed WDM signals to the port 11-2 corresponding to the wavelength λ_1 among the ports 11-2 connected to the WDM device 80b. Further, the optical SW 10f outputs the other distributed WDM signals to the port 11-2 corresponding to the wavelength λ_2 among the ports 11-2 connected to the WDM device 80c. The WDM device 80b filters the WDM signal input from the port corresponding to the wavelength to block the wavelength λ_2 , passes the optical signal with the wavelength λ_1 , and outputs the optical signal to the multiplexing communication transmission line 90. The WDM device 80c filters the WDM signal input from the port corresponding to the wavelength λ_2 to block the wavelength λ_1 , passes the optical signal with the wavelength λ_2 , and outputs the optical signal to the multiplexing communication transmission line 90.

FIG. 10 is a diagram illustrating a case where the optical SW 10f performs multicasting in an uplink direction on a plurality of grounds using a plurality of wavelengths. By including one or more 1xM power splitters 55 in the transmission line 50-1, the plurality of subscriber devices 40 can be connected to the transmission line 50-1 connected to one port 11-1. In FIG. 10, as a plurality of subscriber devices 40a-1, subscriber devices 40a-1-1, 40a-1-2, . . . are connected to one transmission line 50-1. The subscriber devices 40a-1-1, 40a-1-2, . . . use different wavelengths. Here, the subscriber device 40a-1-1 transmits an optical signal with the wavelength 2'i, and the subscriber device 40a transmits an optical signal with the wavelength λ_2 . The optical SW 10f receives, from the port 11-1, an optical signal obtained by multiplexing the optical signal with the wavelength transmitted by the subscriber device 40a-1-1 and the optical signal with the wavelength 2'2 transmitted by the subscriber device 40a-1-2. The optical SW 10f outputs the input optical signal from the port 11-2 to which the folded transmission line is connected and inputs the optical signal transmitted through the folded transmission line from the other port 11-2. The optical SW outputs the input optical signal from the port 11-1 to which the 1xN power splitter 71 is connected. The optical signal distributed by the 1xN power splitter 71 is input to the optical SW 10f from the plurality of ports 11-1.

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The optical SW 10f outputs the optical signal distributed by the power splitter 71 to the port 11-2 corresponding to the wavelength λ_1 and the port 11-2 corresponding to the wavelength λ_2 among the ports 11-2 connected to the WDM device 80b. Further, the optical SW outputs the optical signal distributed by the power splitter 71 to the port 11-2 corresponding to the wavelength λ_1 and the port 11-2 corresponding to the wavelength λ_2 among the ports 11-2 connected to the WDM device 80c. The WDM device 80b filters the optical signal input from the port corresponding to the wavelength λ_1 , passes the optical signal with the wavelength λ_1 , and outputs the optical signal to the multiplexing communication transmission line 90. The WDM device 80b filters the optical signal input from the port corresponding to the wavelength λ_2 , and outputs the optical signal to the multiplexing communication transmission line 90. Similarly, the WDM device 80c filters the optical signal input from the port corresponding to the wavelength λ_1 , passes the optical signal with the wavelength λ_1 , and outputs the optical signal to the multiplexing communication transmission line 90, and filters the optical signal input from the port corresponding to the wavelength λ_2 , passes the optical signal with the wavelength λ_2 , and outputs the optical signal to the multiplexing communication transmission line 90.

FIG. 11 is a diagram illustrating a configuration example of an optical SW 10g that performs WDM transmission and downlink multicasting. The optical SW 10g includes a distribution unit 59 similar to that in FIG. 5. The optical SW 10f illustrated in FIGS. 9 and 10 and the optical SW 10g illustrated in FIG. 11 may include a monitoring unit 65 similar to that in FIG. 6. The wavelength control unit 25 can perform the wavelength changing process in the same way as described above on the subscriber device 40 in which the monitoring unit 65 has detected the abnormality of the communication status.

FIG. 12 is a diagram illustrating a case where the optical SW 10g performs WDM transmission and downlink multicasting. The connection configuration illustrated in FIG. 12 is different from the connection configuration illustrated in FIG. 11 in that the WDM device 80 connected to the plurality of ports 11-2 of the optical SW 10g is replaced with a WDM device 81 connected to the plurality of ports 11-1. One or more subscriber devices 40 are connected to the WDM device 81 on the side opposite to the port 11-1. The optical SW 10g inputs optical signals with a plurality of wavelengths from another ground from the port 11-2 and outputs the optical signals to the port 11-1 to which the folded transmission line of the distribution unit 59 is connected. The optical signals with the plurality of wavelengths are directly branched by the power splitter 72. The optical SW 10d inputs the branched optical signals with the plurality of wavelengths from the plurality of ports 11-2 and outputs the input optical signals to any one of the ports 11-1 connected to the WDM device 81. The WDM device 81 filters and passes an optical signal with the wavelength corresponding to the port 11-1 that has received the optical signal from the input optical signals with the plurality of wavelengths, and outputs the optical signal which has passed to a transmission line connected to the subscriber device 40.

FIG. 13 is a diagram illustrating a configuration example of an optical SW 10h that performs electrical processing on an optical signal. In FIG. 13, the same portions as those of the optical SW 10b illustrated in FIG. 3 are denoted by the same reference numerals, and the description thereof will be omitted. The optical SW 10h is different from the optical

SWs **10a** to **10g** described above in that ports **12-1** and **12-2** are further included. The ports **12-1** and **12-2** are connected to an electrical processing unit **84** via a transmission line **52**. The ports **11-1** and **11-2** may be used as ports connected to the electrical processing unit **84** via the transmission line **52**.

Under the control of the optical SW control unit **26**, the optical SW **10h** outputs the optical signal input from the subscriber device **40** from the port **11-2** or **12-1** according to a combination of the wavelength and the subscriber device **40** that is the transmission source of the optical signal or the port **11-1** that has input the optical signal. Under the control of the optical SW control unit **26**, the optical SW **10h** outputs an output destination of the optical signal input from the port **11-2** from the port **11-1** or **12-1** according to the combination of the wavelength and the port **11-2** to which the optical signal is input.

The optical SW **10h** drops the optical signal on the electrical processing unit **84** by outputting an optical signal from the port **12-1**. The electrical processing unit **84** electrically terminates the dropped optical signal, performs various types of electrical processing such as error correction and line concentration, converts the signal into an optical signal, and inputs the optical signal to the port **12-2** of the optical SW **10h**. The optical SW **10h** outputs the optical signal input from the electrical processing unit **84** from the port **11-1** or **11-2** in accordance with the transmission destination on the path to the communication destination specified in the combination of the port **12-2** and the wavelength. In this way, the electrical processing unit **84** performs electrical processing addition (O-E)-O conversion (where O represents light, and E represents electricity). The electrical processing unit **84** may simply perform O-E-O conversion without performing electrical processing for adding a function. The electrical processing unit **84** can also reduce optical waveform deterioration associated with transmission by performing 3R reproduction (Re-amplification: amplification, Re-timing: timing reproduction, Re-shaping: waveform shaping) or performing 0/1 inversion to use a threshold effect in O-E-O conversion or the like.

The wavelength of the optical signal before the conversion into the electrical signal and the wavelength of the optical signal after the conversion from the electrical signal may be the same or different. The electrical processing unit **84** may multiplex optical signals transmitted from the plurality of subscriber devices **40** at an electrical stage, convert the multiplexed signal into a plurality of optical signals, and branch the plurality of optical signals to separate ports, multiplexers/demultiplexers, or power splitters. The branched optical signal may be further demultiplexed and output to the multiplexer/demultiplexer at a plurality of wavelengths or may be further branched by a power splitter to be multicast. The electrical processing unit **84** may perform only multiplexing or only multicasting in the electrical stage. The process at the electrical stage is appropriate for a case where signals of the transceivers of the subscriber device **40** having a band smaller than that of the transmission line are bundled and treated, for example.

Specifically, the optical SW **10h** is connected to an electrical processing unit **84** that converts optical signals transmitted from the plurality of subscriber devices **40** into electrical signals, multiplexes the electrical signals, processes the multiplexed electrical signals, converts the electrical signals into optical signals with a plurality of wavelengths, and inputs the optical signals to the optical switch **10h**. The optical switch control unit **26** controls the optical switch **10h** such that the plurality of optical signals input from the transmission line **50-1** are output to the electrical

processing unit **84** in accordance with a combination of the plurality of subscriber devices **40** that have transmitted the plurality of input optical signals and the wavelength of the input optical signal, and the signal input from the electrical processing unit **84** is output to the transmission line **50-2** corresponding to the transmission destination on the path to the communication destination specified by the wavelength.

When a port to be an output destination of the optical SW is defined by a combination of a subscriber device that transmits an optical signal and a wavelength, the destination may be different between a direction from the port **11-1** to the port **11-2** and a direction from the port **11-2** to the port **11-1** in order to cause the optical signal to pass through the electrical processing unit **84**.

The electrical processing unit **84** includes an optical/electrical (O/E) conversion unit **85**, a processing execution unit **86**, an E/O (electrical/optical) conversion unit **87**, and a storage unit **88**. The O/E conversion unit **85** converts the optical signal input from the optical SW **10h** into an electrical signal. The processing execution unit **86** includes a processor **861** and an accelerator **862**. The processor **861** is, for example, a general-purpose processor such as a central processing unit (CPU). The accelerator **862** is, for example, a processor such as a graphics processing unit (GPU). The processor **861** and the accelerator **862** perform electrical signal processing on the electrical signal converted by the O/E conversion unit **85** by reading and executing a program from the storage unit **88**. The processing execution unit **86** may perform electrical signal processing of a plurality of functions. Examples of the electrical signal processing are digital signal processing (DSP) for long distance/high speed access, mobile fronthaul processing, error correction, or the like. The E/O conversion unit **87** converts the electrical signal into an optical signal with a wavelength instructed from the optical SW control unit **26** and outputs the optical signal to the optical SW **10h**. The storage unit **88** stores a program used for the processor **861** and the accelerator **862** to perform a function of electrical signal processing.

Since the processing execution unit **86** has a device architecture based on a general-purpose processor, it is possible to add and change electrical signal processing and to replace with various functions other than the transmission function. Since the processing execution unit **86** performs DSP for long distance/high speed access, a dedicated large scale integration (Large-Scale Integration, Large-Scale Integrated Circuit) for long distance/high speed access is not required, and flexible function deployment according to needs can be realized.

The optical SW **10h** may be connected to a plurality of electrical processing units **84**. In this case, the optical SW **10h** includes ports **12-1** and **12-2** connected to the electrical processing units **84**. Each of the electrical processing units **84** may perform different electrical signal processing, or some or all of the electrical processing units may perform the same electrical processing.

The processing execution unit **86** and the storage unit **88** may be realized by using hardware such as an application specific integrated circuit (ASIC), a programmable logic device (PLD), or a field programmable gate array (FPGA).

FIG. **14** is a diagram illustrating an example of connection using the optical SW **10h**. Subscriber devices **40-1**, **40-2**, and **40-3** connected to the optical SW **10h** are, for example, ONUs. A user **46-1** who uses the subscriber device **40-1** is a user who performs long-distance or high-speed communication. One or more communication devices of the user **46-1** are connected to the subscriber device **40-1**. The subscriber device **40-1** communicates with a communication

destination device along a long distance line P1. The mobile base station 46-2 is connected to the subscriber device 40-2. In FIG. 14, the plurality of subscriber devices 40-2 are connected to one transmission line 50-1 by a power splitter 55. The subscriber device 40-2 communicates with the communication destination device along a middle distance line P2. A user 46-3 who uses the subscriber device 40-3 is a user who performs medium-distance or medium-speed communication. One or more communication devices of the user 46-3 communicate with a communication destination device along the middle distance line P3 via the subscriber device 40-3. The optical signals of the long distance line P1, the middle distance line P2, and the middle distance line P3 are wavelength-multiplexed and transmitted along the multiplexing communication transmission line 90 connected to the core NW (network). The electrical processing unit 84 has a DSP function for long-distance/high-speed access, a mobile fronthaul processing function, an error correction function, and the like.

An operation of the optical SW 10*h* will be described with reference to FIGS. 13 and 14. The optical SW 10*h* outputs an uplink optical signal transmitted by the subscriber device 40-1 to the electrical processing unit 84. The O/E conversion unit 85 of the electrical processing unit 84 converts the input optical signal into an electrical signal. The processing execution unit 86 performs DSP processing for long-distance/high-speed access on the converted electrical signal. The E/O conversion unit 87 converts the electrical signal subjected to the DSP processing into an optical signal and outputs the optical signal to the optical SW 10*h*. The converted wavelength may be the same as or different from the wavelength when input to the electrical processing unit 84. The optical SW 10*h* outputs the optical signal input from the electrical processing unit 84 from the port 11-2 to the multiplexing communication transmission line 90.

The optical SW 10*h* receives a downlink optical signal transmitted along the multiplexing communication transmission line 90 and addressed to the subscriber device 40-1. The optical SW 10*h* outputs the input downlink optical signal from the port 12-1 to the electrical processing unit 84 according to a combination of the input port 51-2 and the wavelength. The O/E conversion unit 85 of the electrical processing unit 84 converts the input optical signal into an electrical signal, and the processing execution unit 86 performs DSP processing for long-distance/high-speed access on the converted electrical signal. The E/O conversion unit 87 converts the electrical signal subjected to the DSP processing into an optical signal and outputs the optical signal to the optical SW 10*h*. The converted wavelength of the optical signal may be the same as or different from the wavelength at the time of inputting to the electrical processing unit 84. The optical SW 10*h* outputs the optical signal input from the electrical processing unit 84 to the port 11-1 connected to the subscriber device 40-1.

In the foregoing example, the configuration in which the electrical processing unit 84 modulates the multiplexed electrical signal with the same signal and converts the modulated electrical signal into optical signals with a plurality of wavelengths has been described. The electrical processing unit 84 may be configured to convert the multiplexed or demultiplexed electrical signals into one or more optical signals (optical signals modulated with the same signal or different signals) with one or more wavelengths.

The optical signal transmitted and received by the subscriber device 40-2 is also processed in the same way as the optical signal transmitted and received by the subscriber device 40-1, as described above. Here, the processing execu-

tion unit 86 performs mobile fronthaul processing on the optical signal transmitted and received by the subscriber device 40-2. The processing execution unit 86 determines signal processing to be performed on the electrical signal based on any information included in the electrical signal.

On the other hand, the optical SW 10*h* outputs an uplink optical signal input from the subscriber device 40-3 from the port 11-2 to the multiplexing communication transmission line 90. The optical SW 10*h* inputs a downlink optical signal addressed to the subscriber device 40-3 transmitted along the multiplexing communication transmission line 90 and outputs a downlink optical signal to the port 11-1 connected to the subscriber device 40-3 according to a combination of the wavelength and the port 11-2 to which the optical signal is input.

FIG. 15 is a diagram illustrating a configuration example of an optical SW 10*i* that monitors an optical signal before WDM transmission is performed. In the drawing, the same portions as those of the optical SW 10*e* illustrated in FIG. 6 are denoted by the same reference numerals, and the description thereof will be omitted. The difference from FIG. 6 is that the monitoring unit 60 is provided between the port 11-2 and the WDM device 80. The transmission line 50-2 connected to the port 11-2 includes the monitoring unit 60. In the drawing, only three monitoring units 60 are illustrated. A specific configuration of the monitoring unit 60 is as illustrated in FIG. 2. An optical signal output from the port 11-2 is multiplexed and output by the WDM device 80 via the monitoring unit 60. The WDM device 80 demultiplexes the optical signal received along the multiplexing communication transmission line 90 by the wavelength, and inputs the demultiplexed optical signals to the monitoring unit 60. The monitoring unit 60 generates monitoring information based on a monitoring result and outputs the generated monitoring information. An example of an output destination of the monitoring information is the control unit 20.

Next, an access topology to the optical SW will be described with reference to FIGS. 16 to 25.

FIG. 16 is a diagram illustrating a passive double star (PDS) type access topology using time division multiplexing. As the optical SW 1001, any of the above-described optical SWs 10*a* to 10*i* can be used. The optical SW 1001 includes ports 11-1-1 to 11-1-P and ports 11-2-1 to 11-2-Q. The transmission line 50-1 connected to the port 11-1-*p* (where *p* is an integer equal to or greater than 1 and equal to or less than P) is also referred to as a transmission line 50-1-*p*, and the transmission line 50-2 connected to the port 11-2-*q* (where *q* is an integer equal to or greater than 1 and equal to or less than Q) is also referred to as a transmission line 50-2-*q*. In FIG. 16, the port 11-2-*q* is connected to the ground #*q* by a transmission line 50-2-*q*.

A power splitter 56 is provided in the transmission line 50-1-*p*. To the power splitter 56, N_p (where N_p is an integer equal to or greater than 2) subscriber devices 40-*p* are connected in a star form. The N_p subscriber devices 40-*p* are referred to as subscriber devices 40-*p*-1 to 40-*p*-N_p, and a transmission line 50-1-*p* between the subscriber device 40-*p*-*np* (where *np* is an integer equal to or greater than 1 and equal to or less than N_p) and the power splitter 56 is referred to as 50-1-*p*-*np*. The subscriber devices 40-*p*-1 to 40-*p*-N_p use the same wavelength by time division multiplexing. The wavelength used for an uplink optical signal is different from a wavelength used for a downlink optical signal.

The optical SW 1001 receives downlink optical signals with the wavelength λ_1 addressed to the subscriber devices 40-*p*-1 to 40-*p*-N_p time-division-multiplexed from the port 11-2-*q*. The optical SW 1001 outputs an input downlink

optical signal from the port 11-1- p which is the output destination according to a combination of the port 11-2- q and the wavelength λ_1 . The power splitter 56 receives the time-division-multiplexed downlink optical signal from the transmission line 50-1- p , branches the input optical signal, and outputs the optical signal to transmission lines 50-1- p -1 to 50-1- p -N p . The subscriber devices 40- p -1 to 40- p -N p receive time-division-multiplexed optical signals and select downlink optical signals addressed to the own devices from the received optical signals.

The subscriber devices 40- p -1 to 40- p -N p transmit uplink optical signals time-division-multiplexed with the same wavelength λ_2 by time division multiple access (TDMA). The power splitter 56 receives an uplink optical signal with a wavelength λ_2 from each of the transmission lines 50-1- p -1 to 50-1- p -N p , time-division-multiplexes the input optical signal, and outputs the optical signal to the transmission line 50-1- p . The optical SW 1001 outputs the time-division-multiplexed uplink optical signal from the port 11-2- q according to the combination of the port 11-1- p and the wavelength λ_2 .

Note that the PDS type access topology can be applied to any one or more of the transmission lines 50-1-1 to 50-1-P.

FIG. 17 is a diagram illustrating a PDS type access topology using wavelength multiplexing. As an optical SW 1002, any of the above-described optical SWs 10a to 10i can be used. The optical SW 1002 is connected to one or more WDM devices 81. The WDM device 81 multiplexes downlink optical signals with different wavelengths output from the plurality of ports 11-1 and outputs the multiplexed optical signal to the multiplexing communication transmission line 91. The WDM device 81 demultiplexes the uplink wavelength-multiplexed optical signal received along the multiplexing communication transmission line 91 and inputs the demultiplexed optical signals to different ports 11-1. A power splitter 56 is provided in the multiplexing communication transmission line 91. N (where N is an integer equal to or greater than 2) subscriber devices 40 are connected to the power splitter 56 in a star form. The subscriber device 40 and the power splitter 56 are connected by a transmission line 92. The plurality of subscriber devices 40 connected to the power splitter 56 transmit and receive optical signals of different wavelengths.

In FIG. 17, the ports 11-1- p to 11-1-(p +N) of the optical SW 1002 are connected to the WDM device 81 via the transmission line 50-1 (where p and N are integers equal to or greater than 1, and p +N is an integer equal to or less than P). Subscriber devices 40- p to 40-(p +N) are connected to the power splitter 56.

The optical SW 1002 inputs a downlink optical signal addressed to the subscriber device 40-(p + n) with the wavelength $\lambda_{1(q+n)}$ from the port 11-2-(q + n) (where q is an integer equal to or greater than 1 and n is an integer of 0 or more and equal to or less than N). FIG. 17 illustrates an example in a case where $q=1$. The optical SW 1002 routes the downlink optical signal with the wavelength $\lambda_{1(1+n)}$ input from the port 11-2-(q + n) to the port 11-1-(q + n) which is the output destination corresponding to a combination of the port 11-2-(p + n) and the wavelength $\lambda_{1(1+n)}$. Accordingly, the optical SW 1002 routes the downlink optical signal having the wavelength λ_{11} input from the port 11-2-1 to the port 11-1- p , and routes the downlink optical signal with the wavelength λ_{12} input from the port 11-2-2 to the port 11-1-(p +1).

The WDM device 81 multiplexes the downlink optical signals with the wavelengths λ_{11} to λ_{1N} output from the ports 11-1- p to 11-1-(p +N) and outputs the multiplexed optical signal to the multiplexing communication transmission line

91. The power splitter 56 receives the wavelength-multiplexed downlink optical signal from the multiplexing communication transmission line 91, branches the received downlink optical signal as it is, and outputs the optical signal to the transmission lines 92 to the subscriber devices 40- p to 40-(p +N). The subscriber devices 40- p to 40-(p +N) receive the wavelength-multiplexed downlink optical signal, and select the downlink optical signal with the wavelength used by the own device from the received optical signal.

The subscriber device 40-(p + n) transmits an uplink optical signal with a wavelength $\lambda_{2(1+n)}$. The power splitter 56 receives an uplink optical signal from each of the subscriber devices 40- p to 40-(p +N) via the transmission line 92, performs wavelength multiplexing on the input uplink optical signal of each of the wavelengths λ_{21} to $\lambda_{2(1+N)}$, and outputs the multiplexed optical signal to the multiplexing communication transmission line 91. The WDM device 81 receives the wavelength-multiplexed uplink optical signal from the multiplexing communication transmission line 91 and performs wavelength separation. The WDM device 81 inputs uplink optical signals with the wavelength $\lambda_{2(1+n)}$ to the port 11-1-(p + n). The optical SW 1002 outputs the uplink optical signal with the wavelength $\lambda_{2(1+n)}$ from the port 11-2-(q + n) which is the output destination corresponding to a combination of the input port 11-1-(p + n) and the wavelength $\lambda_{2(1+n)}$. Accordingly, the uplink optical signal with the wavelength λ_{21} transmitted by the subscriber device 40- p is input from the port 11-1- p and output from the port 11-2-1. The uplink optical signal with the wavelength λ_{22} transmitted by the subscriber device 40-(p +1) is input from the port 11-1-(p +1) and is output from the port 11-2-2. As illustrated in FIG. 18, a WDM device may be disposed at a subsequent stage of the optical SW.

FIG. 18 is a diagram illustrating a PDS type access topology in which wavelength multiplexing is used and a WDM device is disposed at a subsequent stage of an optical SW. As the optical SW 1003, any of the above-described optical SWs 10a to 10i can be used. A port 11-2- q (where q is an integer between 1 and Q) of the optical SW 1003 is connected to a WDM device 97 via a transmission line 50-2- q . The WDM device 97 is connected to a ground # n (n is an integer equal to or greater than 1 and equal to or less than N) via a transmission line 50-2- q - n . A power splitter 56 is provided in the transmission line 50-1- p connected to the port 11-1- p of the optical SW 1003. N subscriber devices 40- p -1 to 40- p -N are connected to the power splitter 56 in a star configuration.

The WDM device 97 inputs a downlink optical signal addressed to the subscriber device 40- p - n with the wavelength λ_{1n} transmitted from the ground # n from the transmission line 50-2- q - n . The WDM device 97 inputs, to the optical SW 1003, the wavelength-multiplexed signal obtained by multiplexing the downlink optical signals of λ_{11} to λ_{1N} input from the ground #1 to the ground #N. The optical SW 1003 outputs the downlink wavelength-multiplexed signal input from the port 11-2- q from the port 11-1- p which is the output destination. The power splitter 56 branches the wavelength-multiplexed signal input from the transmission line 50-1- p and outputs the branched signals to the transmission lines 50-1- p -1 to 50-1- p -N. The subscriber devices 40- p -1 to 40- p -N receive the wavelength-multiplexed signals and select a downlink optical signal addressed to the own devices from the received optical signals. Accordingly, the subscriber device 40- p - n receives the optical signal with the wavelength λ_{1n} from the ground # n .

The subscriber device **40-p-n** transmits an uplink optical signal with a wavelength λ_{2n} . The power splitter **56** inputs uplink optical signals with wavelengths **221** to **22N** from the respective subscriber devices **40-p-1** to **40-p-N** via transmission lines **50-1-p-1** to **50-1-p-N**. The power splitter **56** outputs a wavelength-multiplexed signal obtained by wavelength-multiplexing uplink optical signals with the wavelengths **221** to **22N** to the transmission line **50-1-p**. The optical SW **1003** receives, from the port **11-1-p**, the wavelength-multiplexed signals obtained by wavelength-multiplexing the uplink optical signals with the wavelengths λ_{21} to λ_{2N} . The optical SW **1003** outputs the uplink wavelength-multiplexed signal from the port **11-2-q** which is the output destination to the transmission line **50-2-q**. The WDM device **97** receives the wavelength-multiplexed uplink optical signal from the transmission line **50-2-q** and performs wavelength separation. The WDM device **97** outputs an uplink optical signal with having the wavelength λ_{2n} to the transmission line **50-2-n** connected to the ground #n. Accordingly, the optical signal with the wavelength λ_{2n} transmitted by the subscriber device **40-p-n** is transmitted to the ground #n.

FIG. **19** is a diagram illustrating a bus type access topology using time division multiplexing. As the optical SW **1004**, any of the above-described optical SWs **10a** to **10i** can be used. The access topology illustrated in FIG. **19** is different from the access topology illustrated in FIG. **16** in that a plurality of bus type subscriber devices **40-p-1** to **40-p-Np** are connected to the transmission line **50-1-p**. One or more power splitters **55** are provided in the transmission line **50-1-p**. The power splitter **55** to which the subscriber device **40-p-n** (where n is an integer equal to or greater than 1 and equal to or less than Np-1) is connected is referred to as a power splitter **55-n**.

The subscriber devices **40-p-1** to **40-p-Np** use the same wavelength by time division multiplexing. The wavelength used for an uplink optical signal is different from a wavelength used for a downlink optical signal. The transmission line **50-2-1** connected to the ground #1 transmits time-division-multiplexed downlink optical signals with the wavelength addressed to the subscriber devices **40-p-1** to **40-p-Np**. The optical SW **1004** receives, from the port **11-2-1**, the time-division-multiplexed downlink optical signal with the wavelength λ_1 transmitted through the transmission line **50-2-1**. The optical SW **1004** routes the input downlink optical signal to the port **11-1-p** which is the output destination according to a combination of the port **11-2-1** (or ground #1) and the wavelength λ_1 . The optical SW **1004** outputs the time-division-multiplexed downlink optical signal with the wavelength λ_1 from the port **11-1-p** to the transmission line **50-1-p**. The power splitter **55-n** branches the time-division-multiplexed downlink optical signal from the transmission line **50-1-p** and outputs the branched downlink optical signal to the subscriber device **40-p-n**. The subscriber devices **40-p-1** to **40-p-Np** receive time-division-multiplexed downlink optical signals and select downlink optical signals addressed to the own devices from the received downlink optical signals.

The subscriber devices **40-p-1** to **40-p-Np** transmit uplink optical signals time-division-multiplexed with the same wavelength λ_2 by TDMA. Each power splitter **55-n** time-division-multiplexes an uplink optical signal with the wavelength λ_2 input from the subscriber device **40-p-n** on the uplink optical signal transmitted along the transmission line **50-1-p**. The optical SW **1004** inputs the time-division-multiplexed uplink optical signal from the port **11-1-p**, routes the signal to the port **11-2-1** which is an output

destination according to a combination of the port **11-1-p** and the wavelength and outputs the signal to the transmission line **50-2-1** connected to the ground #1.

The bus type access topology can be applied to any one or more of the transmission lines **50-1-1** to **50-1-P**.

FIG. **20** is a diagram illustrating a bus type access topology using wavelength multiplexing. As an optical SW **1005**, any of the above-described optical SWs **10a** to **10i** can be used. The access topology illustrated in FIG. **19** is different from the access topology illustrated in FIG. **17** in that a plurality of bus type subscriber devices **40-p** to **40-(p+N)** are connected to the multiplexing communication transmission line **91**. The subscriber devices **40-p** to **40-(p+N)** transmit and receive optical signals with different wavelengths. One or more power splitters **55** are provided in the multiplexing communication transmission line **91**. The power splitter **55** to which the subscriber device **40-(p+n)** (where n is an integer of 0 to N-1 and N is an integer equal to or greater than 1) is connected will be referred to as a power splitter **55-(p+n)**.

Like the optical SW **1002** illustrated in FIG. **17**, the optical SW **1005** inputs a downlink optical signal addressed to the subscriber device **40-(p+n)** with the wavelength $\lambda_{1(1+n)}$ from the port **11-2-(q+n)** (where q is an integer equal to or greater than 1 and n is an integer equal to or greater than 0 and equal to or less than N). FIG. **20** illustrates an example in a case where q=1. The optical SW **1005** routes the downlink optical signal of the wavelength $\lambda_{1(1+n)}$ input from the port **11-2-(q+n)** to the port **11-1-(p+n)** which is the output destination corresponding to the combination of the port **11-2-(q+n)** and the wavelength $\lambda_{1(1+n)}$.

The WDM device **81** multiplexes the downlink optical signals with the wavelengths λ_{11} to λ_{1N} output from the ports **11-1-p** to **11-1-(p+N)** and outputs the multiplexed optical signal to the multiplexing communication transmission line **91**. The power splitter **55-(p+n)** branches the wavelength-multiplexed downlink optical signal from the multiplexing communication transmission line **91** and outputs the branched downlink optical signal to the subscriber device **40-(p+n)**. The subscriber devices **40-p** to **40-(p+N)** receive the wavelength-multiplexed downlink optical signals and select the downlink optical signals addressed to the own devices from the received downlink optical signals.

The subscriber device **40-(p+n)** transmits an uplink optical signal with a wavelength $\lambda_{2(1+n)}$. Each power splitter **55-(p+n)** multiplexes the uplink optical signal with the wavelength $\lambda_{2(1+n)}$ input from the subscriber device **40-(p+n)** to the uplink optical signal with the wavelengths $\lambda_{2(2+n)}$ to λ_{2N} transmitted along the multiplexing communication transmission line **91**. The WDM device **81** receives the wavelength-multiplexed uplink optical signal from the multiplexing communication transmission line **91** and separates the optical signal into uplink optical signals with wavelengths λ_{21} to λ_{2N} . The WDM device **81** inputs an uplink optical signal with a wavelength $\lambda_{2(1+n)}$ to the port **11-1-(p+n)**. Like the optical SW **1002** illustrated in FIG. **16**, the optical SW **1005** outputs the uplink optical signal with the wavelength $\lambda_{2(1+n)}$ from the port **11-1-(q+n)** which is the output destination corresponding to a combination of the input port **11-2-(p+n)** and the wavelength $\lambda_{2(1+n)}$. Accordingly, the uplink optical signal with the wavelength λ_{21} transmitted by the subscriber device **40-p** is input from the port **11-1-p** and output from the port **11-2-1**. The uplink optical signal with the wavelength λ_{22} transmitted by the subscriber device **40-(p+1)** is input from the port **11-1-(p+1)**

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and is output from the port 11-2-2. As illustrated in FIG. 21, a WDM device may be disposed at a subsequent stage of the optical SW.

FIG. 21 is a diagram illustrating a bus type access topology in which wavelength multiplexing is used and a WDM device is disposed at a subsequent stage of an optical SW. As an optical SW 1006, any of the above-described optical SWs 10a to 10i can be used. In description of FIG. 21, the same portions as those of FIG. 18 will be denoted by the same reference numerals. A port 11-2-q (where q is an integer between 1 and Q) of the optical SW 1006 is connected to the WDM device 97 via the transmission line 50-2-q. The WDM device 97 is connected to a ground #n (where n is an integer equal to or greater than 1 and equal to or less than N, and N is an integer equal to or greater than 2) via the transmission line 50-2-q-n. One or more power splitters 55 are provided in the transmission line 50-1-p connected to the port 11-1-p (where p is an integer equal to or greater than 1 and equal to or less than P) of the optical SW 1006. The power splitter 55 to which the subscriber device 40-p-n is connected is referred to as a power splitter 55-n.

The WDM device 97 inputs a downlink optical signal addressed to the subscriber device 40-p-n with the wavelength λ_{1n} transmitted from the ground #n from the transmission line 50-2-q-n. The WDM device 97 inputs, to the optical SW 1006, a wavelength-multiplexed signal obtained by multiplexing downlink optical signals with wavelengths λ_{11} to λ_{1N} input from each of the ground #1 to the ground #N. The optical SW 1006 outputs the downlink wavelength-multiplexed signal input from the port 11-2-q from the port 11-1-p which is the output destination. The power splitter 55-n branches the downlink wavelength-multiplexed signal from the transmission line 50-1-p and outputs the branched downlink wavelength-multiplexed signal to the subscriber device 40-p-n. The subscriber devices 40-p-1 to 40-p-N select downlink optical signals addressed to the own devices from the received downlink wavelength multiplexing signals. Accordingly, the subscriber device 40-p-n receives the optical signal with the wavelength λ_{1n} from the ground #n.

The subscriber device 40-p-n transmits an uplink optical signal with a wavelength λ_{2n} . Each power splitter 55-n wavelength-multiplexes the uplink optical signal with the wavelength λ_{2n} input from the subscriber device 40-p-n into the uplink optical signal transmitted through the transmission line 50-1-p. The optical SW 1006 receives, from the port 11-1-p, a wavelength-multiplexed signal obtained by wavelength-multiplexing the uplink optical signals with the wavelengths λ_{21} to λ_{2N} . The optical SW 1006 outputs the uplink wavelength-multiplexed signal from the port 11-2-q which is output destination to the transmission line 50-2-q. The WDM device 97 receives a wavelength-multiplexed signal from the transmission line 50-2-q and performs wavelength separation. The WDM device 97 outputs an uplink optical signal with the wavelength λ_{2n} to the transmission line 50-2-q-n connected to the ground #n. Accordingly, the optical signal with the wavelength λ_{2n} transmitted by the subscriber device 40-p-n is transmitted to the ground #n.

FIG. 22 is a diagram illustrating a loop type access topology. As the optical SW 1007, any of the above-described optical SWs 10a to 10i can be used. Some ports 11-1-p1 to 11-1-pN (where $p1 < pN$, p1 is an integer equal to or greater than 1, and pN is an integer equal to or less than P) of the optical SW 1007 are connected to a WDM access ring network 31 that transmits optical signals with a plurality of wavelengths. Optical signals with several wavelengths

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used in the WDM access ring network 31 are transmitted to the communication destination subscriber device 40 or an upper NW via the optical SW 1007. In the topology illustrated in FIG. 22, the WDM multiplexer/demultiplexer is not used, and the facing subscriber devices 40 are connected by two transmission lines to perform communication.

The WDM access ring network 31 is a network in which R Add/Drop nodes 32 are connected by a transmission line 53. FIG. 21 illustrates an example of R=4. The R Add/Drop nodes 32 will be referred to as Add/Drop nodes 32-1 to 32-R, and a transmission line 53 between an Add/Drop node 32-r (where r is an integer equal to or greater than 1 and equal to or less than R) and an Add/Drop node 32-(r+1) will be referred to as a transmission line 53-r. However, the Add/Drop node 32-(r+1) is regarded as an Add/Drop node 32-1. The Add/Drop node 32-1 is connected to a port 11-1-pn (where pn is an integer equal to or greater than p1 and equal to or less than pN) of the optical SW 1007 via a transmission line 50-1-pn.

The Add/Drop node 32 includes a demultiplexing unit 33, an optical SW 34, and a multiplexing unit 35. The demultiplexing unit 33 of the Add/Drop node 32-r (where r is an integer equal to or greater than 2 and equal to or less than R) demultiplexes a wavelength-multiplexed optical signal input from a transmission line 53-(r-1), and outputs an optical signal obtained by the demultiplexing to the optical SW 34. The optical SW 34 is connected to one or more subscriber devices 40. In the drawing, only one subscriber device 40 connected to the optical SW 34 is illustrated. The optical SW 34 drops an optical signal with a wavelength corresponding to the own node among the optical signals input from the demultiplexing unit 33. The optical receiver 43 of the subscriber device 40 receives an optical signal dropped by the optical SW 34. The optical SW 34 receives the optical signal transmitted from the optical transmitter 42 of the subscriber device 40 and outputs the received optical signal and the optical signals that have not been dropped to the multiplexing unit 35. The multiplexing unit 35 of the Add/Drop node 32-r multiplexes the optical signals input from the optical SW 34 and outputs the multiplexed optical signal to the transmission line 53-r. The optical SW 34 of the Add/Drop node 32-1 drops an optical signal with a wavelength corresponding to the own node among the optical signals demultiplexed by the demultiplexing unit 33 and inputs the optical signal with each wavelength to the transmission line 50-1-pn₁ (where pn₁=1, 3, 5, . . . , p(N-1)). The port 11-1-pn₁ of the optical SW 1007 inputs the optical signal dropped by the Add/Drop node 32-1 from the transmission line 50-1-pn₁. The optical SW 34 of the Add/Drop node 32-1 inputs an optical signal output from a port 11-1-pn₂ (where pn₂=2, 4, 6, . . . , pN) by the optical SW 1007 from the transmission line 50-1-pn₂ and outputs the input optical signal and the optical signals that have not been dropped to the multiplexing unit 35.

Accordingly, ONU #1 that is the subscriber device 40 connected to the Add/Drop node 32-4 of the WDM access ring network 31 and ONU #2 that is the subscriber device 40 connected to the port 11-2-1 and the port 11-2-2 of the optical SW 1007 communicate as follows.

ONU #1 transmits an optical signal with the wavelength to the Add/Drop node 32-4. The multiplexing unit 35 of the Add/Drop node 32-4 multiplexes the optical signal with the wavelength λ_1 input by the optical SW 34 and the optical signal not dropped by the optical SW 34, and outputs the multiplexed signal to the Add/Drop node 32-1. The optical SW 34 of the Add/Drop node 32-1 drops the optical signal with the wavelength λ_1 demultiplexed by the demultiplexing

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unit 33 and outputs the optical signals that have not been dropped to the multiplexing unit 35. The port 11-1-p1 of the optical SW 1007 inputs the optical signal having the wavelength λ_1 dropped by the Add/Drop node 32-1 from the transmission line 50-1-p1. The optical SW 1007 outputs the optical signal with the wavelength input from the port 11-1-p1 from the port 11-2-1. The optical receiver 43 of ONU #2 receives the optical signal with the wavelength λ_1 transmitted along the transmission line 50-2-1.

The optical transmitter 42 of ONU #2 transmits a downlink optical signal with the wavelength λ_2 . The port 11-2-2 of the optical SW 1007 inputs the optical signal transmitted by ONU #2 from the transmission line 50-2-2. The optical SW 1007 outputs the downlink optical signal with the wavelength λ_2 input from the port 11-2-2 from the port 11-1-p2. The optical SW 34 of the Add/Drop node 32-1 inputs the optical signal with the wavelength λ_2 output from the optical SW 1007 from the transmission line 50-1-p2, and outputs the input optical signal and the optical signals that have not been dropped to the multiplexing unit 35. The optical signal with the wavelength λ_2 is input to the Add/Drop node 32-4 via the Add/Drop nodes 32-2 and 32-3. The optical SW 34 of the Add/Drop node 32-4 drops the optical signal with the wavelength λ_2 . The optical receiver 43 of ONU #1 receives the optical signal with the wavelength λ_2 dropped by the Add/Drop node 32-4.

FIG. 23 is a diagram illustrating a loop type access topology using a WDM multiplexer/demultiplexer. As an optical SW 1008, any of the above-described optical SWs 10a to 10i can be used. The access topology illustrated in FIG. 23 is different from the access topology illustrated in FIG. 22 in that the optical SW 1008 and the WDM access ring network 31 are connected via the WDM device 81 and the WDM device 89.

The Add/Drop node 32-1 of the WDM access ring network 31 and the WDM device 89 are connected by transmission lines 93-1 to 93-N (N is an integer equal to or greater than 2). The WDM device 89 receives the uplink optical signal having the wavelength λ_{n1} from the transmission line 93- n_1 (where $n_1=1, 3, 5, \dots, N-1$) and outputs a multiplexed signal obtained by multiplexing the received uplink optical signals to the multiplexing communication transmission line 91. The WDM device 89 demultiplexes the downlink wavelength-multiplexed optical signal received via the multiplexing communication transmission line 91 and inputs the demultiplexed downlink optical signal with the wavelength to the transmission line 93- n_2 (where $n_2=2, 4, 6, \dots, N$).

The WDM device 81 demultiplexes the uplink wavelength-multiplexed optical signal received via the multiplexing communication transmission line 91, and inputs the demultiplexed uplink optical signal with the wavelength λ_{n1} to the port 11-1- n_1 . The WDM device 81 receives the downlink optical signal with the wavelength λ_{n2} output from each of the ports 11-1 to pn_2 , multiplexes the received downlink signals, and outputs the multiplexed signal to the multiplexing communication transmission line 91.

Accordingly, ONU #1 that is the subscriber device 40 connected to the Add/Drop node 32-4 of the WDM access ring network 31 and ONU #2 that is the subscriber device 40 connected to the port 11-2-1 and the port 11-2-2 of the optical SW 1008 communicate as follows. The case of $N=18$ will be described as an example.

ONU #1 transmits an uplink optical signal with the wavelength λ_1 to the Add/Drop node 32-4. Another ONU transmits an uplink optical signal with the wavelengths λ_3 and λ_5 to the Add/Drop node 32-4. The optical SW 34 of the Add/Drop node 32-4 receives optical signals with wave-

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lengths λ_1 , λ_3 , and λ_5 . The multiplexing unit 35 of the Add/Drop node 32-4 multiplexes the optical signals with the wavelengths λ_1 , λ_3 , and λ_5 input by the optical SW 34 and the optical signal not dropped by the optical SW 34, and outputs the multiplexed signal to the Add/Drop node 32-1. The optical SW 34 of the Add/Drop node 32-1 drops the optical signals with the wavelengths λ_i , demultiplexed by the demultiplexing unit 33 and outputs the optical signals that have not been dropped to the multiplexing unit 35. The WDM device 89 outputs, to the multiplexing communication transmission line 91, the multiplexed signal obtained by multiplexing the uplink optical signals with the wavelengths λ_1 , λ_3 , $\lambda_5, \dots, \lambda_{17}$ input from the transmission lines 93-1, 93-3, 93-5, $\dots$, and 93-17, respectively.

The WDM device 81 receives the wavelength-multiplexed uplink optical signal from the multiplexing communication transmission line 91 and performs wavelength separation. The WDM device 81 inputs the uplink optical signals with the wavelengths λ_1 , λ_3 , $\lambda_5, \dots, \lambda_{17}$ to the ports 11-1-p1, 11-1-p3, 11-1-p5, $\dots$, and 11-1-p17 of the optical SW 1008, respectively. The optical SW 1008 outputs an upward optical signal with a wavelength from the output destination port 11-2-1. The optical receiver 43 of ONU #2 receives the optical signal with the wavelength λ_1 transmitted along the transmission line 50-2-1.

The optical transmitter 42 of ONU #2 transmits a downlink optical signal with the wavelength λ_2 . The port 11-2-2 of the optical SW 1008 inputs the optical signal transmitted by ONU #2 from the transmission line 50-2-2. The optical SW 1008 outputs the downlink optical signal with the wavelength λ_2 input from the port 11-2-2 from the port 11-1-p2. Further, the optical SW 1008 outputs downlink optical signals with wavelengths and λ_{18} input from the ports 11-2-4, 11-2-6, $\dots$, and 11-2-18, respectively, from the ports 11-1-p4, 11-2-p6, $\dots$, and 11-2-p18.

The WDM device 81 outputs, to the multiplexing communication transmission line 91, the wavelength-multiplexed signal obtained by multiplexing the downlink optical signals with the wavelengths λ_2 , λ_4 , $\lambda_6, \dots$, and λ_{18} output from the ports 11-1-p2, 11-1-p4, 11-1-p6, $\dots$, and 11-1-p18, respectively. The WDM device 89 separates the wavelength-multiplexed signal transmitted along the multiplexing communication transmission line 91, and outputs the downlink optical signals of the wavelengths λ_2 , λ_4 , $\lambda_6, \dots$, and λ_{18} obtained by the separation to the transmission lines 93-2, 93-4, 93-6, $\dots$, and 93-18, respectively. The optical SW 34 of the Add/Drop node 32-1 inputs the optical signals having the wavelengths λ_2 , λ_4 , $\lambda_6, \dots$, and λ_{18} output from the WDM device 89 from the transmission lines 93-2, 93-4, 93-6, $\dots$, and 93-18, respectively, and outputs the input optical signal and the optical signals that have not been dropped to the multiplexing unit 35. The multiplexing unit 35 multiplexes the optical signals input from the optical SW 34 and outputs the multiplexed optical signals to the transmission line 53-1.

The demultiplexing unit 33 of the Add/Drop node 32-2 demultiplexes the optical signal input from the transmission line 53-1 and outputs the signals to the optical SW 34. The optical SW 34 drops optical signals with wavelengths λ_{14} , λ_{16} , and λ_{18} corresponding to the own node. The optical signals with the wavelengths λ_{14} , λ_{16} , and λ_{18} are transmitted to the optical receiver 43 of the subscriber device 40 corresponding to the wavelengths. The optical SW 34 of the Add/Drop node 32-2 receives optical signals of the wavelengths λ_{13} , λ_{15} , and λ_{17} transmitted by the optical transmitter 42 of each subscriber device 40, and outputs the input optical signals and the optical signals that have not been

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dropped to the multiplexing unit 35. The multiplexing unit 35 multiplexes the optical signals input from the optical SW 34 and outputs the multiplexed optical signal to the transmission line 53-2.

The Add/Drop node 32-3 also operates like the Add/Drop node 32-2. However, the optical SW 34 of the Add/Drop node 32-3 drops the optical signals with the wavelengths λ_8 , λ_{10} , and λ_{12} corresponding to the own node, and inputs the optical signals with the wavelengths λ_7 , λ_9 , and λ_{11} . The demultiplexing unit 33 of the Add/Drop node 32-4 demultiplexes the wavelength-multiplexed optical signal input from the transmission line 53-3 and outputs the signal to the optical SW 34. The optical SW 34 of the Add/Drop node 32-4 drops the optical signals having wavelengths λ_2 , λ_4 , and λ_6 corresponding to the own node. The optical receiver 43 of ONU #1 receives the optical signal with the wavelength λ_2 dropped by the optical SW 34 of the Add/Drop node 32-4.

FIG. 24 is a diagram illustrating an access topology in which two access surfaces form one loop. As optical SWs 1009a and SW 1009b, any of the above-described optical SWs 10a to 10f can be used. The optical SWs 1009a and SW 1009b are collectively referred to as an optical SW 1009. The two ports 11-1 of the optical SW 1009 are connected to both ends of one transmission line 54. One or more power splitters 57 are connected to the transmission line 54. The power splitter 57 is connected to the optical transmitters 42 of the one or more subscriber devices 40 via a multiplexer 82 and the optical SW 95, and is connected to the optical receivers 43 of the one or more subscriber devices 40 via a demultiplexer 83 and the optical SW 96. Each of the subscriber devices 40 transmits and receives optical signals with different wavelengths.

The transmission line 54 connected to the optical SW 1009a is referred to as a transmission line 54a, the two ports 11-1 connected to the transmission line 54a are referred to as ports 11a-1-p1 and 11a-1-p2, the transmission line 54 connected to the optical SW 1009b is referred to as a transmission line 54b, and the two ports 11-1 connected to the transmission line 54b are referred to as ports 11b-1-p1 and 11b-1-p2. N (where N is an integer equal to or greater than 1) power splitters 57 connected to the transmission line 54a are referred to as power splitters 57a-1 to 57a-N, and M (where M is an integer equal to or greater than 1) power splitters 57 connected to the transmission line 54b are referred to as power splitters 57b-1 to 57b-M. The multiplexer 82 and the demultiplexer 83 connected to the power splitters 57a-n (where n is an integer equal to or greater than 1 and equal to or less than N) are referred to as a multiplexer 82a-n and a demultiplexer 83a-n, respectively. The multiplexer 82 and the demultiplexer 83 connected to the power splitters 57b-m (where m is an integer equal to or greater than 1 and equal to or less than M) are referred to as a multiplexer 82b-m and a demultiplexer 83b-m, respectively. The optical SW 95 connected to the multiplexer 82a-n is referred to as an optical SW 95a-n, and the optical SW 96 connected to the demultiplexer 83a-n is referred to as an optical SW 96a-n. The optical SW 95 connected to the multiplexer 82b-m is referred to as an optical SW 95b-m, and the optical SW 96 connected to the demultiplexer 83b-m is referred to as an optical SW 96b-m.

Optical SW 1009a and optical SW 1009b are connected by transmission lines 54c and 54d. The port 11-2 of the optical SW 1009a connected to the transmission line 54c is referred to as a port 11a-2-q1, and the port 11-2 of the optical SW 1009a connected to the transmission line 54d is referred to as a port 11a-2-q2. The port 11-2 of the optical SW 1009b

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connected to the transmission line 54c is referred to as a port 11b-2-q1, and the port 11-2 of the optical SW 1009b connected to the transmission line 54d is referred to as a port 11b-2-q2.

In the above configuration, the optical SW 95b-m outputs optical signals with different wavelengths transmitted by the optical transmitter 42 of each of the subscriber devices 40 to the port corresponding to each wavelength of the multiplexer 82b-m. The multiplexer 82b-m inputs optical signals with different wavelengths transmitted by the optical transmitters 42 of the subscriber devices 40 via the optical SW 95b-m and outputs a wavelength-multiplexed optical signal obtained by multiplexing the input optical signals. The power splitter 57b-m multiplexes the wavelength-multiplexed optical signal output from the multiplexer 82b-m into the wavelength-multiplexed optical signal transmitted along the transmission line 54b in a direction from the port 11b-1-p2 to the port 11b-1-p1, and outputs the multiplexed optical signal.

The port 11b-1-p1 of the optical SW 1009b receives the wavelength-multiplexed optical signal from the transmission line 54b and outputs the wavelength-multiplexed optical signal from the port 11b-2-q1. The port 11a-2-q1 of the optical SW 1009a inputs the wavelength-multiplexed optical signal output from the port 11b-2-q1 of the optical SW 1009b from the transmission line 54c. The optical SW 1009a outputs the wavelength-multiplexed optical signal input from the port 11a-2-q1 to the transmission line 54a from the port 11a-1-p1.

The power splitter 57a-n branches the wavelength-multiplexed optical signal transmitted along the transmission line 54a in a direction from the port 11a-1-p1 to the port 11a-1-p2, and outputs the branched wavelength-multiplexed optical signal to the demultiplexer 83a-n. The demultiplexer 83a-n demultiplexes the wavelength-multiplexed optical signal received from the power splitter 57a-n, and outputs the demultiplexed optical signal to the optical SW 96a-n from a port corresponding to a wavelength. The optical SW 96a-n outputs the optical signal with each wavelength input from the demultiplexer 83a-n to the optical receiver 43 of the subscriber device 40 that receives the optical signal with the wavelength.

On the other hand, the optical SW 95a-n output optical signals with different wavelengths transmitted by the optical transmitters 42 of the respective subscriber devices 40 to ports corresponding to the wavelengths of the multiplexers 82a-n. The multiplexers 82a-n input optical signals with different wavelengths transmitted by the optical transmitters 42 of the subscriber devices 40 via the optical SWs 95a-n, and output wavelength-multiplexed optical signals obtained by multiplexing the input optical signals. The power splitter 57a-n multiplexes the wavelength-multiplexed optical signal output from the multiplexer 82a-n into the wavelength-multiplexed optical signal transmitted through the transmission line 54a in a direction from the port 11a-1-p1 to the port 11a-1-p2, and outputs the multiplexed optical signal.

The port 11a-1-p2 of the optical SW 1009a receives the wavelength-multiplexed optical signal from the transmission line 54a and outputs the wavelength-multiplexed optical signal from the port 11a-2-q2. The port 11b-2-q2 of the optical SW 1009b inputs the wavelength-multiplexed optical signal output from the port 11a-2-q2 of the optical SW 1009a from the transmission line 54d. The optical SW 1009b outputs the wavelength-multiplexed optical signal input from the port 11b-2-q2 from the port 11b-1-p2 to the transmission line 54b.

The power splitter **57b-m** branches the wavelength-multiplexed optical signal transmitted along the transmission line **54b** in a direction from the port **11b-1-p2** to the port **11b-1-p1**, and outputs the branched wavelength-multiplexed optical signal to the demultiplexer **83b-m**. The demultiplexer **83b-m** demultiplexes the wavelength-multiplexed optical signal received from the power splitter **57b-m** and outputs the demultiplexed optical signal to the optical SW **96b-m** from a port corresponding to the wavelength. The optical SW **96b-m** outputs the optical signal with each wavelength input from the demultiplexer **83b-m** to the optical receiver **43** of the subscriber device **40** that receives the optical signal with the wavelength.

Although FIG. **24** illustrates a case where the optical signal is transmitted counterclockwise, the optical signal may be transmitted clockwise, and a pair of left and right dual cores may be used for redundancy.

FIG. **25** is a diagram illustrating a line type access topology. As an optical SW **1100**, any of the above-described optical SWs **10a** to **10i** can be used. The access topology illustrated in FIG. **25** is different from the access topology illustrated in FIG. **16** in that the monitoring unit **60** is connected between the subscriber device **40-p** and the optical SW **1100**. The monitoring unit **60** monitors the optical signal output from the subscriber device **40-p** or the optical signal output from the port **11-1-p**.

Next, a connection configuration in a case where the number of user connections increases will be described. FIG. **26** is a diagram illustrating an example in which scalability of the optical SW is required. FIG. **26** illustrates N (N is an integer equal to or greater than 1) optical SWs **1010-1** to **1010-N**. As the optical SWs **1010-1** to **1010-N**, any of the optical SWs **10a** to **10i** described above can be used. FIG. **26** illustrates an example of $N=4$. In the drawing, ONU $\#np$ serving as the subscriber device **40** is connected to the port **11-1-p** of the optical SW **1010-n** (where n is an integer equal to or greater than 1 and equal to or less than N). The port **11-2-q** of the optical SW **1010-n** is connected to an uplink. The uplink is a transmission line **50-2** connected to a higher network.

When the number of users becomes enormous and the number of ONUs increases, the optical SW **1010** may exceed an allowable size. In the present embodiment, even in such a case, a function similar to that in a case where the number of users is small, for example, connection in which any uplink is selected or optical folding to any subscriber device is realized in the connection configuration illustrated in FIG. **27** or **28**.

FIG. **27** is a diagram illustrating an example of optical SW scalability by a mesh configuration. Some of the ports **11-1** of the optical SW **1010** are connected to the ONU by the transmission line **50-1**, and some of the ports **11-2** are connected to the uplink transmission line **50-2**. Further, some of the ports **11-1** of the optical SW **1010** and some of the ports **11-2** of the other optical SW **1010** are connected by a transmission line **50-3**. In the drawing, one optical SW **1010** is connected to all the other optical SWs **1010**.

The plurality of ports **11-1** of the optical SW **1010** are referred to as ports **11-1-1**, **11-1-2**, **11-1-3**, . . . , **11-1-p1**, **11-1-p2**, and **11-1-p3**. The plurality of ports **11-2** of the optical SW **1010** are referred to as ports **11-2-1**, **11-2-2**, **11-2-3**, . . . , **11-2-q1**, **11-2-q2**, and **11-2-q3**.

In FIG. **27**, the ports **11-1-1**, **11-1-2**, **11-1-3**, . . . of the optical SW **1010-n** (where n is an integer equal to or greater than 1 and equal to or less than N) are connected to ONUs $\#n1$, $\#n2$, $\#n3$, . . . , and the ports **11-2-1**, **11-2-2**, **11-2-3**, . . . are connected to the transmission lines **50-2** of uplinks

$\#n1$, $\#n2$, $\#n3$, Further, the optical SWs **1010-n** are connected to the ports **11-1** of all the other optical SW **1010-j** (where $j \neq n$ and j is an integer equal to or greater than 1 and equal to or less than N) by some of the ports **11-2**. For example, the port **11-2-q1** of the optical SW **1010-1** is connected to the port **11-1-p1** of the optical SW **1010-2**, the port **11-2-q2** of the optical SW **1010-1** is connected to the port **11-1-p1** of the optical SW **1010-3**, and the port **11-2-q3** of the optical SW **1010-1** is connected to the port **11-1-p1** of the optical SW **1010-4**. The port **11-2-q1** of the optical SW **1010-2** is connected to the port **11-1-p1** of the optical SW **1010-1**, the port **11-2-q2** of the optical SW **1010-2** is connected to the port **11-1-p2** of the optical SW **1010-3**, and the port **11-2-q3** of the optical SW **1010-2** is connected to the port **11-1-p2** of the optical SW **1010-4**. The optical SW **1010-n** may be connected to some of the ports **11-1** of the optical SW **1010-j** among all the other optical SWs **1010-j** (where $j \neq n$, and j is an integer equal to or greater than 1 and equal to or less than N) by some of the ports **11-2**.

For example, when ONU $\#11$ transmits an uplink optical signal with the wavelength λ_1 addressed to uplink $\#41$, the optical SW **1010-1** outputs the optical signal input from the port **11-1-1** from the port **11-2-q3**. The port **11-1-p1** of the optical SW **1010-4** receives the optical signal with the wavelength λ_1 output from the port **11-2-q3** of the optical SW **1010-1**, and outputs the optical signal from the port **11-2-1**.

When ONU $\#12$ transmits the uplink optical signal with the wavelength λ_2 addressed to ONU $\#31$, the optical SW **1010-1** outputs the optical signal input from the port **11-1-2** from the port **11-2-q2**. The port **11-1-p1** of the optical SW **1010-3** receives the optical signal output from the port **11-2-q2** of the optical SW **1010-1**. The optical SW **1010-3** performs folded communication similar to the optical SW **1010b** illustrated in FIG. **3** for the optical signal with the wavelength λ_2 input from the port **11-1-p1**, and outputs the optical signal from the port **11-1-1**.

In FIG. **27**, only the uplink optical signal is illustrated. When vertical bidirectional communication is performed, a WDM filter (a separation unit) that separates and transmits an uplink optical signal and a downlink optical signal is provided in the transmission lines **50-1**, **50-2**, and **50-3**. Then, the downlink optical signal is connected in a direction opposite to the above-described uplink optical signal.

FIG. **28** is a diagram illustrating another example of the optical SW scalability by a cascade configuration. The configuration illustrated in FIG. **28** is different from the configuration illustrated in FIG. **27** in that an optical SW **1010-n** (where n is an integer equal to or greater than 1 and equal to or less than N) is connected to the port **11-1** of any other optical SW **1010-(n+1)** by some ports **11-2**. The optical SW **1010-(N+1)** is assumed to be the optical SW **1010-1**. Accordingly, the plurality of optical SWs **1010** are connected in series.

In FIG. **28**, the ports **11-1-1**, **11-1-2**, **11-1-3**, . . . of the optical SW **1010-n** (where n is an integer equal to or greater than 1 and equal to or less than N) are connected to ONUs $\#n1$, $\#n2$, $\#n3$, . . . , and the ports **11-2-1**, **11-2-2**, **11-2-3**, . . . are connected to the transmission lines **50-2** of uplinks $\#n1$, $\#n2$, $\#n3$, Further, the port **11-2-q1** of the optical SW **1010-n** is connected to the port **11-1-p1** of the optical SW **1010-(n+1)**, the port **11-2-q2** of the optical SW **1010-n** is connected to the port **11-1-p2** of the optical SW **1010-(n+1)**, and the port **11-2-q3** of the optical SW **1010-n** is connected to the port **11-1-p3** of the optical SW **1010-(n+1)**.

For example, when ONU $\#11$ transmits an uplink optical signal with the wavelength λ_1 addressed to uplink $\#41$, the

optical SW **1010-1** outputs the optical signal input from the port **11-1-1** from the port **11-2-q1**. The port **11-1-p1** of the optical SW **1010-2** receives the optical signal output from the port **11-2-q1** of the optical SW **1010-1** and outputs the optical signal from the port **11-2-q1** in accordance with the wavelength λ_1 . The port **11-1-p1** of the optical SW **1010-3** receives the optical signal output from the port **11-2-q1** of the optical SW **1010-2** and outputs the optical signal from the port **11-2-q1** according to the wavelength λ_1 . The port **11-1-p1** of the optical SW **1010-4** receives the optical signal output from the port **11-2-q1** of the optical SW **1010-3**, and outputs the optical signal from the port **11-2-1** in accordance with the wavelength λ_1 .

When ONU #12 transmits the uplink optical signal with the wavelength λ_2 addressed to ONU #31, the optical SW **1010-1** outputs the optical signal input from the port **11-1-2** from the port **11-2-q2**. The port **11-1-p2** of the optical SW **1010-2** receives the optical signal output from the port **11-2-q2** of the optical SW **1010-1**. The optical SW **1010-2** outputs the optical signal input from the port **11-1-p2** from the port **11-2-q2** in accordance with the wavelength λ_2 . The port **11-1-p2** of the optical SW **1010-3** receives the optical signal output from the port **11-2-q2** of the optical SW **1010-2**. The optical SW **1010-3** performs folded communication similar to the optical SW **10b** illustrated in FIG. 3 on the optical signal input from the port **11-1-p2** according to the wavelength λ_2 and outputs the optical signal from the port **11-1-1**.

In FIG. 28, only the uplink optical signal is illustrated. When vertical bidirectional communication is performed, a WDM filter that separates and transmits an uplink optical signal and a downlink optical signal is provided in the transmission lines **50-1**, **50-2**, and **50-3**. Then, the downlink optical signal is connected in a direction opposite to the above-described uplink optical signal.

Next, a configuration of an optical transceiver will be described with reference to FIGS. 29 to 32. Although a part of the configuration example of the subscriber device **40** connected to the optical SW will be described, the configuration example may be used in a control unit, a monitoring unit, and the like and may be used as a pair for transmission and reception, but may be used for only transmission or only reception depending on a use form. FIGS. 29 and 30 are diagrams illustrating a configuration of a dual-core subscriber device **401**. In FIGS. 29 to 32, in order to describe a plurality of configurations, the same configurations are denoted by the same reference numerals, and description thereof may be omitted.

The subscriber device **401** illustrated in FIG. 29(A) includes an optical transceiver **411**. The optical transceiver **411** includes a wavelength-tunable light source **451** and a tunable wavelength receiver **452**. The wavelength-tunable light source **451** is an example of an optical transmission unit and the tunable wavelength receiver **452** is an example of an optical reception unit.

The wavelength-tunable light source **451** outputs light with a set wavelength. A wavelength set in the wavelength-tunable light source **451** is tunable. The wavelength-tunable light source **451** includes, for example, a tunable wavelength laser diode (LD). The wavelength-tunable light source **451** includes, for example, any one of a combination of a gain medium and a resonator with a tunable resonator length, a combination of a gain medium and a wavelength-selective medium, or a combination of a gain medium, a resonator with a tunable resonator length, and a wavelength-selective medium. For example, the wavelength-tunable light source **451** includes a multiplexer/demultiplexer or a power splitter

and an optical transmitter (Tx) for each wavelength and transmits an optical signal from the optical transmitter with a set wavelength. The multiplexer/demultiplexer is, for example, an AWG. The multiplexer/demultiplexer or the power splitter multiplexes input light and outputs an optical signal with the wavelength. For example, the multiplexer/demultiplexer or the power splitter has a multi-wavelength light source or a broadband light source, and a tunable wavelength filter (tf) that selects and outputs a set wavelength. The tunable wavelength filter passes an optical signal with a set wavelength (tunable) among the input optical signals. The wavelength-tunable light source **451** can output a main signal (or a signal obtained by superimposing a control signal on the main signal) in accordance with, for example, a direct modulation scheme. Alternatively, the wavelength-tunable light source **451** further includes an external modulator, and can output a main signal (or a signal obtained by superimposing a control signal on the main signal) using the external modulator. As the external modulator, a Mach-Zender (MZ), an electro-absorption (EA), a semiconductor optical amplifier (SOA), or the like can be used. The external modulator can have a structure integrated with a light source. The external modulator can be modulated with the main signal and the control signal by the same modulator, or can be separately modulated with the main signal and the control signal by different modulators.

The tunable wavelength receiver **452** includes a tunable wavelength filter and an optical receiver. The tunable filter passes light with a set wavelength (tunable) in the input optical signal. The optical receiver receives the optical signal passing along the tunable wavelength filter. The tunable wavelength filter, that is, a signal with a predetermined wavelength may be selected after the light is received. For example, a beat signal centering on a frequency corresponding to a wavelength difference from a local light with a predetermined wavelength may be selected with a frequency width corresponding to the band of the signal. Depending on the configuration of the optical SW, a multiplexing scheme, or the like, a configuration in which a transmitter that has not a tunable wavelength is used or a configuration in which a tunable wavelength filter or a multiplexer/demultiplexer is not used may be adopted.

A subscriber device **401a** illustrated in FIG. 29(B) includes an optical transceiver **411a**. The optical transceiver **411a** includes a wavelength-tunable light source **451**, a wavelength-tunable filter **453**, and an optical receiver **454**. The wavelength-tunable light source **451** is an example of an optical transmission unit, and the wavelength-tunable filter **453** and the optical receiver **454** are examples of an optical reception unit. The wavelength-tunable filter **453** inputs an optical signal from a transmission line and passes light with a set wavelength to the optical receiver **454**. The wavelength set to the wavelength-tunable filter **453** is tunable. The optical receiver **454** receives the optical signal passed by the wavelength-tunable filter **453**. The subscriber device **40** on the reception side may not use the wavelength-tunable filter **453** depending on the configuration of the optical SW, the multiplexing scheme, or the like.

The subscriber device **401b** illustrated in FIG. 29(C) includes an optical transceiver **411b**. The optical transceiver **411b** includes a light source **455**, a wavelength-tunable filter **456**, and a tunable wavelength receiver **452**. The light source **455** and the wavelength-tunable filter **456** are an example of an optical transmission unit, and the tunable wavelength receiver **452** is an example of an optical reception unit. The light source **455** outputs light with a single wavelength (for example, the wavelength λ_1). That is, the light source **455** is

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not wavelength-tunable. The wavelength-tunable filter 456 receives the optical signal output from light source 455, and outputs the optical signal with the set wavelength to the transmission line. The wavelength set in the wavelength-tunable filter 456 is the wavelength of the optical signal output from the light source 455.

The subscriber device 401c illustrated in FIG. 30(A) includes an optical transceiver 411c. The optical transceiver 411c includes a wavelength-tunable light source 451, a plurality of optical receivers 454-1 to 454-3, and a demultiplexer 457. The wavelength-tunable light source 451 is an example of an optical transmission unit, and the optical receivers 454-1 to 454-3 and the demultiplexer 457 are examples of an optical reception unit. FIG. 30(A) illustrates a case where there are three optical receivers 454-1 to 454-3, but the number of optical receivers 454-1 to 454-3 is not limited. In the following description, the number of the optical receivers 454-1 to 454-3 is not particularly limited. The demultiplexer 457 demultiplexes the optical signal input from the transmission line in accordance with a wavelength. The optical signals demultiplexed by the demultiplexer 457 are input to the optical receivers 454-1 to 454-3. The optical receivers 454-1 to 454-3 receive the optical signals demultiplexed by the demultiplexer 457.

A subscriber device 401d illustrated in FIG. 30(B) includes an optical transceiver 411d. The optical transceiver 411d includes a plurality of light sources 455-1 to 455-3, a multiplexer 458, and a tunable wavelength receiver 452. The light sources 455-1 to 455-3 and the multiplexer 458 are an example of an optical transmission unit, and the tunable wavelength receiver 452 is an example of an optical reception unit. Note that FIG. 30(B) illustrates a case where there are three light sources 455-1 to 455-3, but the number of light sources 455-1 to 455-3 is not limited. Also in the following description, the number of light sources 455 is not particularly limited. The light sources 455-1 to 455-3 transmit optical signals of different wavelengths. The multiplexer 458 multiplexes the plurality of optical signals output from the light sources 455-1 to 455-3 and outputs the multiplexed optical signals to the transmission line.

The subscriber device 401e illustrated in FIG. 30 (C) includes an optical transceiver 411e. The optical transceiver 411e includes a plurality of light sources 455-1 to 455-3, a multiplexer 458, a plurality of optical receivers 454-1 to 454-3, and the demultiplexer 457. The light sources 455-1 to 455-3 and the multiplexer 458 are an example of an optical transmission unit, and the optical receivers 454-1 to 454-3 and the demultiplexer 457 are an example of an optical reception unit. In the subscriber device 401e, optical signals output from the light sources 455-1 to 455-3 are multiplexed by the multiplexer 458 to be output to the transmission line. In the subscriber device 401e, the optical signal input from the transmission line is demultiplexed with a wavelength by the demultiplexer 457, and the demultiplexed optical signals are received by the optical receivers 454-1 to 454-3.

It is preferable to combine a plurality of branch combiners/splitters or multiplexers if wavelengths to be multiplexed/demultiplexed after combination are equal. That is, the demultiplexer 457 and the multiplexer 458 may be combined.

FIGS. 31 and 32 are diagrams illustrating configurations of single-core subscriber devices 402. The subscriber device 402 illustrated in FIG. 31(A) includes an optical transceiver 412. The optical transceiver 412 includes a wavelength-tunable light source 451, a tunable wavelength receiver 452, and a WDM filter 459. The optical transceiver 412 illustrated in FIG. 31(A) is different from the optical transceiver 411

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illustrated in FIG. 29(A) in that the WDM filter 459 is further provided. The WDM filter 459 separates the uplink signal and the downlink signal in accordance with a wavelength. The WDM filter 459 outputs the optical signal generated by the wavelength-tunable light source 451 to the transmission line, and outputs the optical signal input from the transmission line to the tunable wavelength receiver 452. Like the subscriber device 401, the subscriber device 402 further includes an external modulator and can output a main signal (or a signal obtained by superimposing a control signal on the main signal) using the external modulator.

Here, the WDM filter 459 illustrated in FIGS. 31 and 32 may be a power splitter. The WDM filter 459 is appropriate for a case where an insertion loss is sufficiently low with respect to the power splitter and the wavelengths used on the transmission side and the reception side do not overlap. The case where the wavelengths used on the transmission side and the reception side overlap each other is appropriate, for example, in the case of folded communication or the like between the same type of subscriber devices.

In FIGS. 31 and 32, the WDM filter 459 is contained in the optical transceiver 412, but may be outside of the optical transceiver 412 or outside of the subscriber device 40.

The subscriber device 402a illustrated in FIG. 31(B) includes an optical transceiver 412a. The optical transceiver 412a includes a wavelength-tunable light source 451, a wavelength-tunable filter 453, an optical receiver 454, and a WDM filter 459a. The optical transceiver 412a illustrated in FIG. 31(B) is different from the optical transceiver 411a illustrated in FIG. 29(B) in that a WDM filter 459a is further provided. The WDM filter 459a separates the uplink signal and the downlink signal in accordance with a wavelength. The WDM filter 459a outputs the optical signal generated by the wavelength-tunable light source 451 to the transmission line, and outputs the optical signal input from the transmission line to the wavelength-tunable filter 453. The subscriber device 402 on the reception side may not use the wavelength-tunable filter 453 in accordance with the configuration of the optical SW, a multiplexing scheme, or the like.

The subscriber device 402b illustrated in FIG. 31(C) includes an optical transceiver 412b. The optical transceiver 412b includes a light source 455, a wavelength-tunable filter 456, a tunable wavelength receiver 452, and a WDM filter 459b. The optical transceiver 412b illustrated in FIG. 31(C) is different from the optical transceiver 411b illustrated in FIG. 29(C) in that a WDM filter 459b is further provided. The WDM filter 459b separates an uplink signal and a downlink signal in accordance with a wavelength. The WDM filter 459b outputs the optical signal that has passed through the wavelength-tunable filter 456 to the transmission line and outputs the optical signal input from the transmission line to the tunable wavelength receiver 452.

The subscriber device 402c illustrated in FIG. 32(A) includes an optical transceiver 412c. The optical transceiver 412c includes a wavelength-tunable light source 451, a plurality of optical receivers 454-1 to 454-3, a demultiplexer 457, and a WDM filter 459c. The optical transceiver 412c illustrated in FIG. 32(A) is different from the optical transceiver 411c illustrated in FIG. 30(A) in that a WDM filter 459c is further provided. The WDM filter 459c separates the uplink signal and the downlink signal in accordance with a wavelength. The WDM filter 459c outputs the optical signal generated by the wavelength-tunable light source 451 to the transmission line and outputs the optical signal input from the transmission line to the demultiplexer 457.

A subscriber device 402d illustrated in FIG. 32(B) includes an optical transceiver 412d. The optical transceiver

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412*d* includes a plurality of light sources 455-1 to 455-3, a multiplexer 458, a tunable wavelength receiver 452, and a WDM filter 459*d*. The optical transceiver 412*d* illustrated in FIG. 32(B) is different from the optical transceiver 411*d* illustrated in FIG. 30(B) in that a WDM filter 459*d* is further provided. The WDM filter 459*d* separates the uplink signal and the downlink signal in accordance with a wavelength. The WDM filter 459*d* outputs the optical signal multiplexed by the multiplexer 458 to the transmission line and outputs the optical signal input from the transmission line to the tunable wavelength receiver 452.

A subscriber device 402*e* illustrated in FIG. 32(C) includes an optical transceiver 412*e*. The optical transceiver 412*e* includes a plurality of light sources 455-1 to 455-3, a multiplexer 458, a plurality of optical receivers 454-1 to 454-3, a demultiplexer 457, and a WDM filter 459*e*. The optical transceiver 412*e* illustrated in FIG. 32(C) is different from the optical transceiver 411*e* illustrated in FIG. 30(C) in that the WDM filter 459*e* is further provided. The WDM filter 459*e* separates the uplink signal and the downlink signal in accordance with a wavelength. The WDM filter 459*e* outputs the optical signal multiplexed by the multiplexer 458 to the transmission line and outputs the optical signal input from the transmission line to the demultiplexer 457.

It is preferable to combine a plurality of branch combiners/splitters or multiplexers if wavelengths to be multiplexed/demultiplexed after combination are equal. That is, the demultiplexer 457 and the multiplexer 458, the multiplexer 458 and the WDM filter 459*e*, the demultiplexer 457 and the WDM filter 459*e*, and the demultiplexer 457, the multiplexer 458, and the WDM filter 459*e* may be collected.

The configurations of the single-core subscriber device 401 and the double-core subscriber device 402 have been described above. In the present invention, any of the configurations illustrated in FIGS. 29 to 32 may be used as the subscriber device 40.

Next, configuration examples of the monitoring unit 60 and the monitoring unit 65 will be described. First, in the foregoing description, a signal is branched and monitored using a power splitter, but a medium in which a gain, an applied voltage, a current, or a resistance changes due to conduction of the signal at a frequency of at least about a carrier wave of the AMCC, for example, a semiconductor amplifier or the like, may be incorporated in a transmission line without branching. The wavelength-multiplexed signal may also be monitored without being branched by using a medium that has high wavelength dependency on changes in a gain, a voltage, a current, resistance, or the like using a superlattice structure such as multiple quantum well (MQW) or a photonic crystal.

Hereinafter, a configuration example of the monitoring unit 60 and the monitoring unit 65 will be described with reference to FIGS. 33 to 37 in the case of branching. In FIGS. 33 to 37, the monitoring unit 60 will be described as an example. Although FIGS. 33 to 37 illustrate examples of monitoring in an uplink direction and a downlink direction, the monitoring unit 60 may monitor only one direction output from the subscriber device 40, for example, from the viewpoint of monitoring any one device, for example, the subscriber device 40. In the case of an example of only one direction, both directions are obtained, for example, by replacing a transmitter and a receiver with a combination of a transmitter and a receiver connected to a transceiver, a power splitter, or a multiplexer/demultiplexer.

Although the receiver and the transmitter in proximity to the power splitter 61, the multiplexer/demultiplexer, and the

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like branched from the transmission line are described. The receiver and the transmitter may be arranged far away, for example, via one or a plurality of optical SWs, and the receiver and the transmitter of the control unit may be connected to the receiver and the transmitter of the monitoring unit 60 on the assumption that the receiver and the transmitter of the control unit also serve as the receiver and the transmitter of the monitoring unit 60.

The monitoring unit 60 monitors an optical signal. At the time of monitoring, an optical receiver or an optical transceiver is arranged near the branch combiner/splitter or the multiplexer/demultiplexer or at a location where an optical signal can be connected. The branch combiner/splitter or the multiplexer/demultiplexer branches the optical signal, and the monitoring unit 60 performs monitoring by a received optical signal, and further receives a control signal superimposed on the received optical signal.

The monitoring unit 60 includes a branch combiner/splitter or a multiplexer/demultiplexer provided at a port, a transmission line, or a connection point thereof on the input side, the passing side, or the output side of the optical SW, and a blocking unit that blocks a main signal depending on a case. For example, a multiplexer/demultiplexer may be disposed at the port 11-2 on the ground side, the transmission line 2, or a connection point thereof. An uplink optical signal from the optical SW and a downlink optical signal input from another ground or an upper network may be separated in accordance with a wavelength of the input optical signal and may be output to the optical SW. The disposition of the blocking unit in the case of including the blocking unit is preferably in a multiplexed or merged state, for example, in a case where the disposition is set or changed in both upper and lower directions, and is preferably in a demultiplexed or branched state, for example, in a case where the disposition is separately set or changed in upper and lower directions.

The monitoring unit 60 may be connected via an optical SW different from the optical SW to which the subscriber device 40 to be controlled is connected. For example, an optical signal separated from the transmission line through which the branch combiner/splitter transmits the optical signal of the subscriber device 40 to be controlled is input to the optical SW connected to the monitoring unit 60 separately from the connection of the subscriber device 40, is multiplexed as necessary, is transmitted to the optical SW connected to the monitoring unit 60, and is connected to the monitoring unit 60. Alternatively, the optical signal is directly input to the optical SW connected to the control unit or the monitoring unit 60 without being input to the optical SW.

The branch combiner/splitter or the multiplexer/demultiplexer may divide the ports included in the optical SW into a plurality of groups, and may be provided in the port 11-2 belonging to one group in a case where the ports are not connected in the group, the transmission line 2 connected thereto, or the transmission line 2 connected to the port 2. The branching may be performed before or after the wavelength separation. The multiplexer/demultiplexer may be provided closer to the optical SW than the multiplexer/demultiplexer for vertical multiplexing, or may be provided closer to the transmission line. When the multiplexer/demultiplexer is provided on the transmission line side, uplink and downlink optical signals transmitted through the transmission line can be branched by one device. In this case, the branched uplink optical signal and downlink optical signal are input to the port of each optical SW. The optical SW outputs an input optical signal from a port connected to the

monitoring unit 60. Accordingly, the monitoring unit 60 receives the branched optical signal.

A monitoring unit 60a illustrated in FIG. 33(A) illustrates a specific configuration of the monitoring unit 60. The monitoring unit 60a includes the power splitter 61 and a plurality of receivers 62-1 to 62-2. The receivers 62-1 and 62-2 receive the optical signals branched by the power splitter 61. The number of receivers 62-1 to 62-2 corresponding to the number of branches of the power splitter 61 may be provided. FIG. 33(A) illustrates an example in which 2x2 power splitters 61 are used, and a sum of the number of branches (2)—the path (1) of the main signal=1 except for a path of the main signal on the input side and the output side is each one. In the case of a 3x3 power splitters 61, two receivers can be provided. From the viewpoint of a reduction in an influence of reflection and the like, the power splitter 61 of 1x2, 1x3, or the like that does not include the ports not to be monitored may be used. In this case, the receiver 62-1 or 62-2 may set the number of branches (2)—the main signal path (1)=1 and the number of branches (1)—the main signal path (1)=0; the number of branches (3)—the main signal path (1)=2 and the number of branches (1)—the main signal path (1)=0.

Note that a path other than the main signal branched by the power splitter 61 may be branched and received as in the WDM device 63b of FIG. 33(B), or a plurality of power splitters 61 may be installed in the main signal path and may be branched and received.

A monitoring unit 60a illustrated in FIG. 33(A) illustrates a specific configuration of the monitoring unit 60. The monitoring unit 60a includes the power splitter 61 and a plurality of receivers 62-1 to 62-2. The receivers 62-1 and 62-2 receive the optical signals branched by the power splitter 61. The number of receivers 62 is not particularly limited.

A monitoring unit 60b illustrated in FIG. 33(B) illustrates a specific configuration of the monitoring unit 65. The monitoring unit 60b includes a power splitter 61, a plurality of receivers 62-1 to 62-6, and a plurality of WDM devices 63b-1 to 63b-2. The receivers 62-1 to 62-3 are connected to the WDM device 63b-1. The receivers 62-4 to 62-6 are connected to the WDM device 63b-2. The WDM device 63b-1 demultiplexes an optical signal branched by the power splitter 61 and outputs the demultiplexed optical signals to the receivers 62-1 to 62-3. The WDM device 63b-2 demultiplexes an optical signal branched by the power splitter 61 and outputs the demultiplexed optical signals to the receivers 62-4 to 62-6. The number of the WDM devices 63b is not particularly limited.

The monitoring unit 60c illustrated in FIG. 33(C) includes a power splitter 61, a plurality of receivers 62-1 to 62-2, and a WDM device 63c. The WDM device 63c multiplexes the plurality of input optical signals and outputs the multiplexed optical signal to the transmission line 601. The WDM device 63c demultiplexes an optical signal input from the transmission line 601 and outputs the demultiplexed optical signals to transmission lines 602-1 and 602-2.

A monitoring unit 60d illustrated in FIG. 34(A) includes a plurality of power splitters 61-1 to 61-3, a plurality of receivers 62-1 to 62-6, and a WDM device 63d. The transmission line 602-1 is provided with the power splitter 61-1, the transmission line 602-2 is provided with the power splitter 61-2, and the transmission line 602-3 is provided with the power splitter 61-3. The receivers 62-1 and 62-2 are connected to the power splitter 61-1, receivers 62-3 and 62-4 are connected to the power splitter 61-2, and receivers 62-5 and 62-6 are connected to the power splitter 61-3. The WDM

device 63d demultiplexes an optical signal input from the transmission line 601 and outputs the demultiplexed optical signals to the transmission lines 602-1 to 602-3. The WDM device 63d multiplexes the optical signals input from the transmission lines 602-1 to 602-3 and outputs the multiplexed optical signal to the transmission line 601.

In FIG. 34(A), the receivers 62-1 to 62-6 are respectively provided in the transmission lines 602-1 to 602-3 after demultiplexing by the WDM device 63d when viewed from the transmission line 601, or before multiplexing by the WDM device 63d when viewed from the transmission line 602, but the receivers 62 may not be provided in a transmission line that does not require reception.

A monitoring unit 60e illustrated in FIG. 34(B) includes a plurality of power splitters 61-1 and 61-2, a plurality of receivers 62-1 and 62-2, and a plurality of WDM devices 63e-1 and 63e-2. The transmission lines 602-1 and 602-2 are provided between the WDM devices 63e-1 and 63e-2. The power splitter 61-1 is provided in the transmission line 602-1, and the power splitter 61-2 is provided in the transmission line 602-2. The receiver 62-1 is connected to the power splitter 61-1 and the receiver 62-2 is connected to the power splitter 61-2. The WDM device 63e-1 demultiplexes an optical signal input from the transmission line 601 and outputs the demultiplexed optical signals to the transmission lines 602-1 and 602-2. The WDM device 63e-1 multiplexes the optical signals input from the transmission lines 602-1 and 602-2 and outputs the multiplexed optical signal to the transmission line 601. The WDM device 63e-2 demultiplexes the optical signal input from the transmission line 603 and outputs the demultiplexed optical signals to the transmission lines 602-1 and 602-2. The WDM device 63e-2 multiplexes the optical signals input from the transmission lines 602-1 and 602-2 and outputs the multiplexed optical signal to the transmission line 603.

In FIG. 34(B), the receivers 62-1 and 62-2 are provided in the transmission lines 602-1 and 602-2 after demultiplexing and before multiplexing, but the receiver 62 may not be provided in a transmission line that does not require reception.

The monitoring unit may control the subscriber device 40. When the control signal is exchanged with the subscriber device 40, the monitoring unit may set or change a wavelength of the subscriber device 40 like the control unit 20. However, when the setting, particularly, changing the wavelength, an optical signal other than the wavelength set at the time of the change does not reach an unintended destination. Therefore, it is preferable to perform setting such that an output from the target device does not reach the destination until the change is completed.

A configuration controlled by the monitoring unit will be described with reference to FIG. 35.

A monitoring unit 60f illustrated in FIG. 35(A) includes the power splitter 61, a plurality of receivers 62-1 and 62-2, and a modulator 64. The transmission line 601 is provided with the power splitter 61 and the modulator 64. As described above, in the monitoring unit 60f, a modulator is provided in the middle of the path of the main signal, and the main signal is modulated by the modulator 64. As the modulator, modulation may be performed by an amplifier that modulates an amplification factor according to an external modulator or a control signal, gain saturation according to an optical signal from the monitoring unit 60f, intermodulation, nonlinear effects such as four-wave mixing and the Raman effect, or the like.

As long as an extinction ratio of the modulator is high enough not to affect other optical signals, the modulator may also serve as a blocking unit.

Although FIG. 35(A) illustrates the configuration in which the monitoring unit 60f includes the power splitter 61, the receivers 62-1 and 62-2, and the modulator 64, the modulator 64 can be replaced with a device capable of modulating an intensity monitor of an input and an output at a frequency of about a carrier wave of the AMCC, for example, an optical semiconductor amplifier of a multi-electrode.

In the case of modulation with external light, the power splitter 61 also serves as a modulator without separately providing the modulator 64. The power splitter 61 includes a semiconductor optical amplifier, a highly nonlinear fiber (HNLF), a nonlinear optical crystal, or a periodically poled optical element (for example, periodically polarized potassium titanyl phosphate (PPKTP), periodically polarized lithium niobate (PPLN), and periodically polarized lithium tantalate (PPLT)) that facilitates quasi-phase matching for nonlinear interaction, which have large gain saturation and a large nonlinear effect, and acts a signal and external light.

When a signal that has already been modulated by the AMCC is re-modulated, an influence of the previous modulation is assumed. In such a case, a frequency band of the control signal is preferably different from the previous modulated frequency band. For example, when a modulation band of a signal is B to the degree that modulation sidebands of previous modulation and new modulation do not overlap each other, a difference between the frequency bands may be set to be equal to or greater than 2B from the viewpoint of preventing interference, or may be conversely set to about 0.5 times to 0.8 times on the assumption that the signal is cut out by a filter. In addition, the modulation may be performed by another modulator 64 after the inverse modulation illustrated in FIG. 35(B). In this case, since the influence of the previously modulated signal is reduced, the frequency bands may overlap. Modulation may be performed with a signal corresponding to a product of the inverse modulation of the previously modulated signal and the new modulation, and modulation in the plurality of modulators 64 may be performed by modulation of one modulator 64.

In order to reduce the influence of the AMCC on the main signal, the monitoring unit 60f may perform the process described with reference to FIG. 35(B).

As illustrated in FIG. 35(B), the monitoring unit 60f may modulate the main signal modulated by the AMCC using the same modulation scheme as that of the AMCC received by the receiver 62-2 and a reverse signal (in the case of I/O intensity modulation, 0 is superimposed on a signal modulated by 1, 1 is superimposed on a signal modulated by 0, and the modulation has an opposite phase and an intensity that exactly offsets the signal). Accordingly, the influence of the AMCC on the main signal can be reduced.

A monitoring unit 60g illustrated in FIG. 36 includes a plurality of power splitters 61-1 to 61-2, a plurality of receivers 62-1 to 62-2, a plurality of WDM devices 63g-1 to 63g-2, and a plurality of modulators 64g-1 to 64g-2. The transmission lines 602-1 and 602-2 are provided between the WDM device 63g-1 and the WDM device 63g-2. The transmission line 602-1 is provided with a modulator 64g-1 and a power splitter 61-1, and the transmission line 602-2 is provided with a 64g-2 and a power splitter 61-2. The receiver 62-1 is connected to the power splitter 61-1 and the receiver 62-2 is connected to the power splitter 61-2. The WDM device 63g-1 demultiplexes an optical signal input from the transmission line 601 and outputs the demulti-

plexed optical signals to the transmission lines 602-1 to 602-2. The WDM device 63g-1 multiplexes optical signals input from the transmission lines 602-1 to 602-2 and outputs the multiplexed optical signal to the transmission line 601. The WDM device 63e-2 demultiplexes the optical signal input from the transmission line 603 and outputs the demultiplexed optical signals to the transmission lines 602-1 and 602-2. The WDM device 63e-2 multiplexes the optical signals input from the transmission lines 602-1 and 602-2 and outputs the multiplexed optical signal to the transmission line 603. The optical signals input from the transmission lines 602-1 to 602-2 to the WDM device 63e-1 and the WDM device 63e-2 are modulated by modulators 64g-1 and 64g-2.

In FIG. 36, the branched optical signals are modulated after being received by the receivers 62-1 to 62-2. However, when it is not necessary to consider quality of the received optical signals in modulation, for example, deterioration in a signal-to-noise ratio (SN) of a received control signal or the like, the modulated optical signal may be branched by the power splitters 61-1 to 61-2 and may be received by the receivers 62-1 to 62-2.

A configuration in which the monitoring unit uses a transmitter will be described. Like the subscriber device 40 and the control unit 20, the monitoring unit includes optical transceivers, that is, optical transceivers with a tunable wavelength or a non-tunable wavelength, and is connected to a port of the optical SW. If the transmitter is a tunable wavelength transmitter, an optical signal with an any wavelength can be transmitted. The monitoring unit transmits a control signal to the subscriber device 40 by an optical signal. The power splitter of a transmission line for transmitting an optical signal of the subscriber device 40 to be controlled merges the control signal with the optical signal transmitted through the transmission line. In this configuration, even in a state where the subscriber device 40 is performing normal communication, the monitoring unit can receive a connection destination change request or the like from the subscriber device 40, transmit a control signal, and perform wavelength switching or the like to the subscriber device 40. A specific configuration in which the monitoring unit includes a transmitter will be described below.

When the subscriber device 40 performs normal communication, communication with the control unit 20 in a case where the control unit 20 controls the subscriber device 40 by an optical signal directly or via an optical SW or the like cannot be performed. Since the monitoring unit includes the transmitter, it is possible to give an instruction for various settings of the subscriber device 40. That is, the monitoring unit inputs the optical signal inserted and separated into the transmission line through which the optical signal is transmitted to the subscriber device 40 to be controlled by the power splitter.

An example will be described using AMCC. Usually, the AMCC modulates the main signal and the AMCC signal with a signal obtained by superimposing the main signal and the AMCC signal at an electrical stage or further modulates the main signal at an optical stage. In contrast, the transmitter of the monitoring unit inputs and multiplexes an optical signal corresponding to the control signal separately from the main signal. The wavelength of the optical signal corresponding to the control signal is a wavelength that passes through the same path as that of the main signal from the transmitter to a location where at least the control signal corresponding to the AMCC is received. The monitoring unit does not modulate the main signal but performs intensity modulation or the like on the separately input optical

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signal intensity at a frequency of a carrier wave of the AMCC, so that an optical signal obtained by combining the main signal and the control signal corresponding to the AMCC is equivalent to the optical signal modulated at the frequency of the carrier wave of the AMCC.

Here, the example in which the intensity modulation of the control signal is received along with the main signal has been described. In the case of phase modulation or the like, the control signal may be received using delay detection or local light.

Here, when a wavelength difference between the main signal and the control signal corresponding to the AMCC is separated to the extent that beat noise can be ignored, for example, a sum of the line widths of both pieces of light, and if the line widths are equal, the distance is twice or more the line width, demodulation can be performed in the same way as in the case of modulation with the AMCC similarly to an average value of the light intensities of both pieces of light for a time of one bit or one baud of the AMCC. When the modulation sideband at the frequency of the carrier wave in the AMCC and the modulation sideband at the bit rate or the baud rate of the main signal are superimposed, it is difficult to perform demodulation in direct detection with the same modulation as usual. However, for example, coherent reception or the like may be performed, and reception may be performed after removal at an electrical stage through maximum likelihood determination or the like, or the phase of the light of the optical signal of the control signal corresponding to the AMCC may be synchronized with the phase of the light of the optical signal of the main signal. The phase synchronization can be performed even when the AMCC is modulation using the phase. In this case, modulation is performed such that the phase after multiplexing becomes the phase after modulation.

For example, the intensity and wavelengths of the signal light and the control light may be measured by different measurement means or may be measured by the same measurement means such that the optical signal of the main signal and a part of the optical signal input from the transmitter of the monitoring unit to the power splitter can be received by a receiver of the monitoring unit. The latter has an effect that the measurement means is less. For example, an influence of beat noise between pieces of multiplexed light on the main signal may be observed and adjusted. For example, the main signal is measured before the control light is multiplexed. Thereafter, the signal after multiplexing of the control light is equivalent to the signal to which the desired AMCC is applied, or the influence of the beat noise between the pieces of multiplexed light on the main signal is measured. Then, the intensity and wavelength of the optical signal of the control signal are adjusted. In the case of phase synchronization, feedback may be performed in measurement after the multiplexing.

Next, an example of another configuration in which a transmitter is used in the monitoring unit will be described with reference to FIG. 37. Instead of being installed in a port, a transmission line, or a connection point thereof of the optical SW 10, as illustrated in FIG. 37, the control unit 20 may loop using a folded transmission line (the upper side in the drawing) or may cross the optical SW 10 (the lower side in the drawing) like an electrical processing unit to be described below.

For example, a monitoring unit 60h terminates the main signal, and the transmitter transmits an optical signal obtained by superimposing the AMCC on a photoelectrically converted main signal and further modulating the optical signal modulated with the main signal by the AMCC.

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Unlike the monitoring unit 60h, as described above, the monitoring unit 60h does not include a power splitter that combines and branches the main signal, but may be similarly installed in a port, a transmission line, or a connection point thereof on the input side, the passing side, or the output side of the optical SW. In this configuration, since the signal is once terminated, a wavelength to the monitoring unit 60h and a wavelength from the monitoring unit may be different as long as they are wavelengths transmitted through a desired path.

The monitoring unit 60h may not include an optical receiver or an optical transceiver, and the branched optical signal may be transmitted and output to the control unit 20 via the optical SW 10, for example. When the optical signal is output to the control unit 20, the control unit 20 includes an optical receiver or an optical transceiver, and the monitoring unit 60h is a function of setting a path between a power splitter or a multiplexer/demultiplexer provided in a port or a transmission line on an input side, a passing side, or an output side of the optical SW, or a connection point thereof, a path until the control unit 20, and a path of the power splitter or the multiplexer/demultiplexer as necessary. The path may be realized by the optical SW, and the setting may also be performed by the control unit.

The monitoring unit 60h may include an optical receiver or an optical transceiver and may further include a blocking unit. At the time of initial setting, setting change, and abnormality detection, a path from the monitoring unit to an output destination may be blocked or blocked by the blocking unit, and the setting, changing, or blocking of the subscriber device may be performed.

As described above, the monitoring units 60 to 60h monitor optical signals. The monitoring units 60 to 60h perform monitoring by an optical signal branched by a power splitter or a WDM device and further receive a control signal superimposed on the received optical signal. Although FIGS. 33 to 37 illustrate the configuration in which the monitoring units 60 to 60h includes the receiver 62, the monitoring units 60 to 60h may include transmitters.

The monitoring units 60 to 60h may be connected via an optical SW different from the optical SW connected to the subscriber device 40 to be controlled. In this case, the optical signal separated by the power splitter 61 in the transmission line through which the optical signal of the subscriber device 40 to be controlled is transmitted is input to the optical SW different from the optical SW connected to the subscriber device 40 and connected to the monitoring units 60 to 60h. At this time, the signals are multiplexed as necessary, are transmitted to the optical SW connected to the monitoring units 60 to 60h, and are connected to the monitoring units 60 to 60h.

In the following description, the monitoring units 60 to 60h will be described as the monitoring unit 60 unless otherwise particularly distinguished.

Hereinafter, an example of an optical access system using the optical SW that has the above-described functions will be described.

(Configuration Example of Optical Access System 100)

FIG. 38 is a diagram illustrating a configuration example of the optical access system 100. The optical access system 100 includes an optical gateway (GW) 200 and an operation system (OPS) 300. The OPS 300 may be integrated with the control unit 20, and both the OPS 300 and the control unit 20 may be referred to as a control unit or an OPS as a representative. The subscriber device 40 is communicably

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connected to a higher network such as the optical communication network **30** illustrated in FIG. **1** by the optical access system **100**.

The subscriber device **40** is a device on the optical subscriber side. The subscriber device **40** is connected to the optical GW **200** via a transmission line **501**. The transmission line **501** is, for example, an optical fiber. The optical GW **200** is a device in a communication station. The subscriber device **40** denoted by reference sign N1 and the optical GW **200** are connected via, for example, the transmission line **501** or the power splitter **502**. The configuration of the network connected from the subscriber device **40** to the optical GW **200** may be any of various network topologies such as a point-to-point (PtoP) configuration, a PON configuration, a bus type, a mesh type, a ring type, and a multi-ring type. For example, a power splitter **502** or the like may be included in the transmission line **501**, and a plurality of subscriber devices **40** may be connected to one transmission line **501**. The optical GW **200** is connected to another station, a core network, or the like via the transmission line **511** and the transmission line **512**. The transmission line **511** and the transmission line **512** are, for example, optical fibers. The transmission line **511** transmits an uplink signal and the transmission line **512** transmits a downlink signal. The transmission lines **511** and **512** are examples of a multiplexing communication transmission line through which a wavelength-multiplexed optical signal is transmitted. The connection from the optical GW **200** denoted by reference sign N2 to another station or a core network is made by, for example, the transmission line **511** or **512** of an optical fiber, and is made such that the connection between grounds is a full mesh. In this configuration, a case where the optical GW **200** is installed in a station on the ground A and is connected to an optical communication device set in a station on the ground B and an optical communication device installed in a station on the ground C via the optical communication network **30** or the like will be described as an example. The optical communication devices on the grounds B and C to which the optical GW **200** is connected may be the optical GW **200**.

The subscriber device **40** is connected to the optical GW **200** via a transmission line **501**. The subscriber device **40** includes the optical transceiver **41**. The optical transceiver **41** is a wavelength-tunable optical transceiver. The optical transceiver **41** is, for example, an optical transceiver that mutually converts an optical signal and an electrical signal. The subscriber device **40** can select and set a unique wavelength in the optical transceiver **41** in accordance with a transmission/reception destination. The subscriber device **40** sets a wavelength to be used in the optical transceiver **41** in response to the instruction received from the optical GW **200**. M (where M is an integer equal to or greater than 1) subscriber devices **40** connected to the optical GW **200** are referred to as subscriber devices **40-1** to **40-M**.

The optical GW **200** includes an optical SW **210**, a wavelength multiplexer/demultiplexer **220**, a control unit **230**, a multiplexer **241**, a demultiplexer **242**, a branching unit **250**, and a monitoring unit **260**. The monitoring unit **260** may be replaced with any one of the monitoring units **60** to **60/h** described above.

The optical SW **210** has a plurality of input/output ports (hereinafter referred to as a "port"), and connects two or more ports. The optical SW **210** can freely switch an optical path between ports. A port through which an uplink signal is input and output is referred to as an uplink port and a port through which a downlink signal is input and output is

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referred to as a downlink port. Each port of the optical SW **210** is connected to a transmission line.

The wavelength multiplexer/demultiplexer **220** performs vertical demultiplexing in which an uplink signal and a downlink signal are separated in accordance with a wavelength. The wavelength multiplexer/demultiplexer **220** inputs an uplink optical signal transmitted by the subscriber device **40** from the transmission line **501** and outputs the uplink optical signal to the optical SW **210** via a transmission line **521**. The wavelength multiplexer/demultiplexer **220** inputs a downlink optical signal output from the optical SW **210** from a transmission line **522** and outputs the downlink optical signal to the subscriber device **40** via the transmission line **501**.

The control unit **230** is connected to an uplink port and a downlink port to which the subscriber device **40** is not connected among the ports of the optical SW **210**. An uplink port of the optical SW **210** is connected to a port on the transmission side of the control unit **230** by a transmission line **531**. A downlink port of the optical SW **210** is connected to a port on the transmission side of the control unit **230** by a transmission line **533**. The control unit **230** includes a wavelength demultiplexer **231**, an optical receiver (Rx) **232** for each wavelength channel, and a wavelength-tunable transmitter **233**. The wavelength demultiplexer **231** is, for example, an AWG. The wavelength demultiplexer **231** demultiplexes the light input to the port on the reception side via the transmission line **540** for each wavelength. The wavelength demultiplexer **231** outputs the demultiplexed light to the optical receiver **232** that receives an optical signal with the wavelength of the light. The wavelength-tunable transmitter **233** includes a wavelength-tunable laser diode (LD) that generates light with a tunable wavelength. The wavelength-tunable transmitter **233** transmits an optical signal with a tunable wavelength by using light generated by the wavelength-tunable laser diode. The wavelength-tunable transmitter **233** outputs an optical signal in which the generated light is used from the port on the transmission side to the transmission line **533**.

The multiplexer **241** multiplexes the uplink optical signals with different wavelengths output from the plurality of transmission lines **541** by the optical SW **210** and outputs the multiplexed signal to the transmission line **511** connected to another ground. The demultiplexer **242** inputs an optical signal transmitted from any other ground from the transmission line **512** and demultiplexes the input downlink optical signal in accordance with a wavelength. The demultiplexer **242** inputs each of the demultiplexed downlink optical signals to the optical SW **210** via a plurality of transmission lines **542** connected to the uplink port corresponding to the wavelength of the optical signal.

The branching unit **250** is provided in the transmission lines **511** and **512**. The branching unit **250** includes power splitters **251** and **252**. The power splitter **251** branches an uplink optical signal transmitted along the transmission line **511** and inputs the signal to the optical SW **210** via the transmission line **551**. The power splitter **252** branches a downlink optical signal transmitted through the transmission line **512** and inputs the signal to the optical SW **210** via the transmission line **552**.

The monitoring unit **260** includes a wavelength demultiplexer **261** and an optical receiver (Rx) **262** for each wavelength. The wavelength demultiplexer **261** is connected to the optical SW **210** via a transmission line **560**. The optical SW **210** outputs an optical signal input from a port connected to the transmission line **541** or the transmission line **542** to a port connected to the transmission line **560**.

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Accordingly, the wavelength demultiplexer **261** receives the optical signal branched by the branching unit **250**. The wavelength demultiplexer **261** demultiplexes the input optical signal for each wavelength. The wavelength demultiplexer **261** outputs the demultiplexed light to the optical receiver **262** receiving an optical signal with the wavelength of the light. The monitoring unit **260** monitors a state of communication transmitted and received by the subscriber device **40** by using the optical signal received by the optical receiver **262**. The monitoring unit **260** may output a monitoring result to the control unit **230** or the control unit **301** included in the OPS **300**. Here, the control unit **230** and the control unit **301** are used, but the control units may be the same. The monitoring unit **260** may output the monitoring result to the control unit **230**, and the control unit **230** may output the output to the control unit **301**. The control units **230** and **301** may be the same. In that case, the output from the control unit **230** to the control unit **301** may not be output to the outside of the control unit.

The OPS **300** includes a control unit **301** and a management DB **350**. The control unit **301** is connected to the optical GW **200**. The control unit **301** includes a wavelength control unit **310** and an optical SW control unit **320**. The wavelength control unit **310** stores information indicating a wavelength of light used by each user (or each service). The wavelength control unit **310** dynamically allocates a wavelength to be used by each user with reference to this information. The wavelength control unit **310** may be installed in a building different from the optical GW **200** and may be connected to the optical SW **210** and the optical SW control unit **320** via a network. The wavelength control unit **310** manages and controls information which a port connected to the optical SW **210** and which user use which wavelength in real time by sharing each piece of connection information. The control unit **301** may be installed for each optical SW **210** or may be installed for each plurality of optical SWs. As described above, the wavelength control unit **310** performs a process similar to that of the wavelength control unit **25** in FIG. 2.

The control unit **301** is connected to the management database (DB) **350**. The control unit **301** exchanges information regarding the user and the used wavelength with the management DB **350**. The management DB **350** stores a use wavelength and destination information of each user. The destination is represented by, for example, ground A, ground B, or the like. The management DB **350** manages information regarding all users connected to the optical access system **100**.

FIG. 39 is a diagram illustrating an example of a SW connection table. The SW connection table indicates a connection destination of each port of the optical SW **210**. That is, a port through which the optical signal is input and output can be used as information for identifying the subscriber device **40**, the control unit **230**, the branching unit **250**, the monitoring unit **260**, the ground, and the like of a transmission source or a transmission destination of the optical signal.

The wavelength table includes a user wavelength table and an inter-station wavelength table.

FIG. 40 is a diagram illustrating an example of the user wavelength table. The user wavelength table indicates a wavelength which is used for transmission by each user, a wavelength which is used for reception, a free wavelength which is not used for transmission and reception, and a wavelength that cannot be used due to a failure. The management DB **350** may manage the wavelength table for each transmission line connected to the optical SW **210**.

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FIG. 41 is a diagram illustrating an example of the inter-station wavelength table. The inter-station wavelength table indicates a wavelength which is used for communication with each of the other ground, a free wavelength which is not used for communication with each of the other ground, and a wavelength that cannot be used due to a failure in communication with each of the other ground.

Here, an operation when the subscriber device **40** is newly connected will be described. FIG. 42 is a flowchart illustrating an initial setting process of the optical access system **100** when a new subscriber device is connected. An operation of the optical access system **100** when the subscriber device **40-1** is newly connected to the optical GW **200** will be described with reference to FIGS. 38 and 42. It is assumed that a port of the optical SW **210** to which each port of the wavelength demultiplexer **261** (AWG) of the control unit **230** is connected is confirmed in advance by the control unit **230**.

First, user application is performed before the new subscriber device **40-1** is connected. For example, communication between the grounds A and B can be performed by the user application. A service provider registers user information, initial destination information, and the like in the management DB **350** of the OPS **300** based on the user application (step S1). The user information is, for example, information that can obtain a wavelength or the like that can be used by the optical transceiver **41**. The OPS **300** allocates the port of the optical SW **210** to which the subscriber device **40-1** is connected among the free ports of the optical SW **210** with reference to the SW connection table (step S2). Here, at least two ports of an uplink port and a downlink port are allocated in the case of dual cores, and at least one port is allocated in the case of both single-cores. The OPS **300** registers information indicating that the allocated port is connected to the subscriber device **40-1** in the SW connection table. When the control unit **20** controls the subscriber device **400** with an optical signal, the optical SW control unit **320** of the OPS **300** controls the optical SW **210** such that an optical signal is transmitted and received between the port allocated to the target subscriber device **40-1** and the port to which the control unit **230** is connected. When the subscriber device **40** is controlled with an optical signal via the optical SW **210** and the monitoring unit **260**, the optical SW **210** is connected via the monitoring unit **260**. When the control is not performed with the optical signal without being involved with the optical SW **210** and via the monitoring unit **260** or the communication unit, or when the control is not performed with the optical signal, if the port to be connected is not connected by the optical SW **210**, the state may retain, or if the port to be connected is blocked, the state may remain.

When the new subscriber device **40-1** is connected, the subscriber device **40-1** performs an initialization process and transmits a connection request (register request) with an optical signal (step S3).

When a propagation delay or a transmission distance is measured with a register request and when a functional unit receiving a request, for example, the control unit **230**, the monitoring unit **260**, the communication unit, or the like, and a functional unit using a measured value, for example, a control unit, a monitoring unit, a communication unit, or the like different from the received value, a band allocating unit notifying of a transmission permission, a facing device, or the like use the value, the measurement is performed again by a functional unit to be used, or the propagation delay or the transmission distance between the measured functional unit and the functional unit to be used is adjusted.

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and used in consideration of wavelength dispersion, polarization dispersion, or mode dispersion.

The subscriber device **40-1** automatically performs the initialization process before connection or immediately after the connection. The wavelength multiplexer/demultiplexer **220** receives a connection request from the transmission line **501** and outputs a connection request to the optical SW **210** via the transmission line **521**. The optical SW **210** outputs a connection request input from a port connected to the subscriber device **40-1** to an output port to which the control unit **230** is connected when the control unit **20** controls the subscriber device **400** with an optical signal. The input to the reception port of the control unit may be performed via the monitoring unit **260**. The control unit **230** receives the connection request from the reception port via the transmission line **531**. The control unit **230** analyzes the input optical signal, and confirms whether there is a problem in an initially set wavelength or optical power (step **S4**).

When there is a problem in the wavelength or optical power, the control unit **230** transmits a restarting or initialization instruction to the subscriber device **40-1**. After the restarting or the initial setting, the process returns to step **S3** and the subscriber device **40-1** transmits the connection request again.

The control unit **230** analyzes the optical signal received from the subscriber device **40-1** and outputs a connection request to the control unit **301** when it is confirmed that there is no problem. The control unit **301** registers the information regarding the subscriber device **40-1** in the management DB **350**. The connection request includes information regarding a connection source, information regarding a connection destination, and a type of signal to be transmitted. For example, address information such as a medium access control (MAC) address is used as the information regarding the connection source. For example, destination address information is used as the information regarding the connection destination. As the type of the signal to be transmitted, for example, a service, a modulation method, or the like is used. The wavelength control unit **310** registers the connection source information in the management DB **350** based on these pieces of information. Accordingly, identification of a user using the subscriber device **40-1** and the fact that the wavelength that can be used by the subscriber device **40-1** is free are set in the user wavelength table. Further, the wavelength control unit **310** calculates an optimum path between the subscriber device **40-1** and the communication destination such as between the grounds A and B in comparison with connection information stored in the management DB **350**. The wavelength control unit **310** searches vacancy indicated by the inter-station wavelength table in accordance with the calculated path. The wavelength control unit **310** selects a wavelength to be used by the subscriber device **40-1** among free wavelengths and transmits information regarding the selected wavelength to the control unit **230** (step **S5**).

Another subscriber device **40** that is a communication destination of the subscriber device **40-1** is referred to as a communication destination subscriber device **40**. In this case, the wavelength control unit **310** selects a transmission wavelength that is a wavelength used by the subscriber device **40-1** to transmit an optical signal to the communication destination subscriber device **40** and a reception wavelength that is a wavelength used by the subscriber device **40-1** to receive an optical signal from the communication destination subscriber device **40**. The wavelength control unit **310** transmits the selected transmission wavelength and reception wavelength to the control unit **230** as

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wavelengths to be used by the subscriber device **40-1**. When the subscriber device **40-1** performs only transmission to the communication destination subscriber device **40**, the wavelength control unit **310** may not select the reception wavelength. When the subscriber device **40-1** performs only reception from the communication destination subscriber device **40**, the wavelength control unit **310** may not select the transmission wavelength.

The control unit **230** transmits the information regarding the wavelength as follows. The wavelength-tunable transmitter **233** of the control unit **230** transmits a wavelength instruction in which the information regarding the wavelength selected by the wavelength control unit **310** is set, with an optical signal with a wavelength addressed to the subscriber device **40-1**. When the wavelength instruction is transmitted as an optical signal and when the communication unit is used from the port connected to the wavelength-tunable transmitter **233** of the control unit **20** or via the monitoring unit, the optical SW **210** outputs the optical signal input from the port connected to the communication unit to the transmission line **522** connected to the subscriber device **40-1**. The wavelength multiplexer/demultiplexer **220** causes the optical signal input from the optical SW **210** via the transmission line **522** to be incident on the transmission line **501**. The subscriber device **40-1** receives the optical signal transmitted through the transmission line **501**. The wavelength may be branched in the middle of the path or may be different from the setting as long as the wavelength reaches the subscriber device **40**. When there is no multiplexer/demultiplexer or the like that performs demultiplexes in accordance with a wavelength between the control unit **20** and the subscriber device **40**, the wavelength of the signal to be transmitted may not be changed. When the transmission wavelength is set in the instruction by the received optical signal, the subscriber device **40-1** sets a wavelength of the optical transceiver **41** in accordance with the wavelength instruction (step **S6**). That is, the subscriber device **40-1** sets the wavelength of the optical transceiver **41** (the wavelength-tunable light source **451**) to transmit an optical signal using the transmission wavelength set in the wavelength instruction. When the reception wavelength is set in the wavelength instruction, the subscriber device **40-1** sets the optical transceiver **41** (the wavelength-tunable filter **453**) to receive the wavelength signal of the reception wavelength.

When the optical transceiver **41** of the subscriber device **40-1** transmits an optical signal, the optical transceiver transmits a notification signal for notifying that the wavelength has been set with the optical signal of an instructed wavelength. Like the request signal, the notification signal is transmitted to the control unit **230** via the optical SW **210**, the monitoring unit **260**, or the communication unit. Based on the received notification signal, the control unit **230** confirms whether the specified wavelength is set correctly, whether the output power is sufficient, and the like (step **S7**). The control unit **230** confirms whether the value of the notification signal is a value as instructed. Further, when the optical signal is received, the control unit **230** may perform the confirming with a measurement value of the optical signal. When determining that there is no problem as a result of the confirmation, the control unit **230** transmits a permission notification indicating a permission to start communication to the subscriber device **40-1** with the optical signal. The permission notification is transmitted to the subscriber device **40-1**, like the wavelength instruction.

The optical SW control unit **320** transmits the connection information of the optimum port in the optical SW **210** to the optical SW **210** according to the transmission destination of

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the subscriber device **40-1**. The optical SW **210** sets an uplink port and a downlink port of the subscriber device **40-1** in response to an instruction from the optical SW control unit **320** based on the connection information (step **S8**).

The optical access system **100** controls a timing such that path switching in the optical SW **210** is performed after the permission to start communication is transmitted from the control unit **230** to the subscriber device **40-1**. For example, it is assumed that a time required for the path switching of the optical SW **210** is known in advance. In this case, the control unit **230** waits for a time required for the optical SW **210** to receive an instruction to switch the path and actually switch the path until the subscriber device **40-1** receives the permission to start communication and then actually starts communication, and subsequently gives the instructions to start the communication. After the communication is started, the monitoring unit **260** of the GW **200** confirms a communication status between the facing subscriber devices (step **S9**). The monitoring unit **260** notifies the OPS **300** of a confirmation result. When the confirming is NG, the control unit **230** or the OPS **300** performs a cause isolation procedure.

The connection request transmitted by the subscriber device **40-1** and the control signal transmitted by the control unit **230** to the subscriber device **40-1** are optical signals of which a speed is lower than the main signal. As the control signal, for example, a protocol-free control signal (control scheme) typified by AMCC can be used.

The control unit or the OPS **300** instructs the communication destination subscriber device **40** to use a transmission wavelength of the subscriber device **40-1** as a reception wavelength of the communication destination subscriber device **40** and use a reception wavelength of the subscriber device **40-1** as a transmission wavelength of the communication destination subscriber device **40**. For example, in the control unit **301** that controls the optical GW **200** in which the communication destination subscriber device **40** is accommodated, the wavelength control unit **310** instructs the control unit **230** to transmit a wavelength instruction in which the reception wavelength and the transmission wavelength of the communication destination subscriber device **40** are set. The communication destination subscriber device **40** receives the wavelength instruction from the control unit **230** by the control signal, and sets the reception wavelength and the transmission wavelength in the optical transceiver **41** in accordance with the received wavelength instruction. That is, when the transmission wavelength is set in the wavelength instruction, the communication destination subscriber device **40** sets the wavelength of the optical transceiver **41** (the wavelength-tunable light source **451**) so that the optical signal is transmitted with the transmission wavelength. When the reception wavelength is set in the wavelength instruction, the communication destination subscriber device **40** sets the optical transceiver **41** (the wavelength-tunable filter **453**) so that the wavelength signal of the reception wavelength is received.

The optical access system **100** may transmit and receive information to be registered in the management DB **350** in response to the user application between the new subscriber device **40-1** and the control unit **301** without performing the user application in step **S1**. Accordingly, the subscriber device **40-1** can communicate with another subscriber device **40** without performing the user application. Transmission and reception of information between the subscriber device **40-1** and the control unit **301** are performed using, for example, the AMCC via the control unit **230**.

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An operation when a new subscriber device is connected has been described above. Next, a normal communication operation after the connection of the new subscriber device will be described by exemplifying a case where the subscriber device **40-2** in FIG. **38** performs communication.

First, an uplink optical signal describing uplink communication and output from the subscriber device **40-2** is transmitted to the optical GW **200** via the transmission line **501**. The wavelength multiplexer/demultiplexer **220** of the optical GW **200** separates the input optical signal into an uplink optical signal and a downlink optical signal in accordance with a wavelength. The uplink optical signal demultiplexed by the wavelength multiplexer/demultiplexer **220** is input to the optical SW **210** via the transmission line **521**. The optical SW **210** connects a port to which the uplink optical signal has been input from the wavelength multiplexer/demultiplexer **220** to another port corresponding to a path to the transmission destination on the path to the communication destination of the subscriber device **40-2**, and outputs an optical signal. When the wavelength is used as the destination information, the optical SW **210** is connected to another port corresponding to the transmission destination on the path to the communication destination specified by the wavelength allocated to the subscriber device **40-2** and outputs an optical signal. The uplink signal output from the optical SW **210** is multiplexed with an optical signal with a different wavelength transmitted by another subscriber device **40** in the multiplexer **241** to be transmitted to another station (for example, the ground B) via one transmission line **511**. Each multiplexer **241** multiplexes the wavelength channels for each station such as the ground B and the ground C. It is also possible to use the same wavelength between the ground B and the ground C by separating the transmission line **511** with the ground B and the transmission line **511** with the ground C.

Next, the downlink communication will be described. A downlink is communication from the ground B or C toward the subscriber device **40**. The downlink optical signal is transmitted to the optical GW **200** via one transmission line **512**. The demultiplexer **242** of the optical GW **200** demultiplexes the downlink optical signal transmitted through the transmission line **512** in accordance with the wavelength. The demultiplexer **242** inputs the demultiplexed light to the downlink port corresponding to the wavelength of the demultiplexed light via the transmission line **542**. The optical SW **210** connects a port to which the downlink optical signal is input from the demultiplexer **242** to another port corresponding to the wavelength, and outputs an optical signal. The wavelength multiplexer/demultiplexer **220** separates the optical signal input from the optical SW **210** via the transmission line **522** into an uplink optical signal and a downlink optical signal in accordance with a wavelength. The downlink optical signal demultiplexed by the wavelength multiplexer/demultiplexer **220** is input to the subscriber device **40-2** via the transmission line **501**. Wavelength channels transmitted from the optical GW **200** to stations (the ground B, C, and the like) are assumed to be the same wavelength band, but different wavelength bands may be used for the stations.

The monitoring unit **260** of the optical GW **200** receives the light branched by the branching unit **250**. The light branched by the branching unit **250** is an optical signal transmitted and received by each subscriber device **40**. The monitoring unit **260** monitors signals transmitted and received by the subscriber devices **40** by monitoring the received optical signals. When an abnormality such as a wavelength shift, a decrease in an output, or a communica-

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tion abnormality is detected through the monitoring, the monitoring unit 260 transmits an abnormality detection signal to the control unit 301. The optical SW control unit 320 of the control unit 301 controls the optical SW 210 such that the target subscriber device 40 is reconnected to the control unit 230. Then, the control unit 301 performs an allocation process for new wavelengths different from the wavelengths used when the abnormality is detected, like the time at which the subscriber device 40 is newly connected. Accordingly, when an optical signal with a changed wavelength is input from the subscriber device 40, the optical SW 210 connects the input optical signal to the port specified by the wavelength before the change by the subscriber device 40.

(First Configuration Example of Optical Access System 100)

Although the optical GW 200 illustrated in FIG. 38 performs wavelength multiplexing, it is not necessary to perform wavelength multiplexing as illustrated in FIGS. 43 and 44. FIG. 43 is a diagram illustrating a configuration example of the optical access system 101. The optical access system 101 illustrated in FIG. 43 is different from the optical access system 100 illustrated in FIG. 38 in that a GW 201 is included instead of the GW 200. The GW 201 is different from the GW 200 in that a wavelength multiplexer/demultiplexer 243 and a branching unit 250a instead of the multiplexer 241, the demultiplexer 242, and the branching unit 250 are included. The GW 201 is connected to a communication device of a station of another ground via a transmission line 503. Along one transmission line 503, an uplink signal and a downlink signal are transmitted to any ground.

The wavelength multiplexer/demultiplexer 243 separates the input optical signal into an uplink optical signal and a downlink optical signal in accordance with a wavelength. The wavelength multiplexer/demultiplexer 243 separates the uplink optical signal input from the optical SW 210 via the transmission line 543-1 and transmits the signal to another ground or an upper network via the transmission line 503. The wavelength multiplexer/demultiplexer 243 separates a downlink optical signal input from another ground via the transmission line 503 and outputs the signal to the optical SW 210 via the transmission line 543-2.

The branching unit 250a is provided in the transmission line 503. The branching unit 250a includes a power splitter 251a. The power splitter 251a branches uplink and downlink optical signals transmitted through the transmission line 503. The power splitter 251a inputs the branched uplink optical signal to the port of the optical SW 210 via a transmission line 551a and inputs the branched downlink optical signal to the port of the optical SW 210 via a transmission line 551b. The optical SW 210 outputs an optical signal input from a port connected to the transmission line 551a and an optical signal input from a port connected to the transmission line 551b from a port connected to the transmission line 560. Accordingly, the wavelength demultiplexer 261 of the monitoring unit 260 receives the optical signal branched by the branching unit 250a.

(Second Configuration Example of Optical Access System 100)

FIG. 44 is a diagram illustrating a configuration example of an optical access system 102. The optical access system 102 illustrated in FIG. 44 is different from the optical access system 101 illustrated in FIG. 43 in that an optical GW 202 is included instead of the optical GW 201. The optical GW 202 is different from the optical GW 201 in that a wavelength multiplexer/demultiplexer 244, a wavelength multiplexer/demultiplexer 245, and a branching unit 250b are

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included instead of the wavelength multiplexer/demultiplexer 243 and the branching unit 250a.

The wavelength multiplexer/demultiplexer 244 separates an uplink optical signal and a downlink optical signal in accordance with the wavelength. The wavelength multiplexer/demultiplexer 244 inputs the uplink optical signal input from the optical SW 210 via the transmission line 544 to the wavelength multiplexer/demultiplexer 245 via the transmission line 545. The wavelength multiplexer/demultiplexer 244 inputs the downlink optical signal input from the wavelength multiplexer/demultiplexer 245 via the transmission line 546 to the optical SW 210 via the transmission line 544.

The wavelength multiplexer/demultiplexer 245 separates an uplink optical signal and a downlink optical signal in accordance with the wavelength. The wavelength multiplexer/demultiplexer 245 transmits the uplink optical signal input from the wavelength multiplexer/demultiplexer 245 via the transmission line 545 to another ground or an upper network via the transmission line 503. In addition, the wavelength multiplexer/demultiplexer 245 inputs the downlink optical signal received via the transmission line 503 to the wavelength multiplexer/demultiplexer 244 via the transmission line 546.

The branching unit 250b includes a power splitter 251b and a power splitter 252b. The power splitter 251b branches an uplink optical signal transmitted along the transmission line 545. The power splitter 251b inputs the branched uplink optical signal to the port of the optical SW 210 via the transmission line 551b. The power splitter 252b branches the downlink optical signal transmitted along the transmission line 546. The power splitter 252b inputs the branched downlink optical signal to the port of the optical SW 210 via the transmission line 552b. The optical SW 210 outputs an optical signal input from a port connected to the transmission line 551b and an optical signal input from a port connected to the transmission line 552b from a port connected to the transmission line 560. Accordingly, the optical signal branched by the branching unit 250b is input to the wavelength demultiplexer 261 of the monitoring unit 260.

The above-described monitoring unit 260 has a receiver configuration including the wavelength demultiplexer 261 and the optical receiver 262 for each wavelength. The monitoring unit 260 may include a wavelength-tunable optical receiver instead of the receiver configuration. The transceiver of the control unit may include a transmitter that is not wavelength-tunable, or may have a receiver configuration in which a wavelength demultiplexer is not included. An example of such a configuration will be described with reference to FIG. 45.

(Third Configuration Example of Optical Access System 100)

FIG. 45 is a diagram illustrating a configuration example of an optical access system 103. The optical access system 103 illustrated in FIG. 45 is different from the optical access system 100 illustrated in FIG. 38 in that an optical GW 203 is included instead of the optical GW 200. The optical GW 203 is different from the optical GW 200 in that a control unit 235 and a monitoring unit 265 are included instead of the control unit 230 and the monitoring unit 260. The control unit 235 includes an optical receiver 236 and an optical transmitter 237 that is not wavelength-tunable. The monitoring unit 265 includes a wavelength-tunable optical receiver 266. The monitoring unit 265 may be replaced with any of the monitoring units 60 to 60h described above.

(Fourth Configuration Example of Optical Access System 100)

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The monitoring unit may be connected via an optical SW different from the above-described optical SW. An example of such a configuration will be described with reference to FIG. 46. FIG. 46 is a diagram illustrating a configuration example of an optical access system 104. The optical access system 104 illustrated in FIG. 46 is different from the optical access system 103 illustrated in FIG. 45 in that an optical GW 204 is included instead of the optical GW 203. The optical GW 204 is different from the optical GW 203 in that an optical SW 211 is further included and that the monitoring unit 265 is connected to the optical SW 211.

An uplink optical signal separated from the transmission line 511 by the power splitter 251 of the branching unit 250 is input to the optical SW 211 via a transmission line 555, and a downlink optical signal separated from the transmission line 512 by the power splitter 252 is input to the optical SW 211 via the transmission line 555. The optical SW 211 is, for example, a small optical SW or the like. The number of ports of the optical SW 211 is one on the monitoring unit 260 side and are 2M on a side to which an optical signal to be monitored is input. 2M is twice the number M of the subscriber devices 40 connected to the optical GW 204. The monitoring units may be prepared by the number of connected grounds without using the small optical SW, and the signals transmitted to and received from all the grounds may be monitored for each ground.

In the present configuration example, communication is performed between a plurality of subscriber devices connected to the same optical GW using a folded transmission line. In the description below, differences from the configuration example will be mainly explained.

(Fifth Configuration Example of Optical Access System 100)

FIG. 47 is a diagram illustrating a configuration example of an optical access system 105. The optical access system 105 illustrated in FIG. 47 is different from the optical access system 103 illustrated in FIG. 45 in that an optical GW 205 is included instead of the optical GW 203. The optical GW 205 is different from the optical GW 203 in that a multiplexer 247 and a demultiplexer 248 corresponding to the ground A where the optical GW 205 is installed are further included. The multiplexer 247 and the demultiplexer 248 are connected by a transmission line 547. The transmission line 547 is a folded transmission line.

Like the multiplexer 241, the multiplexer 247 multiplexes uplink optical signals with different wavelengths output from the plurality of transmission lines 541 by the optical SW 210 and outputs the multiplexed signal to the transmission line 547. Like the demultiplexer 242, the demultiplexer 248 demultiplexes a downlink optical signal input from the transmission line 547 in accordance with the wavelength. The demultiplexer 248 inputs each of the demultiplexed downlink optical signals to the optical SW 210 via a plurality of transmission lines 542 connected to a downlink port corresponding to the wavelength of the optical signal. The transmission line 547 includes the branching unit 250.

In the foregoing configuration example, the subscriber device connected to the ground A is connected to a port for connecting to the ground B or the ground C via the optical SW. In the present configuration example, another set that is the same as the combination of the multiplexer 241 and the demultiplexer 242 connected to the ground B or the ground C is added. The added set includes the multiplexer 247 and the demultiplexer 248. Then, the output port of the added multiplexer 247 and the input port of the added demultiplexer 248 are connected by a transmission line 547. In this configuration, the signal output from the subscriber device

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40 can be input to the optical SW 210 again. Accordingly, the optical GW 205 folds an optical signal output from a certain subscriber device 40 and causes the optical SW 210 to be incident again as a downlink signal. By connecting this folded signal to another subscriber device 40 in the optical SW 210, folded communication, that is, communication between the subscriber devices 40 connected to the same optical GW 205, becomes possible.

For example, a state in which the subscriber device 40-2 and the subscriber device 40-M communicate with each other will be described. It is assumed that each of K (where K is an integer equal to or greater than 2) uplink ports corresponding to the ground A of the optical SW 210 is connected to the multiplexer 247 by the transmission line 541, and each of K downlink ports corresponding to the ground A of the optical SW 210 is connected to the demultiplexer 248 by the transmission line 542. Then, it is assumed that kth (where k is an integer equal to or greater than 1 and equal to or less than K) uplink port and downlink port among the K downlink ports and uplink ports corresponding to the ground A correspond to a wavelength λ_k . The uplink optical signal with the wavelength λ_1 output from the subscriber device 40-2 is connected to a first uplink port corresponding to the ground A. The input optical signal is folded back along the transmission line 547 and is input to the optical SW 210 again as a downlink optical signal from the first downlink port corresponding to the ground A. The optical SW control unit 320 sets a path in the optical SW 210 so that the optical signal is transmitted to the subscriber device 40-M in accordance with the wavelength. Similarly, the uplink optical signal with the wavelength λ_k output from the subscriber device 40-M is connected to a kth uplink port corresponding to the ground A. The input optical signal is folded back along the transmission line 547 and is input to the optical SW 210 again as a downlink optical signal from the kth downlink port corresponding to the ground A. The optical SW control unit 320 sets a path in the optical SW 210 so that the optical signal is transmitted to the subscriber device 40-2 in accordance with the wavelength. Accordingly, communication is performed between the subscriber device 40-2 and the subscriber device 40-M.

(Sixth Configuration Example of Optical Access System 100)

Another configuration of the present configuration example will be described with reference to FIGS. 48 and 49. FIG. 48 is a diagram illustrating a configuration example of an optical access system 106. The optical access system 106 illustrated in FIG. 48 is different from the optical access system 105 illustrated in FIG. 47 in that an optical GW 206 is included instead of the optical GW 205. The optical GW 206 is different from the optical GW 205 in that the multiplexer 247 and the demultiplexer 248 are not provided, and the uplink port and the downlink port for the ground A of the optical SW 210 are directly connected by the transmission line 548 without wavelength multiplexing, so that a signal is folded back.

(Seventh Configuration Example of Optical Access System 100)

FIG. 49 is a diagram illustrating a configuration example of an optical access system 107. The optical access system 107 illustrated in FIG. 49 is different from the optical access system 105 illustrated in FIG. 47 in that an optical GW 207 is included instead of the optical GW 205. The optical GW 207 is different from the optical GW 205 in that a power splitter 270 is included instead of the demultiplexer 248. The power splitter 270 branches the downlink optical signal input from the multiplexer 247 via the transmission line 547

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into a plurality of signals and inputs the plurality of signals to the optical SW 210 via the plurality of transmission lines 542.

A power splitter may be provided at a subsequent stage of the demultiplexer 248 of the optical GW 205 in FIG. 47. The power splitter branches the optical signal demultiplexed by the demultiplexer 248 into a plurality of signals and inputs the plurality of signals to different ports of the optical SW 210. In this way, multicast communication of folded communication can be performed.

Although the difference from the optical access system 103 has been described above, the difference can also be applied to the optical access systems 100, 101, and 102. (Eighth Configuration Example of Optical Access System 100)

The optical access system of the present configuration example performs multicast communication. In the present configuration example, the difference will be mainly described.

First, multicasting in downlink communication will be described with reference to FIG. 50. FIG. 50 is a diagram illustrating a configuration example of an optical access system 108. The optical access system 108 illustrated in FIG. 50 is different from the optical access system 107 illustrated in FIG. 49 in that an optical GW 208 is included instead of the optical GW 207. The optical GW 208 is different from the optical GW 207 in that a transmission line 549 that connects a folded port of the optical SW 210 is further provided.

A case where a downlink optical signal transmitted from the ground C is multicast will be described. The optical SW control unit 320 controls a port to which a downlink optical signal from the ground C is input such that the port is connected to a folded port to which the transmission line 549 is connected in accordance with the wavelength. Accordingly, the downlink optical signal from the ground C is transmitted along the transmission line 549 and is input again to the optical SW 210 as an uplink signal of the ground A. The optical SW control unit 320 controls the downlink optical signal input from the folded port such that the downlink optical signal is connected to the uplink signal port of the ground A. Accordingly, the optical signal folded back along the transmission line 549 and input to the optical SW 210 is output to the port connected to the multiplexer 247. The multiplexer 247 multiplexes the optical signals output from the optical SW 210 from each of the plurality of transmission lines 541 and outputs the multiplexed optical signals to the transmission line 547. The optical signal output to the transmission line 547 is branched into a plurality of optical signals by the power splitter 270. The power splitter 270 inputs the plurality of branched optical signals to the optical SW 210 as downlink signals of the ground A via the plurality of transmission lines 542. The optical SW 210 outputs the optical signal input from each transmission line 542 to a port connected to the subscriber device 40 in accordance with the wavelength. Accordingly, the downlink signals can be multicast.

(Ninth Configuration Example of Optical Access System 100)

Next, multicasting of uplink communication will be described with reference to FIG. 51. FIG. 51 is a diagram illustrating a configuration example of an optical access system 109. The optical access system 109 illustrated in FIG. 51 is different from the optical access system 103 illustrated in FIG. 45 in that an optical GW 209 is included instead of the optical GW 203. The optical GW 209 is different from the optical GW 203 in that a transmission line

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570 that connects a folded port to the optical SW 210 and a multicast power splitter 271 are further included. The power splitter 271 is connected to the optical SW 210 via the transmission line 572 and the plurality of transmission lines 573.

A case where the uplink optical signal transmitted from the ground A is multicast will be described. The optical SW control unit 320 controls the port to which the uplink optical signal from the ground A is input is connected to the folded port to which the transmission line 570 is connected in accordance with the wavelength. Accordingly, the uplink optical signal from the ground A is transmitted through the transmission line 570 and is input to the optical SW 210 again. The optical SW control unit 320 performs control such that the optical signal input from the folded port is output to the port connected to the power splitter 271. Accordingly, the optical signal that is input to the optical SW 210 and folded along the transmission line 570 is output to the transmission line 572. The optical signal output to the transmission line 572 is branched into a plurality of optical signals by the power splitter 271. The power splitter 271 inputs the plurality of branched optical signals to the optical SW 210 as uplink signals via the plurality of transmission lines 573. The optical SW 210 outputs the optical signal input from each transmission line 573 to a port connected to the ground B or the ground C in accordance with the wavelength. Accordingly, the uplink signals can be multicast.

(Tenth Configuration Example of Optical Access System 100)

Next, a configuration in which point-to-multipoint communication including uplink communication is performed while multicasting of downlink communication is performed will be described with reference to FIG. 52. FIG. 52 is a diagram illustrating a configuration example of an optical access system 110. The optical access system 110 illustrated in FIG. 52 is different from the optical access system 103 illustrated in FIG. 45 in that an optical GW 2010 is included instead of the optical GW 203. The optical GW 2010 is different from the optical GW 203 in that transmission lines 574 and 575 connecting a folded port to the optical SW 210 and power splitters 272 and 273 are further included. The power splitter 272 is connected to the optical SW 210 via a transmission line 581 and a plurality of transmission lines 582. The power splitter 273 is connected to the optical SW 210 via a plurality of transmission lines 583 and a transmission line 584.

A case where a downlink optical signal transmitted from the ground C is multicast will be described. The optical SW control unit 320 controls a port to which a downlink optical signal from the ground C is input is connected to the folded port to which the transmission line 574 is connected in accordance with the wavelength. Accordingly, the downlink optical signal from the ground C is transmitted along the transmission line 574 and is input again to the optical SW 210 as an uplink signal of the ground A. The optical SW control unit 320 performs control such that the downlink optical signal input from the folded port is output to the port to which the power splitter 272 is connected. Accordingly, the optical signal that is input to the optical SW 210 and folded along the transmission line 574 is output to the transmission line 581. The optical signal output to the transmission line 581 is branched into a plurality of optical signals by the power splitter 272. The power splitter 272 inputs the plurality of branched optical signals to the optical SW 210 as downlink signals via the plurality of transmission lines 582. The optical SW 210 outputs the optical signal

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input from each transmission line **582** to a port connected to the subscriber device **40** according to the wavelength. Accordingly, the downlink signals can be multicast.

A case where an uplink optical signal transmitted from the ground A is transmitted to the ground C will be described. The optical SW control unit **320** performs control such that a port to which an uplink optical signal from the ground A is input is connected to a port to which the power splitter **273** is connected in accordance with the wavelength. Accordingly, the uplink optical signals from the ground A is output to the transmission lines **583**. The optical signals output to the plurality of transmission lines **583** are multiplexed by the power splitter **273**. The power splitter **273** inputs the multiplexed optical signal to the optical SW **210** via the transmission line **584**. The optical SW **210** controls the optical signal input from the transmission line **584** such that the optical signal is connected to the folded port to which the transmission line **575** is connected. Accordingly, the optical signal is transmitted along the transmission line **575** and is input to the optical SW **210** again. The optical SW **210** outputs the optical signal input from the transmission line **575** to the multiplexer **241** connected to the ground C in accordance with the wavelength.

As described above, by providing two sets of configurations using a power splitter for multicasting, not only downlink multicast communication but also point-to-multipoint communication including uplink communication can be performed.

(Eleventh Configuration Example of Optical Access System **100**)

In the present configuration example, communication is performed without separating an uplink signal and a downlink signal. In the description below, differences from the configuration example will be mainly explained.

FIG. **53** is a diagram illustrating a configuration example of an optical access system **111**. The optical access system **111** illustrated in FIG. **53** is different from the optical access system **105** illustrated in FIG. **47** in that an optical GW **2011** is included instead of the optical GW **205**. The optical GW **2011** is different from the optical GW **205** in that the wavelength multiplexer/demultiplexer **220** is not included, a wavelength multiplexer/demultiplexer **249** and a branching unit **253** are included instead of the multiplexer **241**, the demultiplexer **242**, and the branching unit **250**, and a wavelength multiplexer/demultiplexer **238** is further included.

The wavelength multiplexer/demultiplexer **249** is connected to the optical SW **210** by a plurality of transmission lines **585**. The wavelength multiplexer/demultiplexer **249** multiplexes the uplink optical signals with different wavelengths output from the plurality of transmission lines **585** by the optical SW **210** and outputs the multiplexed signal to the transmission line **504** connected to any other ground. The wavelength multiplexer/demultiplexer **249** demultiplexes a downlink optical signal input from another ground via the transmission line **504** in accordance with a wavelength. The wavelength multiplexer/demultiplexer **249** inputs each of the demultiplexed downlink optical signals to the optical SW **210** via the plurality of transmission lines **585** connected to an uplink port corresponding to the wavelength of the optical signal.

The branching unit **253** includes a power splitter **254**. The power splitter **254** branches an uplink optical signal and a downlink optical signal transmitted through the transmission line **504**. The power splitter **254** inputs the branched uplink optical signals to the port of the optical SW **210** via the transmission line **586**, and inputs the branched downlink optical signals to the port of the optical SW **210** via the

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transmission line **587**. The optical SW **210** outputs an optical signal input from the transmission line **586** or a port connected to the transmission line **587** to a port connected to the transmission line **560**.

The wavelength multiplexer/demultiplexer **238** is connected to the optical SW **210** by the transmission line **534** and is connected to the control unit **235** by the transmission line **531** and the transmission line **533**. The wavelength multiplexer/demultiplexer **238** separates the input optical signal into an uplink optical signal and a downlink optical signal in accordance with a wavelength. The wavelength multiplexer/demultiplexer **238** outputs the uplink optical signal input from the optical SW **210** via the transmission line **534** to the control unit **235** via the transmission line **531**. The wavelength multiplexer/demultiplexer **238** outputs the downlink optical signal input from the control unit **235** via the transmission line **533** to the optical SW **210** via the transmission line **534**.

As described above, the optical GW **2011** has a configuration in which there is no wavelength multiplexer/demultiplexer between the optical SW **210** and the subscriber device **40** and an uplink signal and a downlink signal are not separated. Accordingly, the number of ports used for the optical SW **210** can be considerably reduced, and an amount of information to be managed can be considerably reduced. As illustrated in FIG. **54**, a portion that separates the optical signal to the monitoring unit **265** may have the configuration illustrated in FIG. **44**.

(Twelfth Configuration Example of Optical Access System **100**)

FIG. **54** is a diagram illustrating a configuration example of an optical access system **112** according to the present embodiment. The optical GW **2012** of the optical access system **112** illustrated in FIG. **54** includes a branching unit **255** instead of the branching unit **253** included in the optical GW **2011** illustrated in FIG. **53**. The branching unit **255** includes a wavelength multiplexer/demultiplexer **256**, a wavelength multiplexer/demultiplexer **257**, a power splitter **258**, and a power splitter **259**.

The wavelength multiplexer/demultiplexer **256** separates an input optical signal into an uplink optical signal and a downlink optical signal in accordance with a wavelength. The wavelength multiplexer/demultiplexer **256** outputs an uplink optical signal input from the wavelength multiplexer/demultiplexer **249** to the wavelength multiplexer/demultiplexer **257** via the transmission line **588**. The wavelength multiplexer/demultiplexer **256** outputs a downlink optical signal input from the wavelength multiplexer/demultiplexer **257** via the transmission line **589** to the wavelength multiplexer/demultiplexer **249**.

The wavelength multiplexer/demultiplexer **257** separates an uplink optical signal and a downlink optical signal in accordance with the wavelength. The wavelength multiplexer/demultiplexer **257** outputs the uplink optical signal input from the wavelength multiplexer/demultiplexer **256** via the transmission line **588** to the transmission line **504**. The wavelength multiplexer/demultiplexer **257** inputs a downlink optical signal received from another ground via the transmission line **504** to the wavelength multiplexer/demultiplexer **256** via the transmission line **589**.

The power splitter **258** branches an uplink optical signal transmitted along the transmission line **588** and inputs the signal to the port of the optical SW **210** via the transmission line **586**. The power splitter **259** branches a downlink optical signal transmitted along the transmission line **589** and inputs the signal to the port of the optical SW **210** via the transmission line **587**. The optical SW **210** outputs an optical

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signal input from the transmission line **586** or a port connected to the transmission line **587** to a port connected to the transmission line **560**.

The optical GW **2011** illustrated in FIG. **53** performs wavelength multiplexing. However, as illustrated in FIG. **55**, a signal to be transmitted to each of the stations (the ground B and the ground C) may be transmitted along individual transmission lines without performing wavelength multiplexing.

(Thirteenth Configuration Example of Optical Access System **100**)

FIG. **55** is a diagram illustrating a configuration example of an optical access system **113**. The optical access system **113** illustrated in FIG. **55** is different from the optical access system **101** illustrated in FIG. **43** in that an optical GW **2013** is included instead of the optical GW **201**. The optical GW **2013** is different from the optical GW **201** in that the wavelength multiplexer/demultiplexer **220** and the wavelength multiplexer/demultiplexer **243** are not included, and the control unit **235**, the wavelength multiplexer/demultiplexer **238**, and the monitoring unit **265** illustrated in FIG. **53** are included instead of the control unit **230** and the monitoring unit **260**. Along a port of the optical SW **210** connected to the transmission line **503**, an uplink optical signal is output and a downlink optical signal is input.

(Fourteenth Configuration Example of Optical Access System **100**)

The branching unit **250a** in the optical GW **2013** may have a configuration illustrated in FIG. **56**. FIG. **56** is a diagram illustrating a configuration example of an optical access system **114**. An optical GW **2014** of the optical access system **114** illustrated in FIG. **56** has a configuration similar to that of the branching unit **255** illustrated in FIG. **54** instead of the branching unit **250a** included in the optical GW **2013** illustrated in FIG. **55**.

(Fifteenth Configuration Example of Optical Access System **100**)

The present configuration example enables a subscriber device during communication to be controlled. In the description below, differences from the configuration example will be mainly explained.

FIG. **57** is a diagram illustrating a configuration example of an optical access system **115**. The optical access system **115** illustrated in FIG. **57** is different from the optical access system **104** illustrated in FIG. **46** in that an optical GW **2015** is included instead of the optical GW **204**. The optical GW **2015** is different from the optical GW **204** in that a monitoring unit **267** is connected to the optical SW **211** instead of the monitoring unit **265**.

The monitoring unit **267** includes a wavelength-tunable receiver **268** and a wavelength-tunable transmitter **269**. The monitoring unit **267** can receive an optical signal with any wavelength by the wavelength-tunable receiver **268** and can transmit an optical signal with any wavelength by the wavelength-tunable transmitter **269**. In addition, the optical GW **2015** includes the control unit **235**. When the subscriber device **40** is connected, the optical GW **2015** performs connection processing (registration, wavelength allocation, or the like) of the subscriber device **40** using the control unit **235**, and starts normal communication.

Here, a state in which the subscriber device **40-1** is connected to the ground B will be considered. The subscriber device **40-1** cannot communicate with the control unit **235** because of a state in which the normal communication is being performed. Accordingly, by providing the monitoring unit **267** connected to the optical SW **211** which is the small optical SW, it is possible not only to monitor a

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communication state of the subscriber device **40-1** but also to give an instruction for various settings of the subscriber device **40-1**. That is, the optical signal separated by the power splitter **251** is output to the optical SW **211** via the transmission line **555**. The optical SW **211** outputs an input optical signal to the monitoring unit **267**. The monitoring unit **267** performs monitoring using the optical signal received from the optical SW **211** by the wavelength-tunable receiver **268**, and further receives a control signal superimposed on the received optical signal. The wavelength-tunable transmitter **269** of the monitoring unit **267** transmits a control signal to the subscriber device **40** by an optical signal. The optical SW **211** outputs the optical signal input from the wavelength-tunable transmitter **269** to a port corresponding to the wavelength. The power splitter **251** multiplexes the control signal input from the optical SW **211** via the transmission line **556** with the optical signal transmitted along the transmission line **512**. In this configuration, even in a state where the subscriber device **40-1** is performing the normal communication, it is possible to receive a connection destination change request or the like from the subscriber device **40-1**, transmit the control signal, and perform wavelength switching or the like on the subscriber device **40-1**.

In the communication of the control signal between the monitoring unit **267** and each subscriber device **40**, a control signal that is slower than the optical main signal between the subscriber devices and can be superimposed on the main signal is used. For example, a technology such as AMCC can be used.

(Sixteenth Configuration Example of Optical Access System **100**)

In the present configuration example, electrical processing is performed on the optical signal extracted from the optical SW. In the description below, differences from the configuration example will be mainly explained.

FIG. **58** is a diagram illustrating a configuration example of an optical access system **116**. The optical access system **116** illustrated in FIG. **58** is different from the optical access system **105** illustrated in FIG. **47** in that an optical GW **2016** is included instead of the optical GW **202**. The optical GW **2016** is different from the optical GW **202** in that an electrical processing unit **600** is connected.

The electrical processing unit **600** converts an optical signal into an electrical signal, performs electrical processing on the electrical signal, and then converts the electrical signal into the optical signal again and outputs the optical signal. The electrical processing unit **600** includes an O/E conversion unit **610**, a processing execution unit **620**, and an E/O converter **630**. The O/E conversion unit **610** corresponds to the O/E conversion unit **85** in FIG. **13**. The O/E conversion unit **610** converts an optical signal input from the optical SW **210** into an electrical signal and outputs the electrical signal to the processing execution unit **620**. The processing execution unit **620** corresponds to the processing execution unit **86** and the storage unit **88** in FIG. **13**. In the processing execution unit **620**, a processor such as a CPU or an accelerator performs electrical processing on the electrical signal converted by the O/E conversion unit **610** by reading a program from a storage unit (not illustrated) and executing the program. In this electrical processing, an electrical signal processing function, a function such as an OLT, and the like are implemented. The signal processing function is, for example, code error correction such as FEC. The E/O converter **630** corresponds to the E/O conversion unit **87** in FIG. **13**. The E/O conversion unit **87** converts the electrical signal into an optical signal and outputs the optical

signal to the optical SW **210**. The O/E conversion unit **610** and the E/O converter **630** are, for example, wavelength-tunable transceivers.

In FIG. **58**, a subscriber device **40-M** is an ONU of a PON. The subscriber device **40-M** is connected to the optical GW **2016** via the transmission line **501** such as an optical fiber and the power splitter **507**. An error correction function, an OLT function, and the like are implemented in the processing execution unit **620** of the electrical processing unit **600**.

The wavelength control unit **310** notifies the processing execution unit **620** of a determination condition for determining a signal to be subjected to electrical processing and a type of electrical processing to be performed on the signal. The processing execution unit **620** stores information regarding the determination condition and the type of electrical processing notification of which is provided by the wavelength control unit **310**.

For example, in step **S5** of FIG. **42**, the wavelength control unit **310** determines whether to perform electrical processing on communication between the subscriber device **40** (hereinafter referred to as a request source subscriber device **40**) that is a transmission source of a connection request and the communication destination subscriber device **40**. The wavelength control unit **310** determines whether to perform the electrical processing according to a distance between the request source subscriber device **40** and the communication destination subscriber device **40** facing each other, a service to be provided to the request source subscriber device **40** or the communication destination subscriber device **40**, or the like. When the electrical processing is performed, the wavelength control unit **310** determines which electrical processing is performed. When the wavelength control unit **310** determines to perform electrical processing (hereinafter referred to as transmission signal electrical processing) on a signal addressed from the request source subscriber device **40** to the communication destination subscriber device **40**, the wavelength control unit **310** allocates a first transmission wavelength and a second transmission wavelength among free wavelengths. When the wavelength control unit **310** determines to perform the electrical processing (hereinafter referred to as reception signal electrical processing) on a signal addressed from the communication destination subscriber device **40** to the request source subscriber device **40**, the wavelength control unit **310** allocates a first reception wavelength and a second reception wavelength among free wavelengths.

The first transmission wavelength is a wavelength for routing an optical transmission signal, which is an optical signal addressed from the request source subscriber device **40** to the communication destination subscriber device **40**, to the electrical processing unit **600**. The second transmission wavelength is a wavelength for routing the transmission signal on which the transmission signal electrical processing is performed by the electrical processing unit **600** to a port corresponding to the communication destination subscriber device **40**. The first reception wavelength is a wavelength for routing a reception signal, which is an optical signal addressed from the communication destination subscriber device **40** to the request source subscriber device **40**, to the electrical processing unit **600**. The second reception wavelength is a wavelength for routing the reception signal on which the reception signal electrical processing is performed by the electrical processing unit **600** to a port corresponding to the request source subscriber device **40**. The first transmission wavelength may be the same as the second trans-

mission wavelength and the first reception wavelength may be the same as the second reception wavelength.

When it is determined that the transmission signal electrical processing is performed, the wavelength control unit **310** sets information regarding the first transmission wavelength as a transmission wavelength in a wavelength instruction to be transmitted to the request source subscriber device **40**. When it is determined that the reception signal electrical processing is performed, the wavelength control unit **310** sets information regarding the second reception wavelength as a reception wavelength in a wavelength instruction to be transmitted to the request source subscriber device **40**.

When it is determined that the transmission signal electrical processing is performed, the OPS **300** gives an instruction to use the second transmission wavelength as a reception wavelength of the communication destination subscriber device **40**. When it is determined that the reception signal electrical processing is performed, the OPS **300** gives an instruction to use the first transmission wavelength as a transmission wavelength of the communication destination subscriber device **40**. For example, in the control unit **301** that controls the optical GW **200** in which the communication destination subscriber device **40** is accommodated, the wavelength control unit **310** instructs the control unit **230** to transmit a wavelength instruction in which the reception wavelength and the transmission wavelength of the communication destination subscriber device **40** are set.

Further, when it is determined that the transmission signal electrical processing is performed, the wavelength control unit **310** generates first instruction information in which a determination condition for determining that the signal is a transmission signal addressed from the request source subscriber device **40** to the communication destination subscriber device **40**, a type of transmission signal electrical processing to be performed on the transmission signal, a first transmission wavelength, and a second transmission wavelength are associated with each other. When it is determined that the reception signal electrical processing is performed, the wavelength control unit **310** generates second instruction information in which a determination condition for determining that the reception signal is addressed from the communication destination subscriber device **40** to the request source subscriber device **40**, a type of the reception signal electrical processing to be applied to the reception signal, the first reception wavelength, and the second reception wavelength are associated with each other. The wavelength control unit **310** transmits an electrical processing execution instruction in which the generated first instruction information and second instruction information are set to the electrical processing unit **600**.

When the transmission signal electrical processing is performed, the optical SW control unit **320** controls the optical SW **210** such that the transmission signal with the first transmission wavelength transmitted by the request source subscriber device **40** is output to the electrical processing unit **600**, and the transmission signal with the second transmission wavelength input from the electrical processing unit **600** is output to the transmission line **541** corresponding to the communication destination subscriber device **40**. When the reception signal electrical processing is performed, the optical SW control unit **320** controls the optical SW **210** such that the reception signal with the first transmission wavelength input from the transmission line **542** corresponding to the communication destination subscriber device **40** is output to the electrical processing unit **600** and the reception signal of the second transmission wavelength input from the electrical processing unit **600** is

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output to the transmission line 522 corresponding to the request source subscriber device 40.

For example, it is assumed that transmission signal electrical processing and reception signal electrical processing are performed on an optical signal between the subscriber device 40-2 and the communication destination subscriber device 40 on the ground C. The transmission signal with the first transmission wavelength transmitted by the subscriber device 40-2 is output to the electrical processing unit 600 via the optical SW 210. The O/E conversion unit 610 converts the transmission signal input from the optical SW 210 into an electrical signal. When it is determined that the determination condition included in the first instruction information is satisfied with reference to predetermined information included in the transmission signal converted into the electrical signal, the processing execution unit 620 performs the transmission signal electrical processing corresponding to the determination condition on the transmission signal. For example, the processing execution unit 620 performs error correction such as forward error correction (FEC). The E/O converter 630 converts the transmission signal of the electrical signal subjected to the error correction by the processing execution unit 620 into an optical signal with the second transmission wavelength indicated by the first instruction information and outputs the optical signal to the optical SW 210. The optical SW 210 outputs the transmission signal with the second transmission wavelength to the transmission line 541 corresponding to the ground C. By performing the error correction, transmission characteristics are improved.

The optical SW 210 outputs the reception signal with the first reception wavelength input from the transmission line 542 corresponding to the communication destination subscriber device 40 on the ground C to the electrical processing unit 600. The O/E conversion unit 610 converts the reception signal input from the optical SW 210 into an electrical signal. When it is determined that the determination condition included in the second instruction information is satisfied with reference to the predetermined information included in the transmission signal converted into the electrical signal, the processing execution unit 620 performs the reception signal electrical processing corresponding to the determination condition on the reception signal. The E/O converter 630 converts the reception signal of the electrical signal subjected to the reception signal electrical processing by the processing execution unit 620 into the optical signal with the second reception wavelength indicated by the second instruction information, and outputs the optical signal to the optical SW 210. The optical SW 210 outputs the transmission signal with the second reception wavelength to the transmission line 522 corresponding to the subscriber device 40-2.

FIG. 59 is a diagram illustrating a configuration example of the optical access system 116 when the electrical processing unit 600 multiplexes signals. The electrical processing unit 600 includes O/E converters 610-1 and 610-2 as the plurality of O/E converters 610.

An uplink optical signal of the subscriber device 40-3 and an uplink optical signal of the subscriber device 40-M are connected to the electrical processing unit 600 via the optical SW 210. An OLT function is implemented in the electrical processing unit 600. The processing execution unit 620 of the electrical processing unit 600 performs processing of an electrical stage of the OLT function. A plurality of subscriber devices 40 is connected to the OLT. The processing execution unit 620 in which the OLT function is implemented collectively manages the subscriber devices 40.

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The O/E conversion unit 610-1 converts the uplink optical signal of the subscriber device 40-3 input from the optical SW 210 into an electrical signal and outputs the electrical signal to the processing execution unit 620. The O/E conversion unit 610-2 converts the uplink optical signal of the subscriber device 40-M input from the optical SW 210 into an electrical signal and outputs the electrical signal to the processing execution unit 620. The processing execution unit 620 combines the uplink electrical signals transmitted from the subscriber device 40-3 and the subscriber device 40-M into one signal and outputs the combined signal to the E/O converter 630. The E/O converter 630 converts the uplink electrical signal output from the processing execution unit 620 into an optical signal in accordance with the wavelength instructed from the control unit 230 and outputs the optical signal to the optical SW 210. The optical SW 210 outputs the uplink optical signal input from the electrical processing unit 600 to the transmission line 541 corresponding to the ground C. As described above, the electrical processing unit 600 receives each of the plurality of optical signals dropped by the optical GW 2016 and converts the reception signal into an electrical signal, multiplexes signals having the same target ground by the multiplexing circuit, converts the signal into an optical signal again, and transmits the optical signal to the optical GW 2016. Accordingly, a transmission rate can be maximized. FIGS. 58 and 59 illustrate an example in which one electrical processing unit is provided, but a plurality of electrical processing units may be provided.

The power splitter 507 between the subscriber device 40 and the optical GW 2016 may be a wavelength multiplexer/demultiplexer. For example, when the optical access system 116 is a WDM-PON, a wavelength demultiplexer is used between the subscriber device 40 and the optical GW 2016. (Seventeenth Configuration Example of Optical Access System 100)

The present configuration example is a mode in which optical SWs on different ground are connected in a ring. In the description below, differences from the configuration example will be mainly explained.

FIG. 60 is a diagram illustrating a configuration example of an optical access system 117. The optical access system 117 has a configuration in which three or more different ground optical SWs 212 are connected in a ring form via the optical communication network 30. In the example illustrated in FIG. 60, the optical access system 117 has a configuration in which an optical SW 212a which is the optical SW 212 of the ground A, an optical SW 212b which is the optical SW 212 of the ground B, and an optical SW 212c which is the optical SW 212 of the ground C are connected in the ring form. A path between the optical SW 212a and the optical SW 212b in the optical communication network 30 is referred to as a path P31, a path between the optical SW 212b and the optical SW 212c in the optical communication network 30 is referred to as a path P32, and a path between the optical SW 212c and the optical SW 212a in the optical communication network 30 is referred to as a path P33. One or more subscriber devices 40a are connected to the optical SW 212a, one or more subscriber devices 40b are connected to the optical SW 212b, and one or more subscriber devices 40c are connected to the optical SW 212c.

As the optical SW 212, the above-described optical SW or optical GW is used. For example, the ground B in FIGS. 6 to 10, 38, and 43 to 59 is defined as a counterclockwise ground in the ring illustrated in FIG. 60, and the ground C in FIGS. 6 to 10, 38, and 43 to 59 is defined as a clockwise

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ground in the ring illustrated in FIG. 60. In this case, connection is made from the optical SW 212a of the ground A to the optical SW 212b of the ground B by the path P31, and connection is made from the optical SW 212b of the ground B to the optical SW 212a of the ground A via the path P32, the optical SW 212c of the ground C, and the path P33. The optical SW 212a of the ground A is connected to the optical SW 212c of the ground C by the path P33, and the optical SW 212c of the ground C is connected to the optical SW 212a of the ground A via the path P32, the optical SW 212b of the ground B, and the path P31.

Therefore, the counterclockwise connection from the optical SW 212a of the ground A to the optical SW 212b of the ground B is set as a backup system, connection by a clockwise path connecting from the optical SW 212a of the ground A to the optical SW 212b of the ground B via the optical SW 212c of the ground C is also possible, and the counterclockwise connection is also possible. Similarly, the clockwise connection from the optical SW 212a of the ground A to the optical SW 212c of the ground C is set as a backup system, connection by a counterclockwise path from the optical SW 212a of the ground A to the optical SW 212c of the ground C via the optical SW 212b of the ground B is also possible, and the counterclockwise connection is also possible.

In addition, as a backup system of the connection between the subscriber devices 40a connected to the optical SW 212a of the ground A, a counterclockwise path through the path P31, the optical SW 212b of the ground B, the path P32, the optical SW 212c of the ground C, and the path P33, or a counterclockwise path through the path P33, the optical SW 212c of the ground C, the path P32, the optical SW 212b of the ground B, and the path P31 can be used.

For example, in FIG. 14, the middle distance line P2 may be a counterclockwise path of the ring, and the middle distance line P3 may be a clockwise path of the ring. Any one of grounds #1 to #q in FIGS. 16 and 19 may be a counterclockwise ground of the ring, and the others may be clockwise grounds of the ring. When FIGS. 27 and 28 illustrate the optical SW 1010 in one GW, any one of uplinks #11 to #43 may be set as a counterclockwise path of the ring, and another uplink may be set as a clockwise path of the ring. Here, the path not selected as the path of the ring may be a path of a ring similarly to the path selected as the path of the ring, may be an oblique line other than the ring, may be connected to the subscriber device 40, or may be connected to another optical SW 1010 illustrated in FIGS. 27 and 28.

As described above, when multicasting is performed as the basic configuration, multicasting in the uplink direction or the downlink direction is realized via the folded transmission line, as illustrated in FIG. 4 or 5. For example, in the example illustrated in FIG. 4, in order to realize the multicasting in the uplink direction, an optical signal output from the port 11-2 of the optical SW 10c is input to the other port 11-2 via the folded transmission line, and the optical SW 10c outputs the input optical signal to the port 11-1 to which the 1xN power splitter 71 is connected. Then, the multicasting in an uplink direction is realized by distributing optical signals output from the ports 11-1 by the power splitter 71 and inputting the optical signals to the plurality of other ports 11-1. However, in such a configuration, it is necessary to use a port for folding when multicasting is performed. Therefore, the number of ports in use may increase. This problem occurs not only when multicast routing is performed but also when broadcast routing is performed. Accordingly, in the following embodiment, a configuration

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for reducing the number of ports when multicast or broadcast routing is performed will be described. In the following description of the embodiment, the description of the control unit 20 and the OPS 300 may be omitted in the drawings. However, in the optical access system, wavelength setting and connection setting of the optical SW by the control unit 20 and the OPS 300 are performed.

First Embodiment

In the following embodiments including the present embodiment, the uplink multicasting from the subscriber device 40a connected to the port 11-1 to the subscriber device 40b or 40c connected to the port 11-2 or the uplink will be mainly described. However, if the port 11-1 and the port 11-2 are interchanged in line symmetry in the center optical SW 210, the uplink or the multicasting from the subscriber device 40b or 40c connected to the port 11-2 to the subscriber device 40a connected to the port 11-1 or the like is performed. Even in broadcasting, a flow of processing is similar to that in multicasting.

FIG. 61 is a diagram illustrating a configuration example of an optical GW 200a in an optical access system of a first embodiment. In FIG. 61, a case where uplink optical signals transmitted from the subscriber devices 40a-1 and 40a-2 located at the ground A are multicast will be described. Here, in description, as illustrated in FIG. 61, it is assumed that ports connected by a dotted line are connected inside the optical SW 210a. This connection between the ports is made by the optical SW control unit 320 (not illustrated).

The optical GW 200a folds an optical signal output from the port 11-2 without passing through the optical SW 210a-1 and inputs the optical signal to the power splitter 71 connected to the port 11-1. Accordingly, multicasting is possible. FIG. 61 illustrates two configurations, that is, a configuration in which an optical signal passes through another optical switch and a configuration in which an optical signal does not pass through another optical switch when the optical signal is folded without passing through the optical SW 210a-1. Each of the configurations will be described below.

(Configuration in which an Optical Signal does not Pass Through Another Optical Switch)

In the configuration in which an optical signal does not pass through another optical switch, the optical GW 200a includes an optical SW 210a-1, a transmission line 2101, and a power splitter 71-1. The transmission line 2101 connects a port (for example, the port 11-2-1) on the output side of the optical SW 210a-1 and the power splitter 71-1 via the outside of the optical SW 210a-1. The passing of the outside of the optical SW 210a-1 means that the optical signal does not pass through the inside (between ports) of the optical SW 210a-1. That is, the transmission line 2101 is a transmission line that is provided outside of the optical SW 210a-1 and directly connects the port 11-2-1 and the power splitter 71-1. The transmission line 2101 is, for example, an optical fiber. The power splitter 71-1 divides an input optical signal into a plurality of signals and inputs the plurality of divided optical signals to different ports 11-1. A new power splitter may be provided between the power splitter 71-1 and the port 11-1. In this case, the power splitter 71-1 divides the input optical signal into a plurality of signals, and inputs some or all of the plurality of divided optical signals to the port 11-1 via the power splitter.

A flow of a process in the case of a configuration in which the optical GW 200a does not pass through another optical switch will be described. In FIG. 61, the optical GW 200a

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outputs, from the port 11-2-1, an optical signal transmitted by the subscriber device 40a-1 and input to the port 11-1-1. The optical signal output from the port 11-2-1 is input to the power splitter 71-1 connected to the plurality of ports 11-1-*n* to 11-1-(*n*+2) via the transmission line 2101. For example, in FIG. 61, the power splitter 71-1 is connected to ports 11-1-3 to 11-1-5.

The power splitter 71 divides the input optical signals and inputs the optical signals to the plurality of ports 11-1-3 to 11-1-5. The optical SW 210a-1 outputs the optical signals input from the plurality of ports 11-1-3 to 11-1-5 to the different ports 11-2-3 to 11-2-5, respectively. Bidirectional communication is also possible. The optical signal in the downlink direction is routed in a direction reverse to the uplink direction.

In this case, the uplink direction is one-to-many communication, and the downlink direction is many-to-one communication. In this case, the power splitter 71 is used as a branch combiner/splitter. The power splitter 71 used here has *H*×*I* input/output ports. *H* and *I* are integers equal to or greater than 1, and a relation of *H*≤*I* is satisfied. In the case of *H*=1 and *I*=2, for example, the power splitter 71 branches the optical signal in the uplink direction and multiplexes the optical signal in the downlink direction. In the example of FIG. 61, the power splitter 71 branches the input optical signal, inputs the optical signals to the plurality of ports 11-1-3 to 11-1-5, multiplexes the optical signals output from the plurality of ports 11-1-3 to 11-1-5, and outputs the multiplexed optical signal to the transmission line 2101.

In order to realize the above processing, when a multicast transmission request is given from the subscriber device 40a-1, the optical SW control unit 320 connects the port 11-1-1 to which the subscriber device 40a-1 is connected and the port 11-2-1 to which the power splitter 71-1 for performing multicasting is connected. Accordingly, the optical signal transmitted from the subscriber device 40a-1 is output from the port 11-2-1. Further, the optical SW control unit 320 connects the ports 11-1-3 to 11-1-5 to which the power splitter 71-1 is connected and the port 11-2 (In FIG. 61, 11-2-3 to 11-2-5) corresponding to the transmission destination on the path to the communication destination. Accordingly, each optical signal branched by the power splitter 71-1 is output from the port 11-2 to which the destination subscriber device 40b is connected. In such a configuration, the optical signal is folded to enable multicasting.

In the case of downlink multicasting, the following applies.

The optical GW 200a includes an optical SW 210a-1, a transmission line, and a power splitter. The transmission line connects a port (for example, the port 11-1-1) on the output side of the optical SW 210a-1 and the power splitter via the outside of the optical SW 210a-1. The transmission line is a transmission line that directly connects the port 11-1-1 and the power splitter. The transmission line is, for example, an optical fiber. The power splitter divides the input optical signal into a plurality of signals and inputs the plurality of divided optical signals to the different ports 11-2. The optical SW 210a-1 outputs the optical signal input to the port 11-2-1 from the port 11-1-1. The optical signal output from the port 11-1-1 is input to a power splitter connected to the plurality of ports 11-2-*n* to 11-2-(*n*+2) via a transmission line. For example, the power splitter is connected to the ports 11-2-3 to 11-2-5.

The power splitter distributes the input optical signal and inputs the optical signals to the plurality of ports 11-2-3 to 11-2-5. The optical SW 210a-1 outputs the optical signals

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input from the plurality of ports 11-2-3 to 11-2-5 to the different ports 11-1-3 to 11-1-5, respectively. (Configuration in which Optical Signal Passes Through Another Optical Switch)

In the configuration in which an optical signal passes through another optical switch, the optical GW 200a includes an optical SW 210a-1, an optical SW 210a-2, a transmission line 2102, and a power splitter 71-2. The optical SW 210a-2 is connected to a port (for example, the port 11-2-*q*) on the output side of the optical SW 210a-1. The transmission line 2102 connects the optical SW 210a-2 and the power splitter 71-2 via the outside of the optical SW 210a-1. The optical SW 210a-2 outputs the optical signal output from the port 11-2 on the output side of the optical SW 210a-1 to the power splitter 71-2 via the transmission line 2102. The power splitter 71-2 divides the input optical signal into a plurality of signals and inputs the plurality of divided optical signals to the different ports 11-1. A new power splitter may be provided between the power splitter 71-2 and the port 11-1. In this case, the power splitter 71-1 divides the input optical signal into a plurality of signals, and inputs some or all of the plurality of divided optical signals to the port 11-1 via the power splitter. The optical GW 200a may have only one or both of the configuration in which an optical signal does not pass through another optical switch and the configuration in which an optical signal passes through another optical switch.

A flow of a process in the case of the configuration in which the optical GW 200a passes through another optical switch will be described. In FIG. 61, the optical GW 200a outputs the optical signal transmitted by the subscriber device 40a-2 and input to the port 11-1-*p* from the port 11-2-*q*. The optical signal output from the port 11-2-1 is input to the optical SW 210a-2. The optical SW 210a-2 outputs the optical signal output from the port 11-2-*q* to the power splitter 71-2 connected to the plurality of ports 11-1-(*P*-2) to 11-1-*P* via the transmission line 2102.

The power splitter 72 distributes the input optical signal and inputs the optical signals to the plurality of ports 11-1-(*P*-2) to 11-1-*P*. The optical SW 210a-1 outputs the optical signals input from the plurality of ports 11-1-(*P*-2) to 11-1-*P* to different ports 11-1-(*Q*-2) to 11-1-*Q*, respectively. Bidirectional communication is also possible. The optical signal in the downlink direction is routed in a direction reverse to the uplink direction. In this case, the uplink direction is one-to-many communication, and the downlink direction is many-to-one communication. In this case, the power splitter 71 is used as a branch combiner/splitter. The power splitter 71 used here has *H*×*I* input/output ports. In the case of *H*=1 and *I*=2, for example, the power splitter 71 branches the optical signal in the uplink direction and multiplexes the optical signal in the downlink direction. In the example of FIG. 61, the power splitter 71 branches the input optical signal, inputs the optical signals to the plurality of ports 11-1-(*Q*-2) to 11-1-*Q*, multiplexes the optical signals output from the plurality of ports 11-1-(*Q*-2) to 11-1-*Q*, and outputs the multiplexed optical signal to the transmission line 2102.

In order to realize the above process, when a multicast transmission request is given from the subscriber device 40a-2, the optical SW control unit 320 connects the port 11-1-*p* to which the subscriber device 40a-2 is connected and the port 11-2-*q* to which the power splitter 71-2 for performing multicasting is connected. Accordingly, the optical signal transmitted from the subscriber device 40a-2 is output from the port 11-2-*q*. Further, the optical SW control unit 320 connects the ports 11-1-(*P*-2) to 11-1-*P* to which

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the power splitter 71-2 is connected and the port 11-2 (in FIG. 61, 11-2-(Q-2) to 11-2-Q) corresponding to the transmission destination on the path to the communication destination. Accordingly, each optical signal branched by the power splitter 71-2 is output from the port 11-2 to which the destination subscriber device 40c is connected. In such a configuration, the optical signal is folded back to enable multicasting to the destination.

In the case of downlink multicasting, the following applies.

The optical GW 200a includes an optical SW 210a-1, an optical SW 210a-2, a transmission line, and a power splitter. The optical SW 210a-2 is connected to a port (for example, the port 11-1 q) on the output side of the optical SW 210a-1. The transmission line connects the optical SW 210a-2 and the power splitter via the outside of the optical SW 210a-1. The optical SW 210a-2 outputs the optical signal output from the port on the output side of the optical SW 210a-1 to the power splitter via the transmission line. The power splitter divides the input optical signal into a plurality of signals and inputs the plurality of divided optical signals to the different ports 11-2. The optical GW 200a may have only one or both of the configuration in which an optical signal does not pass through another optical switch and the configuration in which an optical signal passes through another optical switch.

According to the optical GW 200a that has the foregoing configuration, the port of the optical SW 210a-1 is not used as a folded path when multicast or broadcast routing is performed. Accordingly, even when routing is performed, the number of paths inside the optical SW 210a-1 is reduced by one path, that is, one port for each of the ports 11-1-p and 11-2-q. Therefore, it is possible to transmit an optical signal according to a transmission destination on a path to a communication destination while reducing the number of use ports when the multicast or broadcast routing is performed in the optical SW 210a-1.

In the first embodiment, the power splitter is mainly used, but a multiplexer/demultiplexer may be used in the case of multicasting demultiplexed for each wavelength (effectively, unicasting demultiplexed in accordance with a wavelength).

Second Embodiment

In the first embodiment, the configuration in which the optical signal output from the port 11-2 once via the optical SW is folded via the outside of the optical SW has been described. In this case, in order to fold the optical signal, it is necessary to transmit the optical signal between the ports of the optical SW once. Therefore, the corresponding number of ports cannot be used. Therefore, in the second embodiment, a configuration for reducing the number of ports when multicasting is performed as compared with the first embodiment will be described. Even in broadcasting, a flow of processing is similar to that in multicasting.

FIG. 62 is a diagram illustrating a configuration example of an optical GW 200b in the optical access system according to a second embodiment. In FIG. 62, a case of multicasting uplink optical signals transmitted from the subscriber devices 40a-1 and 40a-2 located at the ground A will be described. Here, in the description, as illustrated in FIG. 62, it is assumed that ports connected by a dotted line are connected inside the optical SW 210b. This connection between the ports is made by the optical SW control unit 320 (not illustrated).

The optical GW 200b distributes the optical signal output from the subscriber device 40a-1 into a plurality of signals

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and inputs the plurality of distributed optical signals to different ports 11-1. Accordingly, it is possible to reduce the number of sets of ports for folding back optical signals and perform multicasting. The optical GW 200b includes power splitters 71-1 and 71-2. The optical GW 200b may include either the power splitter 71-1 or the power splitter 71-2, or may include both the power splitters.

In FIG. 62, the power splitter 71-1 divides the optical signal output from the subscriber device 40a-1 into a plurality of signals, and inputs the plurality of divided optical signals to different ports 11-1 (in FIG. 62, ports 11-1-3 to 11-1-5). The power splitter 71-1 is connected to the subscriber device 40a-1 via the transmission line 2103. The power splitter 71-2 divides the optical signal output from the subscriber device 40a-2 into a plurality of signals, and inputs the plurality of divided optical signals to different ports 11-1 (in FIG. 62, 11-1-(P-2) to 11-1-P). The power splitter 71-2 is connected to the subscriber device 40a-2 via the transmission line 2104. New power splitters may be provided between the power splitter 71-1 and the port 11-1 and between the power splitter 71-2 and the port 11-1. In this case, the power splitters 71-1 and 71-2 divide the input optical signal into a plurality of signals and input some or all of the plurality of divided optical signals to the port 11-1 via the power splitter.

As described above, in the second embodiment, the power splitter 71 and the power splitter 72 are provided at the preceding stage at which the optical signal is input to the port 11-1, and the optical signal output from the subscriber device 40a is multicast. The optical SW 210b outputs the optical signals input from the plurality of ports 11-1 to the different ports 11-2. Bidirectional communication is also possible. The optical signal in the downlink direction is routed in a direction reverse to the uplink direction. In this case, the power splitter 71 is used as a branch combiner/splitter. The power splitter 71 used here has H×I input/output ports. In the case of H=1 and I=2, for example, the power splitter 71 branches the optical signal in the uplink direction and multiplexes the optical signal in the downlink direction. In the example of FIG. 62, the power splitter 71 branches the input optical signal, inputs the optical signal to the plurality of ports 11-1-3 to 11-1-5, multiplexes the optical signals output from each of the plurality of ports 11-1-3 to 11-1-5, and outputs the multiplexed optical signal to the subscriber device 40a-1.

In order to realize the foregoing process, when a multicast transmission request is given from the subscriber device 40a, the optical SW control unit 320 connects the port 11-1 to which the power splitter 71 to which the subscriber device 40a that is the transmission source of the transmission request is connected is connected and the port 11-2 corresponding to the transmission destination on the path to the communication destination. Accordingly, the optical signal transmitted from the subscriber device 40a is branched by the power splitter 71, and each of the plurality of branched optical signals is output from the transmission destination port 11-2 on the path to the communication destination. In such a configuration, it is possible to perform multicasting to a destination while reducing the number of ports.

According to the optical GW 200b that has the foregoing configuration, the basic configuration or the folded path as described in the first embodiment is not used to perform multicasting. Therefore, it is possible to reduce the number of ports used for the folded path. For example, in the second embodiment, as compared with the first embodiment, the number of paths inside the optical SW 210b is further reduced by one path, that is, by one port of each of the ports

11-1-*p* and 11-2-*q*. Therefore, it is possible to transmit an optical signal in accordance with a transmission destination on the path to the communication destination while reducing the number of use ports in the optical SW **210b**.

In the second embodiment, the power splitter is mainly used, but a multiplexer/demultiplexer may be used in the case of multicasting demultiplexed for each wavelength (effectively, unicasting demultiplexed in accordance with a wavelength).

Third Embodiment

In the second embodiment, although the number of ports can be reduced, it is necessary to select whether to make connection to the power splitter or connection to an unconnected port of the power splitter when the subscriber device **40a** is connected. As described above, in the second embodiment, when a communication source starts or stops multicasting, the port to be connected is changed. In a third embodiment, a configuration for suppressing a change in a port will be described. In the configurations, the above-described optical SW is replaced with a set of cascade-connected optical SWs.

FIG. **63** is a diagram illustrating a configuration example of an optical GW **200c** in the optical access system of the third embodiment. In FIG. **63**, to facilitate explanation, the optical SW that has a $(11-1-P) \times (11-2-Q)$ configuration will be described as a symmetric SW of $S \times S$ or the like (where S is an integer equal to or greater than 1), but the optical SW can be easily expanded even in the case of an asymmetric optical SW such as a SW which has the $(11-1-P) \times (11-2-Q)$ configuration.

The optical GW **200c** illustrated in FIG. **63** includes a first optical switch **210c-1**, a second optical switch **210c-2**, and power splitters **450-1** to **450-M**. The first optical switch **210c-1** and the second optical switch **210c-2** are cascade-connected. In the cascade configuration of the optical GW **200c** in the third embodiment, an $S \times S$ first optical switch **210c-1** and a $((S-T)+T \times k) \times ((S-T)+T \times k)$ second optical switch **210c-2** are used. Here, S is the number of subscriber devices **40a**, T is the number of ports to be branched, and k is the number of branches for each port. The number of ports to be branched is the number of ports to which the power splitter is connected.

When the number of branches k is different for each port, $T \times k$ is replaced with a sum of the number of branches of the ports. An output of the branching T ports in the $S \times S$ optical SW **210c-1** is directly connected to ports of the remaining $(S-T)$ ports $((S-T)+T \times k) \times ((S-T)+T \times k)$ optical SW **210c-2** via the power splitters **450-1** to **450-M**.

Advantageous effects obtained in the third embodiment will be described.

An SW sizes of the two optical SWs illustrated in FIG. **63** are as follows: a first stage: S^2 and a second stage: $((S-T)+T \times k)^2 = (S+T \times (k-1))^2$. Therefore, a maximum size $(S+T \times k)^2$ decreases to $(S+T \times (k-1))^2$, and a total number of SW fabrics does not decrease at $(S+T \times k)^2 - \{(S)^2 + ((S+T \times (k-1))^2)\} = S(2S-2k+1) - S^2$.

Although all the ports are connected, there may be a free port or a port connected to an electrical processing unit or the like. Here, a configuration in a case where the electrical processing unit and the like are connected in the third embodiment will be described. In the optical GW **200c**, the electrical processing units are connected by optical SW (for example, the optical switch **210c-1**) near the subscriber device **40a** in a case where the optical GW passes through the optical GW before multicasting or without multicasting,

and the electrical processing units are connected by other optical SW (for example, the optical switch **210c-2**) in a case where the optical GW passes through the optical GW after multicasting. It is preferable to pass through before multicasting in a case where forward error correction (FEC), encryption, or the like is collectively processed, and it is preferable to pass through after multicasting in a case where FEC, encryption, or the like is changed for each destination.

In the third embodiment, bidirectional communication is also possible. The optical signal in the downlink direction is routed in a direction reverse to the uplink direction. In this case, the power splitter **450** is used as a branch combiner/splitter. The power splitter **450** used here has $H \times I$ input/output ports. In the case of $H=1$ and $I=2$, the power splitter **450**, for example, branches the optical signal in the uplink direction and multiplexes the optical signal in the downlink direction. In the example of FIG. **63**, the power splitter **450** branches the input optical signal, inputs the optical signal to a plurality of ports of the optical SW **210c-2**, multiplexes the optical signals output from each of the plurality of ports of the optical SW **210c-2**, and outputs the multiplexed optical signal to the port 11-2 (for example, port 11-2-($Q-1$)) of the optical SW **210c-1**.

Fourth Embodiment

FIG. **64** is a diagram illustrating a configuration example of an optical GW **200d** in an optical access system of the third embodiment. In FIG. **64**, for simplification of description, the optical SW which has the $(11-1-P) \times (11-2-Q)$ configuration has been described as a symmetric SW such as $S \times S$, but can also be easily expanded to the case of an asymmetric optical SW such as a SW which has the $(11-1-P) \times (11-2-Q)$ configuration.

The optical GW **200d** illustrated in FIG. **64** includes the first optical switch **210c-1**, the second optical switch **210d-2**, and power splitters **450-1** to **450-M**. The optical GW **200d** is different from the optical GW **200c** in that a second optical switch **210d-2** is included instead of the second optical switch **210c-2**. In the cascade configuration of the optical GW **200d** in the fourth embodiment, an $S \times S$ optical SW and a $(T \times k) \times (T \times k)$ optical SW are used.

When the number of branches k is different for each port, $T \times k$ is replaced with a sum of the number of branches of the ports. An output corresponding to branched T ports in the $S \times S$ optical SW **210c-1** is connected to the second optical switch **210d-2** via the power splitters **450-1** to **450-M**. An output of the remaining $(S-T)$ ports does not pass through the second optical switch **210d-2**.

Advantageous effects obtained in the fourth embodiment will be described.

The SW sizes of the two optical SWs illustrated in FIG. **64** are S^2 at the first stage and $(T \times k)^2$ at the second stage. Accordingly, a maximum size $(S+T \times k)^2$ decreases to $\text{MAX}(S^2, (T \times k)^2)$, and the total number of SW fabrics decreases by $(S+T \times k)^2 - \{(S)^2 + (T \times k)^2\} = 2STk$.

Although all the ports are connected, there may be a free port or a port connected to an electrical processing unit or the like. Here, a configuration in a case where the electrical processing unit and the like are connected in the fourth embodiment will be described. In the optical GW **200d**, the electrical processing unit is connected by an optical SW (for example, the optical switch **210c-1**) close to the subscriber device **40a**, and the electrical processing unit is connected to another optical SW (for example, the optical switch **210d-2**) in a case where each path is passed after multicasting. It is preferable to pass through before multicasting in a case

where forward error correction (FEC), encryption, or the like is collectively processed, and it is preferable to pass through after multicasting in a case where FEC, encryption, or the like is changed for each destination.

In the fourth embodiment, bidirectional communication is also possible. The optical signal in the downlink direction is routed in a direction reverse to the uplink direction. In this case, the power splitter 450 is used as a branch combiner/splitter. The power splitter 450 used here has H×I input/output ports. In the case of H=1 and I=2, the power splitter 450, for example, branches the optical signal in the uplink direction and multiplexes the optical signal in the downlink direction. In the example of FIG. 64, the power splitter 450 branches the input optical signal, inputs the optical signal to the plurality of ports of the optical SW 210d-2, multiplexes the optical signals output from the plurality of ports of the optical SW 210d-2, and outputs the multiplexed optical signal to the port 11-2 (for example, port 11-2-(Q-1)) of the optical SW 210c-1.

Fifth Embodiment

In the basic configuration, as illustrated in FIGS. 3 to 5, the configuration in which the power splitter or the WDM device is not provided in the folded transmission line, and the ports are directly connected to fold the optical signal has been described. In a fifth embodiment, a variation of the folded configuration of an optical signal will be described. For example, the folded transmission line may be configured by a network that is folded via a network such as an optical SW of an intermediate layer such as a Folded Clocks network. The folded transmission line may be configured such that some of the ports of some of the input/output SWs of the Folded Clocks network to the intermediate layer SW and some of the ports of the intermediate layer SW are on a 11-2-Q side, and some of the ports of some of the input/output SWs to the intermediate layer SW and some of the ports of the intermediate layer SW are on a 11-2-Q side. Accordingly, the optical SW enables folded communication.

FIG. 65 is a diagram illustrating a configuration example of an optical GW 200e in an optical access system according to the fifth embodiment. In FIG. 65, three configurations (first to third configurations) will be described as variations of the configuration of folding of the optical signal. (First Configuration of Folding of Optical Signal)

In the first configuration of the folding back of the optical signals, the optical GW 200e includes an optical SW 210e-1 and a WDM device 80-1. The subscriber devices 40a-1 to 40a-2 are connected to the ports 11-1-1 to 11-1-2 of the optical GW 200e by transmission lines 50-1-1 to 50-1-2. The WDM device 80-1 is connected to the ports 11-2-1 to 11-2-2 of the optical GW 200e. The subscriber devices 40a-1 to 40a-2 transmit and receive optical signals with different wavelengths. The WDM device 80-1 multiplexes the optical signals output from the ports 11-2-1 and 11-2-2 and outputs the multiplexed optical signal to the folded transmission line 51a-1. The folded transmission line 51a-1 is connected to the port 11-2-3. Such a configuration enables folding of the optical signals.

(Second Configuration of Folding of Optical Signal)

In the second configuration of the folding of the optical signal, the optical GW 200e includes an optical SW 210e-1, a plurality of WDM devices 80-2 to 80-3, and a plurality of monitoring units 60-1 and 60-2. The subscriber devices 40a-p to 40a-(p+1) are connected to the ports 11-1-p to 11-1-(p+1) of the optical GW 200e by transmission lines

50-1-p to 50-1-(p+1). The WDM device 80-2 is connected to the ports 11-2-q to 11-2-(q+1) of the optical GW 200e.

The subscriber devices 40a-p to 40a-(p+1) transmit and receive optical signals with different wavelengths. The WDM device 80-2 multiplexes the optical signals output from the ports 11-2-1 and 11-2-2 and outputs the multiplexed optical signal to the folded transmission line 51a-2. The folded transmission line 51a-2 is connected to the WDM device 80-3. The WDM device 80-3 demultiplexes the input optical signal and outputs the demultiplexed optical signals to the monitoring units 60-1 and 60-2. The monitoring units 60-1 and 60-2 monitor the optical signals transmitted through the transmission lines. Such a configuration enables folding of the optical signals.

(Third Configuration of Folding of Optical Signal)

In the third configuration of the folding of the optical signal, the optical GW 200e includes an optical SW 210e-1 and an optical SW 210e-2. A subscriber device 40a-P is connected to a port 11-1-(P-1) of the optical GW 200e through a transmission line 50-1-P. The optical SW 210e-2 is connected to the port 11-2-(Q-1) of the optical GW 200e. The subscriber devices 40a-P transmit and receive optical signals. The optical SW 210e-1 outputs the optical signal input from the port 11-1-(P-1) from the port 11-2-(Q-1). The optical signal output from the port 11-2-(Q-1) is output to the optical SW 210e-2. The optical SW 210e-2 outputs the input optical signal to the port 11-2-Q. Such a configuration enables folding of the optical signals.

According to the optical GW 200e configured as described above, a configuration of the folded communication other than the configuration illustrated in the basic configuration can be applied. Accordingly, the degree of freedom of combination is improved, and convenience can be improved.

Next, a configuration for reducing the used ports of the optical SW accommodating the subscriber device 40 will be presented. Here, the optical SW 1010 illustrated in FIGS. 27 and 28 will be described as an example. However, the optical SW can be similarly applied to other optical SWs (for example, the optical SW 10, the optical SW 210, and the like) described above.

In the configuration illustrated in FIGS. 27 and 28, for example, when viewed from the subscriber device 40 (in FIGS. 27 and 28, the ONU) on the port 11-1 side as a transmission line, it clearly corresponds to an increase in the number of connections of output paths connected to the port 11-2. Therefore, when the optical SW 1010 is connected to the subscriber device 40 on the port 11-1 side of another optical SW 1010 (for example, optical SW 1010-2 to 1010-4), as described above, and the optical SW 1010 is connected in the full mesh type illustrated in FIG. 27, the port 11-1 (for example, the port 11-1-1 (connected to ONU #11) and the port 11-2 (for example, the port 11-2-q1 (connected to *a)) of the optical SW 1010-1, the port 11-1 and the port 11-2 (for example, the port 11-1-p1 (connected to *a) of the optical SW 1010-2 to which the facing subscriber device 40 is connected on the port 11-1 side, a port 11-2-x (connected to the folded transmission line) and the port 11-2 and the port 11-2 (for example, a port 11-2-y and a port 11-2-z) connected to the folded transmission line, and the port 11-1 to which the opposing ONU is connected (for example, the port 11-1-1 (connected to ONU #21)) and 6 ports are occupied. The same applies to a case where a folded transmission line is installed in the optical SW 1001.

In the case of the ring-type connection illustrated in FIG. 28, the port 11-1 and the port 11-2 (for example, the port 11-1-1 (connected to ONU #11) and a port 11-2-q1 (con-

connected to *a)) of the optical SW 1010-1, the port 11-1 and the port 11-2 (for example, the port 11-1-p1 (connected to *a)) of the optical SW 1010-2 passing therethrough, the port 11-2-q1 (connected to *d), the port 11-1 and the port 11-2 (for example, the port 11-1-p1 (connected to *d)) of the optical SW 1010-3 to which the facing subscriber device 40 is connected on the port 11-1 side, the port 11-2-x (connection with the folded transmission line), the port 11-2 and the port 11-2 (for example, a port 11-2-y and a port 11-2-z) connected to the folded transmission line, and the port 11-1 (for example, the port 11-1-1 (connection with ONU #21) to which the opposing ONU is connected, and eight ports are occupied. The same applies to a case where the folded transmission line is installed in the optical SW 1010-1 or the optical SW 1010-2.

In view of the foregoing description, a configuration for reducing the used ports of the optical SW accommodating the subscriber device 40 will be described with reference to FIG. 66. In FIG. 66, four configurations are illustrated as configurations for reducing used ports of the optical SW. (First Configuration)

As illustrated in FIG. 66(A), in the first configuration, the optical SW 1010-1 to the optical SW 1010-3 and the optical SW 1200 are illustrated. In the first configuration, from the viewpoint of an improvement in the degree of freedom of connection for changing a ratio of the ports of the optical SW passed along a transmission line across the optical SW, the transmission line across the optical SW is assumed to be the optical SW 1200. The configuration of the optical SW 1200 is basically similar to that of the other optical SWs 1010. When the transmission line across the optical SW is the optical SW, the used ports of the optical SW accommodating the subscriber device 40 can be reduced.

Means for reducing used ports of the optical SW accommodating the subscriber device 40 will be described. In FIG. 27, the transmission lines connected across each of the optical SWs 1010-1 to 1010-4 are as one set, but the transmission lines across from the single optical SW 1010 are occupied and connected to an ONU of the same optical SW. For example, in the example of FIG. 27, when ONU #11, ONU #12, and ONU #13 are connected to ONU #21, ONU #22, or ONU #23, 22 ports, for example, 6 ports, 8 ports, or 8 ports, are occupied via other optical SW. On the other hand, as illustrated in FIG. 66(A), when the transmission line across the optical SW is the optical SW 1200, only 18 ports are used as the ports of the optical SW 1010-1 and the optical SW 1010-2 that accommodate the subscriber device 40. The optical SWs 1010-1 to 1010-4 are kinds of plurality of optical switches. In this description, the optical SW 1010-1 will be described as a kind of first optical switch among the plurality of optical switches, and the optical SW 1010-2 will be described as a kind of second optical switch among the plurality of optical switches.

More specifically, the optical SW 1200 is connected to the port 11-1 and the port 11-2 of the optical SW 1010-1 and is further connected to the port 11-1 and the port 11-2 of the optical SW 1010-2 and the port 11-1 and the port 11-2 of the optical SW 1010-3. In this way, the optical SW 1200 is connected to the first port (For example, port 11-1) of the plurality of optical switches and the second port (for example, the port 11-2) on the side different from the first port (for example, the side different from the side on which the port 11-1 is provided among the plurality of optical switches). The first port and the second port illustrated here are exemplary, and the port 11-1 may be the second port and the port 11-2 may be the first port. The same applies to the following description. The optical SW 1200 connects the

port 11-2 on the output side of the plurality of optical SWs 1010 and the port 11-1 on the input side of the plurality of optical SWs 1010. For example, the optical SW 1200 connects the port 11-2 on the output side of the optical SW 1010-1 and the port 11-1 on the input side of the optical SW 1010-2. The optical SW control unit 320 controls the optical SW 1200 such that the path inside the optical SW 1200 becomes a path oriented to a transmission destination on the path to the communication destination. In such a configuration, as illustrated in FIG. 27, it is possible to reduce the number of ports passing through other optical SWs 1010-2 and 1010-3.

(Second Configuration)

As illustrated in FIG. 66(B), in the second configuration, the optical SW 1010-1 to the optical SW 1010-3 are illustrated. In the second configuration, the folded transmission line is not connected to the same optical SW, but is connected between the optical SWs connected by the facing subscribers. For example, the connection is made by the port 11-1-1 that connects the optical SW 1010-1 and ONU #11, a port 11-2-x of the optical SW 1010-1, a folded transmission line, a port 11-2-y of the optical SW 1010-2, and a port 11-1-1 that connects the optical SW 1010-2 and ONU #21. That is, in the second embodiment, the ports 11-2 on the same side of the different optical SWs 1010 are connected by the folded transmission line. In the example illustrated in FIG. 66(B), the port 11-2 of the optical SW 1010-1 and the port 11-2 of the optical SW 1010-2 are connected by the folded transmission line, the port 11-2 of the optical SW 1010-2 and the port 11-2 of the optical SW 1010-3 are connected by the folded transmission line, and the port 11-2 of the optical SW 1010-1 and the port 11-2 of the optical SW 1010-3 are connected by the folded transmission line. In this case, since four ports are used, it can be understood that two ports can be reduced in use as compared with use of six ports.

(Third Configuration)

As illustrated in FIG. 66(C), in the third configuration, optical SW 1010-1 to optical SW 1010-3 and optical SW 1200 are illustrated. In the third configuration, the optical SW 1200 is used as the folded transmission line. When the optical SW 1200 is an SW that has a configuration of $(11-1-P) \times (11-2-Q)$, a single optical SW (for example, the optical SW 1010-1) accommodating the subscriber device 40 is connected to each of the ports 11-1 and 11-2 of the optical SW 1200, which is a transmission line across the optical SW, and is connected from the port 11-1 to the port 11-2 and from the port 11-2 to the port 11-1. When different optical SWs accommodating the subscriber device 40 are connected to each other, a transmission line across the optical SW is formed. When connected is made through the same optical SW, a folded transmission line in the same SW is formed. That is, in the third configuration, some of the ports 11-2 on the output side of the optical SW 1010-1 to the optical SW 1010-3 are connected to the ports 11-1 of the optical SW 1200, and the others of the port 11-2 on the output side of the optical SW 1010-1 to the optical SW 1010-3 are connected to the ports 11-2 of the optical SW 1200. Accordingly, the optical SW 1200 connects the ports 11-2 on the output side of the optical SW 1010-1 to the optical SW 1010-3. The optical SW control unit 320 controls the optical SW 1200 such that the path inside the optical SW 1200 connects the ports 11-2 on the output side of the optical SW 1010-1 to the optical SW 1010-3.

The second or third configuration may be combined with the transmission line across the mesh type optical SW of FIG. 27 or the ring type optical SW of FIG. 28 or the first configuration.

(Fourth Configuration)

As illustrated in FIG. 66(D), in the fourth configuration, the optical SW 1010-1 to the optical SW 1010-3 and the optical SW 1200 are illustrated. The fourth configuration is an extension of the third configuration. In the third configuration, at least some of the uplinks are installed not on the port 11-2 side but on the port 11-1 side, and are connected to the uplinks via a folded transmission line at the time of connection to the uplinks. That is, in the fourth configuration, the ports 11-1 of the optical switches 1010-1 to 1010-3 that are not connected to the optical SW 1200 are connected to the uplink. This configuration is appropriate when the folding is mainly performed. For example, when the subscriber devices 40 are connected to each other in a full mesh or a form close to the full mesh, the number of the ports 11-2 is less than the number of the ports 11-1. In such a case, when the optical SW is a P×P symmetric SW, the ports 11-1 side to which subscribers are connected are left, the optical SW is diverted to the uplink side and the ports are effectively used.

Next, as other aspects (modifications) of the monitoring unit 60 illustrated in FIGS. 33 to 36, configurations in which the monitoring unit includes a transmitter will be specifically described with reference to FIGS. 67 to 69. FIGS. 67 to 69 are configuration diagrams of other aspects (modifications) of the monitoring unit 60. Although the monitoring unit 60 will be described as an example in FIGS. 67 to 69, the monitoring unit 260 may be used instead of the monitoring unit 60, and the configuration of the monitoring unit 260 may be the configurations illustrated in FIGS. 67 to 69. FIGS. 67 and 68 illustrate examples in which a transmitter and a receiver are connected to the transmission line on the same side in opposite directions as a configuration in which a signal is transmitted to a target to be monitored. FIG. 69 illustrates a configuration in which the transmitter and the receiver are connected in the same direction in the case of merging into the transmission line on the opposite side of the input of the main signal in the same direction as the main signal.

A monitoring unit 75a illustrated in FIG. 67(A) includes a power splitter 61, a receiver 62, and a transmitter 76. The monitoring unit 75a illustrated in FIG. 67(A) is different from the monitoring unit 60a illustrated in FIG. 33(A) in that one transmitter 76 and one receiver 62 are included. The transmitter 76 transmits an optical signal input via the power splitter 61 to a monitor target (for example, the control unit 20 or the OPS 300). As illustrated in FIG. 67(A), the transmitter 76 and the receiver 62 are connected to the same transmission line in opposite directions.

A monitoring unit 75b illustrated in FIG. 67(B) includes a power splitter 61, a plurality of receivers 62-1 to 62-3, a plurality of WDM devices 63b-1 to 63b-2, and a plurality of transmitters 76-1 to 76-3. The monitoring unit 75b illustrated in FIG. 67(B) is different from the monitoring unit 60b illustrated in FIG. 33(B) in that three receivers 62-1 to 62-3 and three transmitters 76-1 to 76-3 are included instead of the six receivers 62-1 to 62-6. The transmitters 76-1 to 76-3 are connected to the WDM device 63b-1. The receivers 62-1 to 62-3 are connected to the WDM device 63b-2. The WDM device 63b-1 demultiplexes the optical signal branched by the power splitter 61 and outputs the demultiplexed optical signals to the transmitters 76-1 to 76-3. As illustrated in FIG.

67(B), the transmitters 76-1 to 76-3 and the receivers 62-1 to 62-3 are connected to the same transmission line in opposite directions.

A monitoring unit 75c illustrated in FIG. 67(C) includes a power splitter 61, a receiver 62, a WDM device 63c, and a transmitter 76. The monitoring unit 75c illustrated in FIG. 67(C) is different from the monitoring unit 60c illustrated in FIG. 33(C) in that a receiver 62 and a transmitter 76 are included instead of the two receivers 62-1 to 62-2.

A monitoring unit 75d illustrated in FIG. 68(A) includes a plurality of power splitters 61-1 to 61-3, a plurality of receivers 62-1 to 62-3, a WDM device 63d, and a plurality of transmitters 76-1 to 76-3. The monitoring unit 75d illustrated in FIG. 68(A) is different from the monitoring unit 60d illustrated in FIG. 34(A) in that three receivers 62-1 to 62-3 and three transmitters 76-1 to 76-3 are included instead of the six receivers 62-1 to 62-6. The power splitter 61-1 is provided in the transmission line 606-1, and the receiver 62-1 and the transmitter 76-1 are connected to the power splitter 61-1. The power splitter 61-2 is provided in the transmission line 606-2, and the receiver 62-2 and the transmitter 76-2 are connected to the power splitter 61-2. The power splitter 61-3 is provided in the transmission line 606-3, and the receiver 62-3 and the transmitter 76-3 are connected to the power splitter 61-3.

A monitoring unit 75e illustrated in FIG. 68(B) includes a plurality of power splitters 61-1 to 61-2, a receiver 62, a plurality of WDM devices 63e-1 to 63e-2, and a transmitter 76. The monitoring unit 75e illustrated in FIG. 68(B) is different from the monitoring unit 60e illustrated in FIG. 34(B) in that the receiver 62 and the transmitter 76 are included instead of the two receivers 62-1 to 62-2. The power splitter 61-1 is provided in the transmission line 602-1, and the transmitter 76 is connected to the power splitter 61-1. The power splitter 61-2 is provided in the transmission line 602-2, and the receiver 62 is connected to the power splitter 61-2.

A monitoring unit 77a illustrated in FIG. 69(A) includes a power splitter 61, a receiver 62, and a transmitter 76. The monitoring unit 77a illustrated in FIG. 69(A) is different from the monitoring unit 60a illustrated in FIG. 33(A) in that a receiver 62 and a transmitter 76 are included instead of the two receivers 62-1 to 62-2. As illustrated in FIG. 69(A), the transmitter 76 and the receiver 62 are connected to the same transmission line in the same direction.

A monitoring unit 77b illustrated in FIG. 69(B) includes a power splitter 61, a plurality of receivers 62-1 to 62-4, a plurality of WDM devices 63b-1 to 63b-2, a plurality of transmitters 76-1 to 76-4, and a plurality of power splitters 78-1 to 78-4. The monitoring unit 77b illustrated in FIG. 69(B) is different from the monitoring unit 60b illustrated in FIG. 33(B) in that four receivers 62-1-1 to 62-4, four transmitters 76-1 to 76-4, and four power splitters 78-1 to 78-4 are included instead of the six receivers 62-1 to 62-6. The power splitters 78-1 and 78-2 are connected to the WDM device 63b-1, and the power splitters 78-3 and 78-4 are connected to the WDM device 63b-2. The transmitter 76-1 and the receiver 62-1 are connected to the power splitter 78-1, the transmitter 76-2 and the receiver 62-2 are connected to the power splitter 78-2, the transmitter 76-3 and the receiver 62-3 are connected to the power splitter 78-3, and the transmitter 76-4 and the receiver 62-4 are connected to the power splitter 78-4. As illustrated in FIG. 67(B), the transmitters 76-1 to 76-4 and the receivers 62-1 to 62-4 are connected to the same transmission line in the same direction of the same side in a set of the transmitter 76 and the receiver 62. In FIG. 69(B), the transmitters 76-2 and

76-4 are provided outside of the monitoring unit 77b, but may be provided inside the monitoring unit 77b.

A monitoring unit 77c illustrated in FIG. 69(C) includes a power splitter 61, a plurality of receivers 62-1 and 62-2, a WDM device 63c, and a plurality of transmitters 76-1 to 76-2. The monitoring unit 77c illustrated in FIG. 69(C) is different from the monitoring unit 60c illustrated in FIG. 33(C) in that two transmitters 76-1 and 76-2 are newly provided. The monitoring unit 77c illustrated in FIG. 69(C) has a different connection relationship with the receivers 62-1 and 62-2 of the monitoring unit 60c illustrated in FIG. 33(C). Specifically, in the monitoring unit 77c, the receivers 62-1 and 62-2 and the transmitters 76-1 and 76-2 are connected to the power splitter 61, the receiver 62-1 and the transmitter 76-1 are connected to the transmission line on the same side, and the receiver 62-2 and the transmitter 76-2 are connected to the transmission line on the same side.

The number of the power splitters 61, the receivers 62, the WDM devices 63, and the transmitters 76 illustrated in FIGS. 67 to 69 is exemplary, and may be changed in accordance with a situation.

The AMCC signal superimposed on the main signal will be described. Since the main signal and the AMCC signal are superimposed, the main signal and the AMCC signal are transferred with an optical signal with the same wavelength. The main signal is, for example, a signal such as a common public radio interface (CPRI) such as an on-off keying (OOK) signal of 10 Gbit/s (gigabits per second). The AMCC signal is superimposed on the main signal with, for example, a carrier wave which is not superimposed on an electrical main signal and has a carrier frequency such as 1 MHz or 500 kHz. A modulation scheme is intensity modulation, phase modulation, or the like. In the superimposition, for example, an electrical main signal of 10 GHz and an electrical AMCC signal of 1 MHz are combined by a power combiner to be modulated as the combined signal, and thus a main signal on which the AMCC signal is superimposed is generated. The superimposed AMCC signal can be separated from the main signal.

In the electrical region, the AMCC signal and the main signal use different frequencies. The AMCC signal has a narrower band than the main signal. For example, as described in the OITDA standard TP 20 Fiber optic active components and devices-Performance standards-GPON transceivers (Reference Literature 1: <http://www.oitda.or.jp/main/st/TP20-1.pdf>) or ITU-T G. 958 Appendix I, when the reference of the identical code succession tolerance is assumed to be 72 bits, a lower limit of GE-PON of 1.25 Gbit/s may be sufficiently lower than about 20 MHz, for example, half thereof or sufficiently higher than 1.25 GHz, for example, twice thereof, or sufficiently lower than 720 kHz of the lower limit of STM-0 of 51.84 Mbit/s of the low-speed signal, for example, half thereof or sufficiently higher than 51.84 MHz, for example, twice thereof.

As the carrier frequency, another frequency that does not overlap with an electrical main signal, such as 500 kHz, may be used. As a modulation scheme, another modulation scheme such as phase modulation may be used.

In each of the above embodiments, when the subscriber device 40 and the control unit 20 have a single-core bidirectional transceiver, the power splitters used for both the upper and lower directions may be connected via the same path of the optical SW 210 in the single-core bidirectional state, or may be separated into dual cores and connected via the separate paths. When the subscriber device 40 and the control unit 20 have a dual-cores bidirectional transceiver, they may be bundled in a single-core bidirectional manner

and then connected via the same path of the optical SW 210, or may be connected via separate paths in a dual-cores state. When one of the subscriber device 40 and the control unit 20 has a single-core bidirectional transceiver and the other thereof has a bidirectional transceiver, the bidirectional directions of the dual cores may be bundled to form a single core and connected via the same path of the optical SW 210, or the single-core bidirectional direction may be separated into dual cores and connected via the separate paths.

The power splitters used in both upper and lower directions are appropriate when wavelength bands that can be used do not overlap in the uplink direction and the downlink direction like, for example, the 1.3 micron band and the 1.55 micron band. When at least one of the subscriber device 40 and the control unit 20 transmits and receives an optical signal to and from a single-core bidirectional optical transceiver, they may be connected by a set of ports of the optical SW or may be connected by two sets of ports and multiplexed in opposite directions by a multiplexer/demultiplexer (for example, a WDM device or a WDM filter) or a power splitter. In each transmission and reception, different sets of ports may be used to connect one optical transmitter and the other optical receiver, and the other optical transmitter and the one optical receiver. This is appropriate when a dual-cores optical transceiver is used. In this case, different paths can be used on the transmission side and the reception side.

A plurality of sets may be connected to the transmission side or a plurality of sets may be connected to the reception side. In this case, connection with a plurality of devices or functional units or control thereof can be achieved. In the demultiplexing of connections and control, a transmission source or a destination port, a wavelength, a device, or a functional unit, for example, an identifier for identifying a subscriber device, for example, a MAC address, can be used.

The optical SW, the port, the transmission line, or the connection point thereof may include a power splitter or a multiplexer/demultiplexer between the optical SW and a transmission line connected to another ground, the optical SW, or an upper network. The multiplexer/demultiplexer multiplexes optical signals with different wavelengths output from different subscriber devices or the like from a plurality of ports of the optical SW, and outputs the multiplexed optical signals to another ground or a transmission line connected to the optical SW or an upper network. The multiplexer/demultiplexer demultiplexes an optical signal transmitted from any other ground, the optical SW, or the upper network in accordance with a wavelength, and inputs the optical signal to the optical SW from a port corresponding to the wavelength. In each of the above embodiments, the multiplexer/demultiplexer and the power splitter have been used for description. However, when the multiplexing/demultiplexing is not performed in accordance with a wavelength, the multiplexer/demultiplexer may be a power splitter. In the case of only the multiplexing or only demultiplexing in accordance with a wavelength, only the merging multiplexing regardless of a wavelength, and only branching regardless of a wavelength, a multiplexer, a demultiplexer, a merging unit, or a branch splitter may be used, respectively.

The power splitter is appropriate when an uplink signal and a downlink signal are multiplexed in, for example, time division multiplexing manner other than a wavelength division multiplexing manner and when wavelength bands used for the uplink signal and the downlink signal at least partially overlap. In the latter case, it is sufficient to perform filtering in an optical receiver, a transmission line before arrival at the optical receiver, between the power splitter and

the transmission line, or the like. When the optical receiver passes through a plurality of sections, it is desirable to perform filtering before multiplexing so that other signals are not influenced at the time of multiplexing with other signals.

When a wavelength to be filtered by a multiplexer/demultiplexer or the combination of a power splitter and a filter can be changed, a subscriber device may change a wavelength according to a wavelength to be transmitted and received or the control unit may change the wavelength.

Each configuration described above with reference to FIGS. 1 to 69 may be modified as follows. The process of steps S1 and S3 to S6 in the flowchart illustrated in FIG. 42 may be performed in other aspects. Step S2 in the flowchart illustrated in FIG. 42 is not performed. Hereinafter, other aspects (modifications) of the process of steps S1 and S3 to S6 will be described.

FIG. 70 is a diagram illustrating a configuration example of the subscriber device 40 and the control unit 301 according to a first modification. In the first modification, not only an optical signal can be transmitted according to a destination, but also the number of times the optical switch is switched at the time of setting a wavelength of an optical signal transmitted from the subscriber device can be reduced.

In the optical access system 100 (an optical communication device), the optical GW 200 includes a monitoring unit 260 for each input path in the transmission line 501 between the optical SW 210 and the subscriber device 40. That is, the monitoring unit 260 is connected to the port 11 on the input side of the optical SW 210. Accordingly, the wavelength control unit 310 is connected to the subscriber device 40 via the monitoring unit 260. The wavelength control unit 310 may not be connected for each subscriber device 40 as long as the wavelength control unit 310 can be connected to each subscriber device 40. For example, even when the wavelength control unit 310 is simultaneously connected to a plurality of subscriber devices 40, the optical SW 21 or the like may be able to select the subscriber device 40 to be connected, or the optical SW 21 or the like may be able to select the path to be connected, as in the initial connection of the PON. In accordance with the configuration of the monitoring unit 260, the monitoring unit 260 and the subscriber device 40 do not necessarily have a one-to-one correspondence. For example, a plurality of input paths may correspond to one monitoring unit 260. In this case, the monitoring unit 260 may monitor an optical signal of the input path in accordance with the setting of the switching of the input path. When the switching of the input path is not set, the monitoring unit 260 may collectively monitor optical signals of the plurality of input paths.

As illustrated in FIG. 70, the control unit 301 is connected to the subscriber device 40 via the monitoring unit 260. Specifically, when the wavelength or the like is allocated, an output of the allocation target device such as a wavelength or the like of the subscriber device 40 or the like is only monitored by the monitoring unit 260 installed on the input side of the optical SW 210 in a state of non-connection to another port of the optical SW 210 in accordance with the setting by the control unit 301. After the allocation of the wavelength or the like, a path to another port in the optical SW 210 is set in accordance with the setting by the control unit 301 with regard to the output of the allocation target device such as the wavelength of the subscriber device 40 or the like.

The optical GW 200 may include a monitoring unit 60 instead of the monitoring unit 260. The optical GW 200 may include a power splitter 502 (a monitoring unit) instead of the monitoring unit 260.

The monitoring unit 260 modulates optical signals transmitted and received by the subscriber device 40 with a control signal, receives some of the optical signals, multiplexes signals modulated with the control signal on the optical signal, transmits some of the optical signals to the wavelength control unit 310, and multiplexes the control signal transmitted from the wavelength control unit with the optical signals. When the monitoring unit 260 receives the optical signal, the monitoring control unit monitors the wavelength or the like based on the partially received optical signal. When the optical signal is transmitted, the wavelength control unit 310 monitors the wavelength or the like of the optical signal of the subscriber device 40 based on the optical signal transmitted from the monitoring unit 260.

In the first modification of the subscriber device 40 and the control unit 301, the wavelength allocation process (wavelength setting) and the path setting process are performed in this order. More specifically, the control unit 301 causes the subscriber device 40 to be in an unconnected state and exchanges data with the subscriber device 40. At the initial stage, since the subscriber device 40 is in the unconnected state, the subscriber device 40 remains. At the time of change, the subscriber device 40 is basically brought into the disconnected state. Only when there is no influence, the control unit 301 may exchange data with the subscriber device 40 while the connection of the subscriber device 40 is maintained or the subscriber device 40 is set to the changed connection state. The optical SW control unit 320 switches the output destination (path) of the optical SW 210 after the wavelength change so that there is no particular influence even if an output destination of the optical SW 210 is switched.

For example, cases where the first port and the first wavelength "λ1" are used before switching and the second port and the second wavelength "λ2" are used after switching are the following cases.

(1) A case where there is no subscriber device 40 using the first port and the second wavelength "λ2."

There is no influence despite switching from the first wavelength "λ1" to the second wavelength "λ2" in the first port. In this case, it is possible to perform wavelength switching before port switching.

(2) A case where there is the subscriber device 40 using the first port and the second wavelength "λ2."

There is an influence when the wavelength is switched from the first wavelength "λ1" to the second wavelength "λ2" in the first port. In this case, it is not possible to perform wavelength switching before port switching.

(3) A case where there is no subscriber device 40 using the second port and the first wavelength "λ1."

There is no influence despite switching from the first wavelength "λ1" to the second wavelength "λ2" in the second port. In this case, it is possible to perform wavelength switching before port switching.

(4) A case where there is the subscriber device 40 using the second port and the first wavelength "λ1":

There is an influence when the wavelength is switched from the first wavelength "λ1" to the second wavelength "λ2" in the second port. In this case, it is not possible to perform wavelength switching before port switching.

In the foregoing "(1)" and "(3) or (4)," the wavelength can be switched before the port switching.

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In the above “(1) or (2)” and “(3),” the wavelength can be switched before the port switching.

In the above “(2)” and “(4),” it is not possible to perform wavelength switching before port switching.

In the wavelength allocation process, the optical SW control unit 320 controls the path in the optical SW 210 such that the optical SW 210 does not transmit the optical signal of the subscriber device 40 which is a wavelength allocation target.

When there is no particular influence despite switching of the output destination of the optical SW 210 before setting of the wavelength (before allocation of the wavelength), the optical SW control unit 320 may switch the output destination of the optical SW 210 before a change in the wavelength. In the wavelength allocation process, a functional unit 15 may stop transmitting the optical signal transmitted from the subscriber device 40. When a blocking unit is provided so that an optical signal is blocked by the blocking unit, if the blocking unit blocks an optical signal, the optical SW control unit 320 can basically exchange with the subscriber device 40. As an exception, the blocking unit may not be able to selectively interrupt the output of the subscriber device 40. For example, when the subscriber device 40 shares a plurality of wavelengths or the like (wavelengths, times, polarized waves orthogonal to each other, modes orthogonal to each other, codes orthogonal to each other, frequencies, core wires (cores), or a combination thereof) by the blocking unit and the other sharing partner of the subscriber device 40 is in operation, when the wavelengths output in the initial setting are uncertain and there is a possibility of the wavelengths or the like (the wavelengths, the times, the polarized waves orthogonal to each other, the modes orthogonal to each other, the codes orthogonal to each other, the frequencies, the core wires (the cores), or a combination thereof) of the other subscriber devices 40 overlapping, when outputs at the time of changing of the wavelengths or the like overlap beyond an allowable time for the other wavelengths or the like in operation, when there is an abnormal case and what is output is unclear. In such cases, after connection is established without influence (for example, after reconnection to a port in no operation), the optical SW control unit 320 exchanges data with the subscriber device 40.

In the wavelength allocation process, the wavelength control unit 310 performs the wavelength allocation process on the subscriber device 40 which is a wavelength allocation target via the monitoring unit 260. For example, the wavelength control unit 310 performs an allocation process for a new wavelength that is the same as or different from the wavelength used when the abnormality is detected on the subscriber device 40 which is a wavelength allocation target.

In the path setting process, the optical switch control unit 320 controls the optical SW 210 such that the optical signal input from the transmission line 501 of the subscriber device 40 which is a wavelength allocation target is output to the port 11 (another port) specified in accordance with the transmission destination. Here, the transmission destination is specified in accordance with a subscriber, the subscriber device 40, the port 11 of the optical SW 210, the wavelength of the optical signal, a combination of the subscriber device 40 and the wavelength of the optical signal, a combination of the port 11 and the wavelength of the optical signal, or the like.

In step S3, when the subscriber device 40 which is a wavelength allocation target is connected, the optical SW control unit 320 disconnects the subscriber device 40. When a blocking unit is provided instead, the blocking unit may

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block the optical signal. Here, in a case where the port 11-1 connected to the monitoring unit 260 is not connected to the other port 11 in the optical SW 210, the optical SW control unit 320 remains in the unconnected state. The subscriber device 40 which is a wavelength allocation target performs an initialization process before connection to the monitoring unit 260. The subscriber device 40 which is a wavelength allocation target may perform the initialization process immediately after connection to the monitoring unit 260. When a wavelength or the like is set in the initialization process or when at least a part of the initialization process such as allocation is performed after reception of a downlink signal or an allocation request permission of the downlink signal, at least a part of the initialization process is performed after connection. The subscriber device 40 which is a wavelength allocation target transmits a connection request (a register request) using an optical signal.

In step S4, the wavelength control unit 310 receives a connection request via the monitoring unit 260 from the subscriber device 40 which is a wavelength allocation target. “Via the monitoring unit 260” means that, for example, the monitoring unit 260 transmits an optical signal branched by the branch combiner/splitter to the wavelength control unit 310. For example, a state of an electrical signal (the monitoring unit 260 includes a reception unit, the reception unit converts the optical signal into an electrical signal, and the reception unit transmits the electrical signal to the connection unit may be achieved. The wavelength control unit 310 analyzes the received optical signal and determines an initially set wavelength. The wavelength control unit 310 may analyze the received optical signal and confirm whether there is a problem in optical power.

In step S6, the wavelength control unit 310 (a control unit) transmits a wavelength instruction indicating information regarding the wavelength determined (selected) by the wavelength control unit 310 to the subscriber device 40 which is a wavelength allocation target. The branch combiner/splitter of the monitoring unit 260 may combine the optical signal of the wavelength control unit 310. The monitoring unit 260 may include a transmission unit, the transmission unit may convert an electrical signal into an optical signal, and the multiplexer/demultiplexer may combine the optical signals. When the wavelength instruction is transmitted using an optical signal via a wavelength-selective element in the path from the monitoring unit 260 to the subscriber device 40, the wavelength instruction is transmitted to the subscriber device 40 using an optical signal with a wavelength addressed to the subscriber device 40. The wavelength instruction is transmitted to the subscriber device 40 which is a wavelength allocation target via the monitoring unit 260. The monitoring unit 260 outputs the optical signal input from the wavelength control unit 310 to the subscriber device 40. The optical signal indicating the wavelength instruction may be branched in the middle of the transmission line 501. When a transmission wavelength is set in the wavelength instruction, the subscriber device 40 which is a wavelength allocation target sets the transmission wavelength of the optical signal of the optical transceiver in response to the wavelength instruction.

In step S7, when the notification signal is transmitted using the optical signal, the subscriber device 40 transmits, to the wavelength control unit 310 via the monitoring unit 260, a notification signal for notifying that the wavelength is set using the optical signal with the wavelength instructed in the wavelength instruction. Based on the notification signal, the wavelength control unit 310 confirms whether the instructed wavelength is correctly set. Based on the notifi-

cation signal, the wavelength control unit 310 may confirm whether the output power of the optical signal is sufficient.

In step S8, the optical SW control unit 320 transmits optimum connection information (path information) of each port 11 of the optical SW 210 to the optical SW 210 in accordance with the transmission destination (a destination) of the optical signal of the subscriber device 40. Based on the connection information, the optical SW 210 sets an uplink port and a downlink port of the subscriber device 40.

FIG. 71 is a flowchart illustrating an initial setting process of the optical access system according to the first modification of the subscriber device 40 and the control unit 302. Reference numerals illustrated in FIG. 71 are similar to the reference numerals illustrated in FIG. 42. A port is connected between steps S1 and S3 illustrated in FIG. 71. In step S4, the present invention is not limited to the AMCC. Before the wavelength is set, it cannot be confirmed that the control signal of the connection request is normal in the connection request because of separate carriers. The control unit, the input monitoring unit, the output monitoring unit, the communication unit, or the subscriber device 40 autonomously confirms the flowing main signal. In step S7, the control unit, the input monitoring unit, the output monitoring unit, the communication unit, or the subscriber device 40 autonomously performs the confirming.

As described above, according to the first modification of the subscriber device 40 and the control unit 301, the optical SW 210 outputs the optical signal input from any one of the plurality of transmission lines 501 to the other 501 transmission line among the plurality of transmission lines 501. The monitoring unit 260 relays the optical signal between the subscriber device 40 and the control unit 301 along the transmission line 501 on the side where the optical signal transmitted from the subscriber device 40 is input to the optical SW 210. The monitoring unit 260 relays the optical signal between the subscriber device 40 and the optical SW 210 along the transmission line 501 on the side where the optical signal is input to the optical SW 210. The wavelength control unit 310 allocates a wavelength to the optical signal of the subscriber device 40 via the monitoring unit 260. After the wavelength is allocated to the optical signal of the subscriber device 40, the optical SW control unit 320 controls the optical SW 210 such that the optical signal input to the optical SW 210 via the transmission line 501 on the input side is output to the other transmission line 501. The optical GW 200 may include a first optical SW 210 and a second optical SW 210 cascade-connected to the first optical SW 210. Before the wavelength is allocated to the optical signal of the subscriber device 40, the optical SW control unit 320 may control the optical SW 210 such that the optical signal input to the optical SW 210 via the transmission line 501 on the input side is not output to another transmission line 501.

Accordingly, according to the first modification of the subscriber device 40 and the control unit 301, it is possible not only to transmit the optical signal in accordance with the destination but also to reduce the number of times the optical switch performs the switching in the setting of the wavelength of the optical signal transmitted from the subscriber device.

Hereinafter, a second modification will be described.

Each configuration described above with reference to FIGS. 1 to 69 may be modified as follows. In this case, for example, the blocking unit is provided at a subsequent stage of the power splitter 61 in the monitoring unit illustrated in FIGS. 33, 67, and 69, at a subsequent stage of the WDM device 63d in the monitoring unit illustrated in FIGS. 34(A)

and 68(A), at a subsequent stage of the WDM device 63e-1 in the monitoring unit illustrated in FIGS. 34(B) and 68(B), at a subsequent stage of the modulator 64 in the monitoring unit illustrated in FIG. 35, or at a subsequent stage of the WDM device 63g-2 in the monitoring unit illustrated in FIG. 36.

The process of steps S1 to S8 in the flowchart illustrated in FIG. 42 may be performed in another manner. Hereinafter, another aspect (a modification) of the process of steps S1 to S8 will be described.

FIG. 72 is a diagram illustrating a configuration example of the subscriber device 40 and the control unit 301 according to the second modification. In the second modification, not only an optical signal can be transmitted in accordance with a destination, but also the number of times the optical switch performs switching in setting of a wavelength of the optical signal transmitted from the subscriber device can be reduced in some cases.

In the optical access system 100 (an optical communication device), the optical GW 200 includes a monitoring unit 260 for each output path in the transmission line 503 on the output side of the optical SW 210. That is, the monitoring unit 260 is connected to the port 11 on the output side of the optical SW 210. Accordingly, the wavelength control unit 310 is connected to the subscriber device 40 via the monitoring unit 260. In accordance with the configuration of the monitoring unit 260, the monitoring unit 260 and the output path do not necessarily have a one-to-one correspondence. For example, when the target subscriber device 40 can be selected as in the initial connection of the PON, or when a path to be connected can be selected by the optical SW 210 or the like, a plurality of output paths and one monitoring unit 260 may correspond to each other. In this case, the monitoring unit 260 may monitor the optical signal of the output path in accordance with the setting of the switching of the output path. When the switching of the output path is not set, the monitoring unit 260 may collectively monitor the optical signals of the plurality of output paths.

As illustrated in FIG. 72, the control unit 301 is connected to the subscriber device 40 via the monitoring unit 260. Specifically, when wavelengths are allocated, an output of a wavelength allocation target device such as the subscriber device 40 is connected to another port in the optical SW 210 in accordance with a setting by the control unit 301. However, the output is only monitored by the monitoring unit 260 installed on the output side, and the transmission along the transmission line 503 is blocked by the blocking unit 263. The blocking is released after the allocation.

The optical GW 200 may include a monitoring unit 60 instead of the monitoring unit 260. The optical GW 200 may include a power splitter 502 (a monitoring unit) instead of the monitoring unit 260.

The monitoring unit 260 transmits the optical signal transmitted and received by the subscriber device 40 to the wavelength control unit 310. The wavelength control unit 310 monitors a wavelength or the like of the optical signal of the subscriber device 40 based on the optical signal transmitted from the monitoring unit 260.

The monitoring unit 260 includes the blocking unit 263. The blocking unit 263 can block transmission of the optical signal in the transmission line 503. The blocking unit 263 blocks the transmission of the optical signal before the path setting process and the wavelength allocation process (wavelength setting) are performed under the control of the wavelength control unit 310. The blocking unit 263 releases

the blocking of the transmission of the optical signal after the path setting process and the wavelength allocation process are performed.

For example, an optical switch can be used as the blocking unit **263**. Alternatively, for example, a semiconductor optical amplifier (SOA) can be used as the blocking unit **263**. In this case, an amplification factor in the semiconductor optical amplifier is set to a value (for example, -10 [dB]) equal to or less than about an extinction ratio of a modulator in a case where outputs of a single path and the subscriber device **40** are combined, a value (for example, -25 [dB] in a case of 32) lower by several outputs in a case where a plurality of outputs are combined, and -35 [dB] or the like in a case where there is a difference of 10 [dB] due to a perspective difference, and thus the transmission of the optical signal is interrupted.

As a position at which the blocking unit **263** is arranged, a position at a subsequent stage of multiplexing or demultiplexing of optical signals is appropriate, for example, in the case of the setting and changing in both upper and lower directions. On the other hand, as a position at which the blocking unit **263** is arranged, a position at a subsequent stage of multiplexing or demultiplexing of optical signals is appropriate, for example, in the case of the separate setting and changing in the upper and lower directions. When a combination of a plurality of outputs is blocked, the number of blocking units **263** is reduced. On the other hand, when the dividing of the output is blocked, the setting can be individually performed and the degree of freedom is improved.

The blocking unit **263** may be a functional unit or a device provided outside of the monitoring unit **260**. When the extinction ratio of the modulator is high enough not to affect other optical signals, the modulator may also be configured to serve as the blocking unit **263**.

Hereinafter, blocking of transmission of an optical signal may be simply referred to as "blocking." Hereinafter, releasing of the blocking of the transmission of the optical signal may be simply referred to as "blocking releasing."

In the first modification of the subscriber device **40** and the control unit **301**, the blocking, the path setting process, the wavelength allocation process (the wavelength setting), and the blocking releasing are performed in this order. In such an order, even if the path setting process is performed, an optical signal before wavelength setting is not transmitted to the outside of the optical GW **200** via the transmission line **503**.

When the optical signal is blocked by the blocking unit, the optical SW control unit **320** can basically communicate with the subscriber device **40** if the blocking unit blocks the optical signal. As an exception, the blocking unit may not be able to selectively interrupt the output of the subscriber device **40**. For example, when the subscriber device **40** shares a plurality of wavelengths or the like (wavelengths, times, polarized waves orthogonal to each other, modes orthogonal to each other, codes orthogonal to each other, frequencies, core wires (cores), or a combination thereof) by the blocking unit and the supply partner of the subscriber device **40** is in operation, when the wavelengths output in the initial setting are uncertain and there is a possibility of the wavelengths or the like (the wavelengths, the times, the polarized waves orthogonal to each other, the modes orthogonal to each other, the codes orthogonal to each other, the frequencies, the core wires (the cores), or a combination thereof) of the other subscriber devices **40** overlapping, when the outputs at the time of changing of the wavelengths or the like overlap beyond an allowable time for the other

wavelengths or the like in operation, or when there is an abnormal case and what is output is unclear. In such cases, after connection is established without influence (for example, after reconnection to a port in no operation), the optical SW control unit **320** exchanges data with the subscriber device **40**.

As an exception, when there is no influence despite non-blocking, the blocking may not be performed. Alternatively, the monitoring unit **260** of the output port may be configured to performing setting so that the connection to an output port that is not affected is made despite non-blocking.

In the path setting process, the optical switch control unit **320** controls the path in the optical SW **210** such that the subscriber device **40** which is a wavelength allocation target is connected to the port **11** (another port) other than the port **11** connected to the monitoring unit **260** of the subscriber device **40** which is a wavelength allocation target. For example, the optical switch control unit **320** controls the optical SW **210** such that the optical signal input from the transmission line **501** of the subscriber device **40** which is a wavelength allocation target is output to the port **11** (another port) specified in accordance with the transmission destination. Here, the transmission destination is specified in accordance with a subscriber, the subscriber device **40**, the port **11** of the optical SW **210**, the wavelength of the optical signal, a combination of the subscriber device **40** and the wavelength of the optical signal, a combination of the port **11** and the wavelength of the optical signal, or the like.

In the wavelength allocation process, the wavelength control unit **310** performs the wavelength allocation process on the subscriber device **40** which is a wavelength allocation target. For example, the wavelength control unit **310** performs an allocation process for a new wavelength that is the same as or different from the wavelength used when the abnormality is detected on the subscriber device **40** which is a wavelength allocation target.

FIG. **73** is a flowchart illustrating an initial setting process of an optical access system according to a second modification. Note that a difference from the initial setting process illustrated in the flowchart of FIG. **42** will be mainly described.

In step **S2**, the control unit **301** controls the monitoring unit **260** installed on the output side of the optical SW **210** such that the blocking unit **263** blocks transmission along the transmission line **503** so that the optical signal of the subscriber device **40** is not transmitted via the transmission line **503**. The optical SW control unit **320** performs a path setting process in the optical SW **210**.

In step **S3**, the wavelength control unit **310** receives a connection request from the subscriber device **40** which is a wavelength allocation target. The subscriber device **40** which is a wavelength allocation target performs the initialization process before connection to the optical SW **210**. The subscriber device **40** which is a wavelength allocation target may perform the initialization process immediately after connection to the optical SW **210**. When a wavelength or the like is set in the initialization process or when at least a part of the initialization process such as allocation is performed after reception of a downlink signal or an allocation request permission of the downlink signal, at least a part of the initialization process is performed after connection. The subscriber device **40** which is a wavelength allocation target transmits a connection request (a register request) using an optical signal.

In step **S4**, the wavelength control unit **310** receives a connection request via the monitoring unit **260** from the subscriber device **40** which is a wavelength allocation target.

“Via the monitoring unit 260” means that, for example, the monitoring unit 260 transmits an optical signal branched by the branch combiner/splitter to the wavelength control unit 310. For example, state of an electrical signal (the monitoring unit 260 includes a reception unit, the reception unit converts the optical signal into an electrical signal, and the reception unit transmits the electrical signal to the connecting unit) may be achieved. The wavelength control unit 310 analyzes the received optical signal and determines an initially set wavelength. The wavelength control unit 310 may analyze the received optical signal and confirm whether there is a problem in optical power.

In step S6, the wavelength control unit 310 (a control unit) transmits a wavelength instruction indicating information regarding the determined (selected) wavelength to the subscriber device 40 which is a wavelength allocation target. The branch combiner/splitter of the monitoring unit 260 may combine the optical signal of the wavelength control unit 310. The monitoring unit 260 may include a transmission unit, the transmission unit may convert an electrical signal into an optical signal, and the multiplexer/demultiplexer may combine the optical signals. When the wavelength instruction is transmitted using the optical signal, the wavelength instruction is transmitted to the subscriber device 40 using the optical signal of the wavelength addressed to the subscriber device 40. The wavelength instruction is transmitted to the subscriber device 40 which is a wavelength allocation target via the monitoring unit 260. The monitoring unit 260 outputs the optical signal input from the wavelength control unit 310 to the subscriber device 40. The optical signal indicating the wavelength instruction may be branched in the middle of the transmission line 501. When a transmission wavelength is set in the wavelength instruction, the subscriber device 40 which is a wavelength allocation target sets the transmission wavelength of the optical signal of the optical transceiver in response to the wavelength instruction.

In step S7, when the notification signal is transmitted using the optical signal, the subscriber device 40 transmits, to the wavelength control unit 310 via the monitoring unit 260, a notification signal for notifying that the wavelength is set using the optical signal with the wavelength instructed in the wavelength instruction. Based on the notification signal, the wavelength control unit 310 confirms whether the instructed wavelength is correctly set. Based on the notification signal, the wavelength control unit 310 may confirm whether the output power of the optical signal is sufficient.

In step S8, when the instructed wavelength is correctly set, the wavelength control unit 310 controls the blocking unit 263 such that the blocking is released. Accordingly, communication between the subscriber device 40 that is the wavelength allocation target and the subscriber device 40 as the communication destination is started.

As described above, in the second modification of the subscriber device 40 and the control unit 301, the optical SW 210 outputs the optical signal input from any one of the plurality of transmission lines 501 to any one of the plurality of transmission lines 503. The monitoring unit 260 relays an optical signal between the optical SW 210 and the subscriber device 40 of a communication destination in the transmission line 503 on the side where the optical signal transmitted from the subscriber device 40 is output from the optical SW 210. The wavelength control unit 310 controls the blocking unit 263 such that transmission of the optical signal output from the monitoring unit 260 is blocked. After the blocking is performed by the blocking unit 263, the optical SW control unit

320 performs a path setting process on the optical signal in the optical SW 210. Next, the wavelength control unit 310 allocates a wavelength to the optical signal of the subscriber device 40 based on the monitoring result of the monitoring unit 260. Next, the wavelength control unit 310 controls the blocking unit 263 such that the blocking of the transmission of the optical signal output from the monitoring unit 260 is released. In this way, the control unit 301 causes the blocking unit 263 to perform the blocking before the allocation of the wavelength and prevents the optical signal input from the transmission line 501 to the optical SW 210 from being output to the other transmission lines 503. After the blocking is performed, the control unit 301 performs the path setting process and the wavelength allocation process (the wavelength setting) in this order. After the wavelength is allocated, the control unit 301 causes the blocking unit 263 to perform releasing of the blocking so that the optical signal input from the transmission line 501 to the optical SW 210 is output to the other transmission line 503. The optical GW 200 may include the first optical SW 210 and the second optical SW 210 cascade-connected to the first optical SW 210.

Accordingly, in the second modification of the subscriber device 40 and the control unit 301, it is possible not only to transmit the optical signal in accordance with the destination but also to reduce the number of times the optical switch performs switching in the wavelength setting of the optical signal transmitted from the subscriber device.

The monitoring unit 60 illustrated in FIGS. 33 to 36 may be configured in another aspect (a modification). Hereinafter, another mode (modification) of the monitoring unit 60 will be described.

FIGS. 74 to 76 are configuration diagrams of another mode (a modification) of the monitoring unit 60. Although the monitoring unit 60 will be described as an example in FIGS. 74 to 76, the monitoring unit 260 may be used instead of the monitoring unit 60, and the configuration of the monitoring unit 260 may be the configuration illustrated in FIGS. 74 to 76.

A monitoring unit 70a illustrated in FIG. 74(A) is another aspect of the monitoring unit 60. The monitoring unit 70a illustrated in FIG. 74(A) includes a power splitter 61, a plurality of receivers 62-1 to 62-2, and a blocking unit 263a. The monitoring unit 70a illustrated in FIG. 74(A) is different from the monitoring unit 60a illustrated in FIG. 33(A) in that a blocking unit 263a is further included. The blocking unit 263a performs the same process as the blocking unit 263 illustrated in FIG. 72.

For example, the blocking unit 263a blocks the transmission of the optical signal before the path setting process and the wavelength allocation process (wavelength setting) are performed under the control of the wavelength control unit 310. For example, the blocking unit 263a releases the blocking of the transmission of the optical signal after the path setting process and the wavelength allocation process are performed under the control of the wavelength control unit 310. In the example illustrated in FIG. 73, the blocking unit 263a performs the foregoing process on the optical signal input via the transmission line 604 or the transmission line 605.

A monitoring unit 70b illustrated in FIG. 74(B) is another aspect of the monitoring unit 65. The monitoring unit 70b illustrated in FIG. 74(B) includes a power splitter 61, a plurality of receivers 62-1 to 62-6, a plurality of WDM devices 63b-1 to 63b-2, and a blocking unit 263b. The monitoring unit 70b illustrated in FIG. 74(B) is different from the monitoring unit 60b illustrated in FIG. 33(B) in that

the blocking unit **263b** is further included. The blocking unit **263b** performs the same process as the blocking unit **263a**.

A monitoring unit **70c** illustrated in FIG. **74(C)** includes a power splitter **61**, a plurality of receivers **62-1** to **62-2**, a plurality of WDM devices **63c**, and a blocking unit **263c**. The monitoring unit **70c** illustrated in FIG. **74(C)** is different from the monitoring unit **60c** illustrated in FIG. **33(C)** in that the blocking unit **263c** is further included. The blocking unit **263c** performs the same process as the blocking unit **263a**. For example, the blocking unit **263c** blocks transmission of the optical signal multiplexed by the WDM device **63c** or releases the blocking of the transmission of the optical signal. For example, the blocking unit **263c** blocks the transmission of the optical signal input via the transmission line **605** or releases the blocking of the transmission of the optical signal.

The monitoring unit **70d** illustrated in FIG. **75(A)** includes a plurality of power splitters **61-1** to **61-3**, a plurality of receivers **62-1** to **62-6**, a WDM device **63d**, and a plurality of blocking units **263d-1** to **263d-3**. The monitoring unit **70d** illustrated in FIG. **75(A)** is different from the monitoring unit **60d** illustrated in FIG. **34(A)** in that blocking units **263d-1** to **263d-3** are further provided. The transmission line **606-1** is provided with the power splitter **61-1** and the blocking unit **263d-1**, the transmission line **602-2** is provided with the power splitter **61-2** and the blocking unit **263d-2**, and the transmission line **602-3** is provided with the power splitter **61-3** and the blocking unit **263d-3**.

The blocking units **263d-1** to **263d-3** perform the same process as the blocking unit **263a**. For example, the blocking units **263d-1** to **263d-3** block the transmission of the optical signals demultiplexed by the WDM device **63d** or release the blocking of the transmission of the optical signals. For example, the blocking unit **263d** blocks the transmission of the optical signal input via the transmission line **606** or releases the blocking of the transmission of the optical signal.

A monitoring unit **70e** illustrated in FIG. **75(B)** includes a plurality of power splitters **61-1** to **61-2**, a plurality of receivers **62-1** to **62-2**, a plurality of WDM devices **63e-1** to **63e-2**, and a plurality of blocking units **263e-1** to **263e-3**. The monitoring unit **70e** illustrated in FIG. **75(B)** is different from the monitoring unit **60e** illustrated in FIG. **34(B)** in that the blocking units **263e-1** to **263e-3** are further included. Transmission lines **607-1** and **607-2** are provided between the WDM device **63e-1** and the WDM device **63e-2**. The transmission line **607-1** is provided with the power splitter **61-1** and the blocking unit **263e-1**, and the transmission line **607-2** is provided with the power splitter **61-2** and the blocking unit **263e-2**. The transmission line **608** connected to the WDM device **63e-1** is provided with a blocking unit **263-3**.

The blocking units **263e-1** to **263e-3** perform the same process as the blocking unit **263a**. For example, the blocking units **263e-1** to **263e-2** block the transmission of the optical signal demultiplexed by the WDM device **63e-1** or the WDM device **63e-2** or release the blocking of the transmission of the optical signal. The blocking unit **263e-3** blocks the transmission of the optical signal multiplexed by the WDM device **63e-1** or releases the blocking of the transmission of the optical signal. Alternatively, the blocking unit **263e-3** blocks the transmission of the optical signal input via the transmission line **609** or releases the blocking of the transmission of the optical signal. In the monitoring unit **70e** illustrated in FIG. **75(B)**, when the blocking is not set simultaneously in the uplink direction and the downlink direction, the optical signal is blocked at a position where

demultiplexing is not performed. When the blocking is set simultaneously in the uplink direction and the downlink direction, the optical signal is blocked at a position where demultiplexing is performed.

A monitoring unit **70f** illustrated in FIG. **76(A)** includes a power splitter **61**, a plurality of receivers **62-1** to **62-2**, a modulator **64**, and a blocking unit **263f**. The monitoring unit **70f** illustrated in FIG. **76(A)** is different from the monitoring unit **60f** illustrated in FIG. **35 (A)** in that the blocking unit **263f** is further included. The transmission line **604** is provided with the power splitter **61**, the blocking unit **263f**, and the modulator **64**.

The blocking unit **263f** performs the same process as the blocking unit **263a**. For example, the blocking unit **263f** blocks the transmission of the optical signal input via the transmission line **604** or releases the blocking of the transmission of the optical signal. The blocking unit **263f** blocks the transmission of the optical signal modulated by the modulator **64** or releases the blocking of the transmission of the optical signal.

A monitoring unit **70g** illustrated in FIG. **76(B)** includes a plurality of power splitters **61-1** to **61-2**, a plurality of receivers **62-1** to **62-2**, a plurality of WDM devices **63g-1** to **63g-2**, a plurality of modulators **64g-1** to **64g-2**, and a plurality of blocking units **263g-1** to **263g-2**. The monitoring unit **70g** illustrated in FIG. **76(B)** is different from the monitoring unit **60g** illustrated in FIG. **36** in that the blocking units **263g-1** to **263g-2** are further included. The transmission lines **607-1** and **607-2** are provided between the WDM device **63g-1** and the WDM device **63g-2**. The transmission line **607-1** is provided with the modulator **64g-1**, the blocking unit **263g-1**, and the power splitter **61-1**, and the transmission line **607-2** is provided with the modulator **64g-2**, the blocking unit **263g-2**, and the power splitter **61-2**.

The blocking units **263g-1** to **263g-2** perform the same process as the blocking unit **263a**. For example, the blocking unit **263g-1** blocks the transmission of the optical signal demultiplexed by the WDM device **63g-2** or releases the blocking of the transmission of the optical signal. The blocking unit **263g-1** blocks the transmission of the optical signal modulated by the modulator **64g-1** or releases the blocking of the transmission of the optical signal. For example, the blocking unit **263g-2** blocks the transmission of the optical signal demultiplexed by the WDM device **63g-1** or releases the blocking of the transmission of the optical signal. The blocking unit **263g-2** blocks the transmission of the optical signal modulated by the modulator **64g-2** or releases the blocking of the transmission of the optical signal. In the monitoring unit **70g** illustrated in FIG. **76(B)**, when the blocking is not set simultaneously in the uplink direction and the downlink direction, the optical signal is blocked at a portion in which demultiplexing is not performed. When the blocking is set simultaneously in the uplink direction and the downlink direction, the optical signal is blocked at a portion in which demultiplexing is performed.

The process in the flowchart illustrated in FIG. **42** may be performed in other aspects. Hereinafter, a flow of process corresponding to a modification of the flowchart of FIG. **42** and a configuration example for realizing such a flow of the process will be described. FIGS. **77** and **78** are diagrams illustrating configuration examples of third and fourth modifications of the subscriber device **40** and the control unit **301**, respectively. FIG. **79** illustrates a modification of the flowchart of FIG. **42**. In description of the flowchart illustrated in FIG. **42**, the subscriber device **40-1** which is a setting

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target has already been connected to the optical SW 210. However, the subscriber device 40-1 to be set may not be connected (connected) to the optical SW 210 at the time when the processing of steps S1 to S7 is performed. In this case, the subscriber device 40-1 and the control unit 301 have a function for communicating with each other.

FIG. 77 is a diagram illustrating a configuration example in the third modification of the subscriber device 40 and the control unit 301. The subscriber device 40 (for example, the subscriber device 40-1) includes a communication unit 44. The control unit 301 includes a communication unit 330.

The communication unit 44 is a communication device used when communicating with the control unit 301 without passing through the optical GW 201 is performed. The communication unit 44 may be configured in any way as long as the communication unit 44 can communicate with the control unit 301. For example, the communication unit 44 is configured using a communication device such as a network interface. The communication unit 44 may be configured using a device that performs wireless communication or may be configured using a device that performs wired communication. The wired communication may be communication performed using an electrical signal.

The communication unit 330 is a communication device used when communication with the subscriber device 40 without passing through the optical SW 210 is performed. The communication unit 330 may be configured in any way as long as the communication unit 330 can communicate with the subscriber device 40. For example, the communication unit 330 is configured using a communication device such as a network interface. The communication unit 330 may be configured using a device that performs wireless communication or may be configured using a device that performs wired communication. The wired communication may be communication performed using an electrical signal.

Next, a flow of a process will be described with reference to FIG. 79. Correspondence between processes illustrated in rectangles in FIG. 79 and step numbers given to the rectangles is the same as the correspondence between the rectangles and the step number illustrated in FIG. 42. The processing order is different between FIGS. 42 and 79.

In the modification, the process of step S3 is performed after the process of step S1. In step S3, the subscriber device 40-1 transmits a connection request (a register request) to the control unit 301 via the communication unit 44. The process of step S4 may be performed using the subscriber device 40 itself or a dedicated port. The process of step S4 may not be performed.

In step S5, the wavelength control unit 310 of the control unit 301 registers information regarding the subscriber device 40-1 (a request of the connection source) and the like in the management DB 350 based on the connection request received via the communication unit 330. The wavelength control unit 310 selects a wavelength to be used by the subscriber device 40-1 from among free wavelengths.

In step S6, the wavelength control unit 310 transmits the wavelength instruction in which the information regarding the selected wavelength is set to the subscriber device 40-1 via the communication unit 330. The subscriber device 40-1 receives the wavelength instruction via the communication unit 44. The subscriber device 40-1 sets the wavelength of the optical transceiver 41 according to the received wavelength instruction. Further, when a reception wavelength is set in the wavelength instruction, the subscriber device 40-1 sets the optical transceiver 41 (the wavelength-tunable filter 453) so that a wavelength signal of the reception wavelength is received. Then, in step S7, the control unit 230 confirms

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whether the specified wavelength is correctly set, whether the output power is sufficient, and the like based on the received notification signal. This confirmation process is not necessarily performed. When the confirmation process is performed, for example, the confirmation process is performed using a dedicated port.

Thereafter, the process of step S2 is performed, and the subscriber device 40 and the optical SW 210 are connected using a port of the optical SW 210 allocated in step S2. This connection may be performed by a human hand or may be performed by the connection device 900. The connection device 900 includes, for example, a connector holding portion that holds a connector located at an end of an optical cable and an actuator that moves the connector holding portion to connect the connector to a port for connection to the optical SW 210. For example, the wavelength control unit 310 of the control unit 301 may transmit a connection instruction to the connection device 900 at a predetermined timing (for example, a timing after a time required until the end of the setting in accordance with the wavelength instruction in the subscriber device 40 has elapsed) after transmission of the wavelength instruction. The transmission process in this case may be performed via the communication unit 330, for example. When the connection instruction is received, the connection device 900 links (connects) the optical cable forming the transmission line 501 of the subscriber device 40 to the optical SW 210. The information indicating the port to which the optical cable is connected may be set in the connection device 900 in advance, for example, or may be included in the connection instruction transmitted from the control unit 301.

When the subscriber device 40-1 and the optical SW 210 are connected, the processing after step S8 is performed. That is, control (a routing process) of the path for the optical SW 210 is performed.

FIG. 78 is a diagram illustrating a configuration example in a fourth modification of the subscriber device 40 and the control unit 301. The control unit 301 includes a communication unit 330.

The communication unit 330 is a communication device used when communication with the subscriber device 40 without passing through the optical SW 210 is performed. The communication unit 330 may be configured in any way as long as communication with the subscriber device 40 can be made using an optical transmission line. For example, the communication unit 330 is configured using a communication device such as an optical network interface. The optical transceiver 41 of the subscriber device 40 (for example, the subscriber device 40-1) is first connected to the communication unit 330 of the control unit 301 via the optical transmission line 501.

Next, a flow of a process will be described with reference to FIG. 79. In the modification, the process of step S3 is performed after the process of step S1. In step S3, the subscriber device 40-1 transmits a connection request (a register request) to the control unit 301 via the optical transceiver 41. The process of step S4 may be performed using the subscriber device itself or a dedicated port. The process of step S4 may be performed or may not be performed.

In step S5, the wavelength control unit 310 of the control unit 301 registers information regarding the subscriber device 40-1 (a request of the connection source) and the like in the management DB 350 based on the connection request received via the communication unit 330. The wavelength control unit 310 selects a wavelength to be used by the subscriber device 40-1 from among free wavelengths.

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In step S6, the wavelength control unit 310 transmits the wavelength instruction in which the information regarding the selected wavelength is set to the subscriber device 40-1 via the communication unit 330. The subscriber device 40-1 receives the wavelength instruction via the optical transceiver 41. The subscriber device 40-1 sets an oscillation wavelength of the optical transceiver 41 in response to the received wavelength instruction. Further, when a reception wavelength is set in the wavelength instruction, the subscriber device 40-1 sets the optical transceiver 41 (the wavelength-tunable filter 453) so that a wavelength signal of the reception wavelength is received. Then, in step S7, the control unit 230 confirms whether the specified wavelength is correctly set, whether the output power is sufficient, and the like based on the received notification signal. This confirmation process is not necessarily performed. When the confirmation process is performed, for example, the confirmation process is performed using a dedicated port.

Thereafter, the process of step S2 is performed, and the subscriber device 40 and the optical SW 210 are connected using a port of the optical SW 210 allocated in step S2. This connection may be performed by a human hand or may be performed by the connection device 900. For example, the wavelength control unit 310 of the control unit 301 may transmit a connection instruction to the connection device 900 at a predetermined timing (for example, a timing after a time required until the end of the setting in accordance with the wavelength instruction in the subscriber device 40 has elapsed) after transmission of the wavelength instruction. The transmission process in this case may be performed via the communication unit 330, for example. When the connection instruction is received, the connection device 900 links (connects) the optical cable forming the transmission line 501 of the subscriber device 40 to the optical SW 210. The information indicating the port to which the optical cable is connected may be set in the connection device 900 in advance, for example, or may be included in the connection instruction transmitted from the control unit 301.

When the subscriber device 40-1 and the optical SW 210 are connected, the processing after step S8 is performed. That is, control (a routing process) of the path for the optical SW 210 is performed.

As described above, in the modification of the subscriber device 40 and the control unit 301, in a state where the subscriber device 40 which is a setting target is not linked (connected) to the optical SW 210, the wavelength is allocated using the communication path without being involved in the optical SW 210. Accordingly, in the modification of the subscriber device 40 and the control unit 301, it is possible to improve the degree of freedom of work and a process related to wavelength allocation to the subscriber device 40.

As described above, in the third and fourth modifications of the subscriber device 40 and the control unit 301, in a state where the subscriber device 40 which is a setting target is not linked (connected) to the optical SW 210, the wavelength is allocated using the communication path without being involved in the optical SW 210. Accordingly, in the third and fourth modifications of the subscriber device 40 and the control unit 301, it is possible to increase the degree of freedom of work and a process related to wavelength allocation to the subscriber device 40. The first to fourth modifications of the subscriber device 40 and the control unit 301 may be combined.

Next, a method of giving a control signal to a main signal will be described.

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The control unit 301 includes a wavelength control unit 310 and an optical SW control unit 320. Hereinafter, a portion described as the control unit 301 may be replaced with the control unit 230 or 235. A portion described as the monitoring unit 260 may be replaced with the monitoring unit 60, 65, 265, or 267. The control signal transmitted and received between the control unit 301 and the subscriber device 40 is preferably a signal that does not depend on a communication protocol of an optical signal (a main signal) between the subscriber device 40 and the like. The control unit 301 sets a control parameter of the subscriber device 40 in the subscriber device 40 by using a control signal given to the main signal. An example in which the control signal (a management control signal) in a time region is used in step S4 illustrated in FIG. 42 will be described. The control unit 301 gives the control signal (the management control signal) to the main signal. The control unit 301 may apply the control signal to the main signal only in the initial setting or setting change. At least a part of the control unit 301 may be provided in the monitoring unit 260. The monitoring unit 260 including at least a part of the control unit 301 may be provided at a preceding stage (an input side) of the optical SW 210 or a subsequent stage (an output side) of the optical SW 210 from the viewpoint of the subscriber device. The monitoring unit 260 in the preceding stage of the optical SW 210 may give the control signal to the main signal when the connection between the port and the port facing the optical SW 210 is unconnected, when the initial setting of the subscriber device 40 or the control unit 301 is set, or when the setting is changed. The monitoring unit 260 at the subsequent stage of the optical SW 210 may give the control signal to the main signal when a port facing the port in the optical SW 210, that is, connection of a port connected to the subscriber device and a port of a connection destination is blocked, or when the initial setting or the setting change is performed in the subscriber device 40 or the control unit 301. The optical SW control unit 320 may be included in the communication unit 330 of the control unit 301.

The optical signal may be converted into an electrical signal inside each of the control unit 301 and the subscriber device 40, and signal processing may be performed on the electrical signal. The process on the electrical signal is appropriate when "a bandwidth of a section between the subscriber device 40 and the subscriber device (a band of the main signal+a band of the control signal) is satisfied" and the main signal and the control signal are contained in the same signal sequence and transmitted. Here, the band of the section between the subscriber device 40 and another subscriber device 40 is a band in a case where the control signal is deleted at the preceding stage of the optical SW. When the control signal is transmitted via the optical SW without being deleted, a band of a section for transmission is obtained. Furthermore, in a case where a control signal is separately given in the middle of the path, the band is obtained by subtracting the corresponding band. The band of the main signal may be a band remaining without being removed by removing the amount of information (for example, an amount of redundant information such as an inter-frame gap (IFG) and a preamble) that can be restored at the transmission destination even if the band is deleted. In that case, after the restorable information is deleted, the deleted information is restored after the transmission. The band remaining without being removed is regarded as a band of the main signal. A process on an optical signal is also possible. The control signal and the main signal, which are overhead as optical signals, may be time-multiplexed. In particular, when the main signal is sparse in terms of time,

the subscriber device **40** delays the main signal as light and inserts the overhead into a gap between the main signals. The subscriber device **40** compresses a time width of a pulse, for example, a return to zero (RZ) signal, which is easily compressed on a time axis, on the time axis to narrow an occupied time region, adds a delay to each pulse little by little, generates a gap of the main signal, and inserts an overhead into the gap. The subscriber device **40** may cause a time-synchronized user device to transmit or not to transmit an unmodulated signal corresponding to overhead time, and the subscriber device **40** may modulate the unmodulated signal. An overhead may be inserted at a non-transmission time. Here, the unmodulated means that the control signal can be written without being modulated, or modulation for transmission with a frame gap or the like, 01 modulation, or the like may be used. Unlike the AMCC in which the main signal cannot be completely extinct at the time of the unmodulated signal or the re-modulation, any extinction ratio is possible. Here, when the modulation is performed again, the control signal cannot be completely extinct. In that case, the extinction ratio is several dB, for example, less than 10 dB. Here, when the control signal is written a plurality of times in the middle of a path, the control signal may be rewritten after a non-modulation region is restored, or the control signal may be removed. Gaps into which non-modulation regions for writing can be inserted (written) may be prepared at a plurality of corresponding locations. In this case, the non-modulation region does not need to be restored or the control signal does not need to be removed because the non-modulation region is inserted (written) into the different region.

FIG. **80** is a diagram illustrating an example of writing and reading of a control signal in which an overhead of a periodic frame is used. The uppermost user data in FIG. **80** is a main signal transmitted (generated) by the subscriber device **40** as a transmission source. The lowermost user data is a main signal received (restored) by the subscriber device **40** as a reception source. A space between the third and fourth stages from the bottom of FIG. **80** is a space between a network-network interface (NNI) of the subscriber device **40** and an NNI of the facing subscriber device **40**.

Any one of (1-1) to (1-4) exemplified below is performed so that a control signal is given to a main signal through time-division multiplexing. Basically, except for a process in the electrical processing unit, a user signal (a main signal) is not processed at an electrical stage between the subscriber device **40**, the optical SW, and the subscriber device **40**. The control signal is processed at each stage of the optical SW.

(1-1) Speed Conversion

When an input/output on a user-network interface (UNI) side of the subscriber device **40** is a digital signal such as 0/1 that can be restored despite a change in a modulation scheme or the like in a network, the subscriber device **40** merges the control signal with the main signal (user data) on the NNI side of the subscriber device **40**, and outputs and inputs the main signal in which the control signal is merged. Here, a bit rate on the NNI side of the subscriber device **40** is equal to or higher than a bit rate on the UNI side. On the NNI side of the subscriber device **40**, the subscriber device **40** performs modulation faster than on the UNI side by the control signal or more. Such a modulation rate difference with respect to the main signal causes a time region (gap) that is not used as the main signal as an overhead. On the NNI side of the subscriber device **40**, the subscriber device **40** inserts (writes) the control signal (header+trailer) into a time region (gap) that is not used as the main signal. The subscriber device **40**, the control unit **301**, or the monitoring unit **260**

may delete (read) the control signal from the gap (a time region) generated in the main signal before the main signal (an optical signal) is received by the communication destination.

In modulation rate conversion, the subscriber device **40** may change encoding such as "8B/10B" to another encoding. The subscriber device **40** may perform scrambling on an encoding result in the modulation rate conversion. The subscriber device **40** may temporarily stop the encoding in the modulation rate conversion. The same applies to the following description.

A delay variation when the control signal is merged into the main signal is desirably equal to or less than a delay variation allowed in the UNI band. In the delay variation in the NNI, a ratio of the NNI band to the UNI band is taken into consideration. For example, when the band ratio of NNI/UNI is 8 and the delay variation allowed in the UNI band in the subscriber device **40** is a 100-bit time, an 800-bit time is an allowable delay, and a management control signal of up to 799 bits can be inserted (written). In the delay variation in the NNI, a ratio of the NNI band to the UNI band is taken into consideration. For example, when the band ratio of NNI/UNI is 8 and the delay variation allowed in the UNI band in the subscriber device **40** is a 100-bit time, a 800-bit time is an allowable delay, and a control signal of up to 799 bits can be inserted. The subscriber device **40** may hold a shift amount of the main signal in which the control signal is merged, and adjust a delay of the main signal when the main signal is output. The same applies to the following method.

The main signal is divided, and the control signal is inserted into a gap between the main signals generated through the division at every periodic time when the control signal is inserted into the main signal. The subscriber device **40** inserts a signal such as padding in a time (gap) in which neither the control signal nor the main signal is transmitted.

When the main signal can be accommodated within a limit where signal omission is allowed, the number of main signals including the time-division multiplexed control signals may be plural. For example, when the main signal is transmitted at 10 Gbit/s and a rate of the main signal is equal to or less than 1.25 Gbit/s, the subscriber device **40** may multiplex 7 pieces of user data at 1.25 Gbit/s. For example, when user data of 1.25 Gbit/s is encoded by "8B/10B", the subscriber device **40** may decode the encoding result and perform 8-multiplexing on the user data of 1 Gbit/s.

(1-2) Compression

The subscriber device **40** inserts (writes) the control signal in a time region in which the main signal (user data) is not transmitted. The subscriber device **40**, the control unit **301**, or the monitoring unit **260** may delete (read) the control signal from the gap (a time region) generated in the main signal before the main signal (an optical signal) is received by the communication destination. A non-transmitted time region is a time region in which the main signal is not burst-transmitted, a time region such as padding, a time region of IFG in Ethernet (registered trademark) or the like, and a time region such as a preamble in Ethernet (registered trademark) or the like. In the time region in which the main signal is transmitted, it is possible to complement the main signal before the subscriber device **40** is connected to customer premises equipment (CPE). When the main signal is transmitted at a periodic time at which the control signal is inserted into the main signal (a periodic time at which an overhead is generated as a gap in the main signal), the subscriber device **40** delays the main signal with respect to

a time at which the control signal is to be inserted, and inserts the control signal (header+trailer) into the main signal.

(1-3) Generation of Non-Signal Region

The subscriber device 40 samples the main signal at a high speed, closes a time interval of the sampled data, and reproduces and outputs the sampled data. The subscriber device 40 inserts (writes) the control signal (header+trailer) in the time of the gap generated in the main signal (time region of non-signal). The subscriber device 40, the control unit 301, or the monitoring unit 260 may delete (read) the control signal from the gap (a time region) generated in the main signal before the main signal (an optical signal) is received by the communication destination. For example, when a control signal of 20 bytes is inserted into the gap for each main signal of 80 bytes, the control unit 301 or the monitoring unit 260 outputs the sampled data at a cycle of $((80+20)/80)$.

In sampling, the subscriber device 40 performs sampling so that the main signal can be restored. For example, when the main signal is phase-modulated, the main signal is sampled so that the phase can be restored. For example, when the main signal is an analog signal, the subscriber device 400 samples the main signal according to a sample rate and an intensity order that can be restored after digital-to-analog conversion. An output side (for example, the subscriber device 40 facing the input side or a host device) restores such a main signal.

(1-4) Compression of Main Signal

The subscriber device 40 compresses the main signal and sequentially shifts an interval of the main signal. The subscriber device 40 inserts (writes) the control signal into the gap generated in the main signal. The subscriber device 40, the control unit 301, or the monitoring unit 260 may delete (read) the control signal from the gap (the time region) generated in the main signal before the main signal (an optical signal) is received by a communication destination. The subscriber device 40 divides the main signal, for example, and accommodates the main signal in a frame. The subscriber device 40 divides the main signal and accommodates the control signal in an overhead (OH) of a frame accommodating the main signal. The frame of the main signal may have an error correction (FEC) region. The overhead (gap) may have the error correction region.

The frame that accommodates the divided main signal (user data) is, for example, an optical transport network (OTN) (G.709) frame. For example, like the OTN (G.709), the overhead may be added to a frame by main signal accommodation (adaptation) by optical channel payload unit-k (OPUk). The overhead may be added to a frame in an end-end path by an optical channel data unit-k (ODUk). The overhead may be added to the frame by a performance monitor. A transmission overhead and an error correction byte may be added to the frame by the optical channel transport unit-k (OTUk). The overhead may be added to the frame sequentially in layers. These may be collectively added to the frame. The error correction region may be added in the middle of the frame instead of the end of the frame.

By using a frame similar to the OTN frame or the OTN frame as a frame that accommodates the divided main signal (user data), consistency with a relay network using the OTN is enhanced.

The bit rate is not limited to the three types of bit rates of "255/238×2488320 kbit/s or 2666057.143 kbit/s (OTN 1)," "225/237×9953280 kbit/s or 10709225.316 kbit/s (OTN 2)," and "255/236×39813120 kbit/s or 43018413.559 kbit/s

(OTU 3)" defined as the "OTU nominal bit rate" or the "nominal OTUk rates", and any bit rate can be used.

An "ODU 4/OTU4 for 100 Gb," an "ODU3e/OTU3e for 10 Gb," an "ODU0 for GbE," an ODUflex, or the like may be used. Considering the wavelength allocation of an optical signal, it is preferable that a device other than the subscriber device 40 insert the control signal into the main signal. The frame may be scrambled on the input side and descrambled on the output side.

Since the frame includes a sufficient timing component in the transmission line, an appropriate bit pattern can be given to a frame by a scrambler in order to prevent 0 or 1 from continuing. An operation of the scrambler may be functionally the same as, for example, a frame-synchronous scrambler that has a sequence length of "65535" at an OTUk rate. A generator polynomial may be $"1+x+x^3+x^{12}+x^{16}"$.

The scrambler may be a frame synchronization type scrambler. For example, the scrambler may initialize data to "FFFF (HEX)" at the first bit of a byte following a last framing byte of the frame. For example, an exclusive OR of the most significant bit of the MFAS byte of the OTN and all the consecutive bits scrambled with the term x^{16} of the scrambler may be calculated, and a calculation result may be output. Scrambling operates on all frames. In the case of an OTN frame, a framing byte (Frame Alignment Signal: FAS) of a frame synchronization region may not be scrambled. The frame may be scrambled after the FEC calculation.

The optical access system 100 (an optical communication device) may include any one of or a combination of the following configurations (2-1) to (2-4) so that the main signal such as the analog modulation is transmitted more freely in a protocol when a modulation rate of the control signal is different from a modulation rate of the main signal.

As described above, the arrangement of signal sequences other than writing, reading, adding, and deleting of control signals and multiplexing/demultiplexing of main signals are performed by the subscriber device 40. Writing, reading, adding, and deleting of the control signal to and from a signal sequence are performed by the monitoring unit 260 or the control unit 301 in addition to the subscriber device 40. (2-1).Dual/Tunable TRx:

In addition to the subscriber device 40, the monitoring unit 260 or the control unit 301 also perform writing, reading, adding, and deleting of a control signal in a signal sequence. The subscriber device 40 performs operations other than writing, reading, adding, and deleting of the control signal (for example, arrangement of a signal sequence and multiplexing/demultiplexing of the main signal). The subscriber device 40, the control unit 301, or the monitoring unit 260 includes different receivers for the user and for the control signal in parallel. The subscriber device 40 may demultiplex the main signal at an optical stage or may demultiplex the main signal at an electrical stage. The subscriber device 40, the control unit 301, or the monitoring unit 260 may write, read, add, and delete a management signal in the optical stage or in the electrical stage.

The subscriber device 40, the control unit 301, or the monitoring unit 260 may switch a state of a receiver in accordance with time. The subscriber device 40, the control unit 301, or the monitoring unit 260 may switch the state of the receiver at the optical stage or may switch the state of the receiver at the electrical stage.

When bit rates or baud rates of the main signal and the control signal are set to an integral multiple, the subscriber device 40 may pad a signal with a higher speed in the main signal and the control signal so that the bit rates or the like becomes an integral multiple.

When the bit rates or the like are not set to an integral multiple, the subscriber device **40** spaces the main signals (user data) such that the time between the main signals is an integral multiple of the rate of the main signal. The subscriber device **40** spaces the control signals such that a time between the control signals is an integral multiple of the rate of the control signal.

The time region that is not used as the control signal in a time region between the main signals and the time region that is not used as the main signal in the time region between the control signals may be left as a signal-free gap (gap) between the main signal and the control signal.

Free time regions of the main signals including the gap and the control signals are preferably equal to or less than a time of consecutive identical codes allowed for each of the main signals and the control signals. It is preferable that each clock shift be an allowable time so that a beat and a baud shift are sufficiently small. In the latter case, a preamble or a burst overhead may be inserted into the gap to enable resynchronization.

When the main signal is an analog signal, like a case where the bit rate or the like is not an integral multiple, the subscriber device **40** spaces the gap between the main signals to the extent that overlapping between the main signal and the control signal is negligible. Further, the subscriber device **40** does not perform clock synchronization for each bit or the like on a digital signal.

(2-2). Adaptive Bit Rate Switching of Control Signal for Subscriber Device Capability/Main Signal Rate:

When a rate of the control signal and a rate of the main signal match each other or have a relationship of an integral multiple, it is preferable that the control signal be switched in accordance with the rate of the main signal so that the rate of the control signal and the rate of the main signal match each other or become an integral multiple. Therefore, the subscriber device **40** may select the control signal at a rate near a rate of the main signal.

(2-3). The subscriber device **40** performs padding on the main signal when communication on the NNI side of the subscriber device **40** is faster than communication on the UNI side. Multi-carriers are used for communication on the NNI side when communication on the NNI side of the subscriber device **40** is slower than communication on the UNI side:

On the UNI side, the subscriber device **40** transmits and receives the main signal in accordance with a signal modulation scheme and a clock, and converts the main signal into a format on the NNI side of the subscriber device **40**. A speed on the NNI side is higher than that of the main signal. Therefore, when a transmission rate and a bandwidth for the main signal remain on the NNI side of the subscriber device **40**, the subscriber device **40** fills the remainder by performing padding on the main signal. The subscriber device **40** of a transmission destination deletes the overhead and padding of the control signal and the like and restores the main signal. When there is no remaining transmission rate or band for the main signal on the NNI side of the subscriber device **40**, the subscriber device **40** divides the main signal into a plurality of carriers and a transmission destination device, for example, the facing subscriber device **40** recombines the carriers to restore the main signal. The two may be combined, or a plurality of signals may be transmitted on a single carrier. A plurality of signals may be transmitted with the plurality of carriers.

(2-4). The control unit **301** first exchanges a type of UNI, analog/digital, a modulation format, a modulation rate, and the like of the main signal with the subscriber device **40**. The

control unit **301** sets a control signal in the subscriber device **40** in accordance with a result of the exchange.

The control unit **301** or the monitoring unit **260** may terminate the optical signal from the subscriber device **40** in initial connection or setting change. In this case, the control signal and the main signal may be demultiplexed similarly to the subscriber device. This is particularly appropriate for control in the initial state after connection to the subscriber device. In this case, the subscriber device **40** may not multiplex the main signal and the control signal and may transmit only the control signal. When only the control signal is transmitted, the time region of the main signal does not need to be secured, and thus there is an effect of accelerating exchange of the control signal.

The control unit **301** or the monitoring unit **260** may write the control signal into the gap of the main signal without terminating the optical signal from the subscriber device **40**.

FIG. **81** is a diagram illustrating a first example of details of writing and reading of a control signal. In the first example, a region where the control signal is not transmitted in a write region **420** is a region where the subscriber device **40** continuously emits light. That is, the subscriber device **40** transmits an unmodulated optical signal as the write region **420** in which the control signal is written. When the monitoring unit **260** writes the control signal after or while amplifying the intensity of the optical signal with a semiconductor optical amplifier (SOA) or the like (for example, when the control signal is represented using ON or OFF), the intensity of the unmodulated optical signal may be the intensity to be turned on after the amplification.

For example, when the optical signal is amplified as in gain saturation or automatic level control (ALC), the control unit **301** or the monitoring unit **260** may modulate the intensity of the optical signal as long as the optical signal can be amplified to a value at which the intensity can be regarded as constant intensity. The control unit **301** or the monitoring unit **260** identifies the write region **420** and transmits the optical signal in which the control signal is written in the write region **420**.

When the written control signal is further transmitted to another optical SW **210** or the like, the control unit **301** of the optical SW **210** or the branch combiner/splitter or the like of the monitoring unit **260** branches the optical signal. The control unit **301** or the monitoring unit **260** may terminate the branched signal and extract the control signal. The control unit **301** or the monitoring unit **260** may extract the control signal from the optical signal without terminating the optical signal by using a change in a gain of the SOA or the like. A reception-side subscriber device **40** facing a transmission source subscriber device **40** may terminate the optical signal and extract the control signal from the optical signal.

The plurality of optical SWs **210** may write the control signal in the write region **420** generated in the gap of the main signal. For example, when the control unit **301** or the monitoring unit **260** of the plurality of optical SWs **210** serving as a path of the optical signal presents a candidate wavelength to the subscriber device **40** and the wavelength is selected based on a result of a logical product of the plurality of candidate wavelengths, it is preferable that the plurality of optical SWs **210** write the control signal in the main signal.

When the plurality of optical SWs **210** write the control signal, the plurality of optical SWs **210** may divide and use the same write region **420**. The plurality of optical SWs **210** may write control signals in different write regions **420**.

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The write region 420 preferably is at a constant cycle so that the control unit 301 or the monitoring unit 260 can easily detect the write region. The control unit 301 or the monitoring unit 260 may write a region such as a preamble into the main signal as the write region 420 so that the main signal and the write region 420 can be identified or synchronized. Alternatively, the transmission source subscriber device 40 may write the write region 420 in the main signal in advance.

In the time region of the write region 420, it is preferable that a time at which 0 or 1 continues is equal to or less than a time allowed for the main signal. It is preferable that a time at which 0 or 1 continues is equal to or less than a time at which synchronization of the main signal is not lost.

A length of a cycle of the write region 420 is preferably equal to or less than a time allowed as a control signal 421. The length of the cycle of the write region 420 is preferably equal to or less than a time at which synchronization of the main signal including the control signal 421 is not lost. When the synchronization is lost, it is preferable that the control unit 301 or the monitoring unit 260 write a preamble or the like for resynchronization in the write region 420.

When the optical signal is amplified to a value at which the intensity can be regarded as constant intensity, such as gain saturation or ALC, the same write region 420 can be repeatedly used. For example, the optical SW 210 amplifies a region of the control signal 421 from the subscriber device 40 to the optical SW 210, the optical SW 210 writes the control signal 421 in the write region 420 again, and the optical SW 210 delivers the written control signal 421 to the next optical SW 210. The next optical SW 210 may amplify the region of the control signal 421 inserted into the main signal, rewrite the control signal 421, and deliver the main signal to the opposing subscriber device 40.

FIG. 82 is a diagram illustrating a second example of details of writing and reading of the control signal. In the second example, a region in which the control signal 422 is not transmitted in the write region 420 of the optical signal is a region which is extinct in advance by the subscriber device 40. When a reception side can identify the extinction of the optical signal, the control signal 422 written by the control unit 301 or the monitoring unit 260 may be multiplexed. For example, when the control signal is written as ON/OFF, the intensity of extinction of the optical signal may be any intensity that can be regarded as OFF. For example, when the modulator, the optical SW 210, or the like blocks the optical signal corresponding to the write region 420 and then the control unit 301 or the monitoring unit 260 multiplexes the control signal in the write region 420, the intensity of the result of the extinction of the optical signal by the modulator or the blocking of the optical signal by the optical SW 210 may be intensity that can be regarded as OFF.

The control unit 301 or the monitoring unit 260 identifies the write region 420, multiplexes the control signal in the write region 420, and transmits the control signal. When the multiplexed control signal is further transmitted to another optical SW 210 or the like, the branch combiner/splitter such as the control unit 301 or the monitoring unit 260 of the optical SW 210 may branch the optical signal, terminate the branched optical signal, and extract the control signal from the optical signal. The control unit 301 or the monitoring unit 260 may extract the control signal from the optical signal without terminating the optical signal based on the change in the gain of the SOA or the like.

The optical SW 210 or the like may extract only the control signal from the optical signal and terminate the extracted control signal. The facing subscriber device 40

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may terminate the optical signal and extract the control signal from the optical signal. The plurality of optical SWs 210 may multiplex the control signals. When the control unit 301 or the monitoring unit 260 of the plurality of optical SWs 210 serving as the path of the optical signal presents the candidate wavelength to the subscriber device 40 and the wavelength is selected based on the result of the logical product thereof, it is preferable that the plurality of optical SWs 210 multiplex the control signals.

When the plurality of optical SWs 210 multiplex the control signals, the plurality of optical SWs 210 may divide and use the same write region 420. The plurality of optical SWs 210 may multiplex control signals in different write regions 420.

It is preferable that the write region 420 be at a constant cycle so that the control unit 301 or the monitoring unit 260 can easily detect the write region 420. The subscriber device 40 may write a region such as a preamble in the frame as the write region 420 so that the main signal and the write region 420 can be identified or synchronized. Alternatively, the transmission source subscriber device 40 may write the write region 420 in the main signal in advance. In order to multiplex the optical signals, it is preferable that the optical signal have a gap between bursts and a burst overhead.

When the control unit 301 or the monitoring unit 260 causes the optical signal to be extinct, the control unit 301 or the monitoring unit 260 may repeatedly use the same write region 420. For example, the control unit 301 or the monitoring unit 260 of a certain optical SW 210 may cause the region of the control signal transmitted from the subscriber device 40 to be extinct, and the control unit 301 or the monitoring unit 260 of the optical SW 210 at the stage or the subsequent stage may multiplex the control signals again to the extinct region. The control unit 301 or the monitoring unit 260 of the next optical SW 210 may cause the region of the control signal to be extinct and multiplex the control signals into the optical signal again. The next optical SW 210 may transmit the optical signal to the facing subscriber device 40.

In the time region of the write region 420, it is preferable that a time at which 0 or 1 continues is equal to or less than a time allowed for the main signal. It is preferable that a time at which 0 or 1 continues is equal to or less than a time at which synchronization of the main signal is not lost.

A length of a cycle of the write region 420 is preferably equal to or less than a time allowed as a control signal 421. The length of the cycle of the write region 420 is preferably equal to or less than a time at which synchronization of the main signal including the control signal 421 is not lost. When synchronization is lost, it is preferable that the subscriber device 40, the control unit 301, or the monitoring unit 260 write a preamble or the like for resynchronization in the write region 420. The protocol-free AMCC may be used.

As described above, the optical SW 210 outputs the optical signal input from any one of the plurality of transmission lines 501 to the other transmission line 501 among the plurality of transmission lines 501. The control unit 301 or the monitoring unit 260 allocates a wavelength corresponding to the subscriber device 40 and the transmission destination on the path to the communication destination to the subscriber device 40 using a control signal, writes or inserts the control signal in the overhead, and notifies the subscriber device 40 of the control signal. The control unit 301 or the monitoring unit 260 controls the optical SW 210 such that the input optical signal is output to another transmission line 501 corresponding to the communication destination. The subscriber device 40 divides an optical

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signal (the main signal) transmitted from a user device (not illustrated). The subscriber device 40 transmits the overhead of each frame including the divided main signal, including room for writing or inserting the control signal. The user device may be the subscriber device 40. The subscriber device 40 divides the input signal, transmits the signal including the control signal in the overhead of the frame including the divided signal, deletes the frame including the overhead including the control signal from the received main signal, combines the divided main signals, and outputs the combined main signal.

The monitoring unit or the control unit writes and inserts a management signal into an allowable region with the overhead of the frame transmitted by the ONU. However, when no frame flows from the facing ONU in initial connection, the monitoring unit or the control unit generates a frame, and writes and inserts the control signal into the overhead of the frame.

When a region for storing the main signal of the frame is a signal such as a frame in which synchronization is not lost, for example, a random pattern or the like, there is an effect of inhibiting synchronization loss. The same value may be used as long as the length is equal to or less than the length of the consecutive identical codes allowed for the receiver. When the overhead is inserted as the optical signal, or the like, it is not necessary to insert the overhead by simulating the optical signal of the main signal portion. That is, only the overhead may be used. The control signal may be ridden on the main signal.

Next, a method of deleting the control signal from the main signal will be described.

In the framing, information deleted as information that can be restored at the transmission destination despite the deletion is restored and inserted into an appropriate position (for example, between frames or the like in the case of IFG).

When the main signal is sparse over time, the main signal is delayed as light. When the overhead is interposed in the gap of the main signal, the interposed overhead is deleted. When a delay variation becomes a problem, a delay is added so that the delay variation is equal to or less than an allowable value to average the delay.

Compression is performed on a time axis with an RZ signal or the like to narrow an occupied time region, a delay is added to each pulse little by little to generate a gap of a main signal. When an overhead is inserted into the gap, the overhead is removed and a delay is added to each pulse little by little to decompress the occupied time region.

It is preferable that a time-synchronized user device returns to the writable region such as non-modulation or a writable original state, and it is preferable that the signal be deleted at a non-transmission time, but the time may be left as it is as long as there is no problem in the user device.

(1-1) Speed Conversion

When the input/output on a user-network interface (UNI) side of the subscriber device 40 is a digital signal such as 0/1 that can be restored despite the change in the modulation scheme or the like in the network, the main signal from which the control signal (header+trailer) has been removed is returned to the speed of the UNI. In the modulation speed conversion, encoding such as "8B/10B" is returned to the original state. When the shift amount is held, the shift is performed to return the delay amount. When a signal such as padding is inserted, the signal is deleted.

(1-2). Decompression

When the inserted control signal (header+trailer) is deleted and the time region such as padding, the time region of the IFG in Ethernet (registered trademark) or the like, the

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time region such as the preamble in Ethernet (registered trademark) or the like is deleted, the control signal is restored.

(1-3). Deletion of Non-Signal Region

The inserted (written) control signal (header+trailer) is excluded, a time interval of the data of the main signal sampled at a high speed is decompressed, and the sampled data is reproduced and output.

(1-4). Decompression of Main Signal

The compressed main signal is decompressed, and an interval of the main signal is sequentially shifted back.

Accordingly, it is possible to relay the optical signal to the destination using the control signal that does not depend on the protocol of the main signal. When, for example, an OTN frame is used as the periodic frame, it is possible to enhance consistency with the relay network using an OTN. The control signal can also be applied to quantum optical communication as long as the main signal is subjected to burst transmission and the main signal (an optical signal) is optically merged, and thus the influence of the control signal on the nonlinear effect on the main signal can be ignored. When communication is performed using the control signal that is a control signal slower than the main signal and can be superimposed on the main signal, the setting change and the monitoring can be performed without affecting the main signal.

The subscriber device 40 performs at least one process among speed conversion, compression, and generation of a non-signal region on the main signal, and thus generates a time region not used as the main signal and writes the control signal in the time region. The subscriber device 40, the control unit 301, or the monitoring unit 260 may delete the control signal from the time region before the main signal is received in the communication destination.

The subscriber device 40 may perform a process of transmitting the optical signal that has a non-modulation region and writing the control signal in the non-modulation region. The subscriber device 40, the control unit 301, or the monitoring unit 260 may perform a process of causing the subscriber device 40 to empty a non-signal region, transmitting the optical signal, and inserting the control signal into the non-signal region. The subscriber device 40, the control unit 301, or the monitoring unit 260 may perform a process of saturating and amplifying the non-modulation region where the control signal is written and overwriting a new control signal on the non-modulation region in which the control signal is deleted. The subscriber device 40, the control unit 301, or the monitoring unit 260 may perform a process of causing the non-signal region into which the control signal is inserted to be extinct and inserting the new control signal into the non-signal region from which the control signal is deleted. The subscriber device 40, the control unit 301, or the monitoring unit 260 may perform a process of writing the control signal in a part of the non-modulation region in a predetermined device on the path of the optical signal and writing the control signal in a non-modulation region where the control signal is not written in another predetermined device on the path of the optical signal. The subscriber device 40, the control unit 301, or the monitoring unit 260 may perform a process of inserting the control signal into a part of the non-signal region in a predetermined device on the path of the optical signal and inserting the control signal into the non-signal region into which the control signal is not inserted in another predetermined device on the path of the optical signal.

The control unit 301 or the monitoring unit 260 writes the control signal in a control signal region of the overhead

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prepared by the facing subscriber device 40 and notified of the control signal. In the initial state, the control unit 301 or the monitoring unit 260 may exchange the main signal without using an overhead. The control unit 301 or the monitoring unit 260 may generate a frame similar to that of the subscriber device 40 and notify of the control signal with the overhead of the frame. Basically, the process of dividing the main signal of the optical signal transmitted from the user device and including the control signal in the overhead of each frame including the divided main signal is a process performed by the subscriber device 40. Only when there is no signal from the facing side, the control unit 301 or the monitoring unit 260 temporarily generates a frame. Since there is no signal to be transmitted and received other than the overhead of the frame, the control unit 301 or the monitoring unit 260 packs a signal (for example, random data or the like) to the extent that scrambling is not lost into the overhead.

The subscriber device 40 or the monitoring unit 260, which is an end point, performs at least one process among speed conversion, compression, and generation of a non-signal region on the main signal to generate a time region not used as the main signal, and writes or inserts the control signal into the time region. If no signal has arrived from the opposing subscriber device 40, the subscriber device 40 may generate a frame with a similar overhead to exchange signals.

An OAM frame may be used as the control signal. This point is one of differences from the example of the control signal using an overhead (OH) of a periodic frame. Hereinafter, a difference from the example of the control signal in which the overhead of the periodic frame is used will be mainly described.

The OAM frame may be, for example, an Ethernet frame such as Ether OAM, an ATM frame, or an OAM used in PON. In order to insert the OAM frame into the optical signal, a configuration of the OAM frame of the PON to which a burst overhead or the like is applied is appropriate, but the configuration is not limited thereto as long as the OAM frame can be transmitted and received.

FIG. 83 is a diagram illustrating an example of writing and reading of a control signal in which an OAM frame is used. In the foregoing "(1-4). Compression of main signal," the following "(1-4a). Compression of main signal" may be performed in the writing and reading of the control signal in which the OAM frame is used. A space between the third and fourth stages from the bottom of FIG. 83 is a space between an NNI of the subscriber device 40 and the NNI of the facing subscriber device 40.

(1-4a). Compression of Main Signal

The subscriber device 40 compresses the optical signals of the main signals and sequentially shifts an interval between the main signals. The subscriber device 40 inserts a control signal (OAM frame) into a gap generated in the main signal. The subscriber device 40 divides the main signal, for example, and accommodates the main signal in a frame. The subscriber device 40 divides the main signal and includes the control signal in the format of an OAM frame between the main signals. The control signal may have an error correction (FEC) region. The OAM frame may have an error correction region.

In order to prevent 0 or 1 from being continuing, an appropriate bit pattern is given to the frame by the scrambler so that the control signal includes a sufficient timing component in the transmission line. The operation of the scram-

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bler may be, for example, a frame synchronization type scrambler. A generator polynomial may be " $1+x+x^3+x^{12}+x^{16}$ ".

The scrambler may be a frame synchronization type scrambler. For example, the scrambler may initialize data to "FFFF (HEX)" at the first bit of a byte following a last framing byte of the frame. For example, like the OTN, an exclusive OR of the term " x^{16} " of the scrambler with the most significant bit and all the consecutive bits including the MFAS byte and being scrambled may be calculated, and a calculation result may be output. Scrambling operates on all frames. In the case of an OTN frame, a framing byte (FAS) of a frame synchronization region may not be scrambled. The frame may be scrambled after the FEC calculation.

As described above, the optical SW 210 outputs the optical signal input from any one of the plurality of transmission lines 501 to the other transmission line 501 among the plurality of transmission lines 501. The subscriber device 40 divides the main signal of the optical signal transmitted from the user device into at least a first partial main signal and a second partial main signal. The subscriber device 40 inserts a control signal (for example, the OAM frame) between the first and second partial main signals. The control unit 301 allocates a wavelength corresponding to a transmission destination on the path to the communication destination with the subscriber device 40 to the subscriber device 40 using the control signal. The optical SW control unit 320 controls the optical SW 210 such that the input optical signal is output to another transmission line 501 corresponding to the communication destination.

The monitoring unit or the control unit writes and inserts the OAM frame or the management signal on the OAM frame in a region where the OAM frame transmitted by the ONU or the OAM frame is allowed to be written and inserted. However, in a state in which no OAM frame flows from the facing ONU in the initial connection, the monitoring unit or the control unit generates an OAM frame, and writes and inserts the control signal into an overhead of the frame. Alternatively, since the signal itself does not flow, the OAM frame is transmitted.

When the time region other than the OAM frame is a signal in which synchronization such as a phase is not lost, for example, a random pattern, there is an effect that synchronization loss can be inhibited. The same value may be used as long as the length is equal to or less than the length of the consecutive identical codes allowed for the receiver.

Next, a method of deleting the control signal from the main signal will be described.

When the OAM frame or the room for the OAM frame is generated, information deleted as information that can be restored at a transmission destination despite deletion is restored and inserted into an appropriate position (for example, between frames or the like in the case of IFG).

When the main signal is sparse in terms of time, the main signal is delayed as light. When an OAM frame or a room for the OAM frame is interposed in a gap between the main signals, the interposed OAM frame or the room for the OAM frame is deleted. When a delay variation becomes a problem, a delay is added so that the delay variation is equal to or less than an allowable value to average the delay.

Compression is performed on the time axis with an RZ signal or the like to narrow an occupied time region, a delay is added to each pulse little by little to generate a gap of the main signal. When the OAM frame or the room for the OAM frame is inserted into the gap, the OAM frame or the room

for the OAM frame is deleted, the delay is added to each pulse little by little to decompress the occupied time region.

It is preferable that a time-synchronized user device returns to the writable region such as non-modulation or a writable original state, and it is preferable that the signal be deleted at a non-transmission time, but the time may be left as it is as long as there is no problem in the user device.

(1-1) Speed Conversion

When the input/output on the UNI (User-Network Interface) side of the subscriber device **40** is a digital signal such as 0/1 that can be restored even if the modulation scheme or the like is changed in the network, the OAM frame and the main signal from which the room for the OAM frame is removed are returned to the UNI speed. In the modulation speed conversion, encoding such as "8B/10B" is returned to the original state. When the shift amount is held, the shift is performed to return the delay amount. When a signal such as padding is inserted, the signal is deleted.

(1-2). Decompression

When the inserted OAM frame and the room for the OAM frame are deleted and the time region such as padding, the time region of the IFG in Ethernet (registered trademark) or the like, the time region such as the preamble in Ethernet (registered trademark) or the like are deleted, the OAM frame and the room therefor are restored.

(1-3). Deletion of Non-Signal Region

The inserted (written) OAM frame and the room for the OAM frame are excluded, a time interval of the data of the main signal sampled at a high speed is decompressed, and the sampled data is reproduced and output.

(1-4). Decompression of Main Signal

The compressed main signal is decompressed, and an interval of the main signal is sequentially shifted back.

Accordingly, it is possible to relay the optical signal to the destination using the control signal that does not depend on the protocol of the main signal. It is possible to enhance consistency with a relay network using burst transmission. If there is a sufficient time interval, it is possible to add error correction (FEC) in part of the link. The control signal can also be applied to quantum optical communication as long as the main signal is subjected to burst transmission and the main signal (an optical signal) is optically merged, and thus the influence of the control signal on the nonlinear effect on the main signal can be ignored. When communication is performed using the control signal that is a control signal slower than the main signal and can be superimposed on the main signal, the setting change and the monitoring can be performed without affecting the main signal.

Further, the main signal and the control signal may be transmitted using different carriers. Accordingly, like the method of dividing the main signal and inserting the control signal between the main signals as described above, the control signal can be configured not to depend on the protocol of the main signal.

As a transmission scheme in which different carriers are used, for example, a wavelength division multiplexing scheme in which the main signal and the control signal are transmitted with different wavelengths can be used. Not only wavelengths but also wavelengths, polarized waves, modes, cores, frequencies, codes, core wires, or combinations thereof may be different. Alternatively, a polarization multiplexing scheme in which the main signal and the control signal are transmitted with polarized waves orthogonal to each other can be used as the transmission scheme. In this case, for example, the main signal may be transmitted

through X-polarization, and the control signal may be transmitted through Y-polarization orthogonal to the X-polarization.

Alternatively, a special division multiplexing scheme can be used as the transmission scheme. For example, the spatial division multiplexing scheme includes a mode division multiplexing scheme in which the main signal and the control signal are transmitted in different modes using a multimode optical fiber. For example, the spatial division multiplexing scheme includes a core division multiplexing system in which the main signal and the control signal are transmitted by different cores using a multi-core optical fiber. A transmission scheme in which the main signal and the control signal are transmitted in different modes and different cores by using a multicore multimode optical fiber may be used. Since the control signal is a different carrier, the control signal may be modulated as the main signal, may be modulated for use in the AMCC, may be transmitted and received with an overhead of a periodic frame, or may be transmitted and received as an OAM frame.

Alternatively, as the transmission scheme, for example, a transmission scheme in which the main signal and the control signal are transmitted with different core lines by using two core lines can be used. Alternatively, as the transmission scheme, for example, a frequency division multiplexing scheme in which the main signal and the control signal are transmitted with different frequencies can be used. The AMCC is also a type of frequency division multiplexing scheme. Alternatively, as the transmission scheme, for example, a code division multiplexing scheme in which the main signal and the control signal are transmitted using different codes can be used.

Hereinafter, a case where a wavelength division multiplexing scheme is used will be described as an example.

FIG. **84** is a block diagram illustrating a configuration example of the subscriber device **40d** that acquires a control signal with a carrier different from the main signal. In FIG. **84**, a branch combiner/splitter, a multiplexer/demultiplexer, a mode multiplexer, a mode separator, a polarization coupler, a polarization splitter, and the like that multiplex/demultiplex the control signal and the main signal that are different carriers are not illustrated, but they are assumed to be included.

The main signal may be subjected to optical-to-electrical-to-optical (OEO) conversion by the UNI of the subscriber device **40d** to be output to the NNI side or may be output as an optical signal to the NNI side. In FIG. **84**, the subscriber device **40d** describes an optical transceiver **41da** exchanging optical signals with the CPE on the UNI side. The exchange between the subscriber device **40d** and the CPE may be performed through riding on the main signal or may be performed by separately providing a transceiver.

The optical transceiver **41da** is an example of a configuration of an optical transmitter and an optical receiver in the subscriber device **40d**. The optical transceiver **41d** includes an optical transmitter (Tx) **42da** and an optical receiver (Rx) **43da**. The optical transmitter (Tx) **42da** includes a main signal optical transmitter (Tx) **42da-1**. The optical receiver (Rx) **43da** includes a main signal optical receiver (Rx) **43da-1**. The optical transceiver **41da** is, for example, a wavelength-tunable optical transceiver and is, for example, a wavelength-fixed optical transceiver. Unlike the NNI side on which the WDM is assumed, when the WDM is not performed on the UNI side, control and device management become easier and cost is inhibited in a case where the optical transceiver is not a wavelength-tunable optical transceiver.

The subscriber device **40d** is connected to the optical SW**10** by, for example, an optical access network such as PON. The subscriber device **40d** includes an optical transceiver **41d**. The optical transceiver **41d** is an example of a configuration of an optical transmitter and an optical receiver in the subscriber device **40d**. The optical transceiver **41d** includes an optical transmitter (Tx) **42d** and an optical receiver (Rx) **43d**.

The optical transmitter (Tx) **42d** includes a main signal optical transmitter (Tx) **42d-1** and a control signal optical transmitter (Tx) **42d-2**. The optical receiver (Rx) **43d** includes a main signal optical receiver (Rx) **43d-1** and a control signal optical receiver (Rx) **43d-2**. The optical transceiver **41d** is, for example, a wavelength-tunable optical transceiver.

The control signal is preferably a signal that does not affect the main signal. Here, the wavelength of the control signal is a wavelength which is sufficiently separated from the wavelength of the main signal. For example, the control signal is a signal with a wavelength that can be removed to the extent that the main signal is not affected at least when the control signal is output from the optical SW**10** to the subscriber device **40d**. For example, the control signal is a signal that has a wavelength interval from the main signal to the extent that the control signal can be removed by the branch combiner/splitter. Alternatively, for example, the control signal is a signal that has a wavelength interval from the main signal to the extent that the control signal be removed by a filter through coherent detection.

When the control signal is transmitted with a polarized wave different from that of the main signal along the same transmission line as that of the main signal, the control signal is transmitted with orthogonal polarized waves or with polarized waves which can be subjected to polarization separation on the reception side. From the viewpoint of non-interference with the main signal, it is preferable that the polarized waves be orthogonal. When the control signal is transmitted in a mode different from the main signal along the same transmission line as the main signal, the control signal is transmitted in a mode orthogonal to the main signal or a mode which is can be separated on the reception side. From the viewpoint of non-interference with the main signal, the modes are preferably orthogonal to each other. When the control signal is transmitted along the same transmission line as the main signal and with a core different from the main signal, the control signal is transmitted with a core in which the inter-core interference is equal to or less than a predetermined value or a core from which interference can be eliminated on the reception side.

For example, when the main signal is a quantum optical signal, at least when the control signal is output from the optical SW**10** to the subscriber device **40d**, the control signal is regarded as a signal that has intensity and a wavelength to the extent that the main signal is not affected by the Raman effect or the like.

The transmission of the main signal and the control signal using the separate carriers may be performed between the subscriber device **40d** and the optical SW**10** and between the subscriber device **40d** and the subscriber device **40d** of a communication destination. Alternatively, transmission of the main signal and the control signal using the above-described separate carriers may be performed only between the subscriber device **40d** and the optical SW**10**. In this case, between the optical SW**10** and the other optical SW**10**, the control signal may be multiplexed into the main signal in a time region, a frequency region, or a wavelength region and transmitted.

The above-described scheme using a signal obtained by superimposing the control signal on the main signal may be used together.

The frame may be scrambled on the input side and may be descrambled on the output side. Since the frame includes a sufficient timing component in the transmission line, an appropriate bit pattern can be given to a frame by a scrambler in order to prevent 0 or 1 from continuing. An operation of the scrambler may be functionally the same as, for example, a frame-synchronous scrambler that has a sequence length of "65535" at an OTUk rate. A generator polynomial may be "1+x+x3+x12+x16".

The scrambler may be a frame synchronization type scrambler. For example, the scrambler may initialize data to "FFFF (HEX)" at the first bit of a byte following a last framing byte of the frame. For example, an exclusive OR of the most significant bit of the MFAS byte of the OTN and all the consecutive bits scrambled with the term x16 of the scrambler may be calculated, and a calculation result may be output. Scrambling operates on all frames. In the case of an OTN frame, a framing byte (Frame Alignment Signal: FAS) of a frame synchronization region may not be scrambled. The frame may be scrambled after the FEC calculation.

In the case of a configuration in which the main signal and the control signal are transmitted with different carriers, step S4 of the above-described flowchart illustrated in FIG. 79 is not limited to the AMCC. Before the wavelength is set, it cannot be confirmed that the control signal of the connection request is normal in the connection request because of separate carriers. After the setting, the main signal may be caused to flow, and the control unit, the input monitoring unit, the output monitoring unit, the communication unit, or the subscriber device **40** may autonomously perform confirmation. In step S7, the control unit, the input monitoring unit, the output monitoring unit, the communication unit, or the subscriber device **40** may autonomously perform confirmation.

In the foregoing configuration, the optical SW **210** outputs the optical signal input from one transmission line **501** among the plurality of transmission lines **501** to another transmission line **501** among the plurality of transmission lines **501**. The wavelength management control unit **25**, the optical GW control unit **301**, or the monitoring device **260** allocates a wavelength corresponding to a communication destination specified based on a combination of the subscriber device **40** and the wavelength of the input optical signal to the subscriber device **40d** by using the control signal. At this time, the wavelength management control unit **25**, the optical GW control unit **301**, or the monitoring device **260** transmits the control signal to the subscriber device **40d** by using a carrier different from the carrier used for transmission of the optical signal (the main signal). The optical SW control unit **320** controls the optical SW **210** such that the input optical signal is output to another transmission line **501** corresponding to the communication destination. The control signal may be transmitted using, for example, a wavelength different from that of the optical signal (the main signal), a different polarized wave, a different mode, a different core, or a different core wire.

In such a configuration, it is possible to relay the optical signal to a destination by using a control signal that does not depend on the protocol of the main signal. Accordingly, it is possible to perform the setting change and monitoring without affecting the main signal.

The control signal is also applicable to quantum optical communication as long as the main signal is subjected to burst transmission and the main signal (the optical signal) is

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optically merged, and thus the influence of the control signal on the non-linear effect on the main signal can be ignored.

Although the wavelength changing process performed by the subscriber device 40 making a request for a change in a wavelength has been described above, the same applies to the wavelength changing process performed by the control unit 20 based on monitoring information or the like.

The control signal is exchanged between the control unit 20 and the subscriber device 40. For example, the subscriber device 40 transmits the connection request to the control unit 20 and the control unit 20 transmits the control signal to the subscriber device 40. For example, the control unit 20 allocates a wavelength used for communication by the subscriber device 40.

The control signal may be monitored by the monitoring unit 60 and may be exchanged between the monitoring unit 60 and the subscriber device 40 and between the monitoring unit 60 and the control unit 20.

The wavelength control unit 310 and the optical SW control unit 320 may be implemented using one information processing device, or may be implemented using a plurality of information processing devices communicably connected via a network.

As illustrated in FIG. 85, the optical GW may be configured as a device (the optical communication device according to the present invention) including one or all of the OPS 300 or the control unit 20, the optical SW, the monitoring unit, the electrical processing unit, the folding transmission line, the branch combiner/splitter (a power splitter), and the multiplexer/demultiplexer (a WDM device or WDM filter).

In each of the foregoing embodiments, the allocation of the wavelength to the subscriber device 40 has been described as an example. However, a wavelength, a time, polarized waves orthogonal to each other, modes orthogonal to each other, codes orthogonal to each other, a frequency, a core, a core wire, a wavelength that is a combination thereof, or the like may be allocated. For example, the same applies to a multiplexer/demultiplexer when allocation is set for a time, a polarized wave, a mode, a code, a core, or core wire if the multiplexer/demultiplexer is replaced with a combination of a branch combiner/splitter and a delay line, a polarization mode coupler, a mode coupler, a demodulator, a branch combiner/splitter between cores, a branch combiner/splitter between core wires, or the like.

The control units 20, 230, and 235, the monitoring units 260, 265, and 267, the wavelength control unit 310, and the optical SW control unit 320 described above may include a CPU, a memory, an auxiliary storage device, and the like connected by a bus, and may implement some or all of the above-described functions by executing a program. Some or all of the functions of the control units 20, 230, and 235, the monitoring units 260, 265, and 267, the wavelength control unit 310, and the optical SW control unit 320 may be realized using hardware such as ASIC, PLD, or FPGA. The program for the control units 20, 230, and 235, the monitoring units 260, 265, and 267, the wavelength control unit 310, and the optical SW control unit 320 may be recorded in a computer-readable recording medium. The computer-readable recording medium is, for example, a portable medium such as a magneto-optical disk, a ROM, or a CD-ROM, or a storage device such as a hard disk incorporated in a computer system. The program may be transmitted via an electrical communication line.

Although the embodiments of the present invention have been described in detail with reference to the drawings, specific configurations are not limited to the embodiments,

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and includes design and the like within a range without departing from the gist of the present invention.

INDUSTRIAL APPLICABILITY

The present invention can be applied to an optical access system technology using an optical switch.

REFERENCE SIGNS LIST

- 1 Optical communication system
- 10, 10a, 10b, 10c, 10d, 10e, 10f, 10g, 34, 95a-1, 95a-2, 95b-1, 95b-2, 96a-1, 96a-2, 96b-1, 96b-2, 210, 210a-1 to 210a-2, 210b, 210c-1 to 210c-2, 210d-2, 210e-1 to 210e-2, 211, 212a, 212b, 212c, 1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008, 1009a, 1009b, 1010-1 to 1010-4 Optical switch
- 11-1, 11-1-1 to 11-1-P, 11-2, 11-2-1 to 11-2-Q Port
- 20 Control unit
- 21, 41, 411, 412 Optical transceiver
- 22, 42, 237 Optical transmitter
- 23, 43, 232, 236, 262 Optical receiver
- 25 Wavelength management control unit
- 26, 320 Optical SW control unit
- 30 Optical communication network
- 31 WDM access ring network
- 32-1 to 32-r Add/Drop node
- 33 Demultiplexing unit
- 35 Multiplexing unit
- 40, 40-1 to 40-M, 40a-1 to 40a-3, 40b-1 to 40b-3, 40c-1 to 40c-3, 40a-1-1, 40a-1-2, 40-p-1 to 40-p-Np, 40-p-N, 40-p to 40-(p+N) Subscriber device
- 44 Communication unit
- 46-1, 46-3 User
- 46-2 Mobile base station
- 50, 50-1, 50-2, 50-1-p to 50-1-(p+N), 50-1-p1 to 50-1-pN, 50-1-p-1 to 50-p-Np, 50-2-1 to 50-2-q, 50-2-(N-1), 50-2-N, 50-2-q-1 to 50-2-q-N, 50-2-(1+N), 53, 54a, 54b, 54c, 54d, 92, 93-1 to 93-N, 501, 503, 504, 511, 512, 521, 522, 531, 533, 534, 540, 541, 542, 543-1, 543-2, 544, 545, 546, 547, 548, 549, 551, 551a, 552, 552b, 555, 560, 561, 562, 563, 570, 571, 572, 573, 574, 575, 581, 582, 583, 584, 585, 586, 587, 588, 589 Transmission line
- 51, 73 Folded transmission line
- 55, 55-1, 55-2, 55-p, 55-(p+1), 56, 57a, 57b, 61, 66, 69, 71, 72, 251, 251a, 251b, 252, 252b, 254, 258, 259, 270, 271, 272, 273, 502, 507 Power splitter
- 58, 59 Distribution unit
- 60, 60a to 60h, 65 Monitoring unit
- 67, 68, 80, 80a, 80b, 80c, 81, 89, 97 WDM device
- 82a-1 to 82a-n, 82b-1 to 82b-m, 241, 247, 458 Multiplexer
- 83a-1 to 83a-n, 83b-1 to 83b-m, 242, 248, 457 Demultiplexer
- 85, 610 O/E converter
- 86, 620 Processing execution unit
- 87, 630 E/O converter
- 88 Storage unit
- 90, 91 Multiplex communication transmission line
- 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117 Optical access system
- 200, 200a to 200e, 201, 202, 203, 204, 205, 206, 207, 208, 209, 2010, 2011, 2012, 2013, 2014, 2015, 2016 Optical gateway
- 220, 238, 243, 244, 245, 249, 256, 257 Wavelength multiplexer/demultiplexer

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230, 235 Control unit
 231, 261 Wavelength demultiplexing unit
 233, 269 Wavelength-tunable transmitter
 250, 250a, 250b, 253, 255 Branching unit
 260, 265 Monitoring unit
 263, 263a to 263g Blocking part
 266 Wavelength-tunable optical receiver
 267 Monitoring unit
 268 Wavelength-tunable receiver
 300 Operation system
 301 Control unit
 310 Wavelength control unit
 330 Communication unit
 350 Management database
 420 Write region
 421 Control signal
 422 Control signal
 452 Tunable wavelength receiver
 453 Wavelength-tunable filter
 459, 459a to 459e WDM filter
 84, 600 Electrical processing unit
 861 Processor
 862 Accelerator
 900 Connection device

The invention claimed is:

1. An optical communication device comprising:
 an optical switch configured to output an optical signal
 input from one transmission line to another transmis-
 sion line among the plurality of transmission lines;
 a wavelength management control unit configured to
 allocate at least three of a wavelength, a time, a
 polarized wave, a mode, a code, a frequency, a core, a
 core wire, and a wavelength that is a combination
 thereof to a subscriber device by using a control signal
 in accordance with the subscriber device and a transfer
 destination on a path to a communication destination,
 and transmits the control signal to the subscriber device
 by using a carrier different from a carrier to which the
 wavelength or the like is allocated; and
 an optical switch control unit configured to control the
 optical switch such that the input optical signal is
 output to the another transmission line corresponding
 to the communication destination.
2. The optical communication device according to claim
 1, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted at
 a wavelength that is different from the optical signal and is
 removable to such an extent that the optical signal is not
 affected when the control signal is output from the optical
 switch.
3. The optical communication device according to claim
 1, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted in
 a polarized wave different from the optical signal, the
 polarized wave being an orthogonal polarized wave or a
 polarized wave that is able to be subjected to polarization
 separation on a reception side.
4. The optical communication device according to claim
 1, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted in
 a mode different from the optical signal, the mode being
 orthogonal to the optical signal or a mode able to be
 separated on a reception side.
5. The optical communication device according to claim
 1, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted
 along a core different from the optical signal, the core having

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an inter-core interference equal to or less than a predeter-
 mined value or the core being a core from which interference
 is able to be eliminated on a reception side.

6. The optical communication device according to claim
 1, wherein the control signal is transmitted along a core wire
 different from the optical signal.

7. An optical communication system comprising: a plu-
 rality of subscriber devices and the optical communication
 device according to claim 1.

8. An optical communication method performed by an
 optical communication device, the method comprising:

an optical switching step of outputting an optical signal
 input from one transmission line to another transmis-
 sion line among the plurality of transmission lines;

- a wavelength management control step of allocating at
 least three of a wavelength, a time, a polarized wave, a
 mode, a code, a frequency, a core, a core wire, and a
 wavelength which is a combination thereof to a sub-
 scriber device by using a control signal in accordance
 with the subscriber device and a transfer destination on
 a path to a communication destination, and transmitting
 the control signal to the subscriber device by using a
 carrier different from a carrier to which the wavelength
 or the like is allocated; and

an optical switch control step of controlling the optical
 switch such that the input optical signal is output to the
 another transmission line corresponding to the commu-
 nication destination.

9. An optical communication device comprising:

one or more processors; and

memory including instructions that, when executed by the
 one or more processors, perform to:

output an optical signal input from one transmission
 line to another transmission line among the plurality
 of transmission lines;

allocate at least three of a wavelength, a time, a
 polarized wave, a mode, a code, a frequency, a core,
 a core wire, and a wavelength that is a combination
 thereof to a subscriber device by using a control
 signal in accordance with the subscriber device and
 a transfer destination on a path to a communication
 destination;

transmit the control signal to the subscriber device by
 using a carrier different from a carrier to which the
 wavelength or the like is allocated; and

control the optical switch such that the input optical
 signal is output to the another transmission line
 corresponding to the communication destination.

10. The optical communication device according to claim
 9, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted at
 a wavelength that is different from the optical signal and is
 removable to such an extent that the optical signal is not
 affected when the control signal is output from the optical
 switch.

11. The optical communication device according to claim
 9, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted in
 a polarized wave different from the optical signal, the
 polarized wave being an orthogonal polarized wave or a
 polarized wave that is able to be subjected to polarization
 separation on a reception side.

12. The optical communication device according to claim
 9, wherein the control signal is transmitted along the same
 transmission line as the optical signal and is transmitted in
 a mode different from the optical signal, the mode being

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orthogonal to the optical signal or a mode able to be separated on a reception side.

13. The optical communication device according to claim **9**, wherein the control signal is transmitted along the same transmission line as the optical signal and is transmitted 5 along a core different from the optical signal, the core having an inter-core interference equal to or less than a predetermined value or the core being a core from which interference is able to be eliminated on a reception side.

14. The optical communication device according to claim 10 **9**, wherein the control signal is transmitted along a core wire different from the optical signal.

15. An optical communication system comprising: a plurality of subscriber devices and the optical communication device according to claim **9**. 15

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